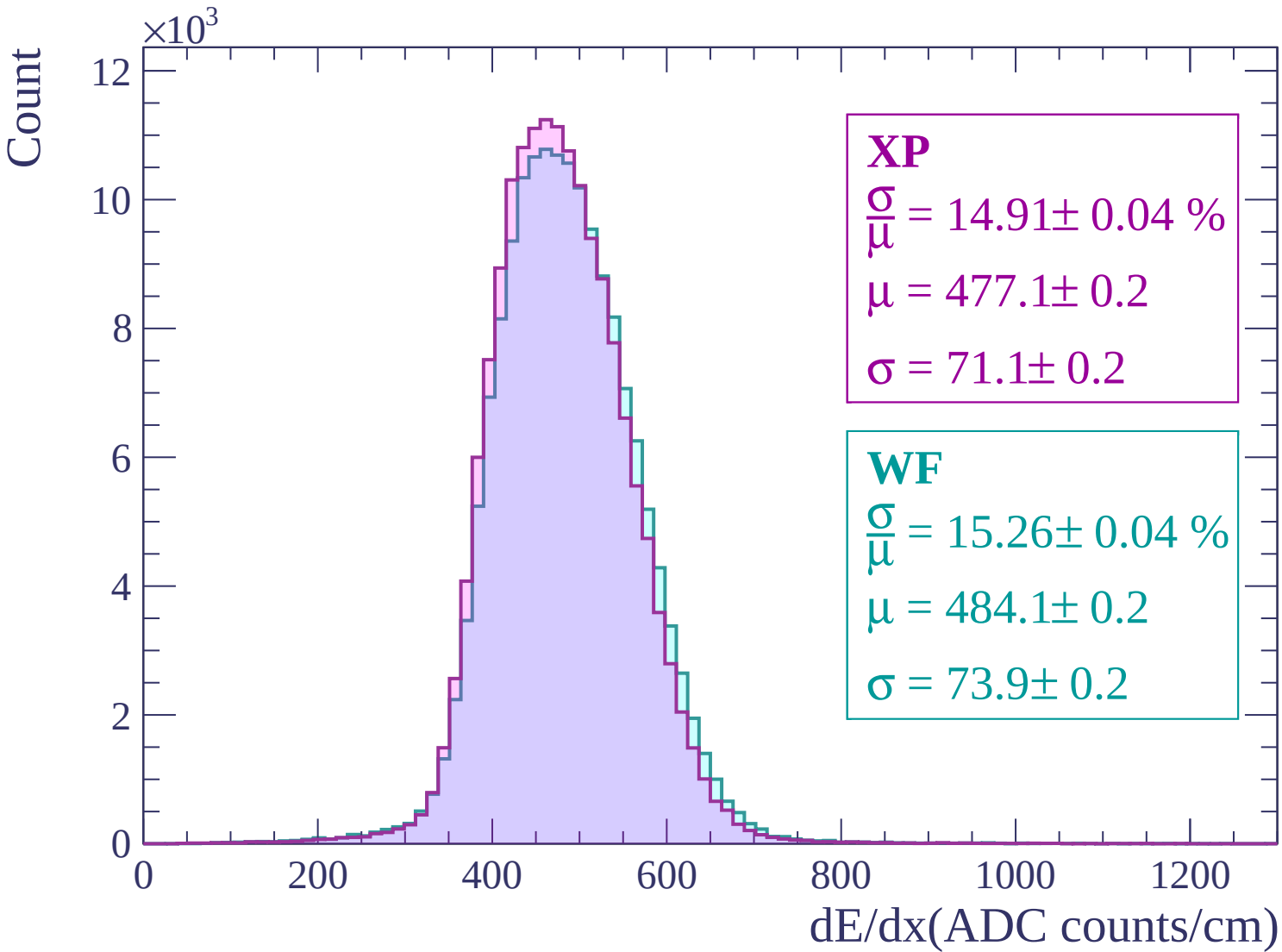
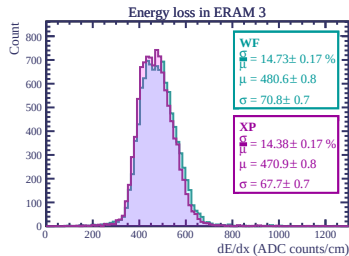
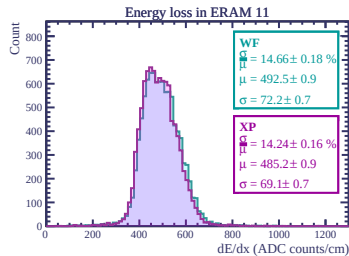
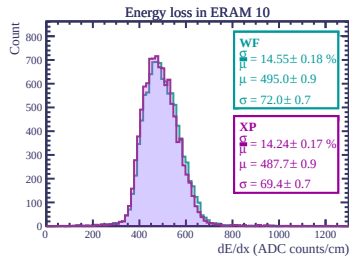
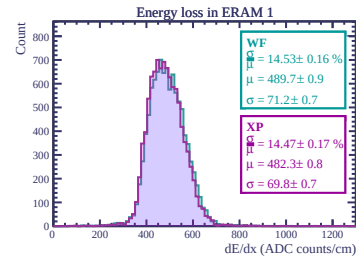
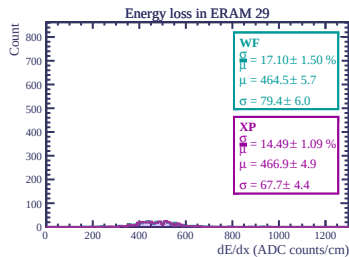
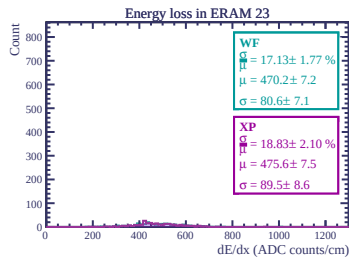
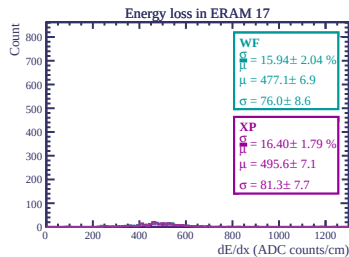
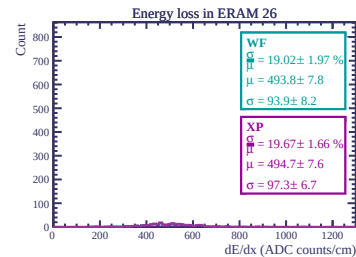
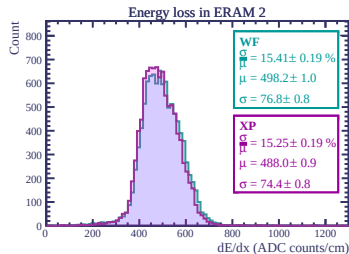
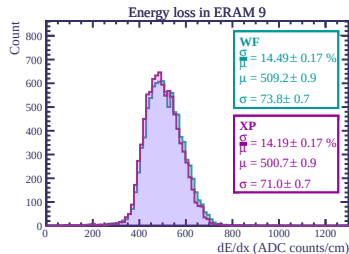
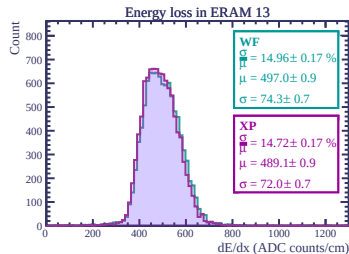
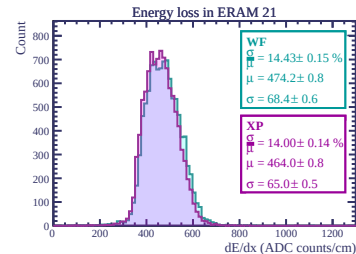
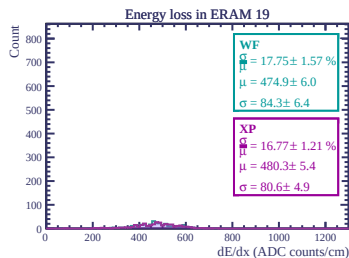
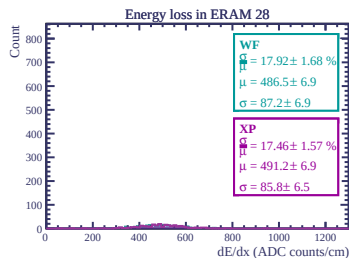
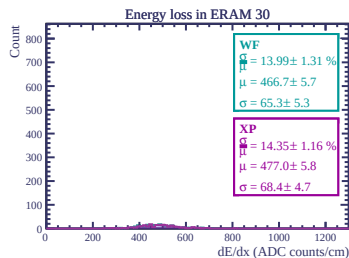
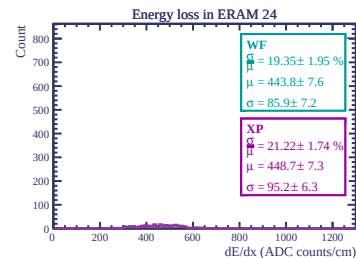
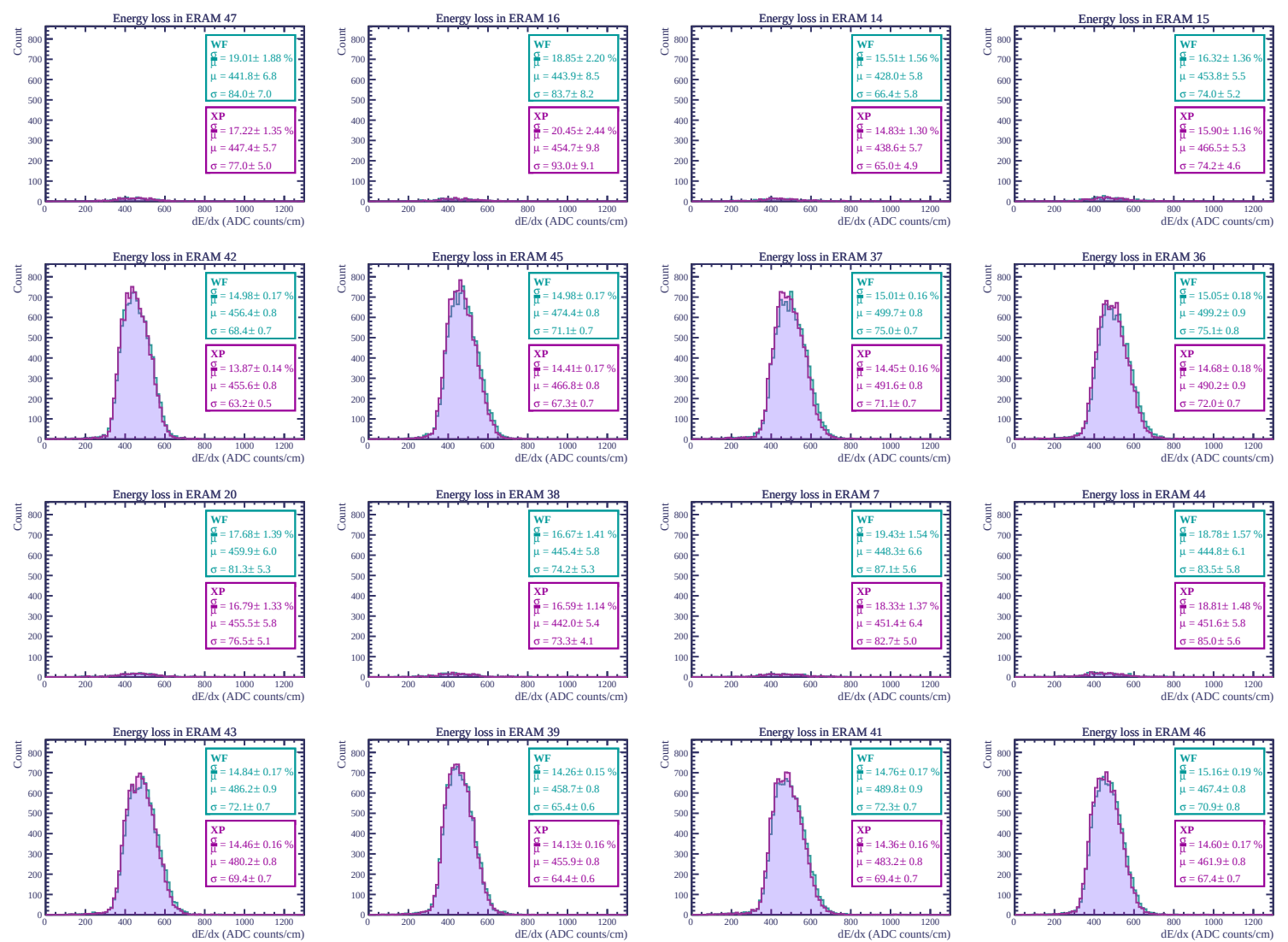
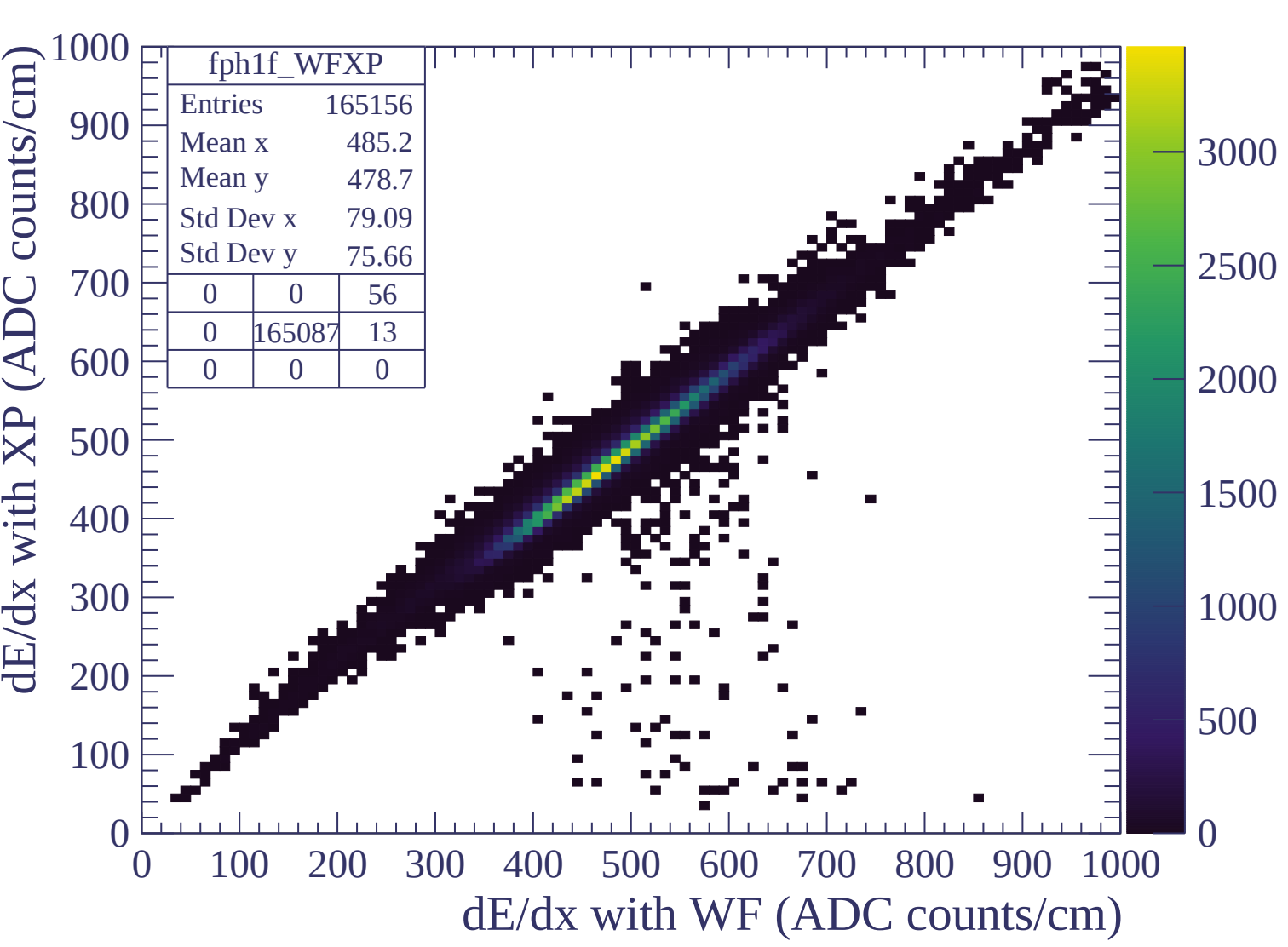
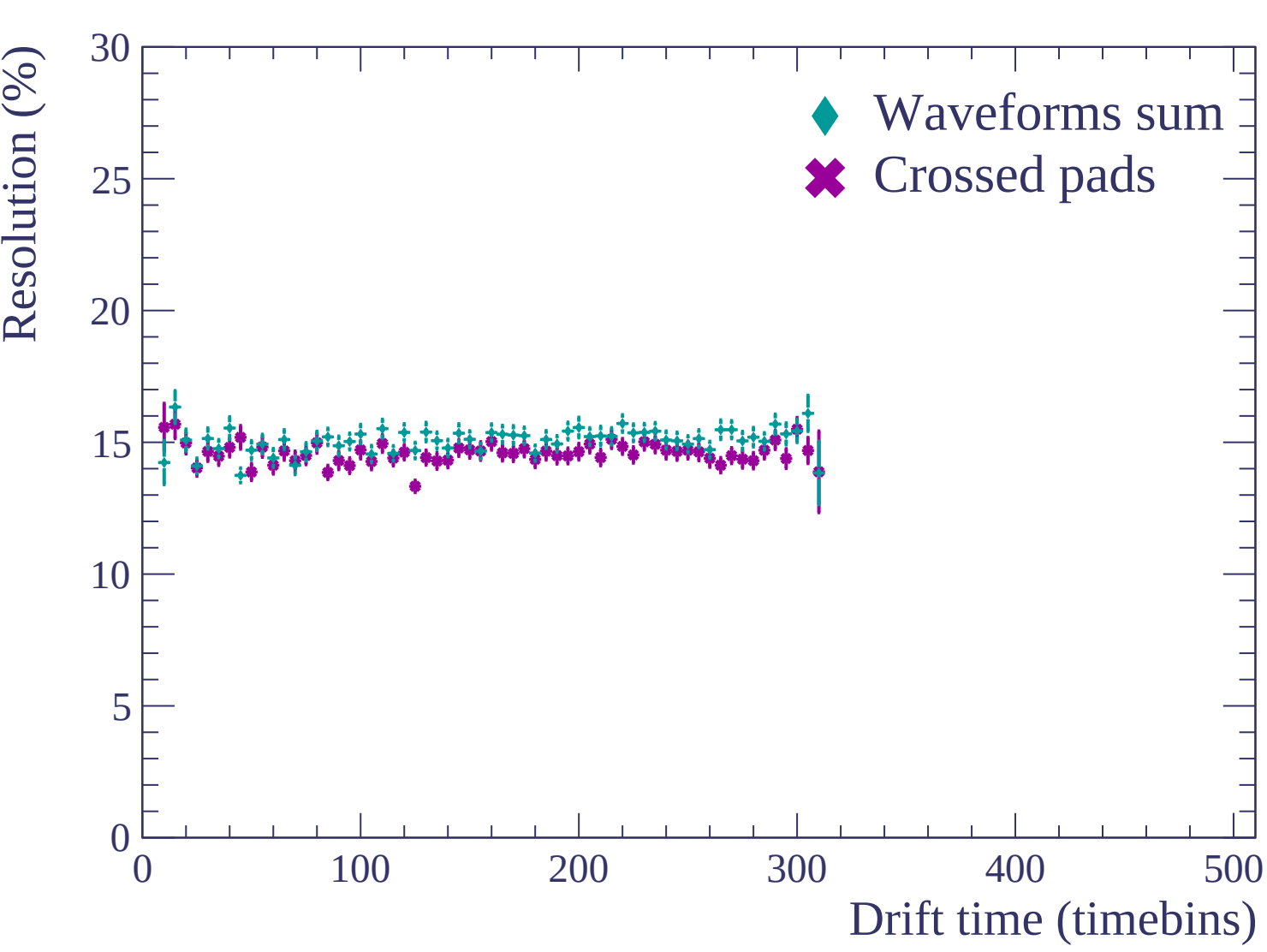


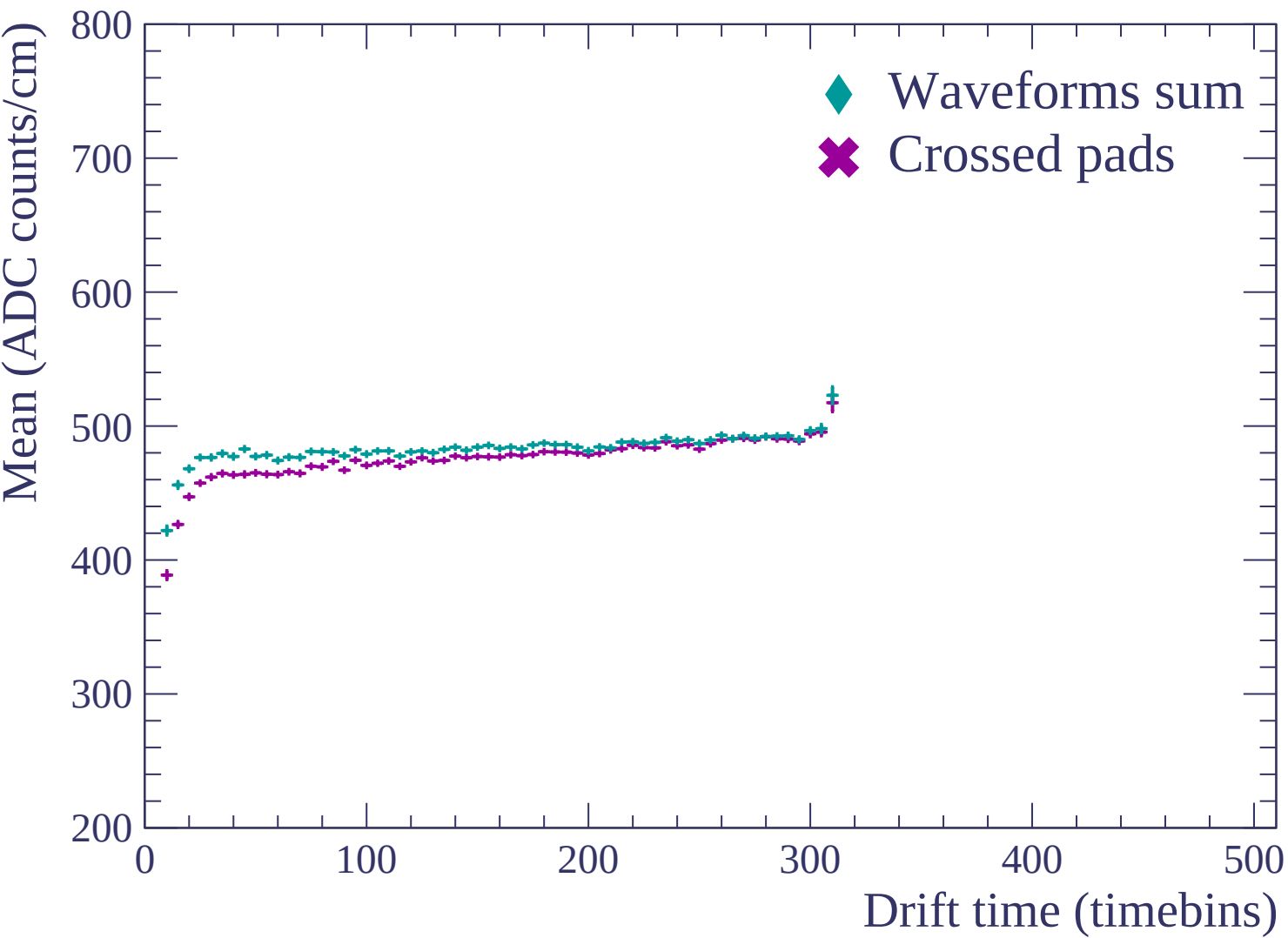


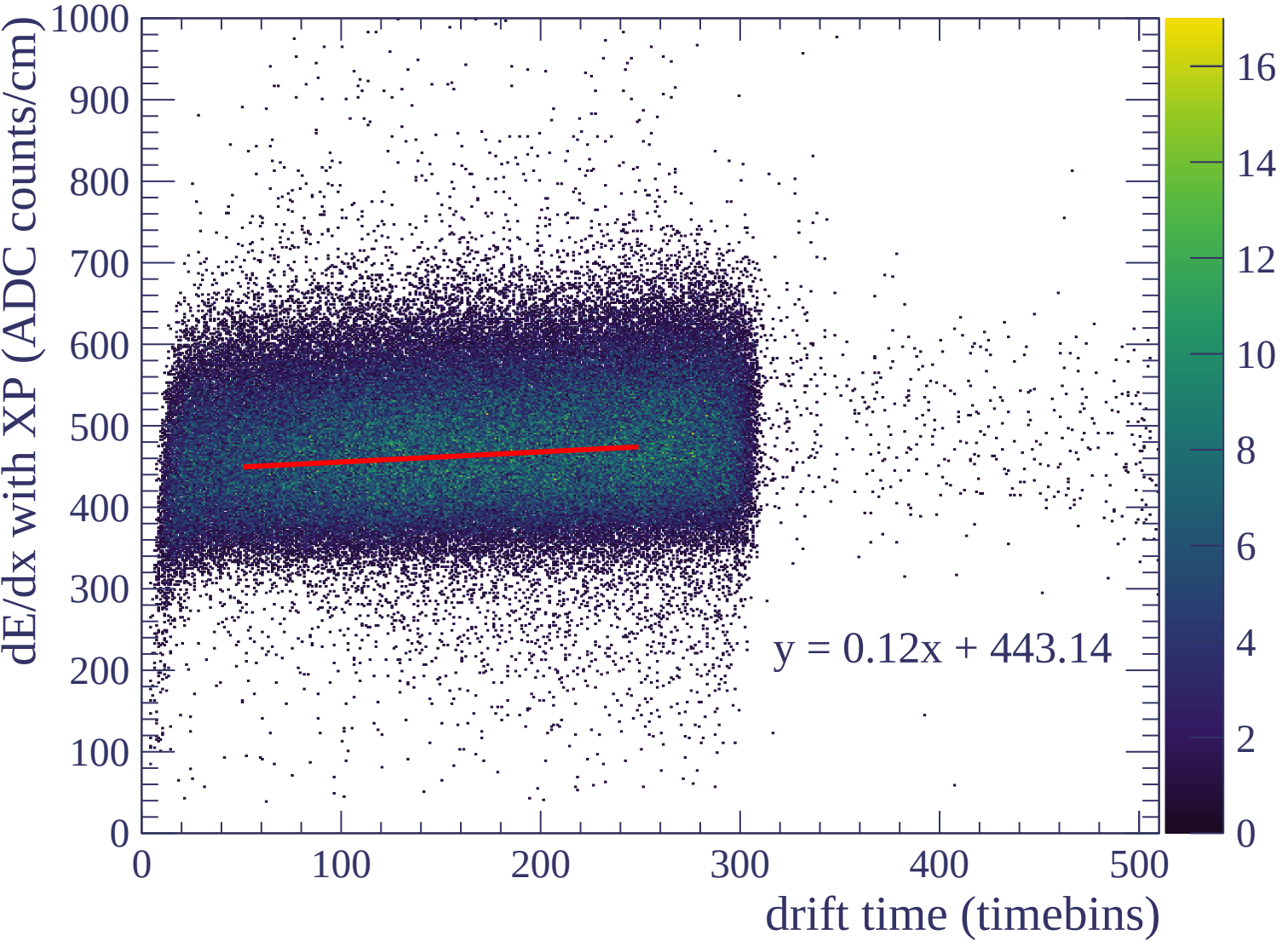


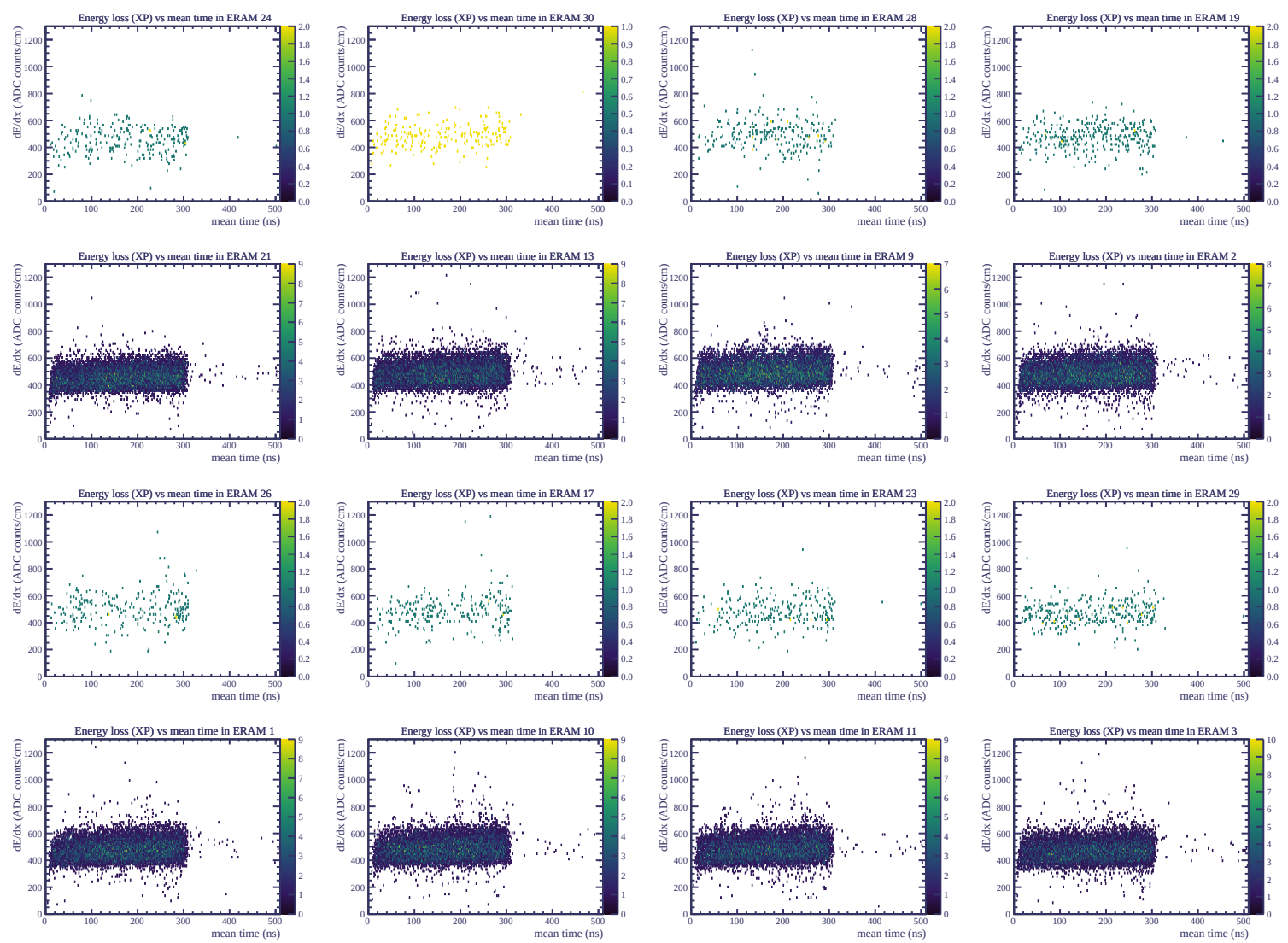


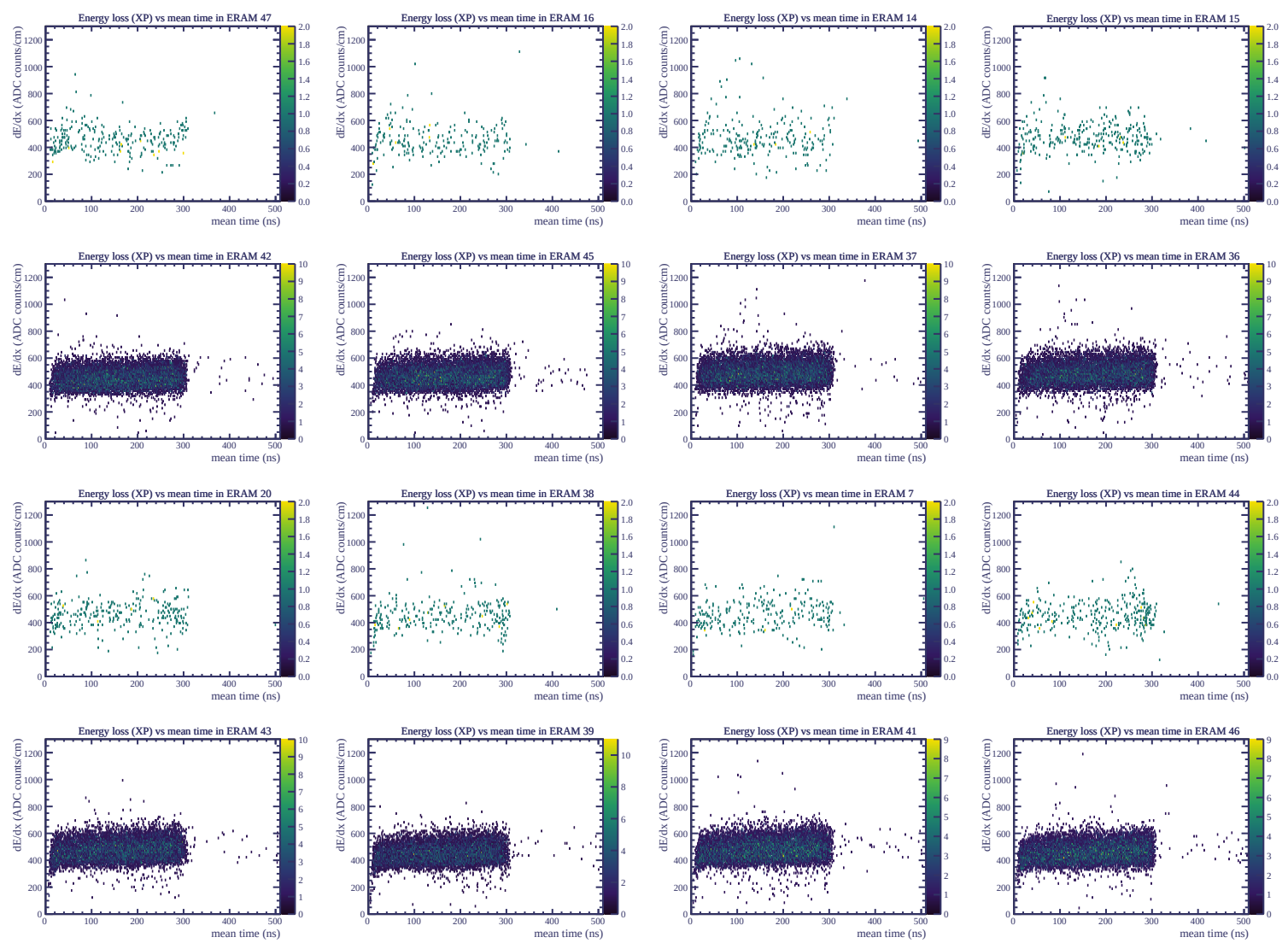


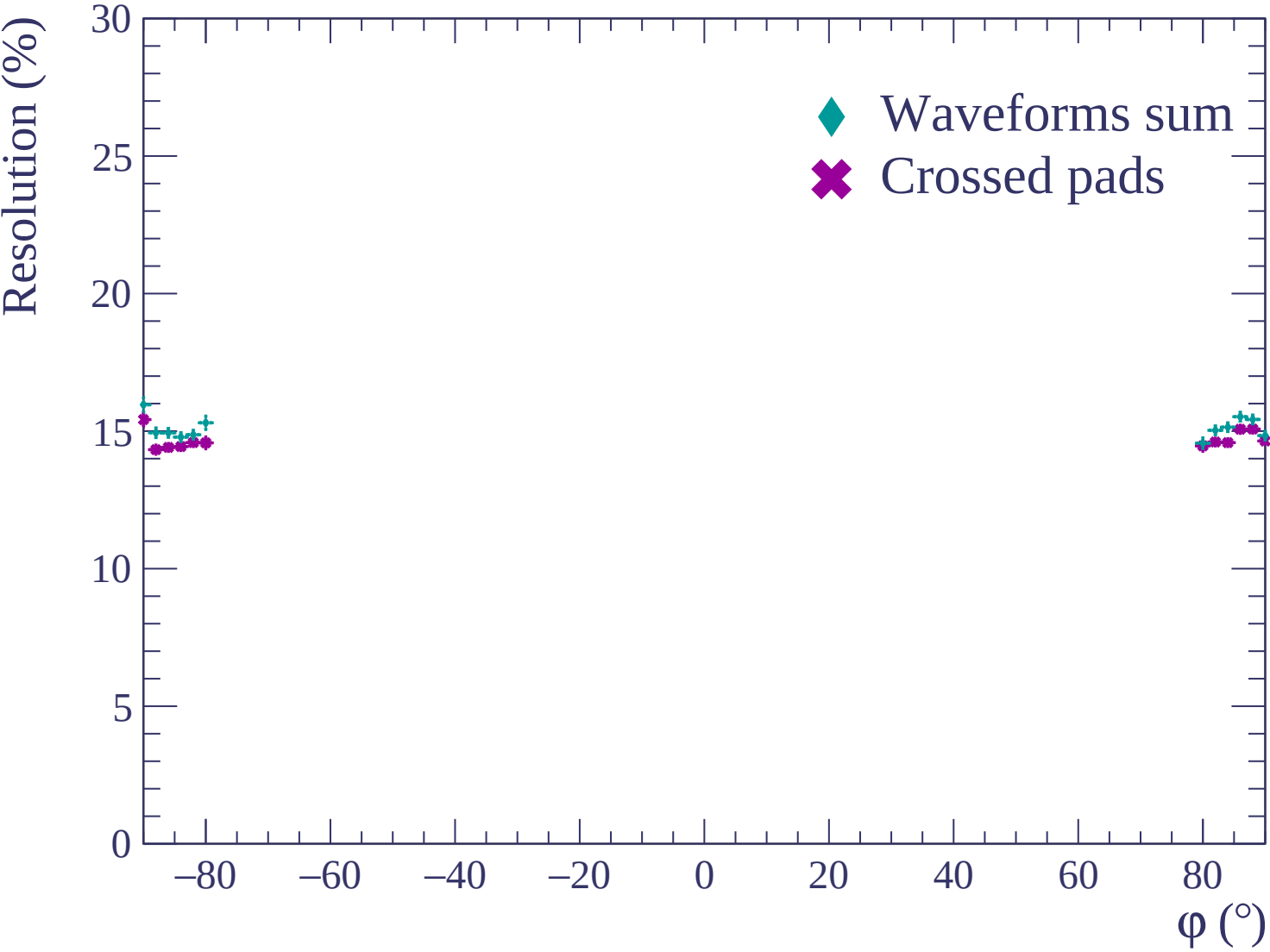


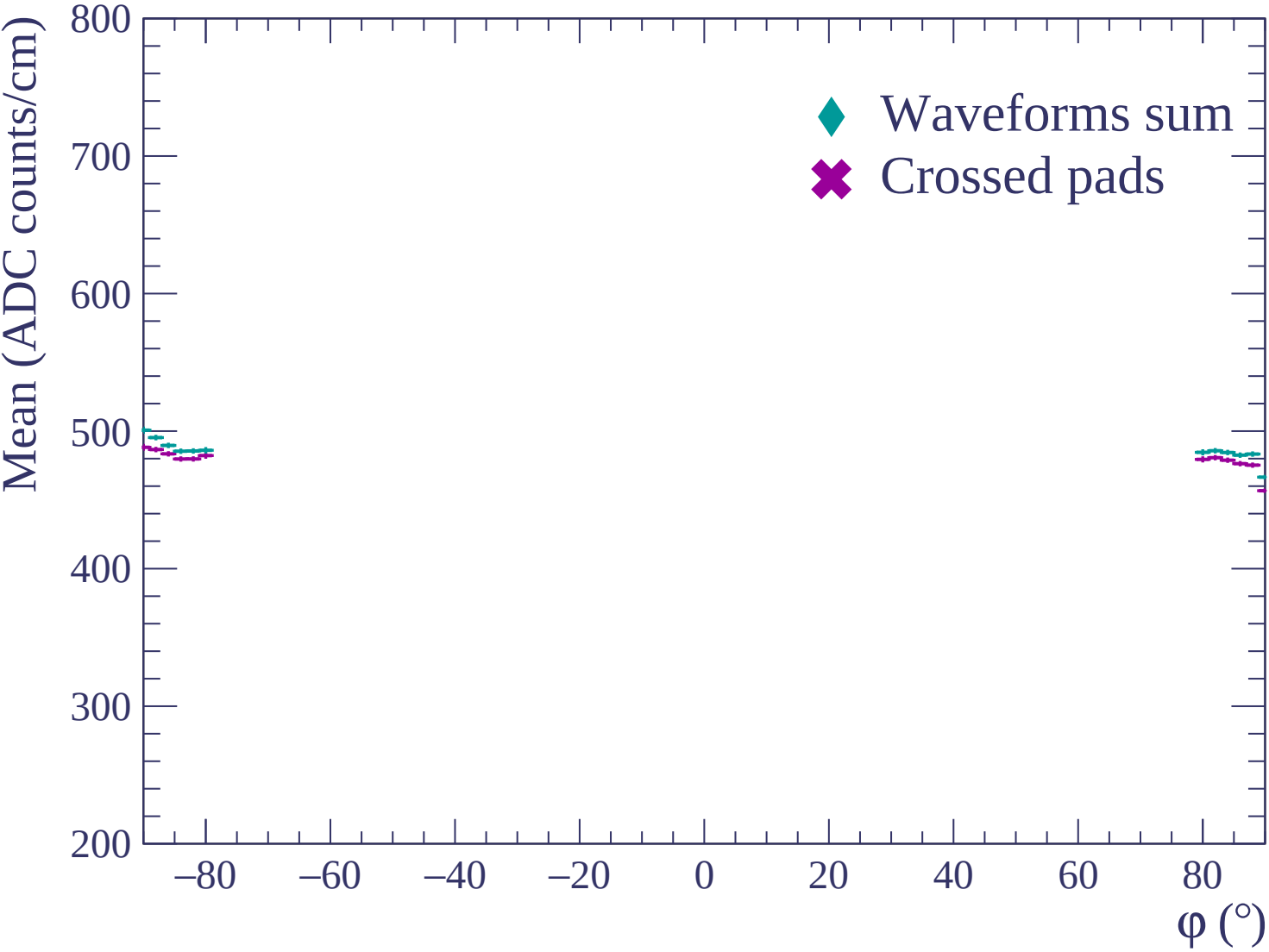


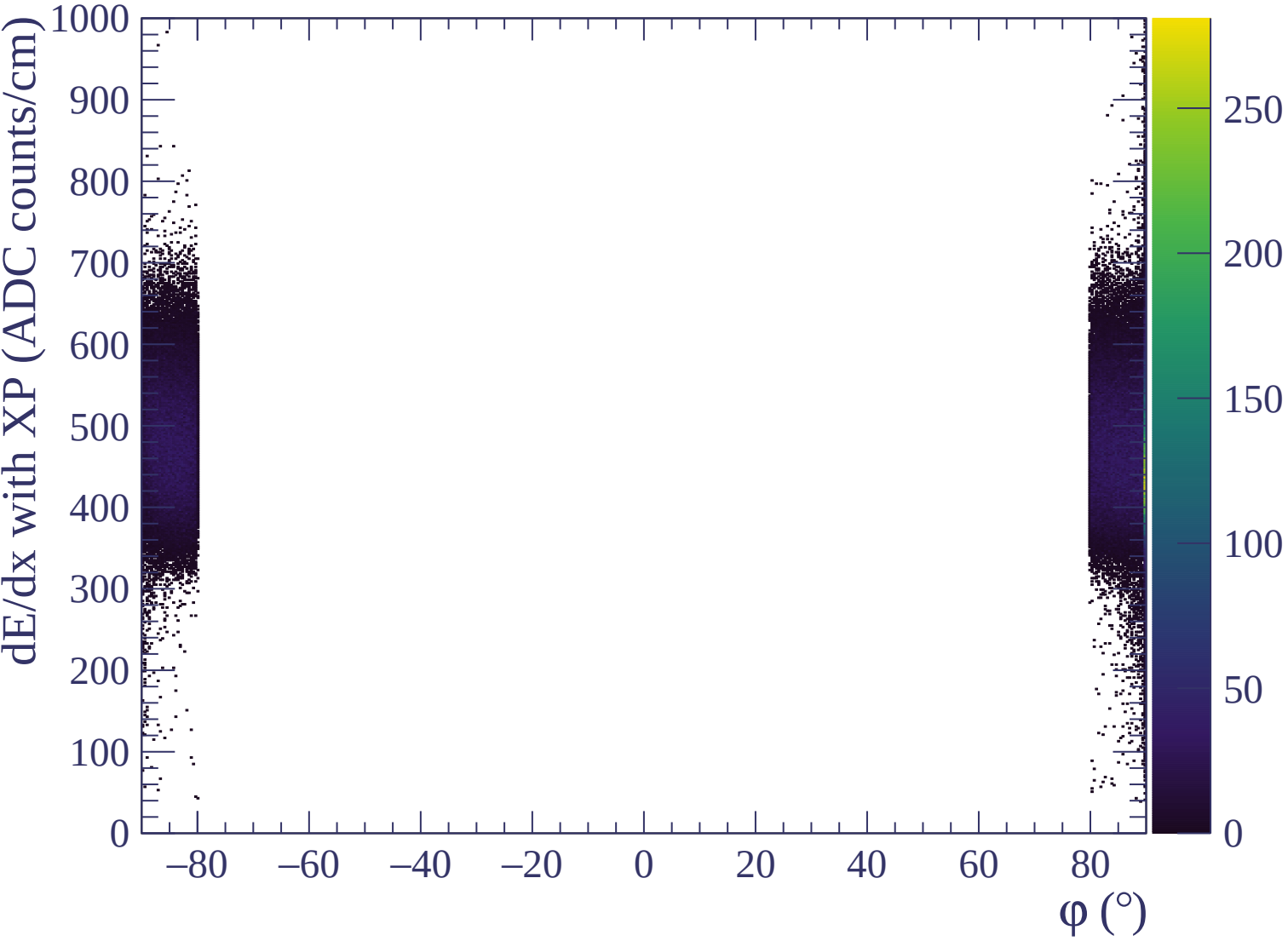


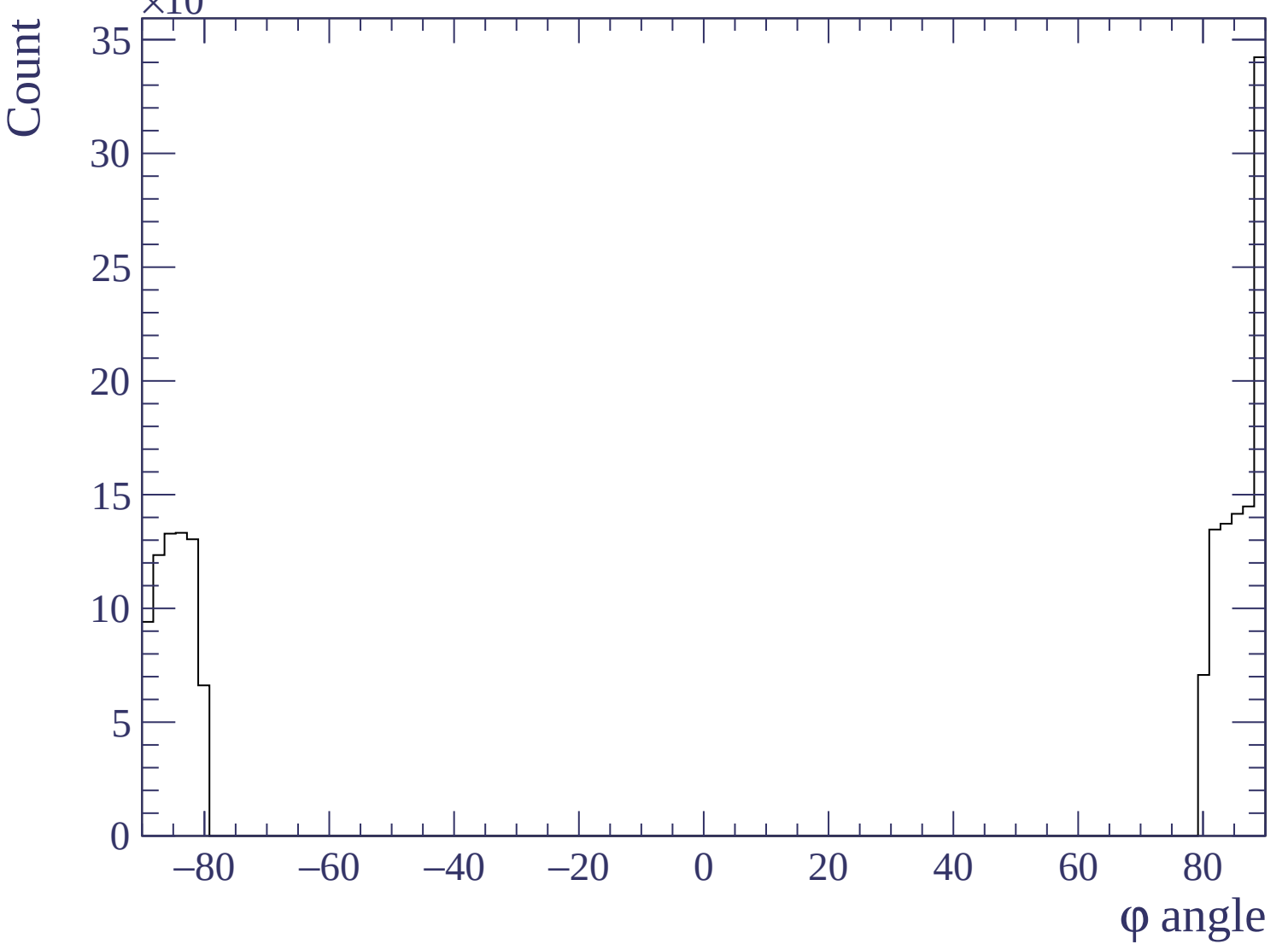


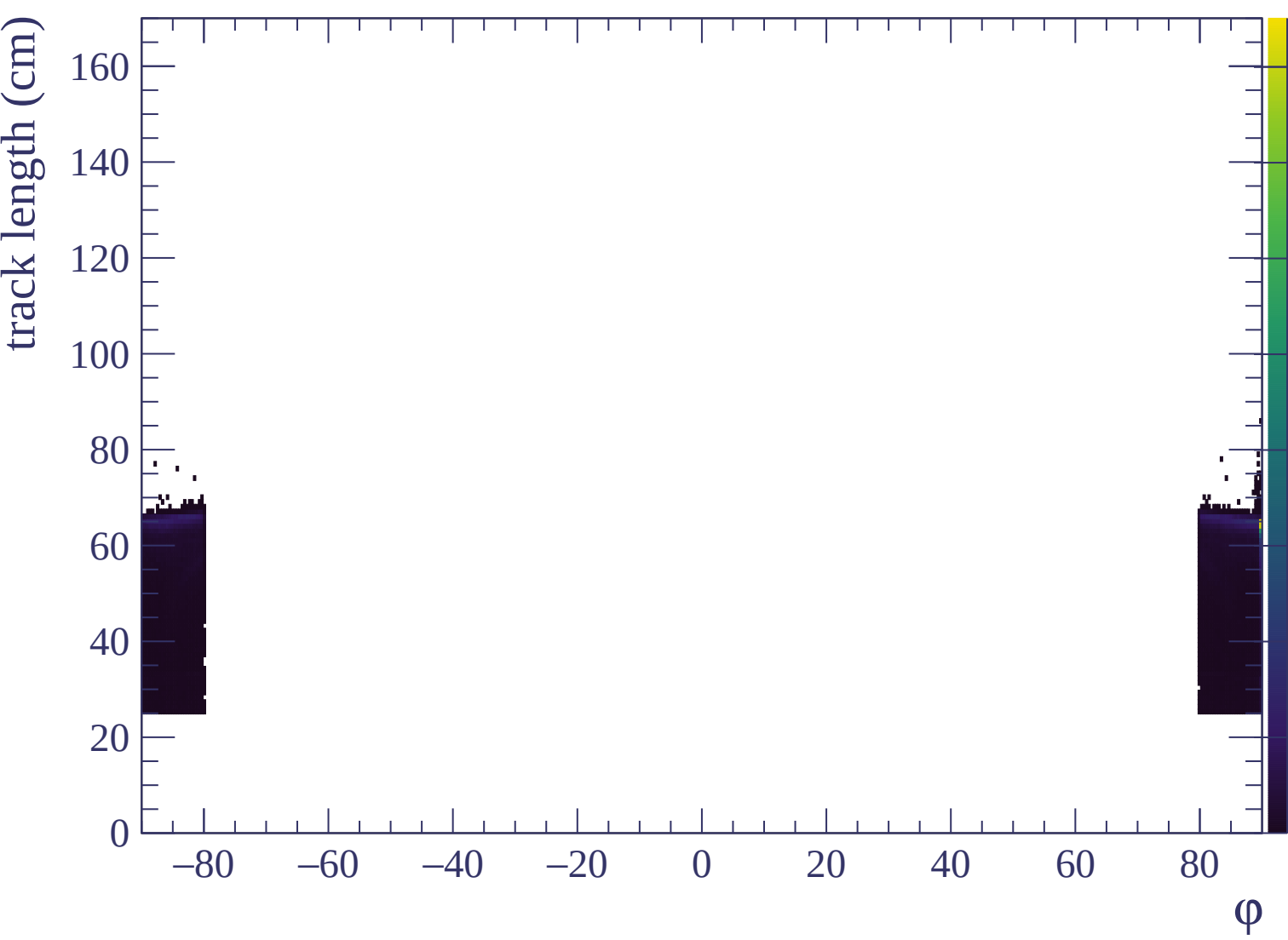


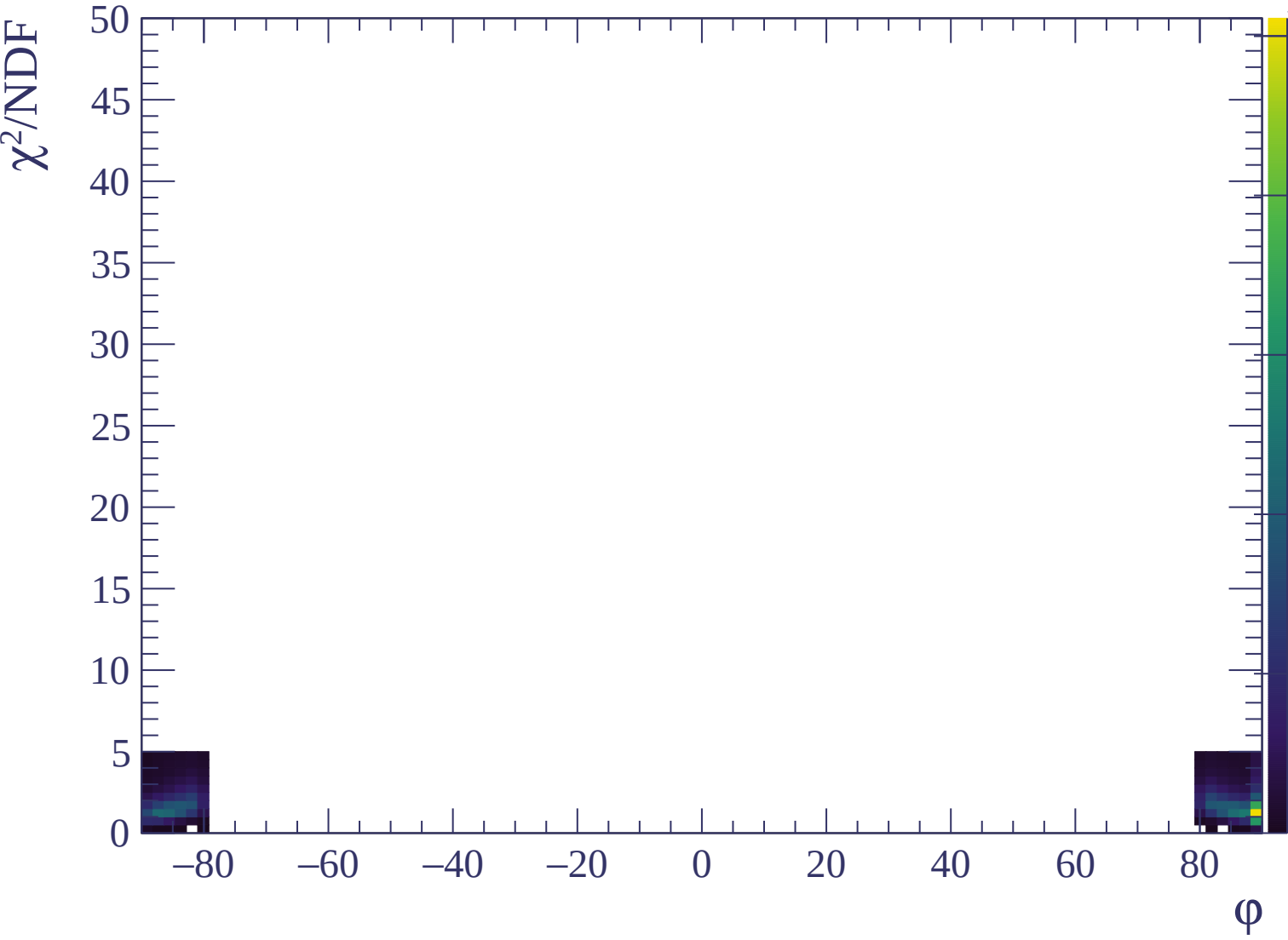


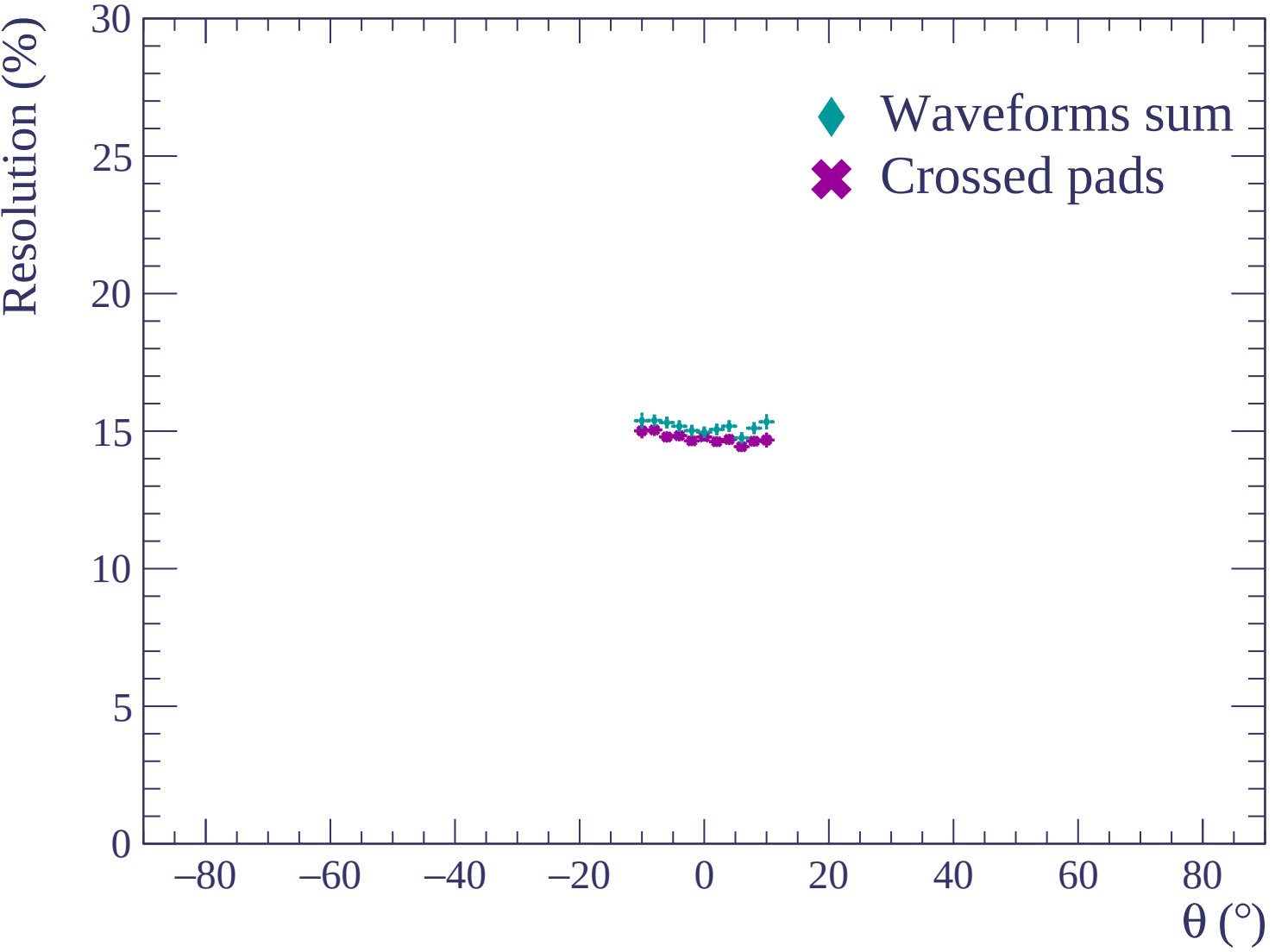


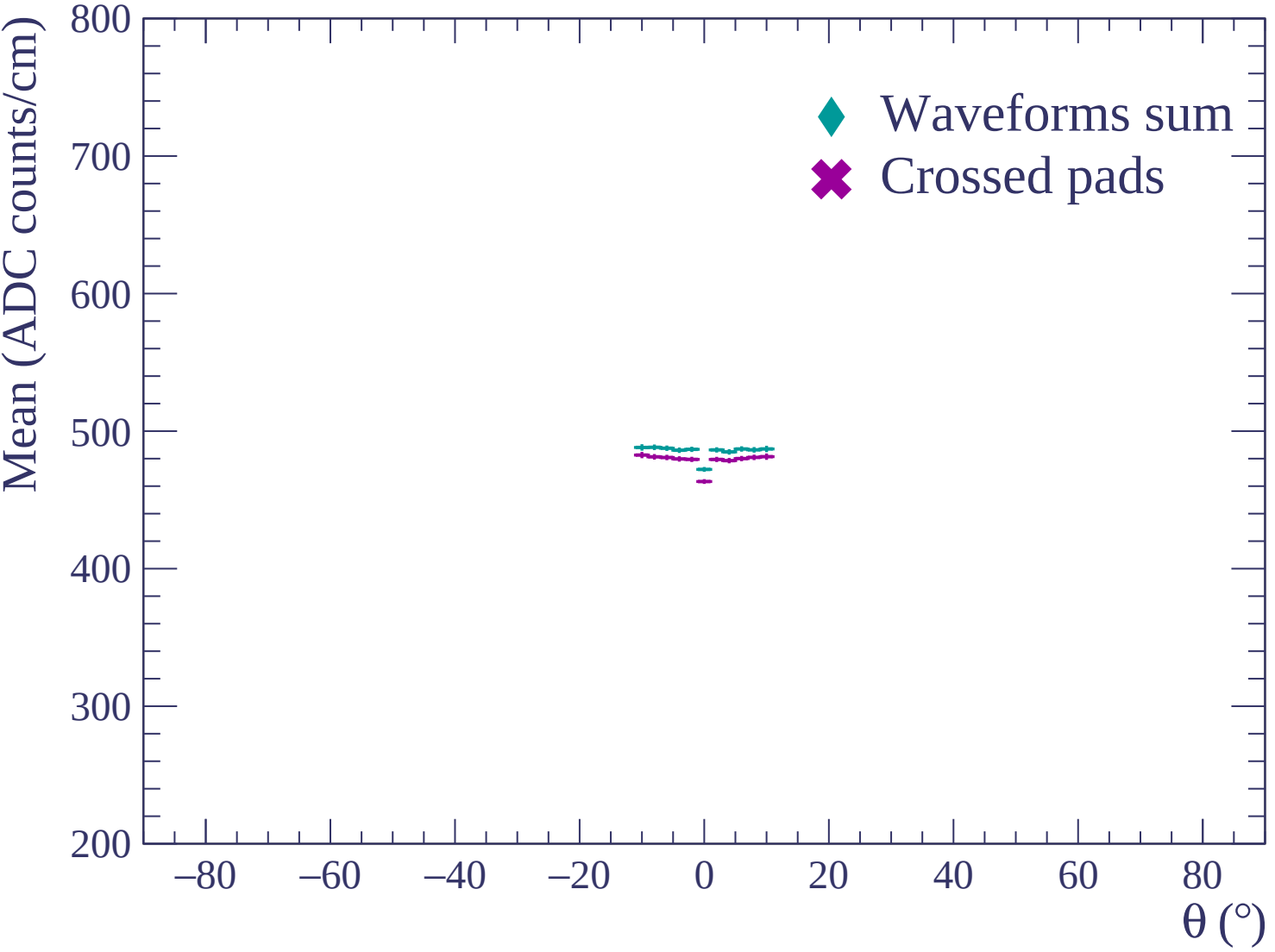


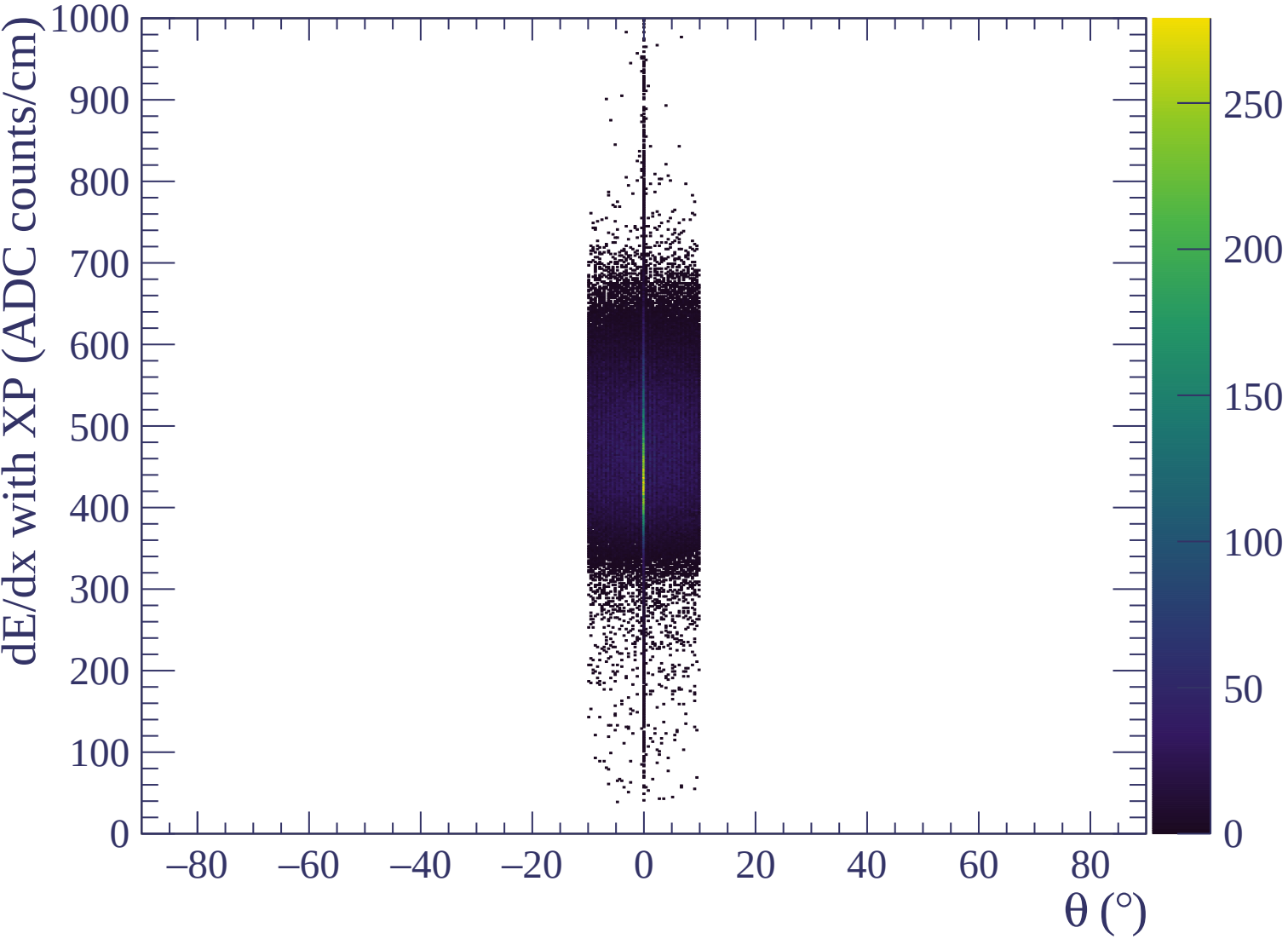


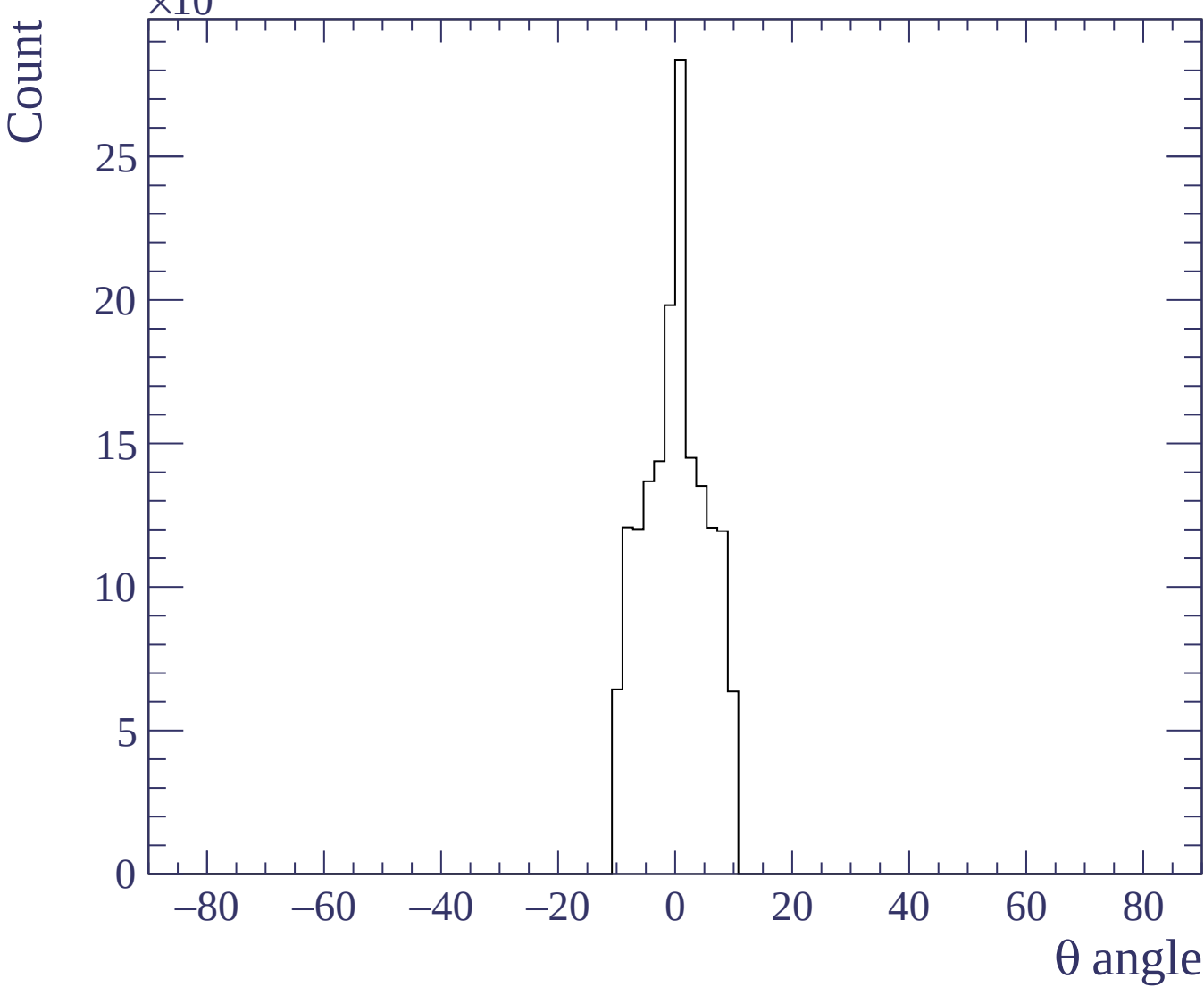


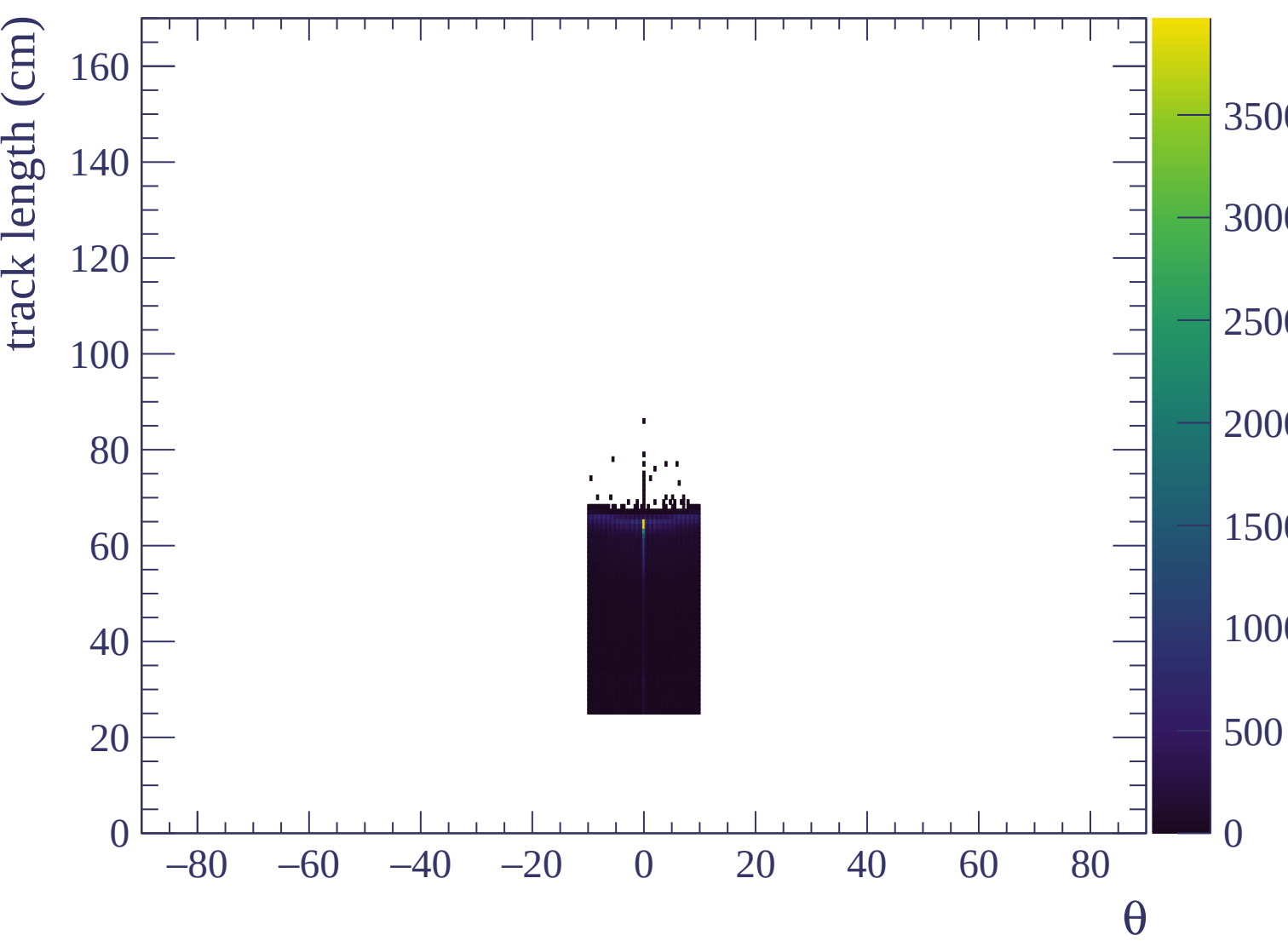


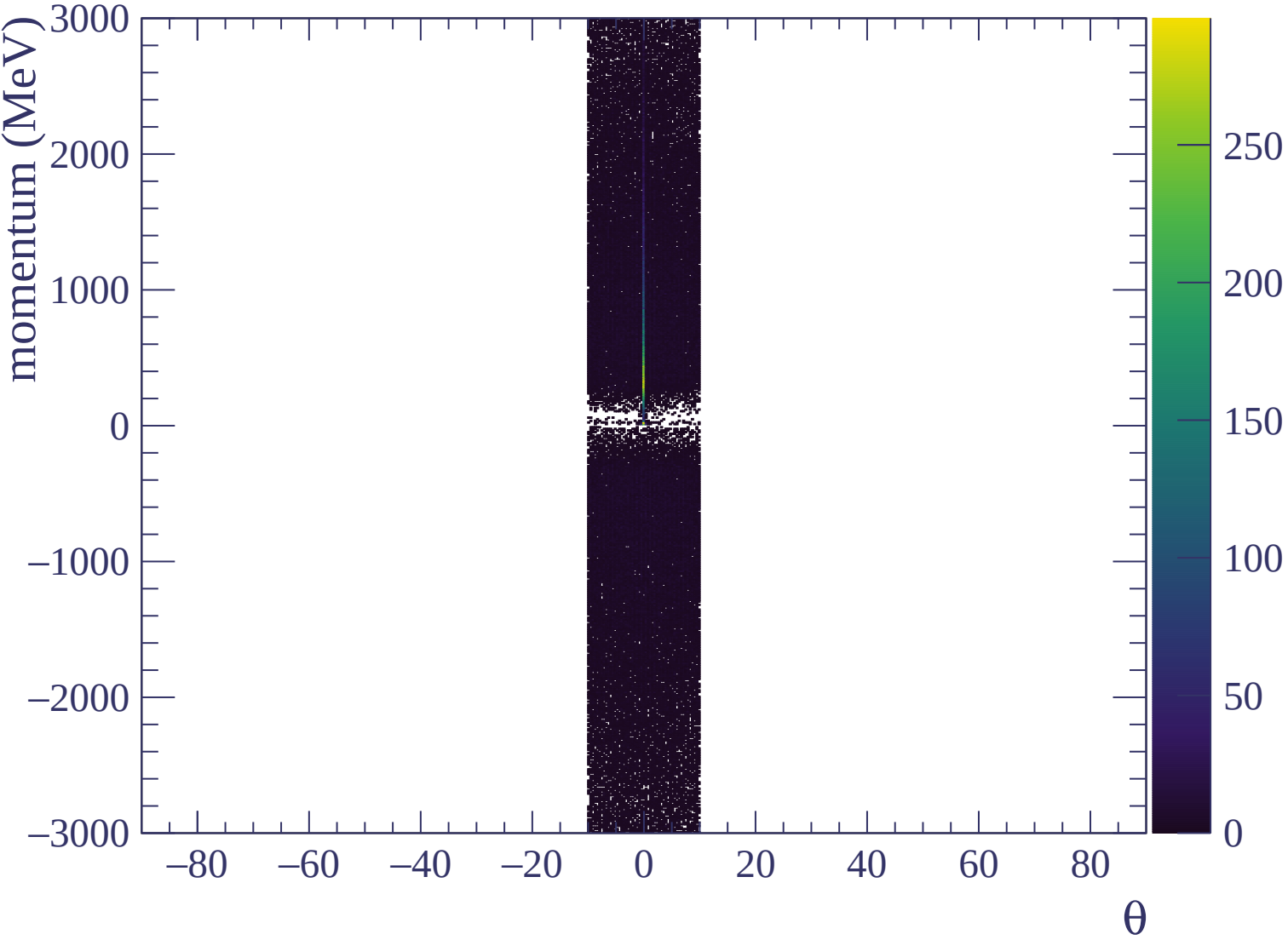


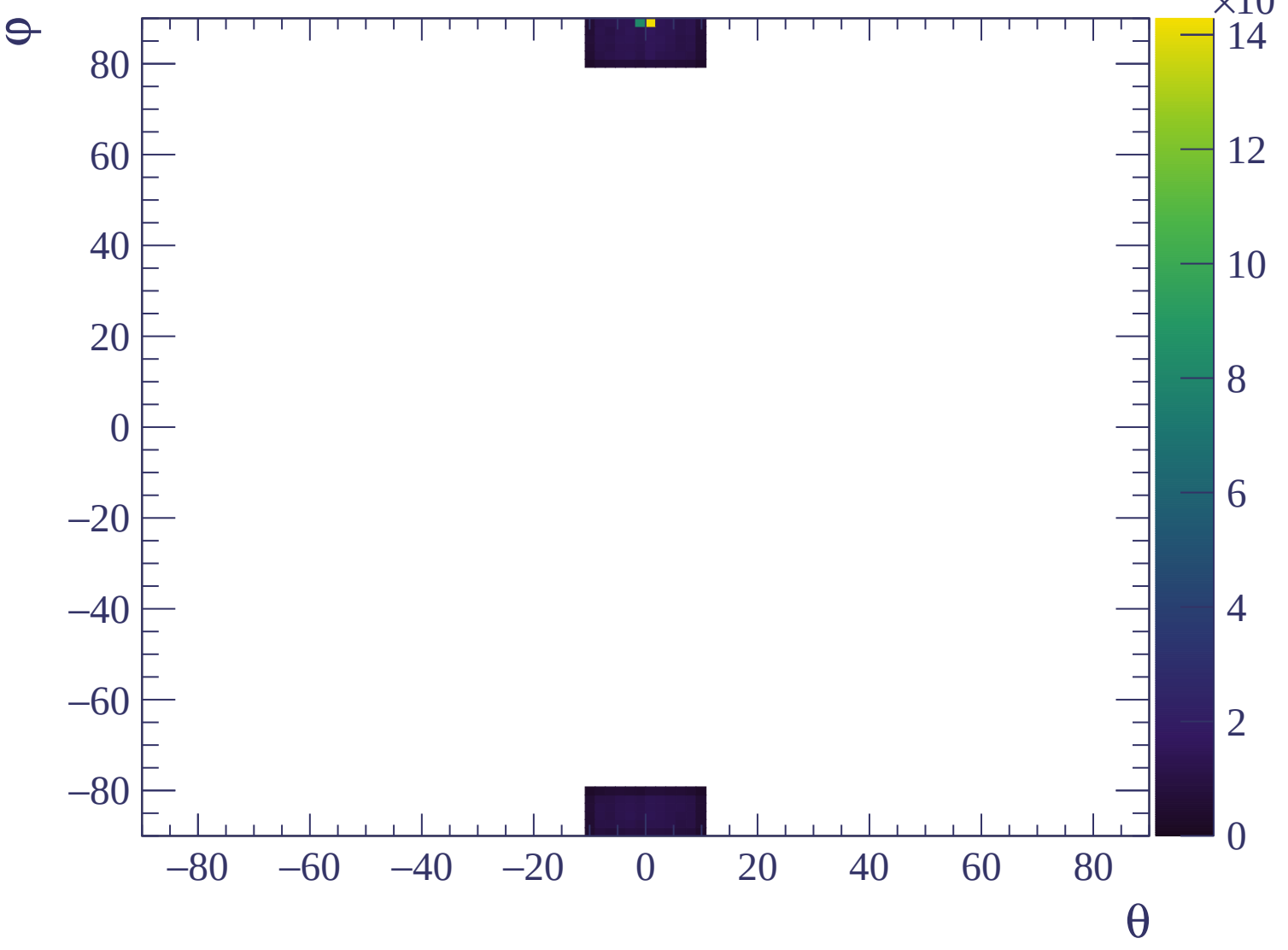


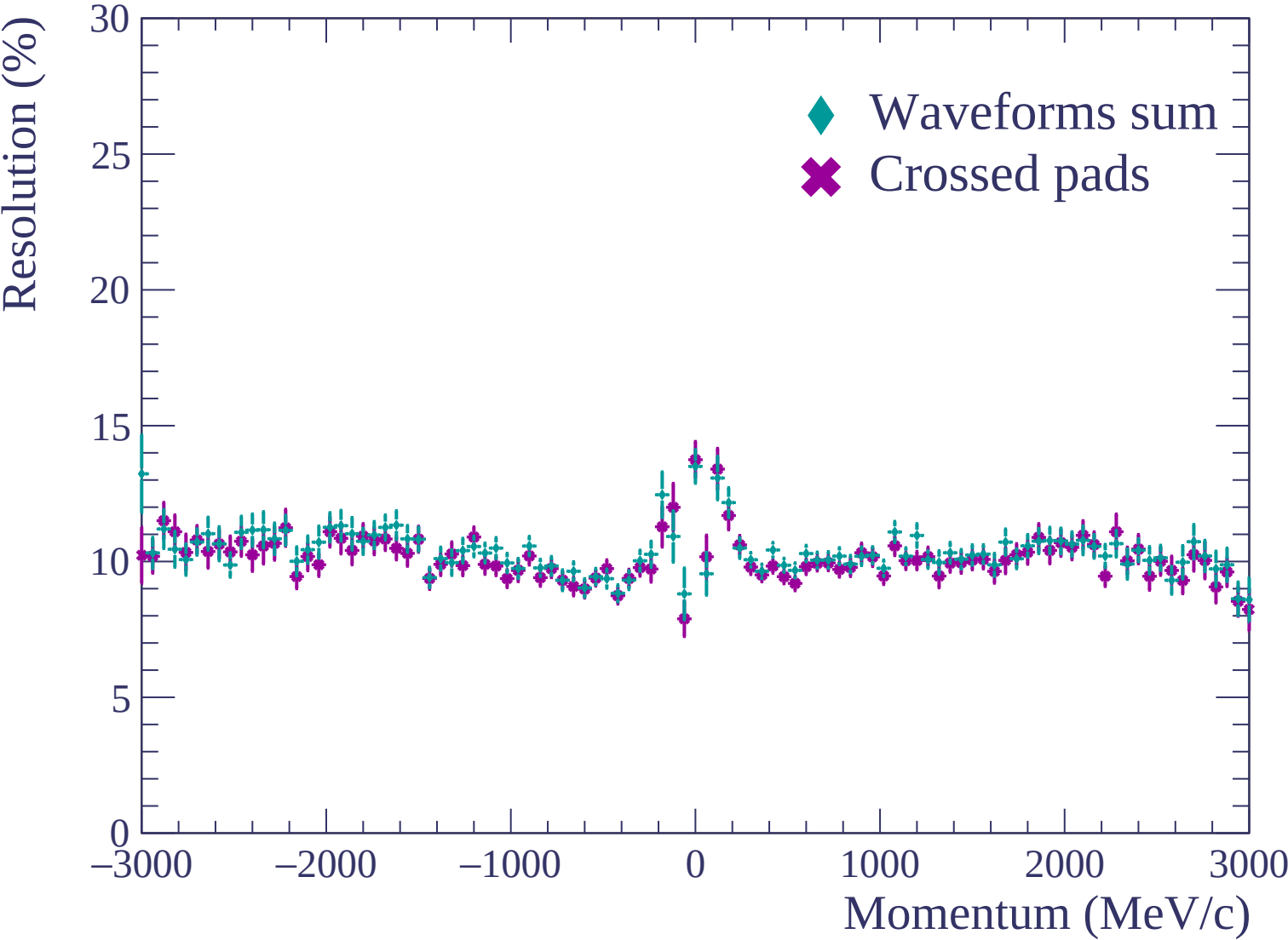


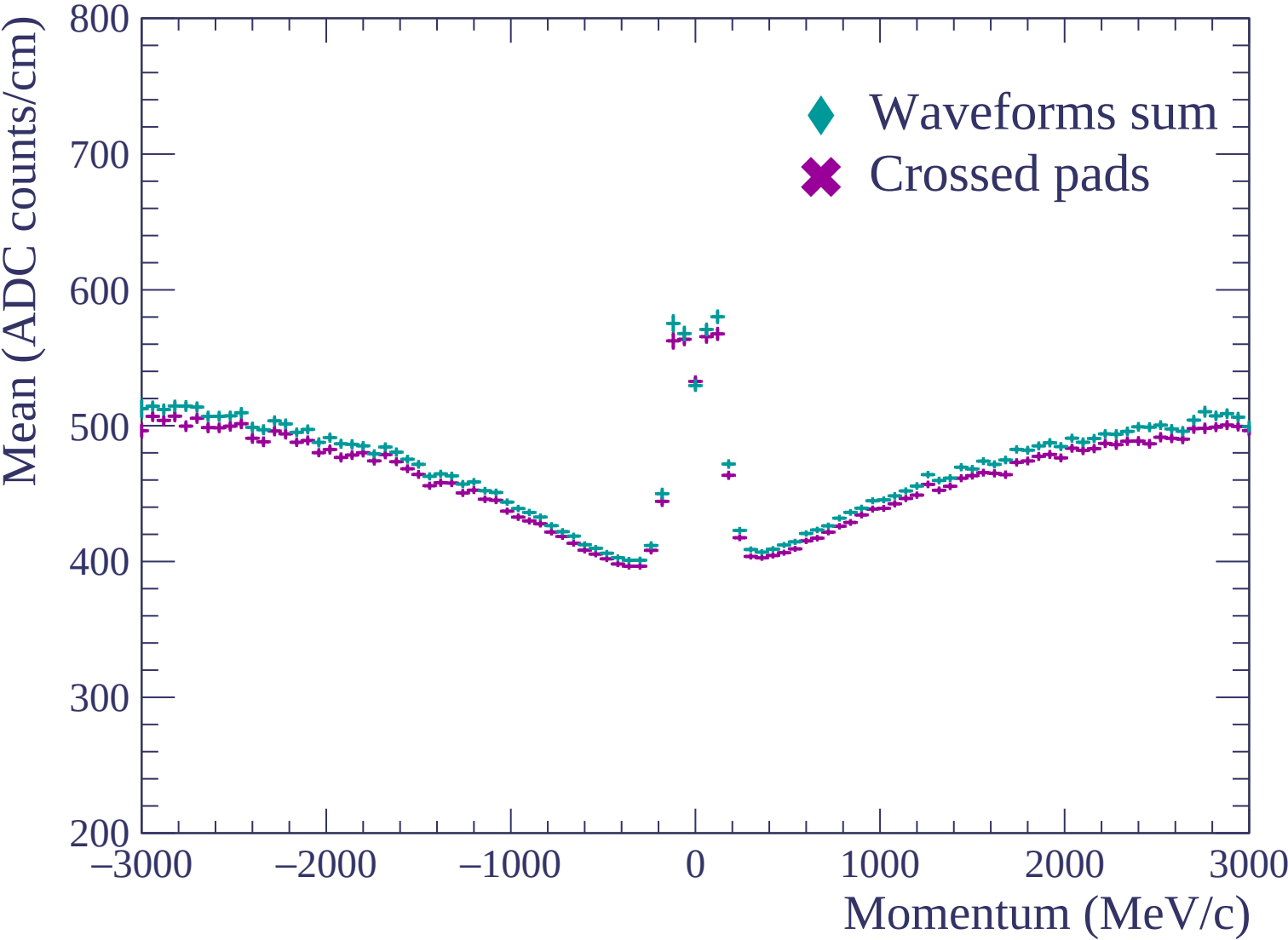


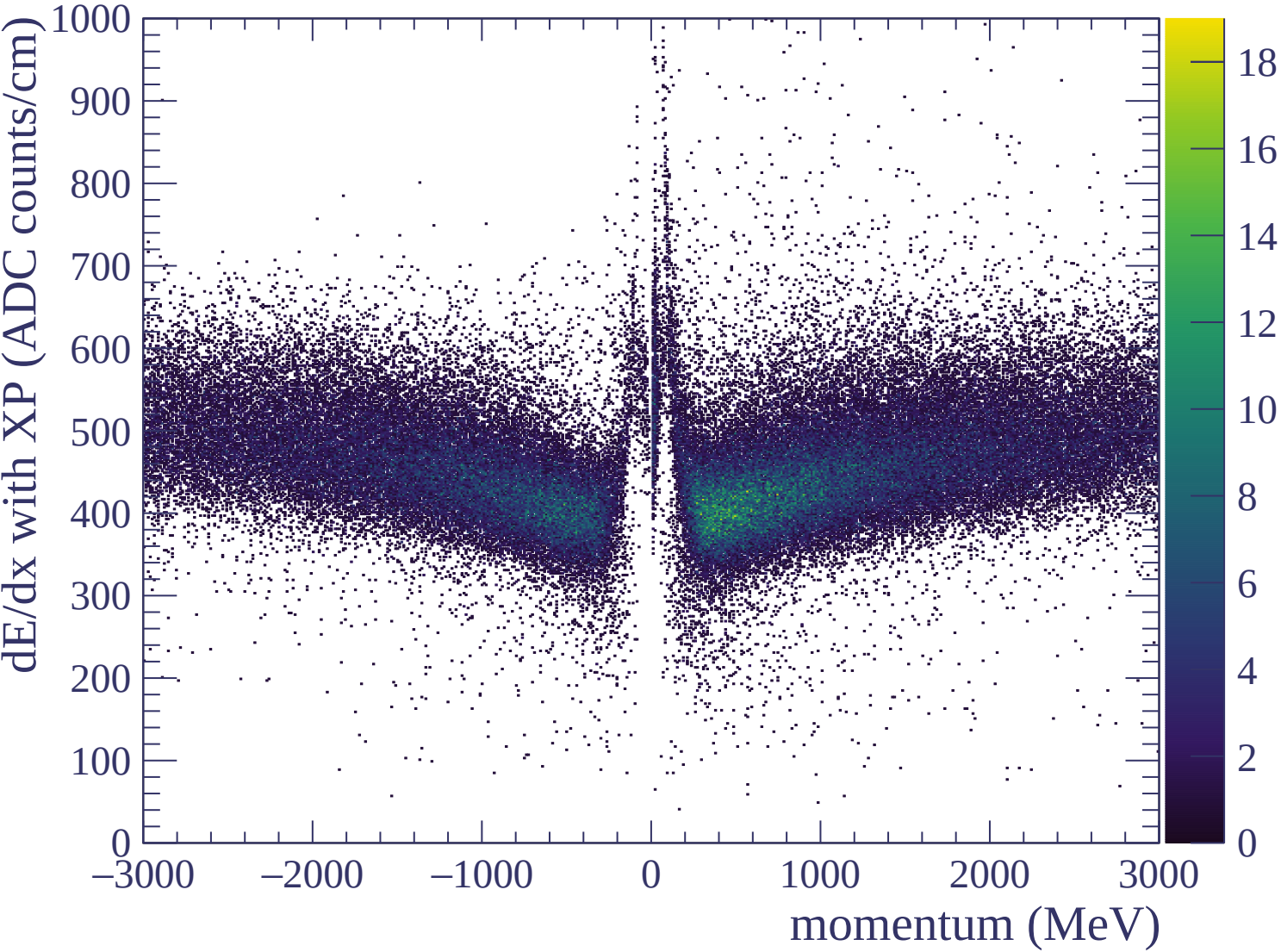


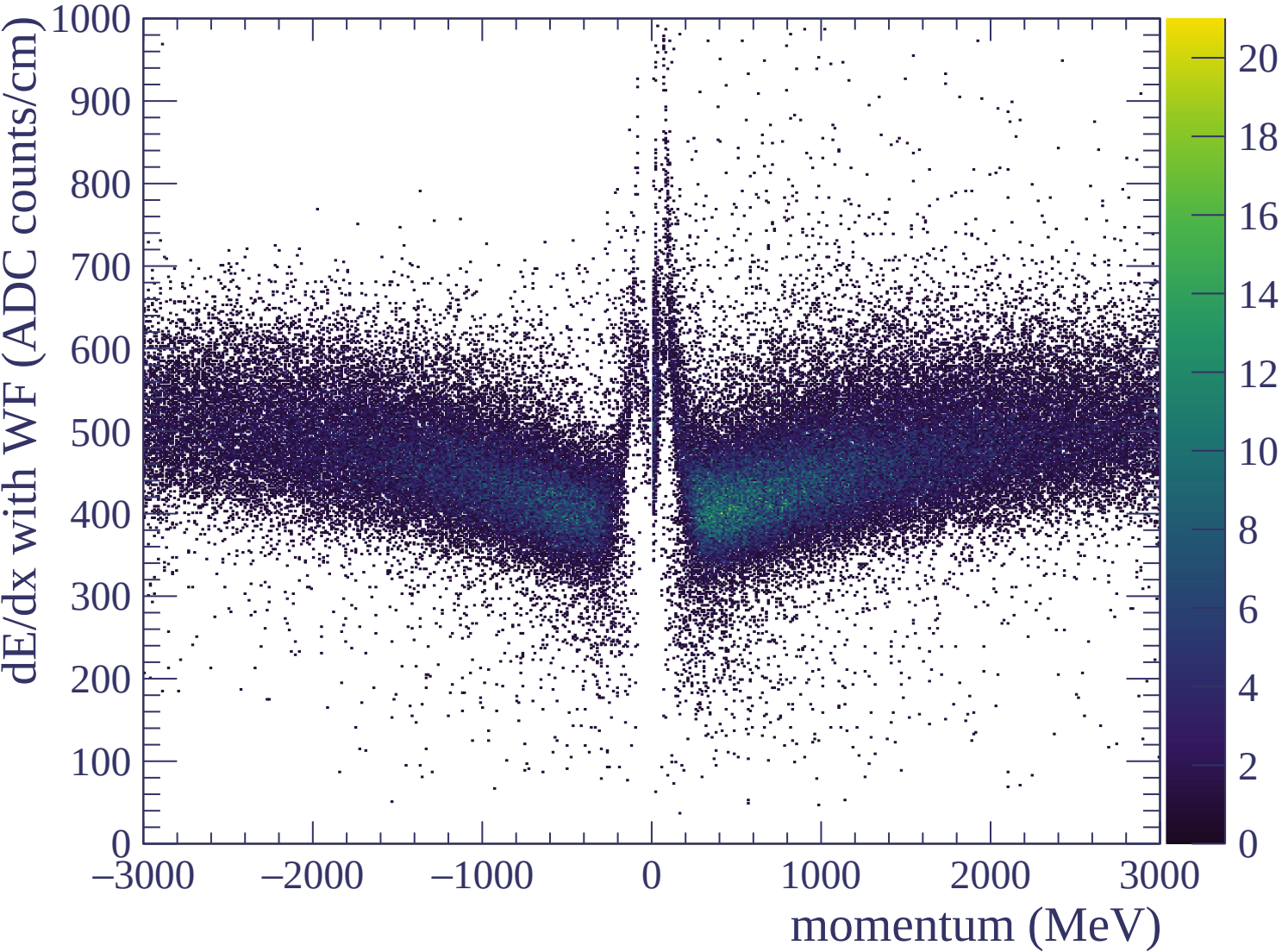


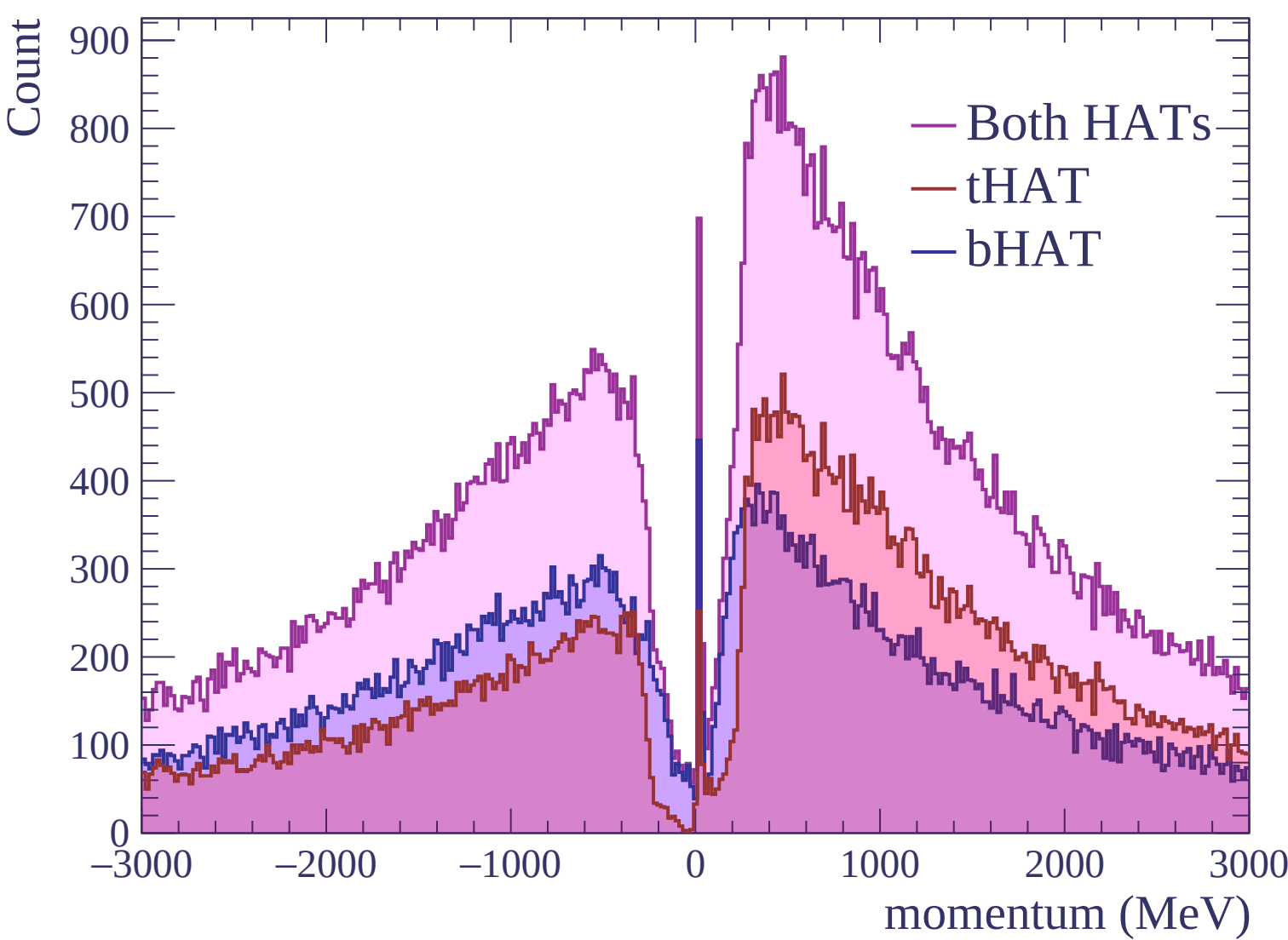


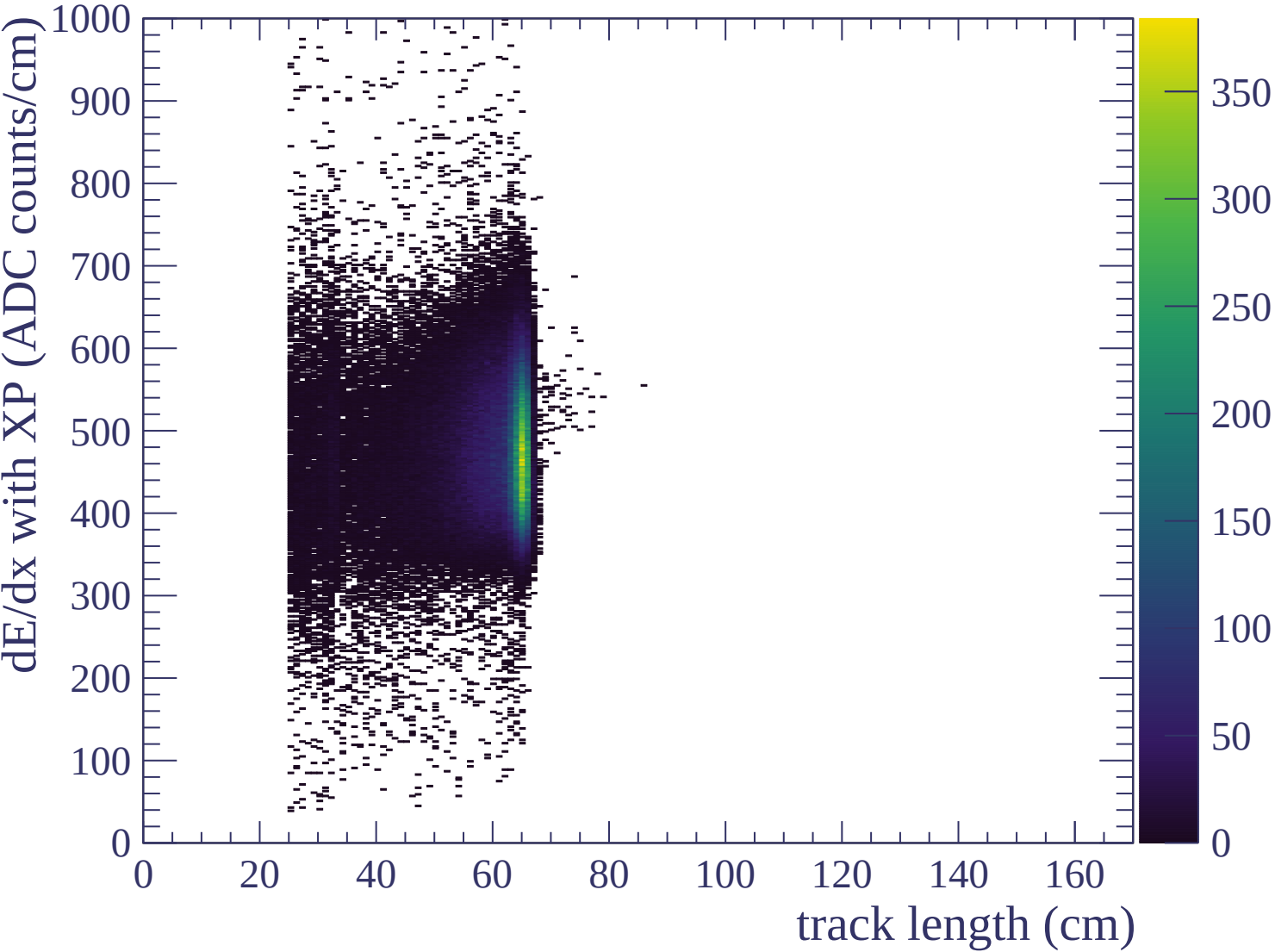


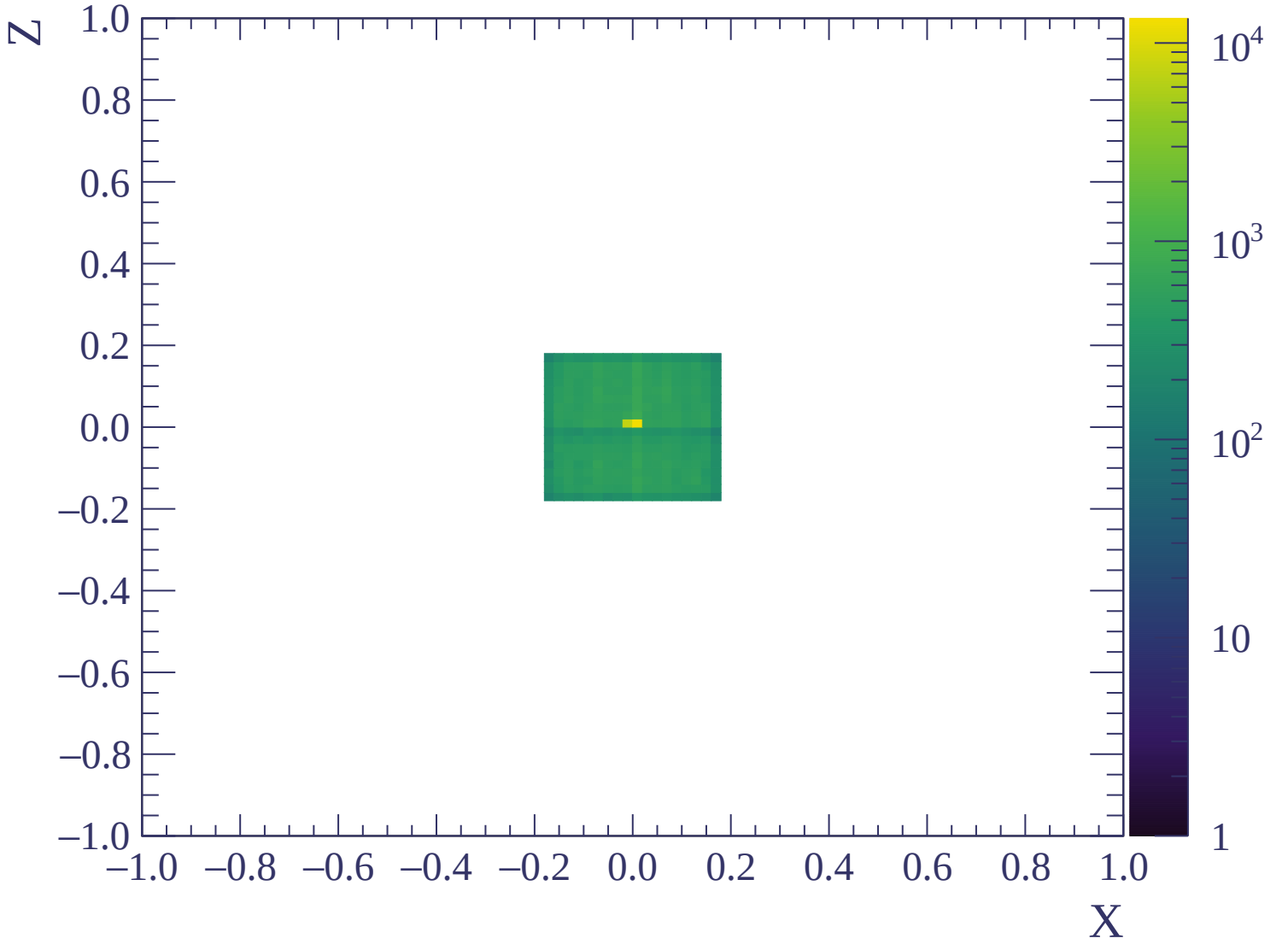


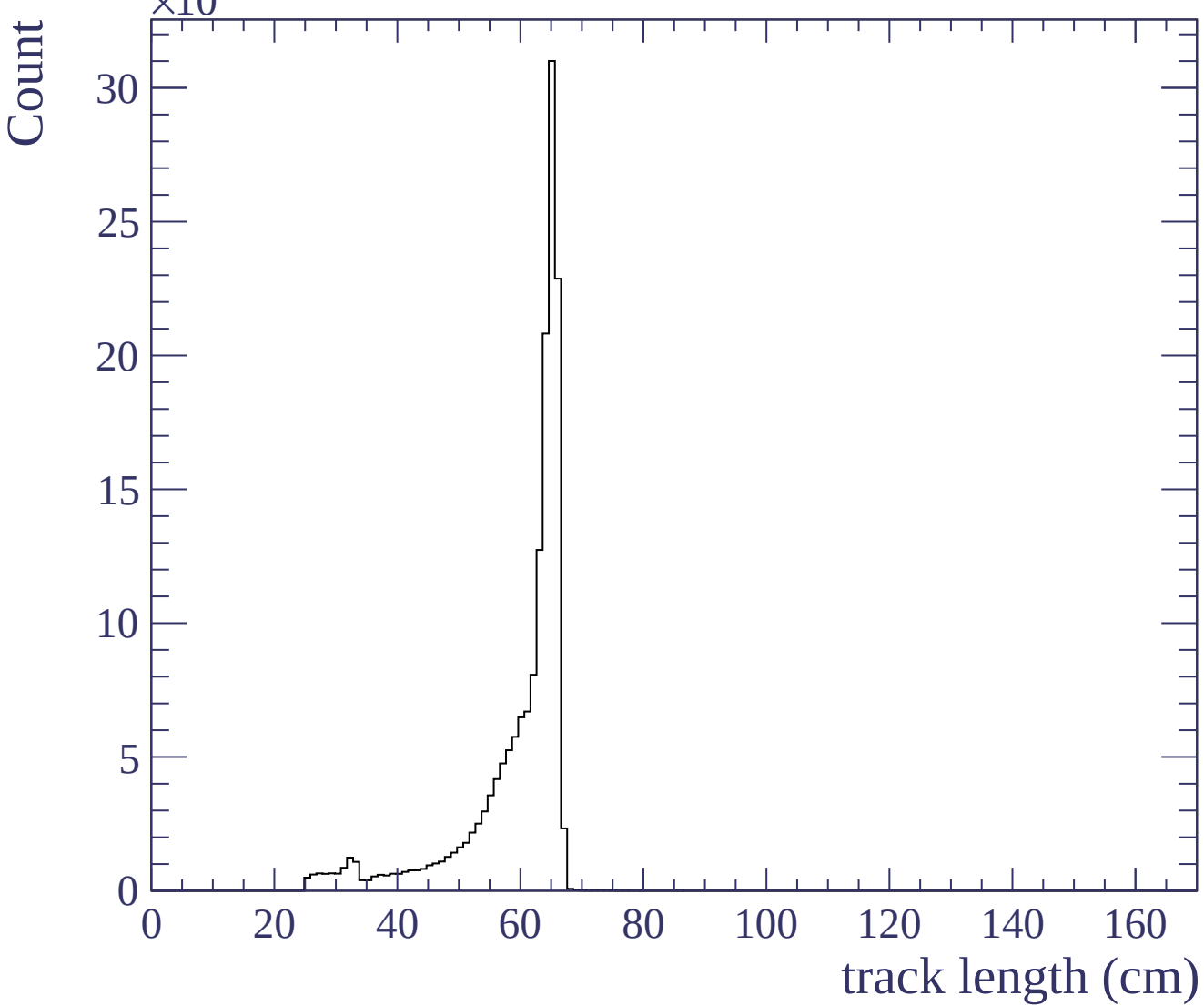


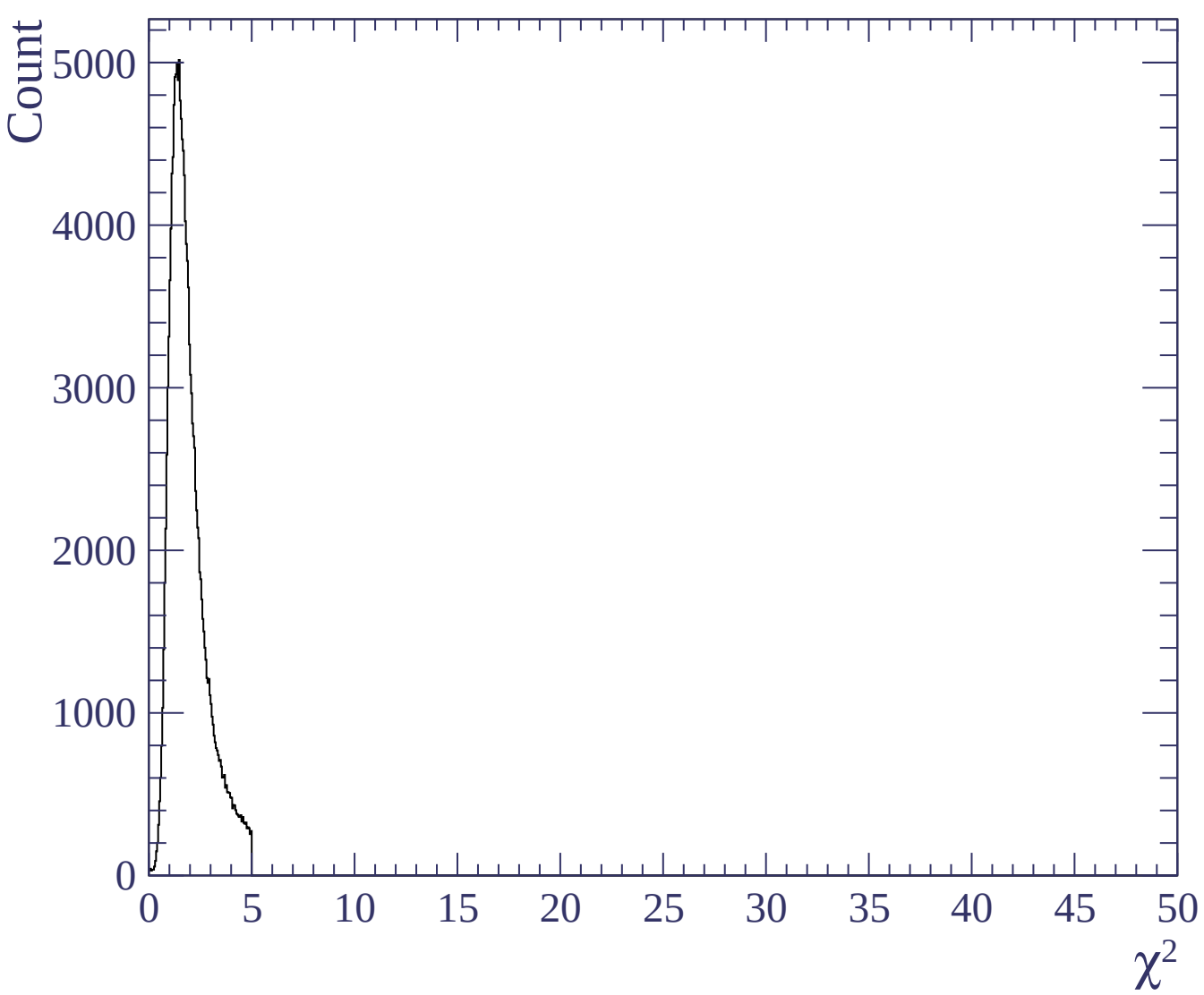


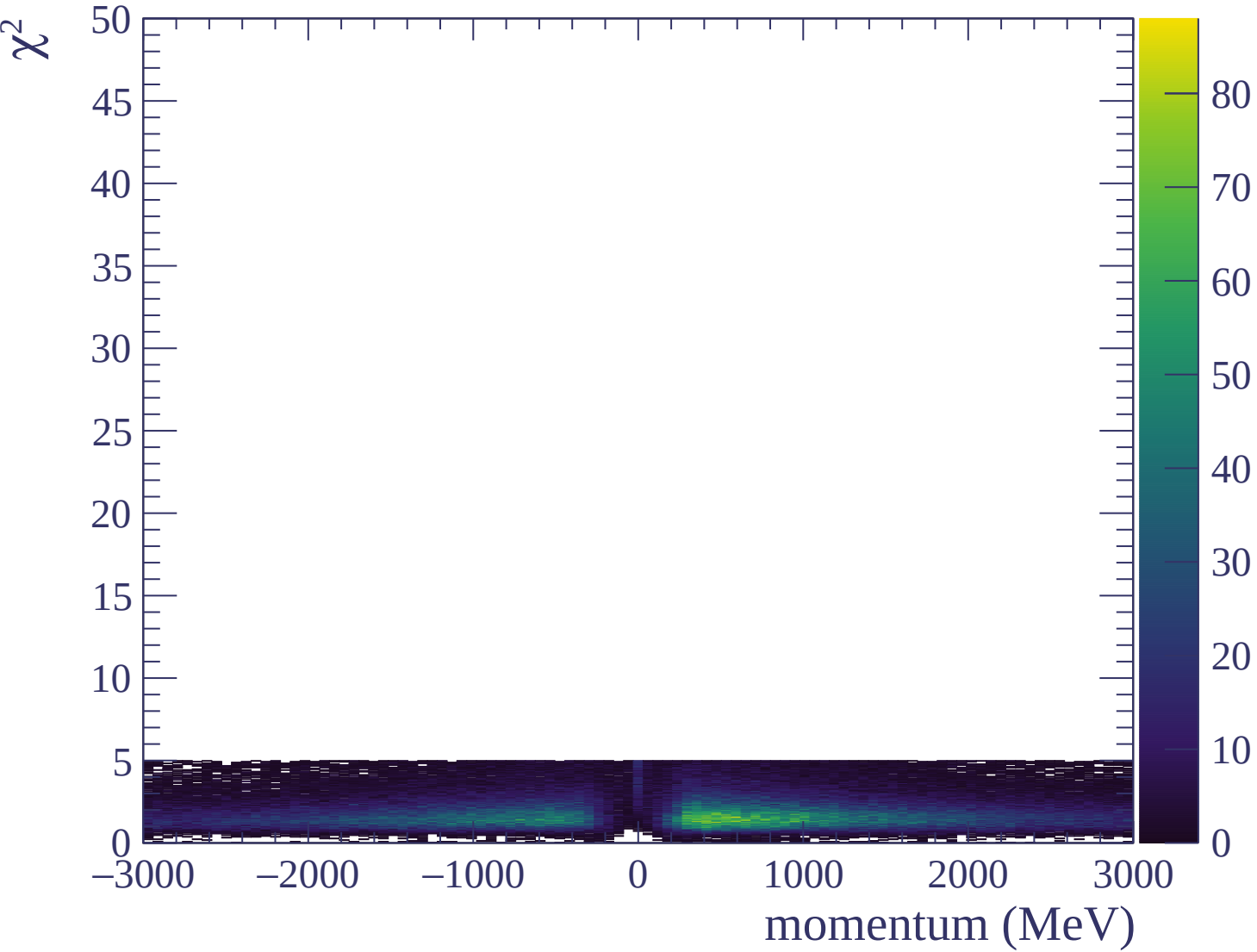


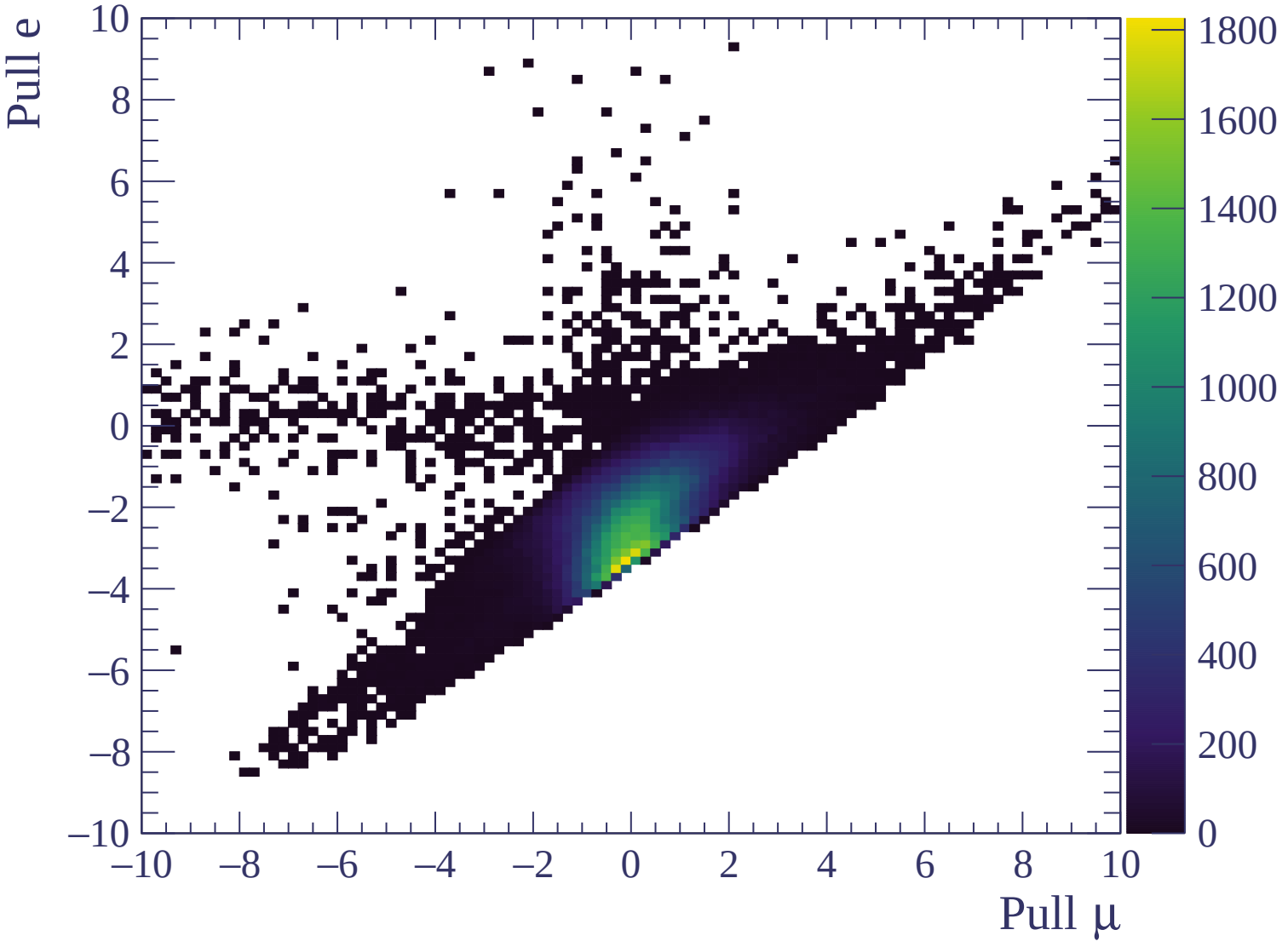


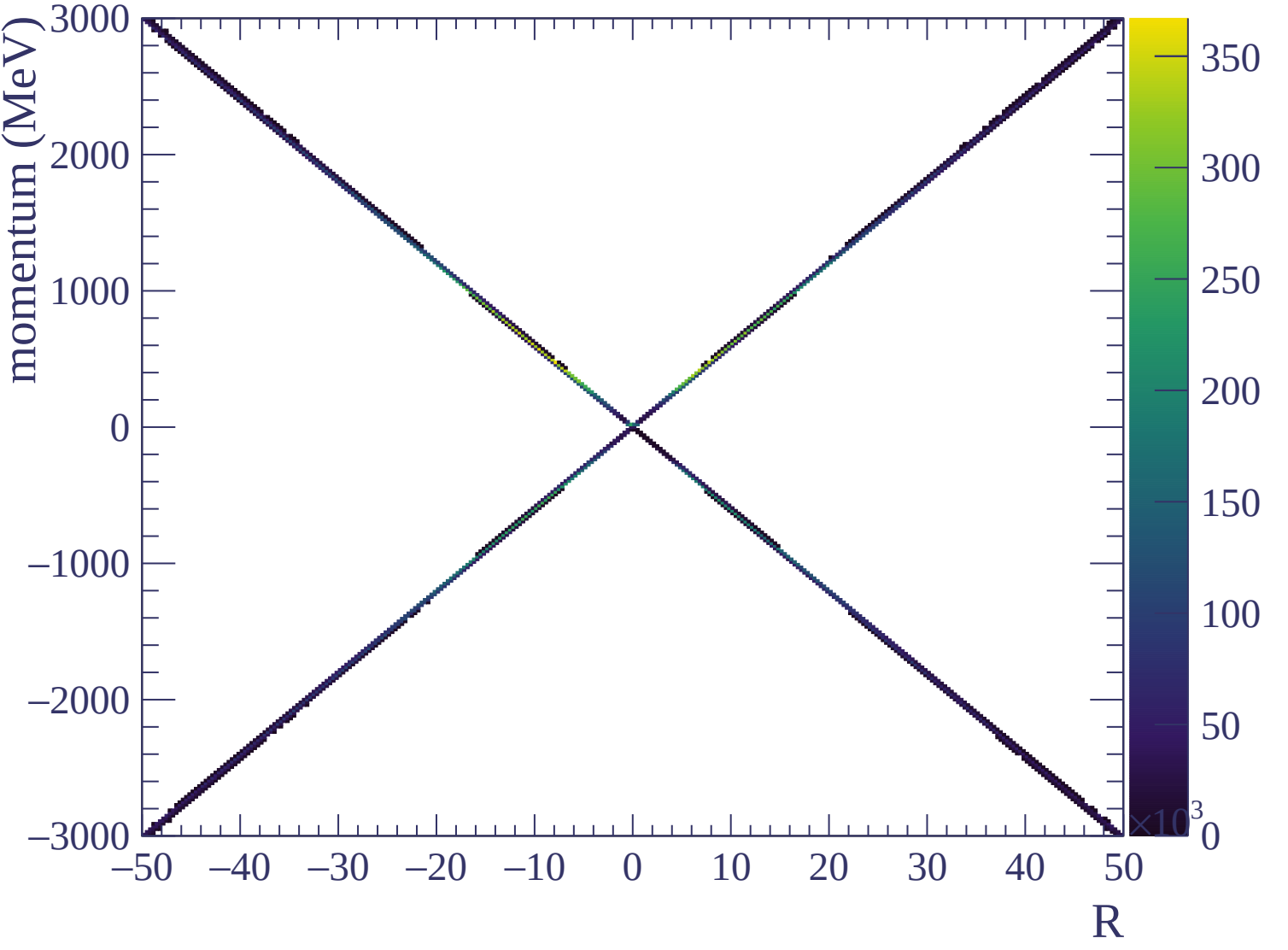


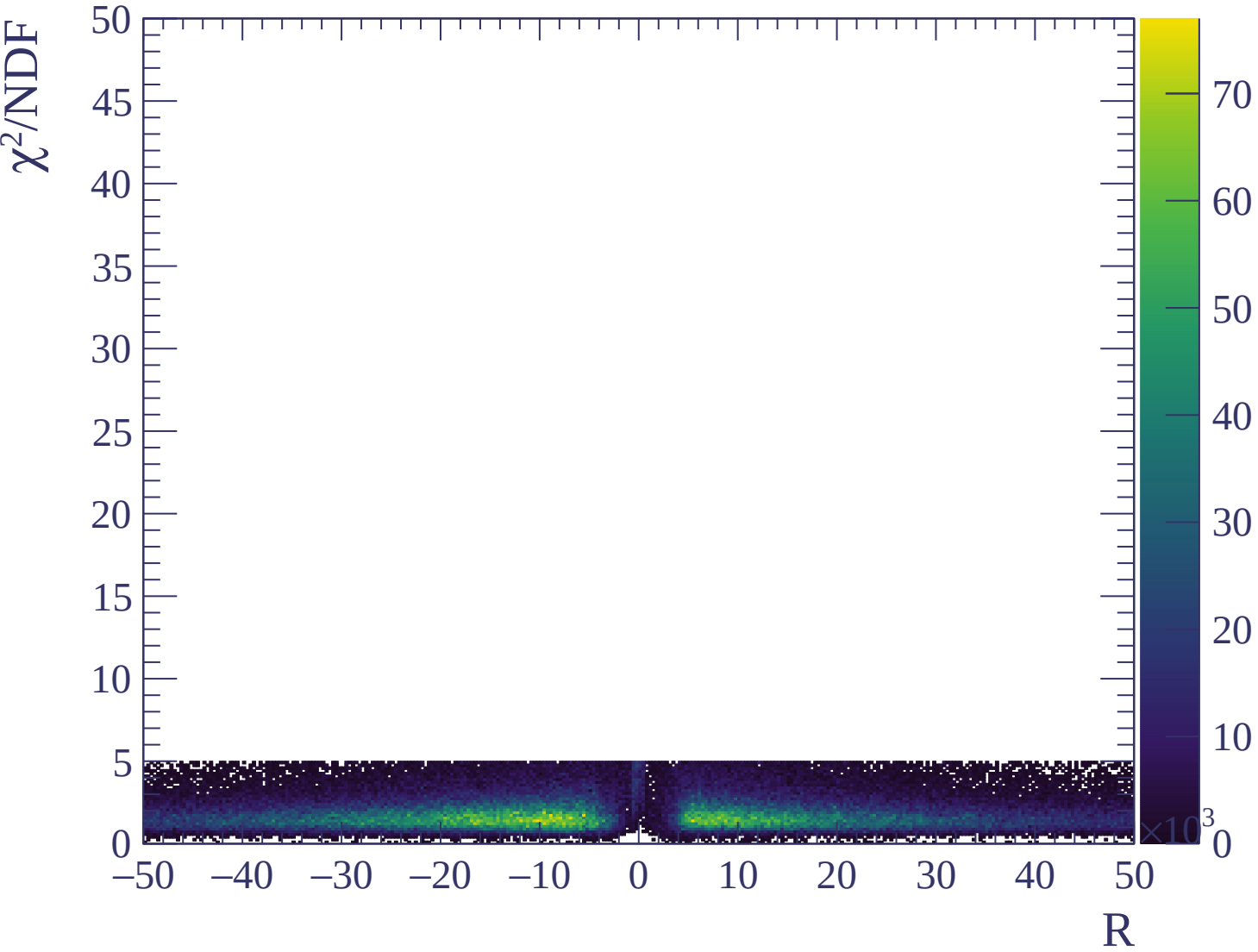






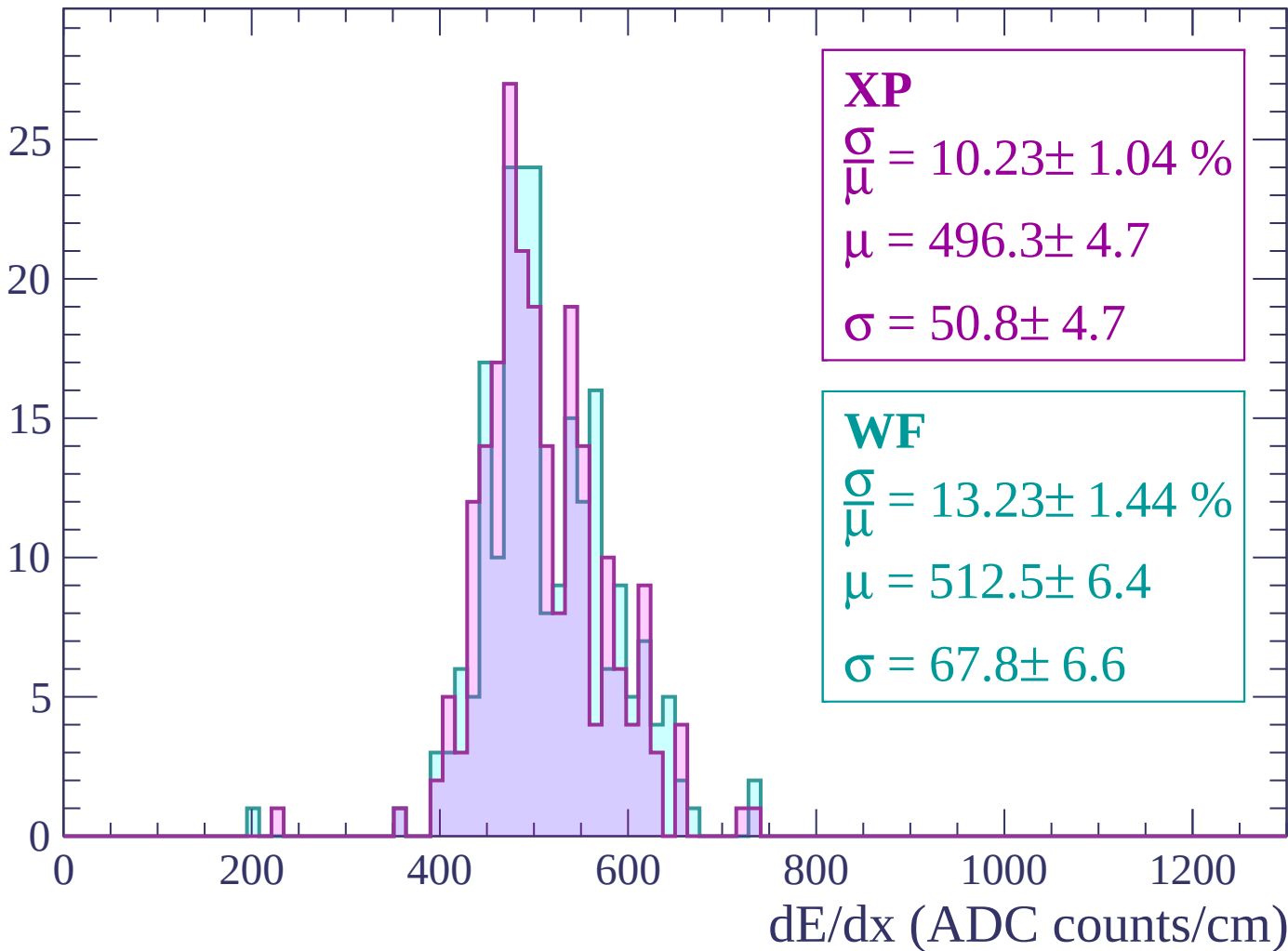






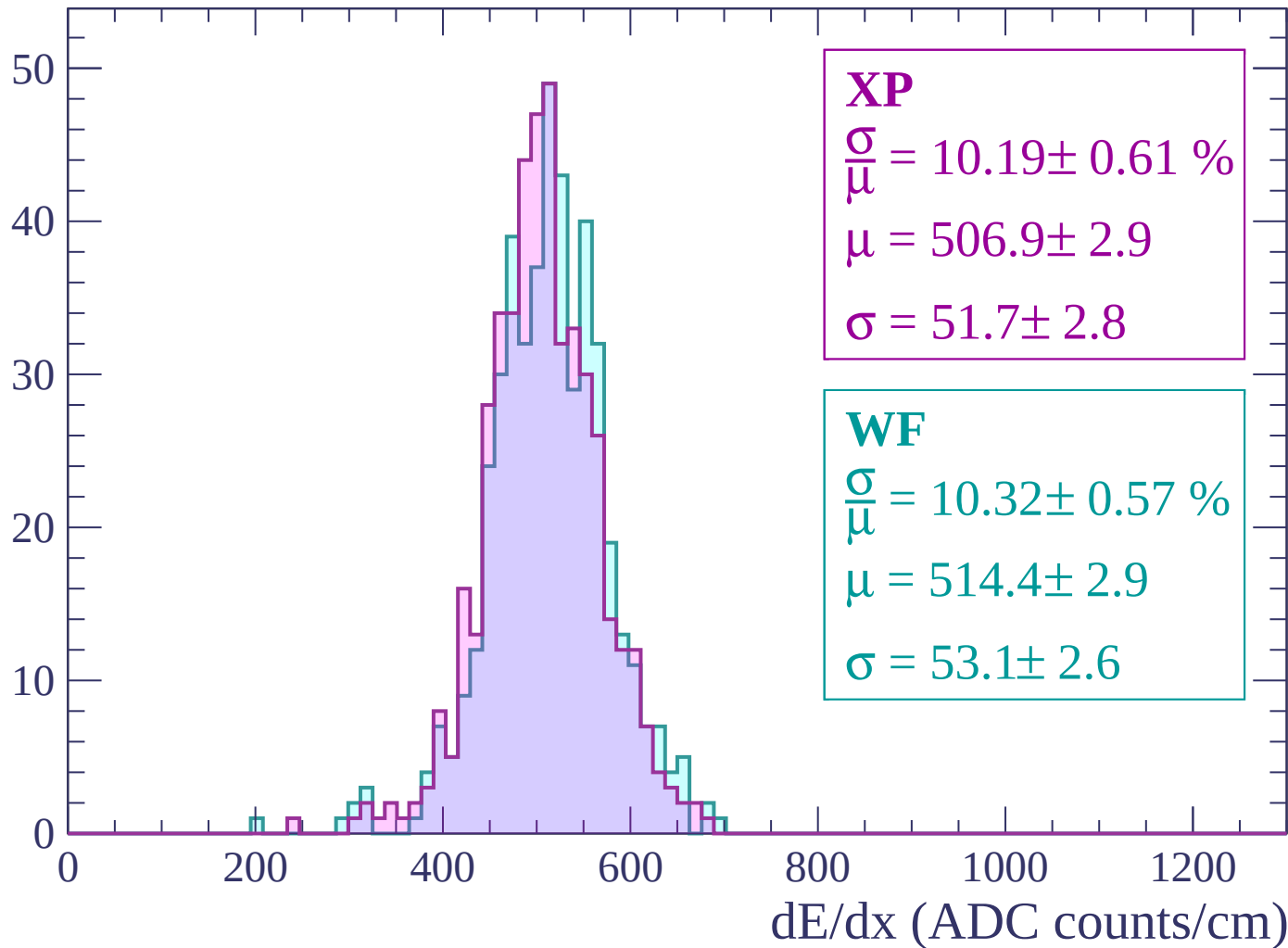
Energy loss | -3000 < p < -2940

Count



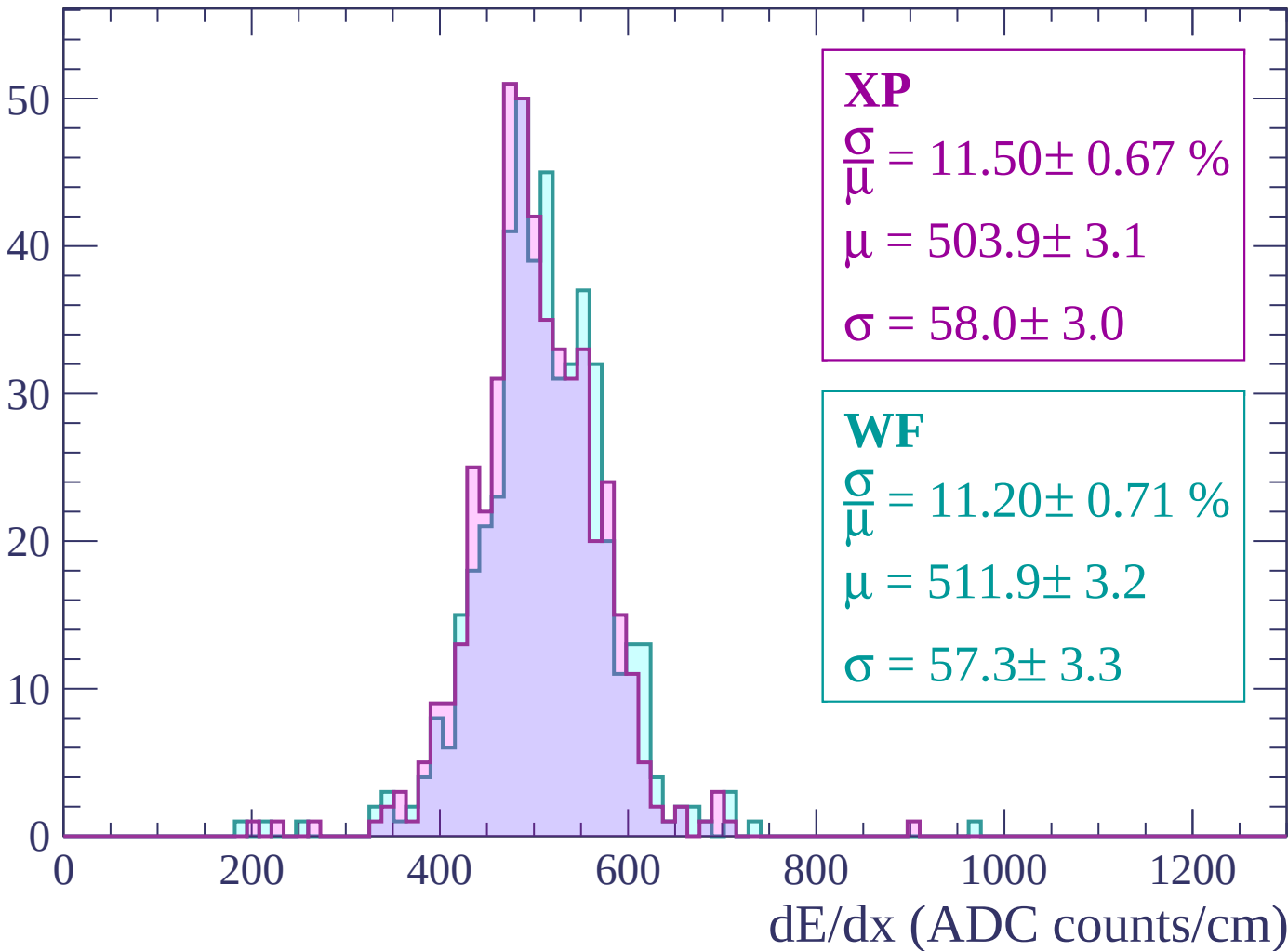
Energy loss | $-2940 < p < -2880$

Count



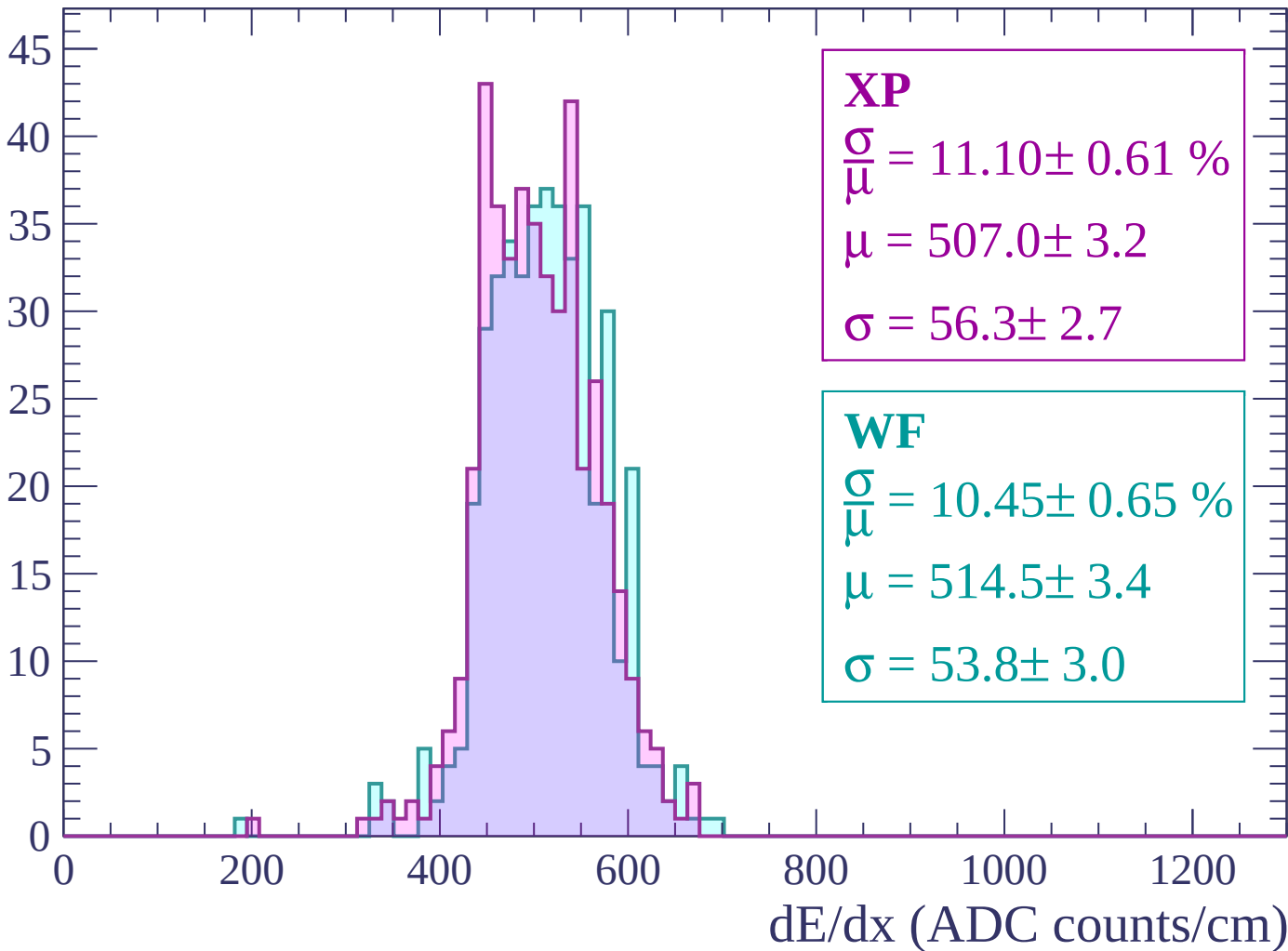
Energy loss | $-2880 < p < -2820$

Count



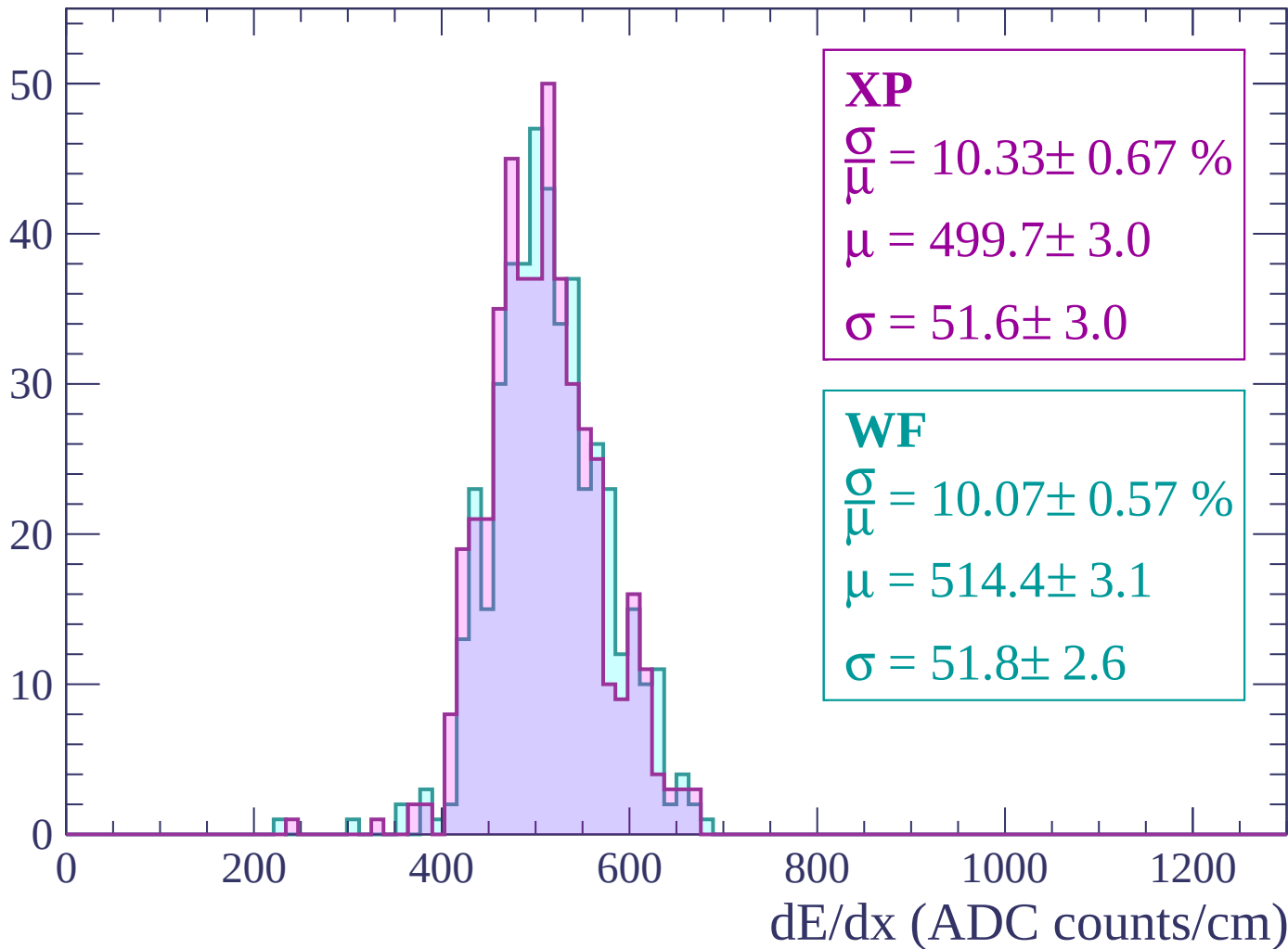
Energy loss | $-2820 < p < -2760$

Count



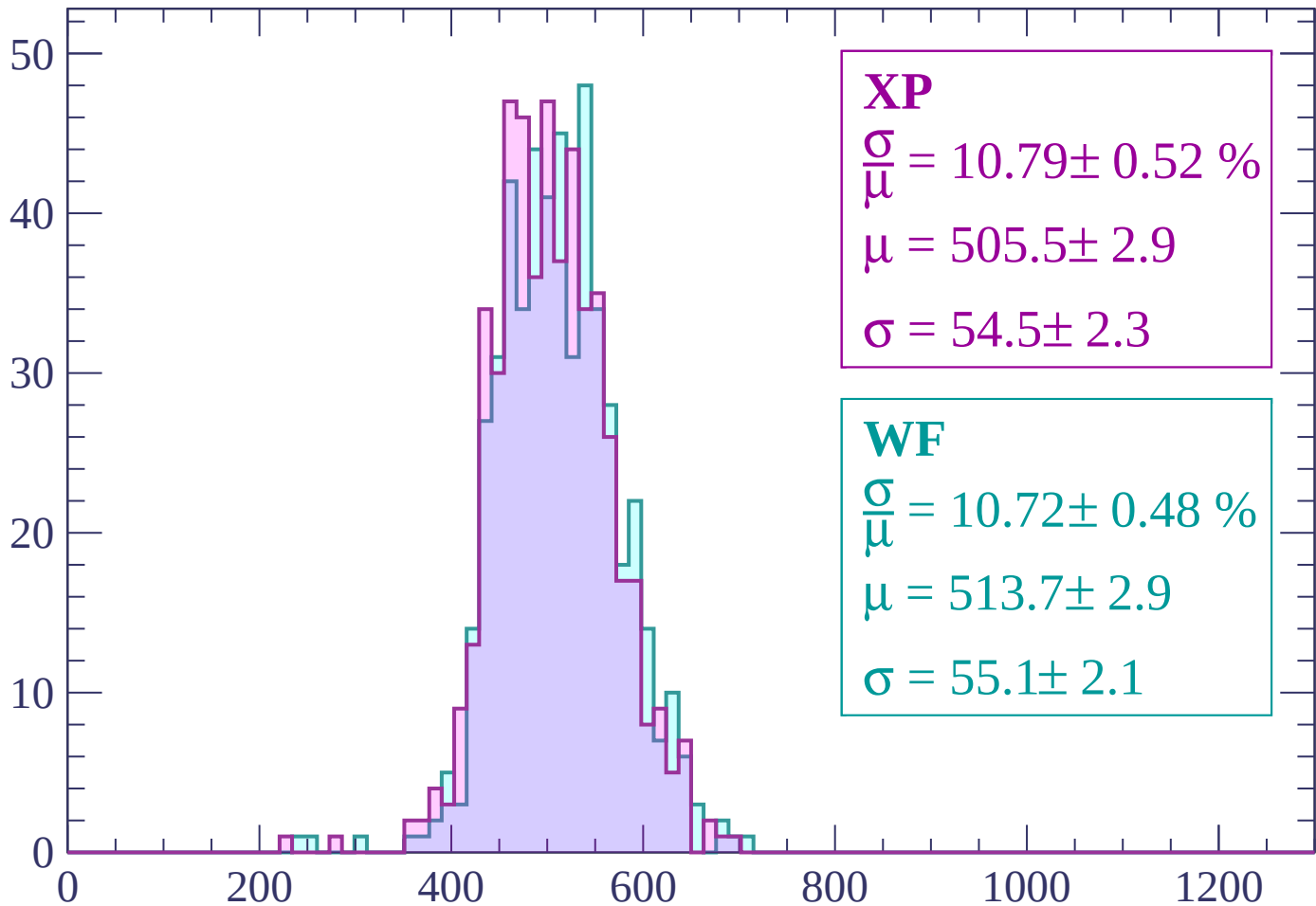
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP

$$\frac{\sigma}{\mu} = 10.79 \pm 0.52 \%$$

$$\mu = 505.5 \pm 2.9$$

$$\sigma = 54.5 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 10.72 \pm 0.48 \%$$

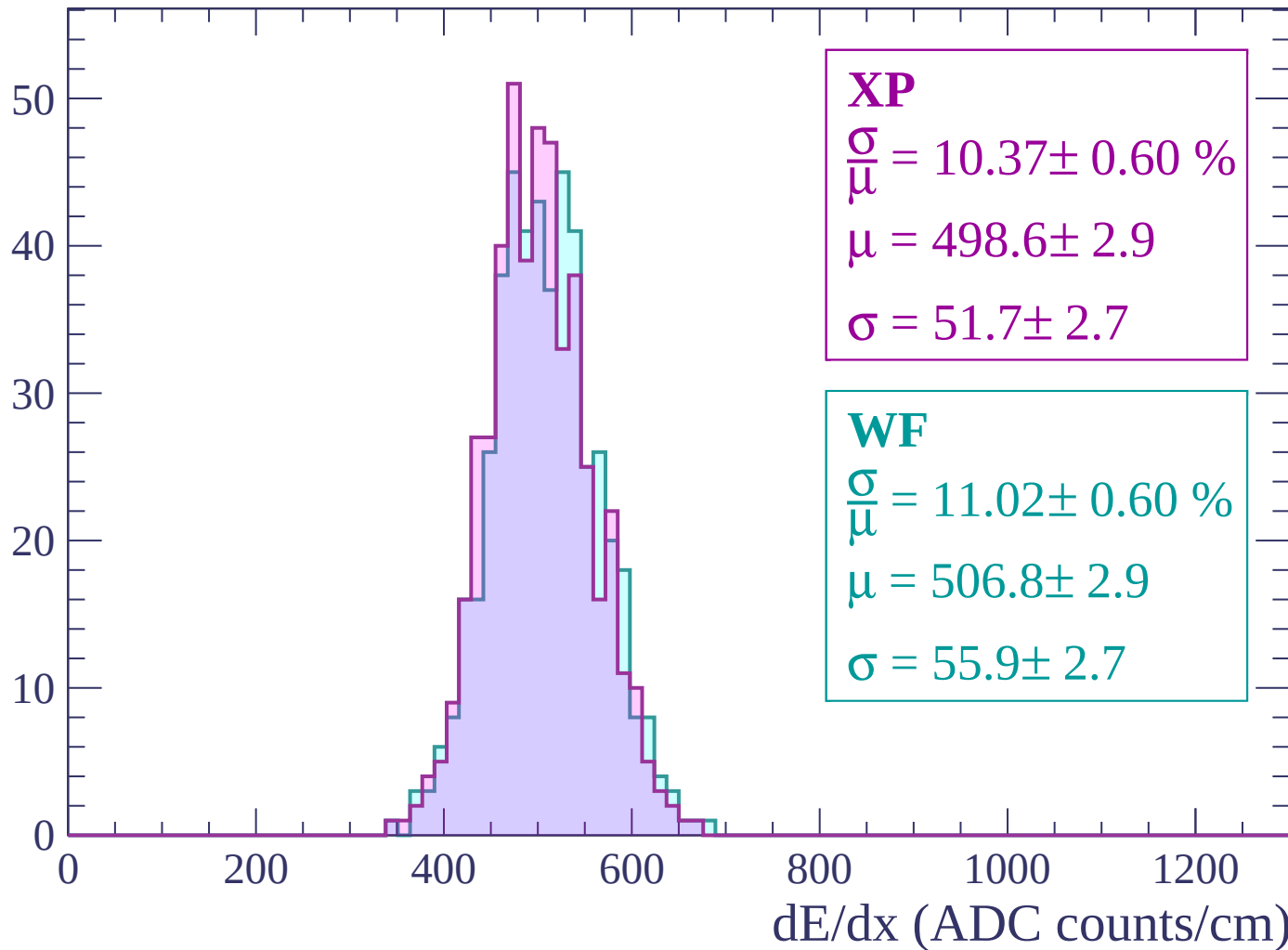
$$\mu = 513.7 \pm 2.9$$

$$\sigma = 55.1 \pm 2.1$$

dE/dx (ADC counts/cm)

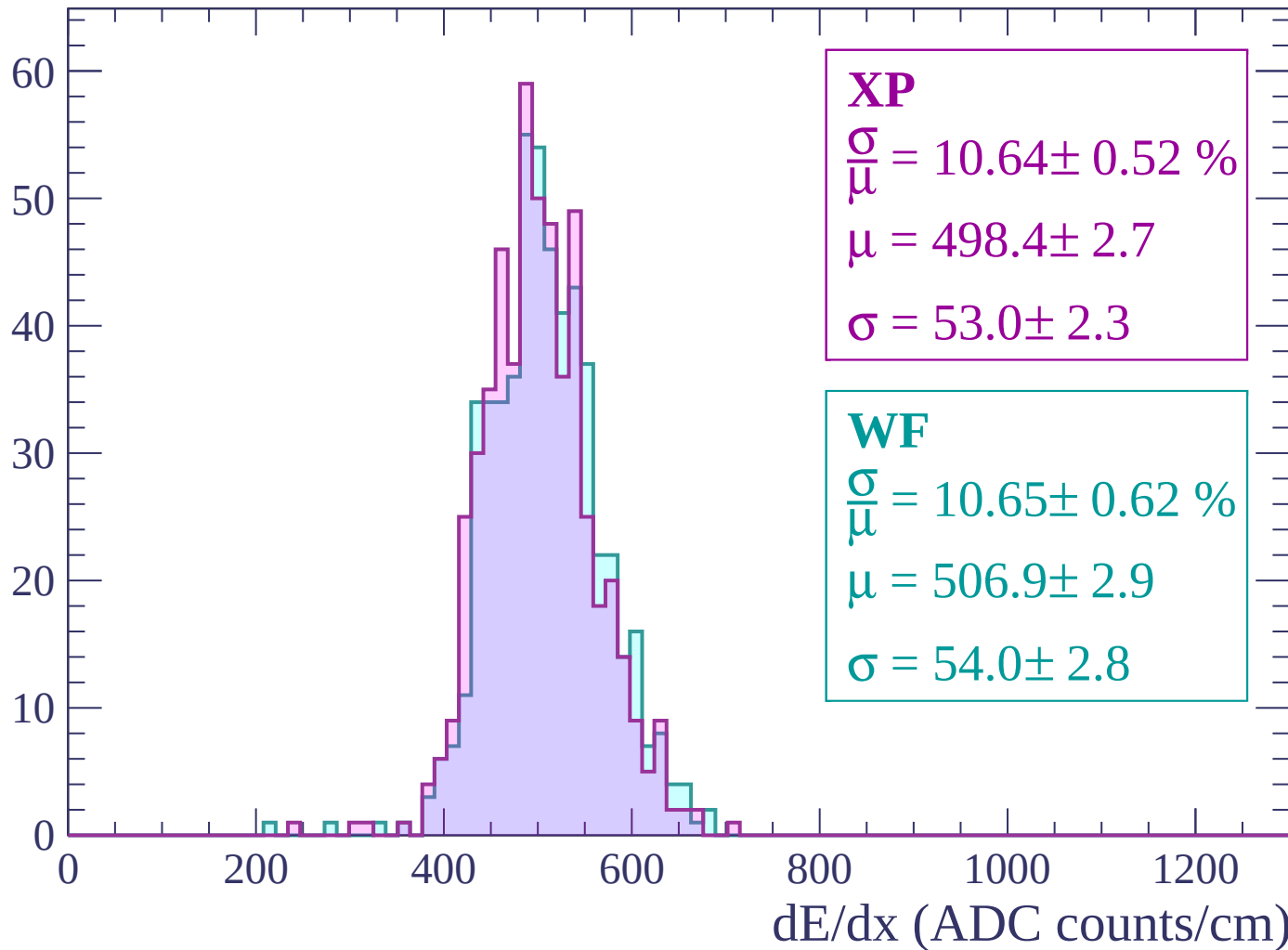
Energy loss | $-2640 < p < -2580$

Count



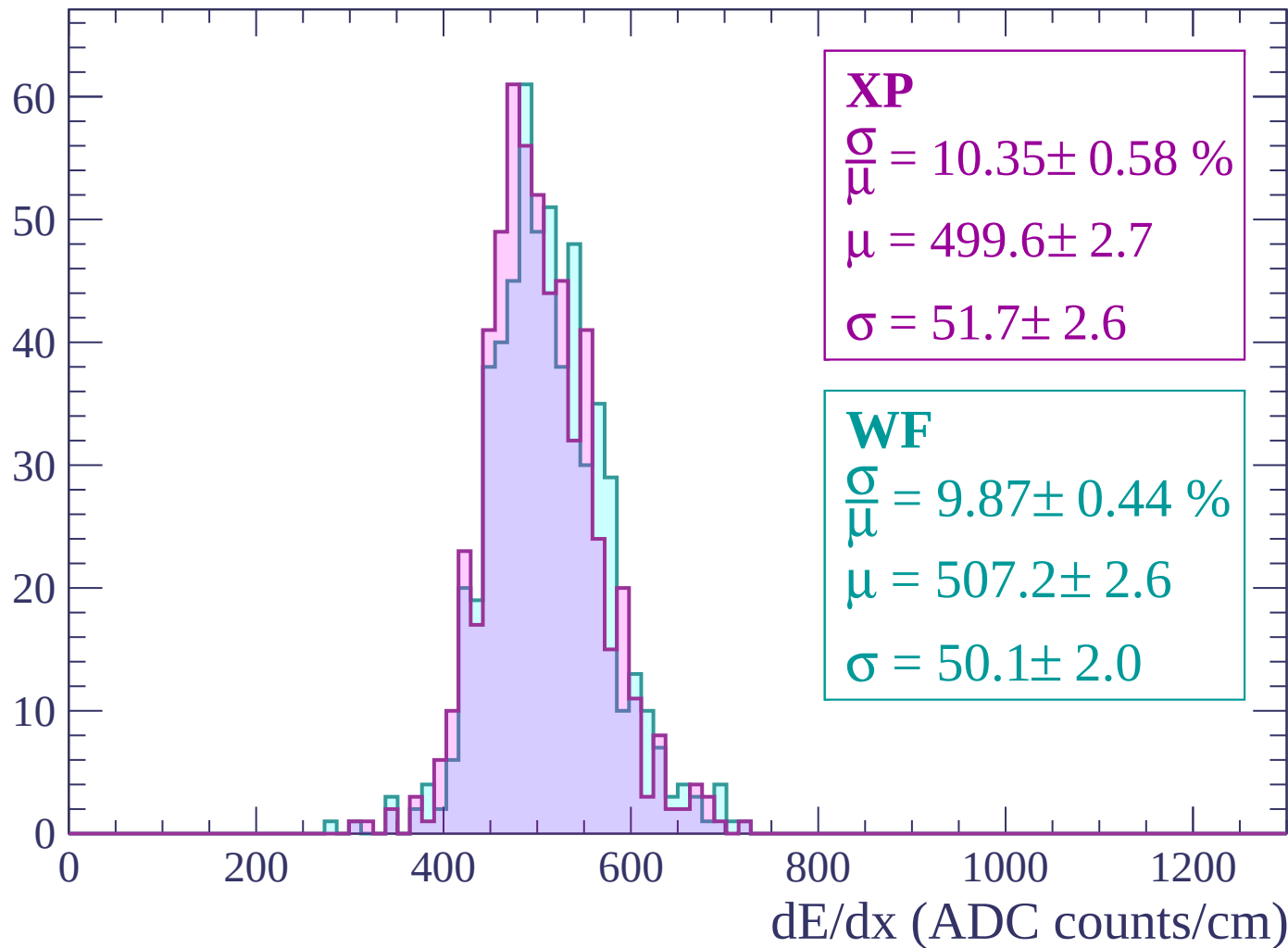
Energy loss | $-2580 < p < -2520$

Count



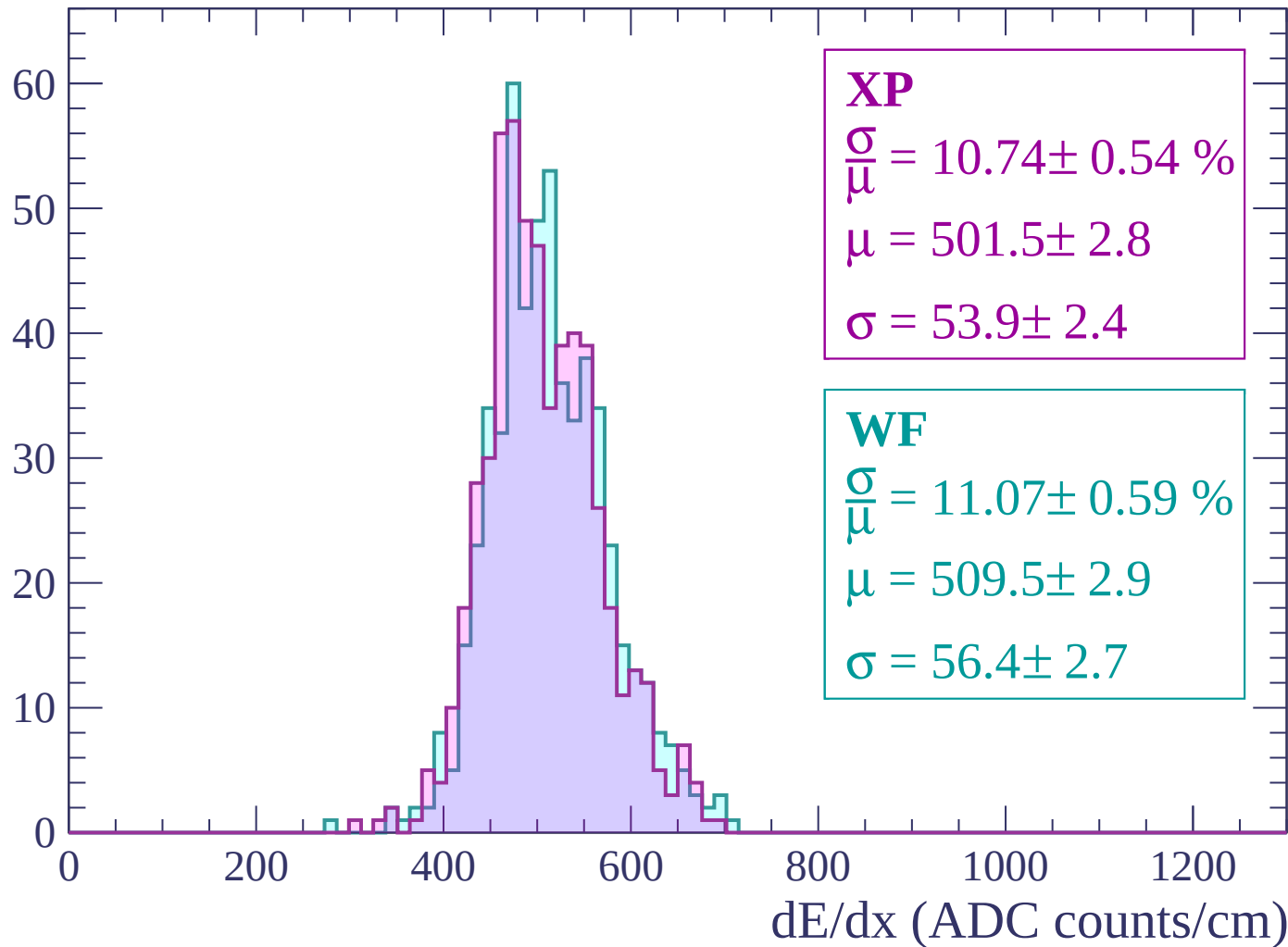
Energy loss | -2520 < p < -2460

Count



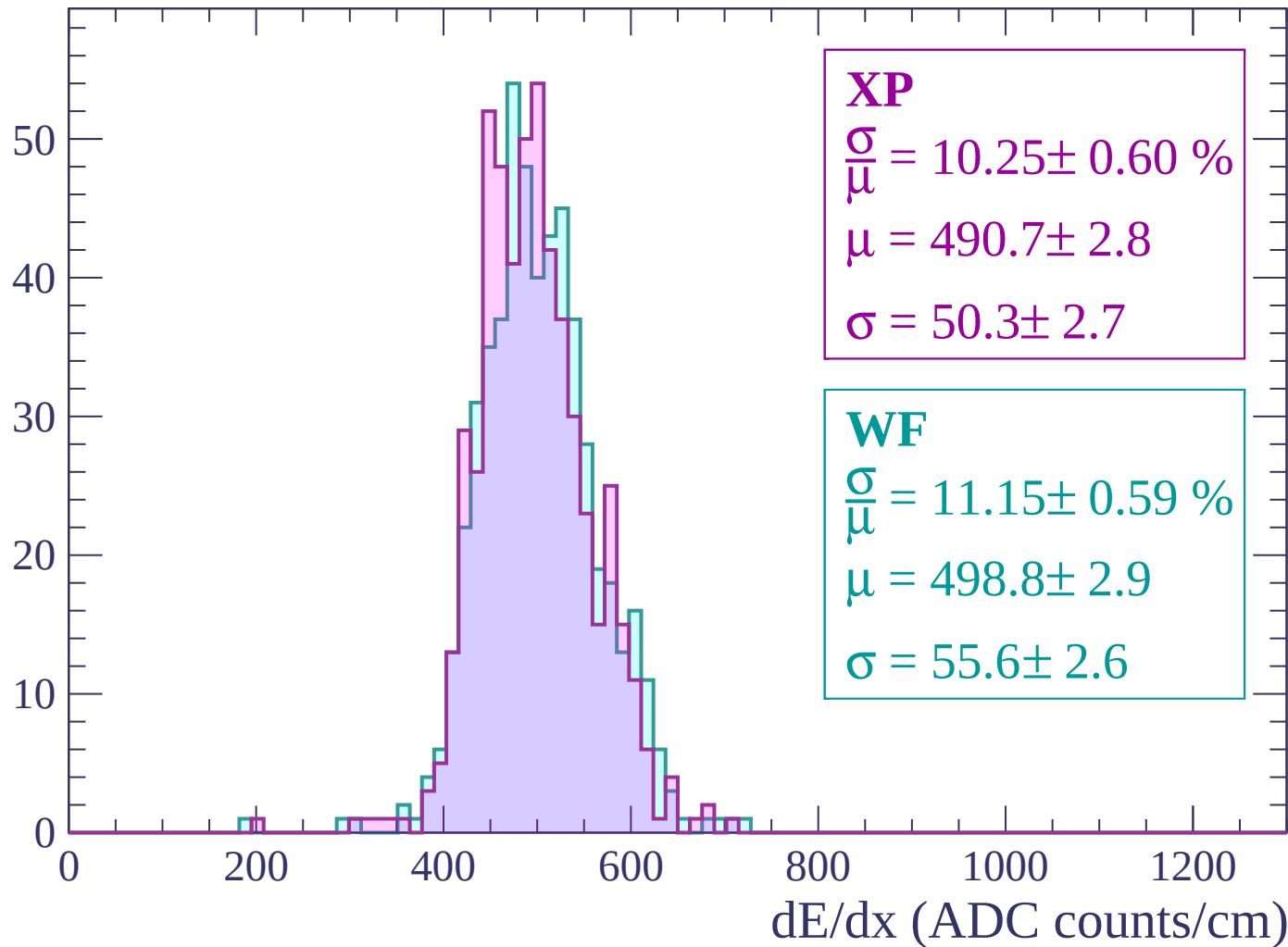
Energy loss | $-2460 < p < -2400$

Count



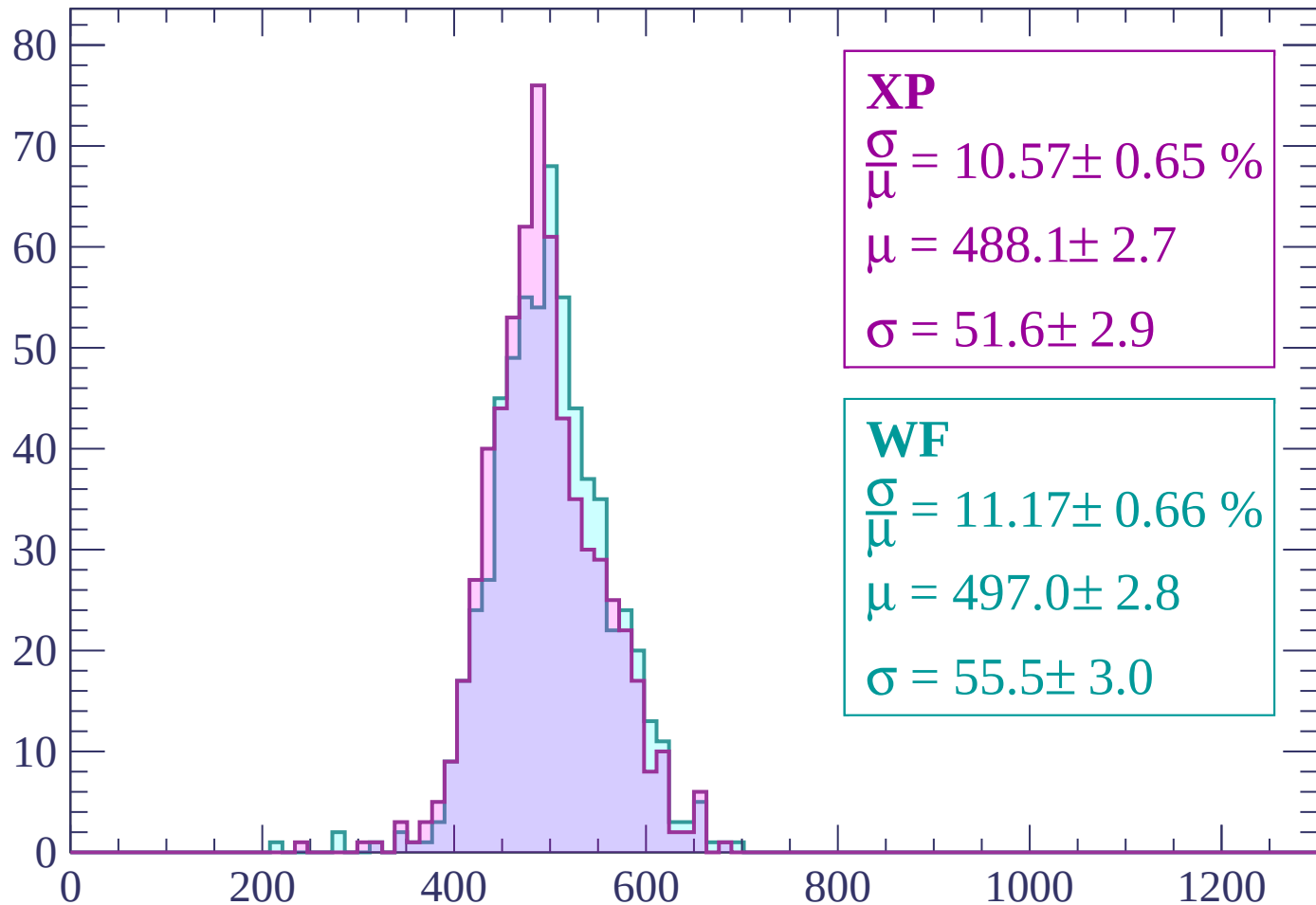
Energy loss | -2400 < p < -2340

Count



Energy loss | $-2340 < p < -2280$

Count



XP

$$\frac{\sigma}{\mu} = 10.57 \pm 0.65 \%$$

$$\mu = 488.1 \pm 2.7$$

$$\sigma = 51.6 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 11.17 \pm 0.66 \%$$

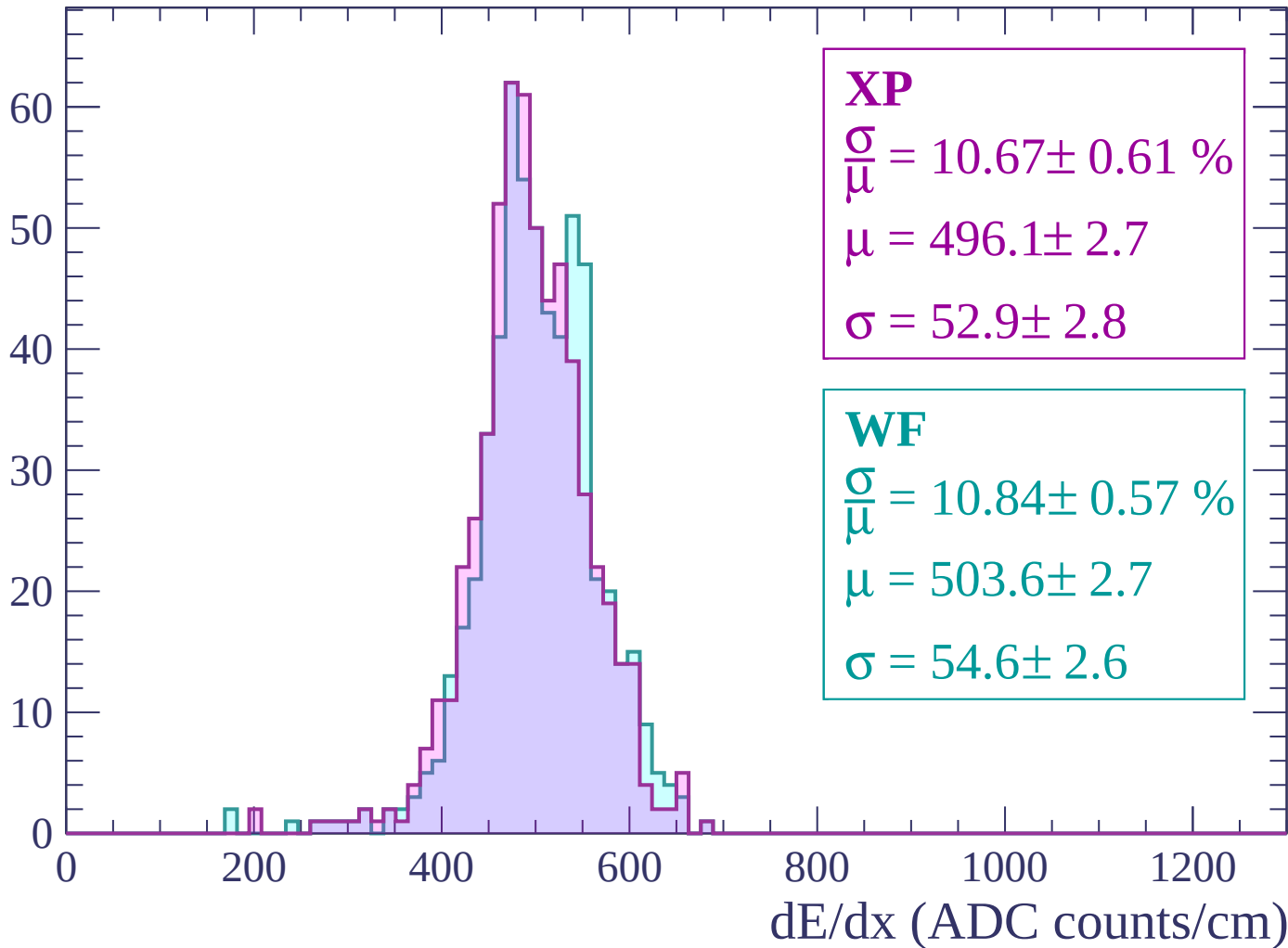
$$\mu = 497.0 \pm 2.8$$

$$\sigma = 55.5 \pm 3.0$$

dE/dx (ADC counts/cm)

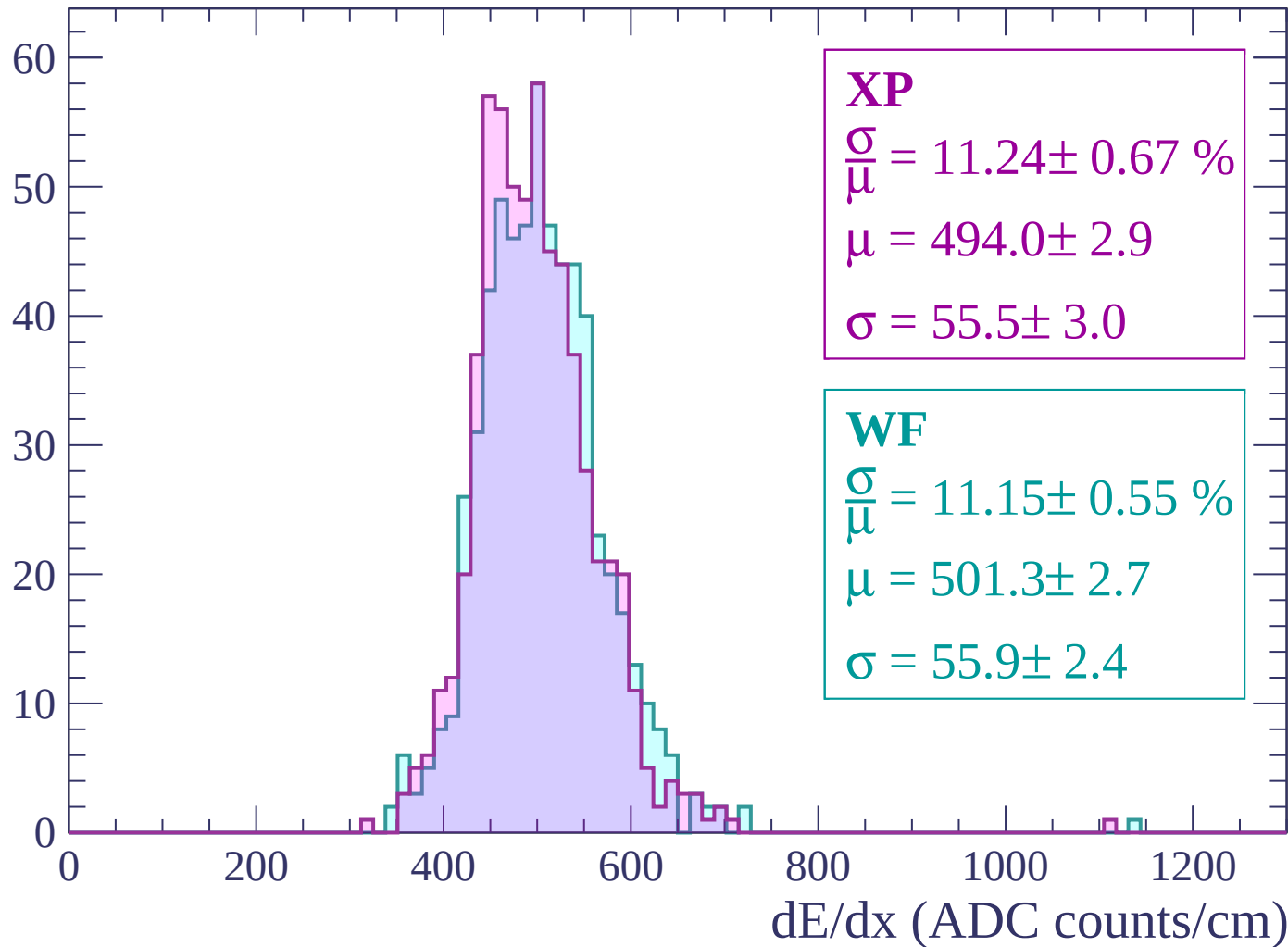
Energy loss | -2280 < p < -2220

Count



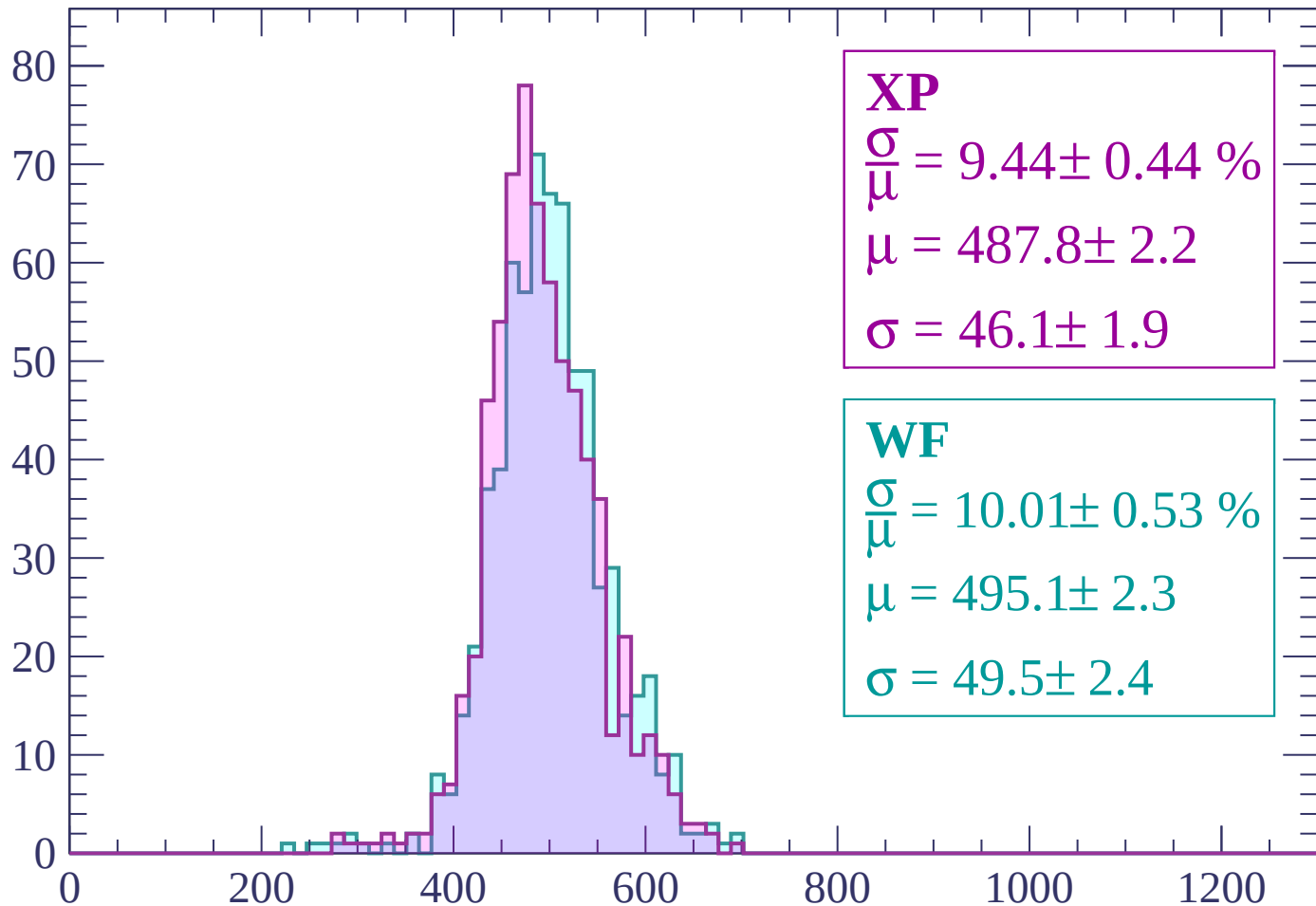
Energy loss | -2220 < p < -2160

Count



Energy loss | -2160 < p < -2100

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.44 \pm 0.44 \%$$

$$\mu = 487.8 \pm 2.2$$

$$\sigma = 46.1 \pm 1.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.01 \pm 0.53 \%$$

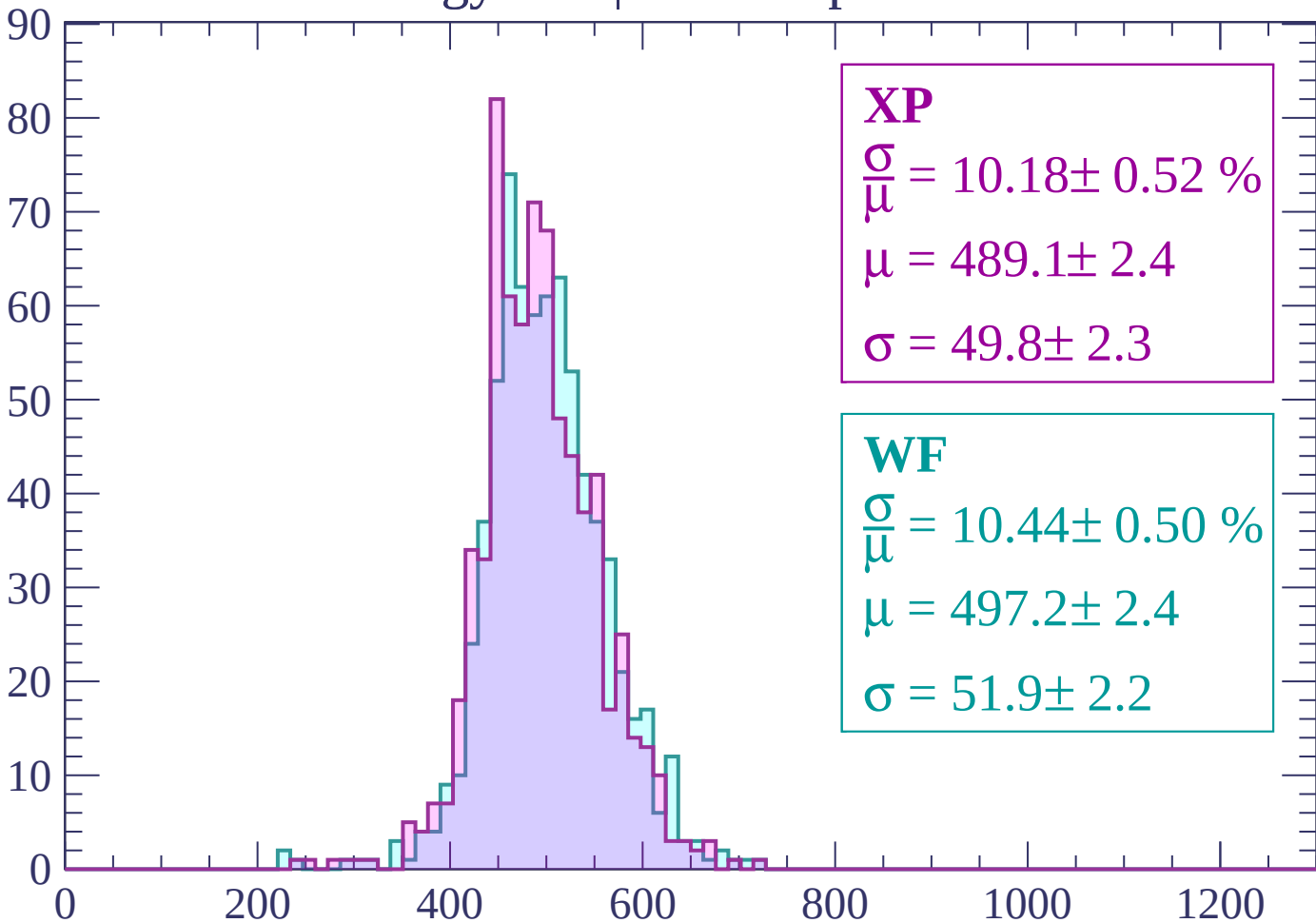
$$\mu = 495.1 \pm 2.3$$

$$\sigma = 49.5 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.52 \%$$

$$\mu = 489.1 \pm 2.4$$

$$\sigma = 49.8 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 10.44 \pm 0.50 \%$$

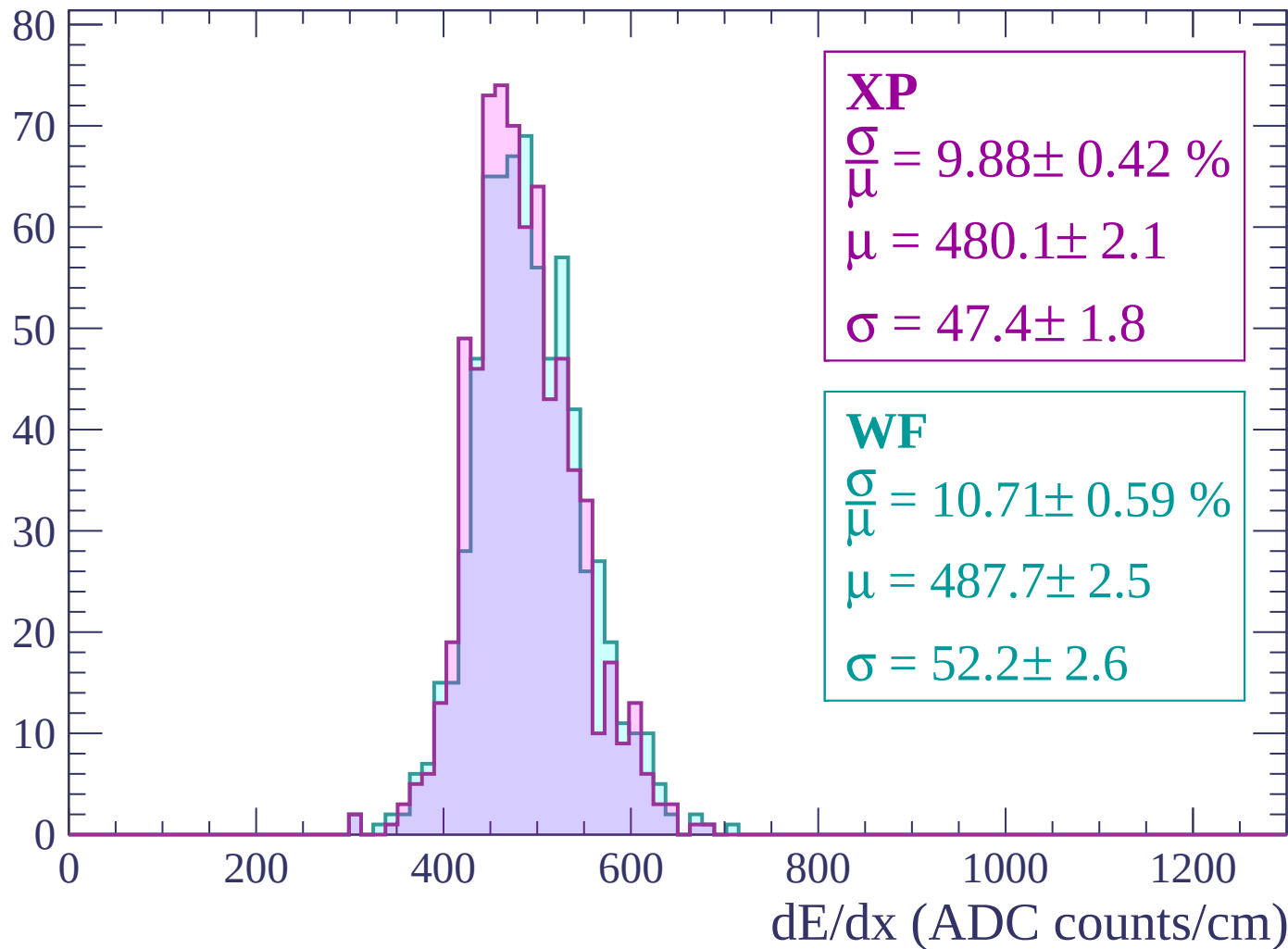
$$\mu = 497.2 \pm 2.4$$

$$\sigma = 51.9 \pm 2.2$$

dE/dx (ADC counts/cm)

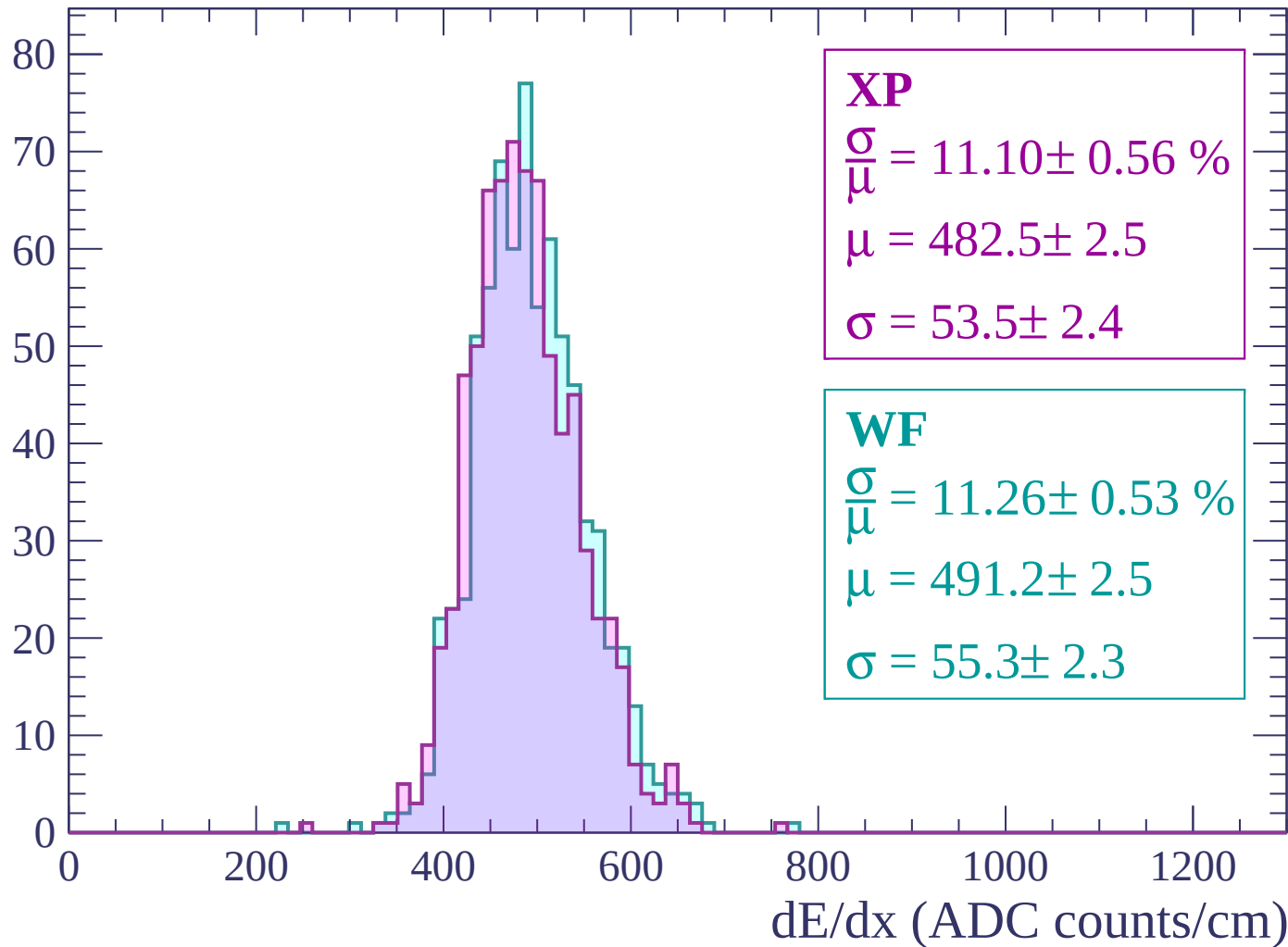
Energy loss | -2040 < p < -1980

Count



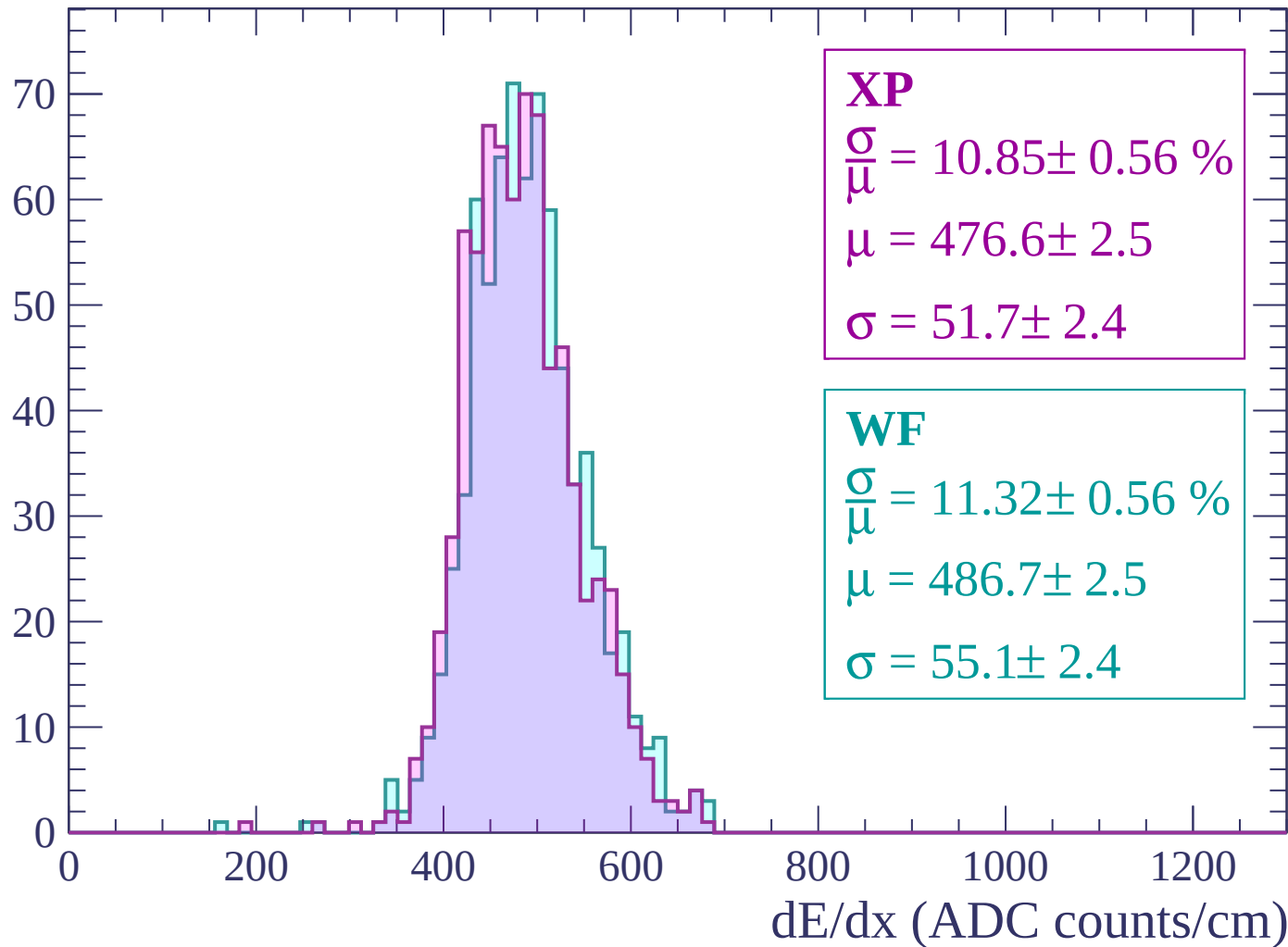
Energy loss | -1980 < p < -1920

Count



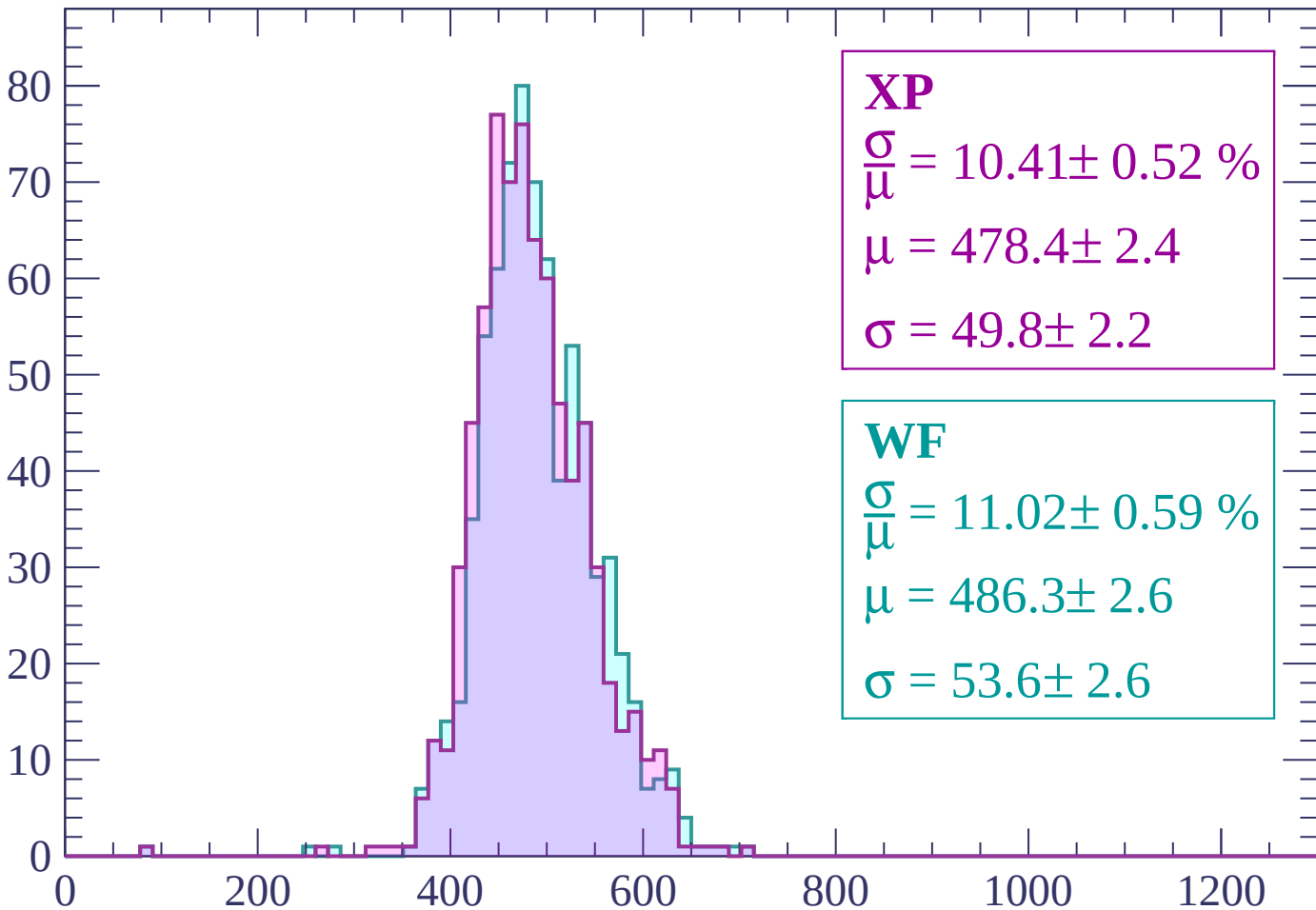
Energy loss | -1920 < p < -1860

Count



Energy loss | $-1860 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 10.41 \pm 0.52 \%$$

$$\mu = 478.4 \pm 2.4$$

$$\sigma = 49.8 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 11.02 \pm 0.59 \%$$

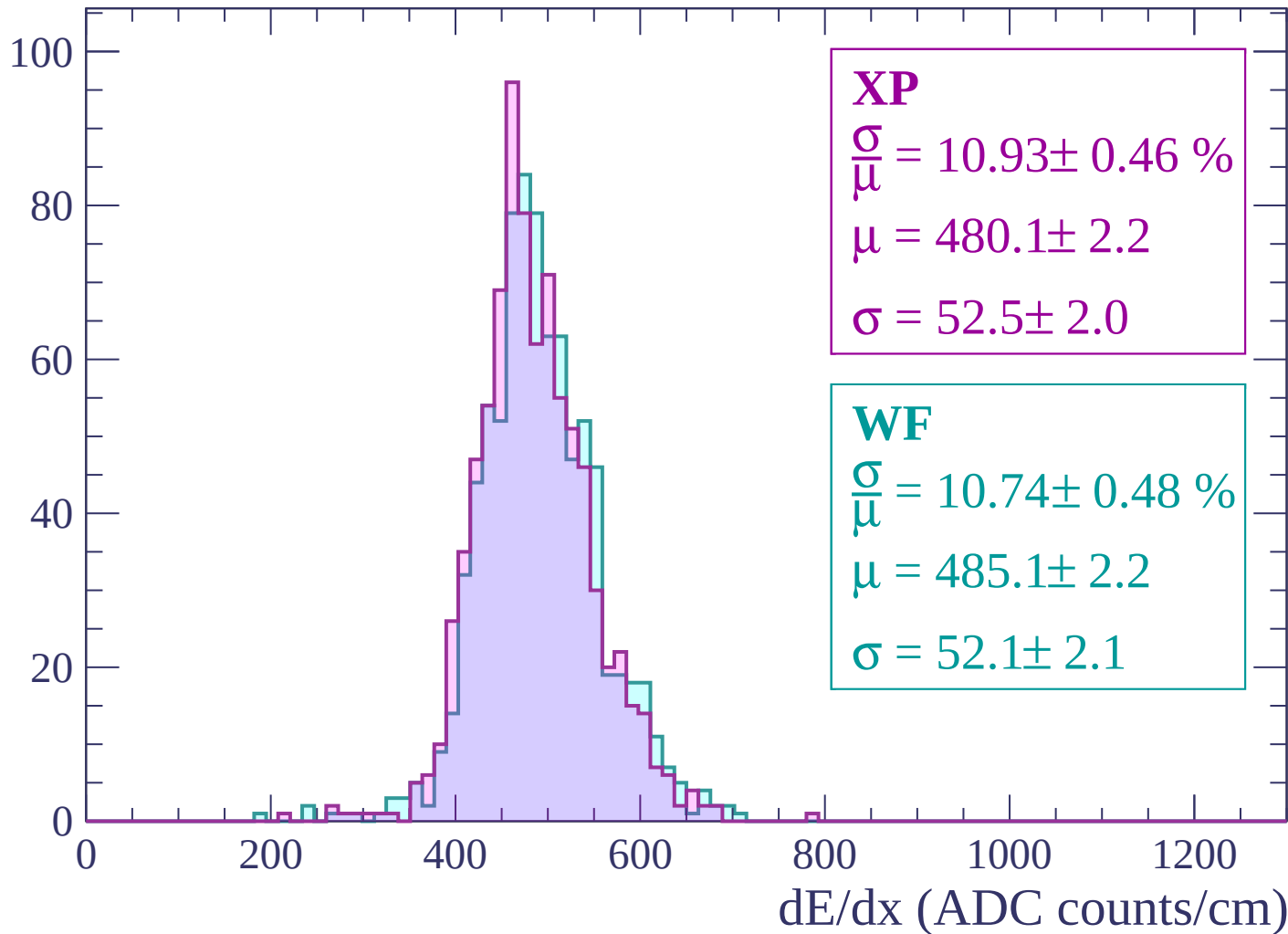
$$\mu = 486.3 \pm 2.6$$

$$\sigma = 53.6 \pm 2.6$$

dE/dx (ADC counts/cm)

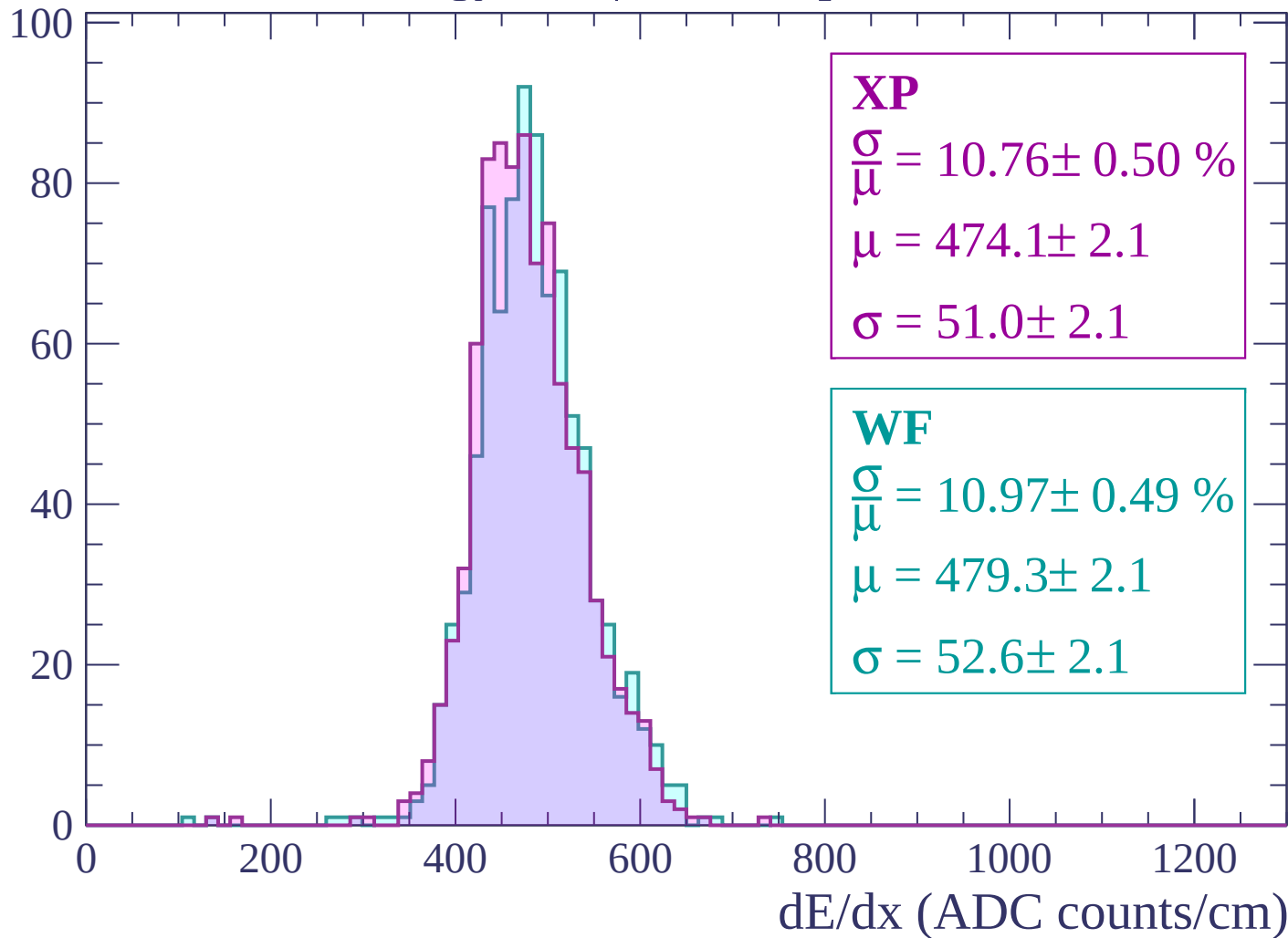
Energy loss | -1800 < p < -1740

Count



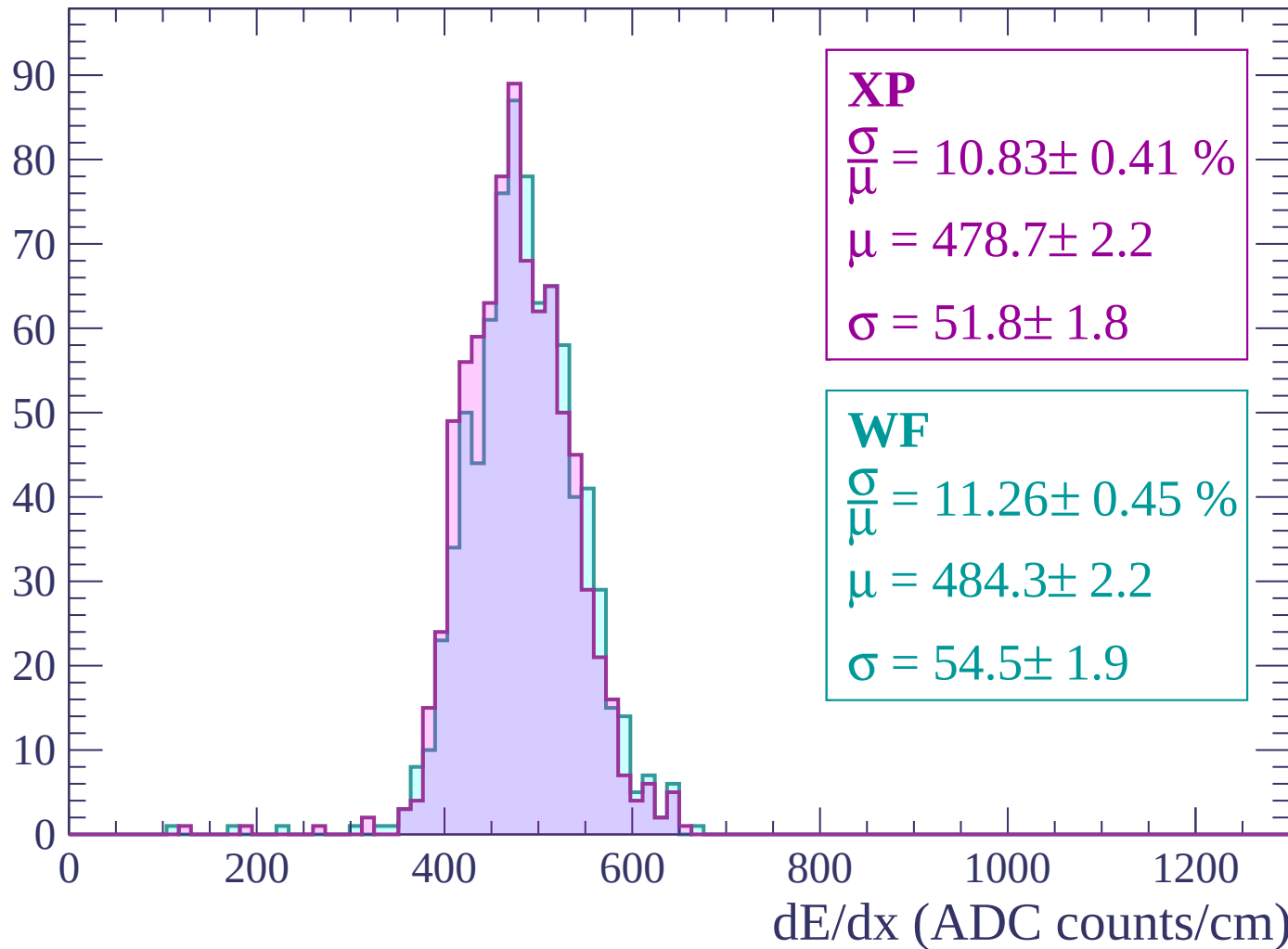
Energy loss | -1740 < p < -1680

Count



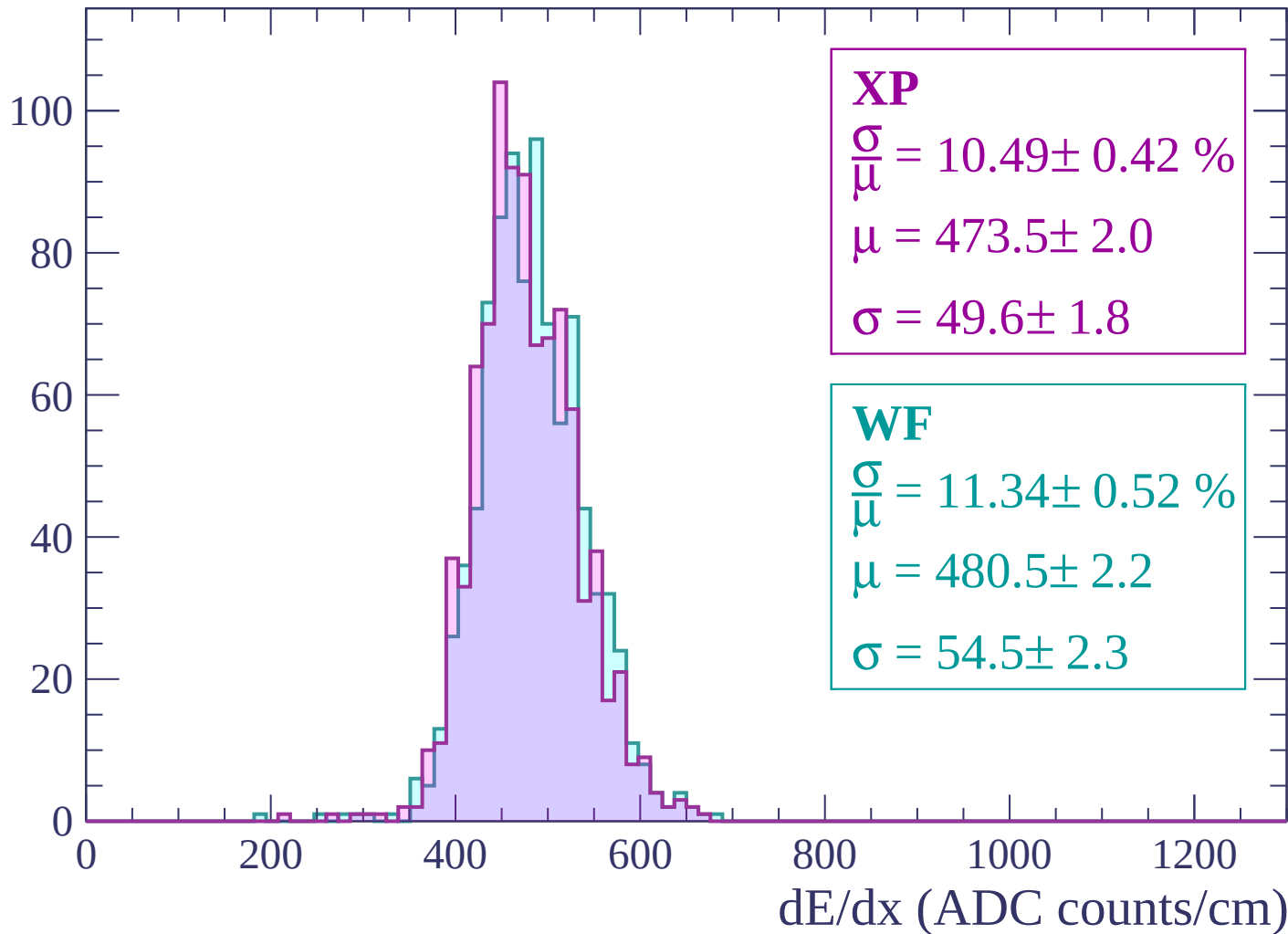
Energy loss | -1680 < p < -1620

Count



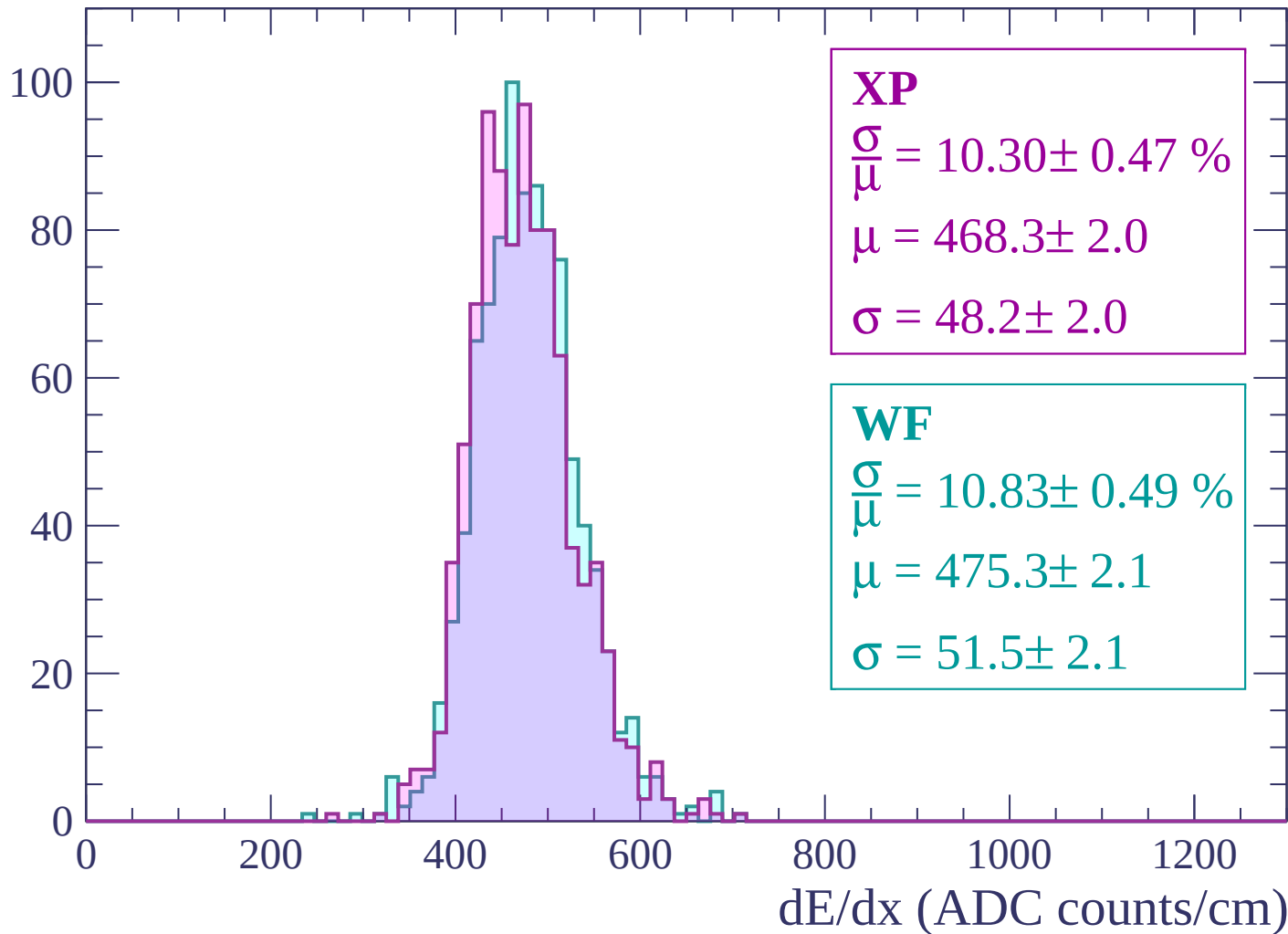
Energy loss | -1620 < p < -1560

Count



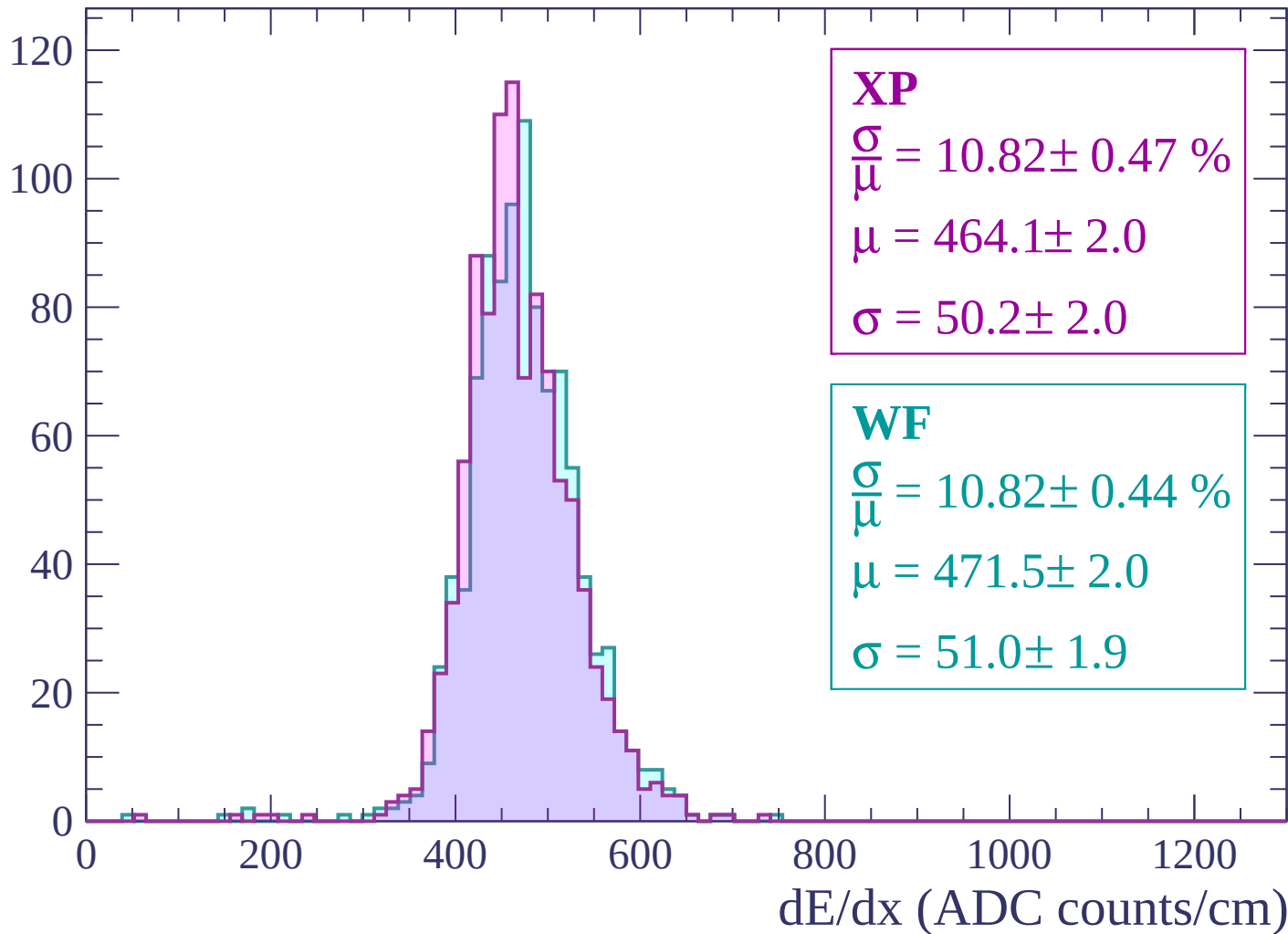
Energy loss | -1560 < p < -1500

Count



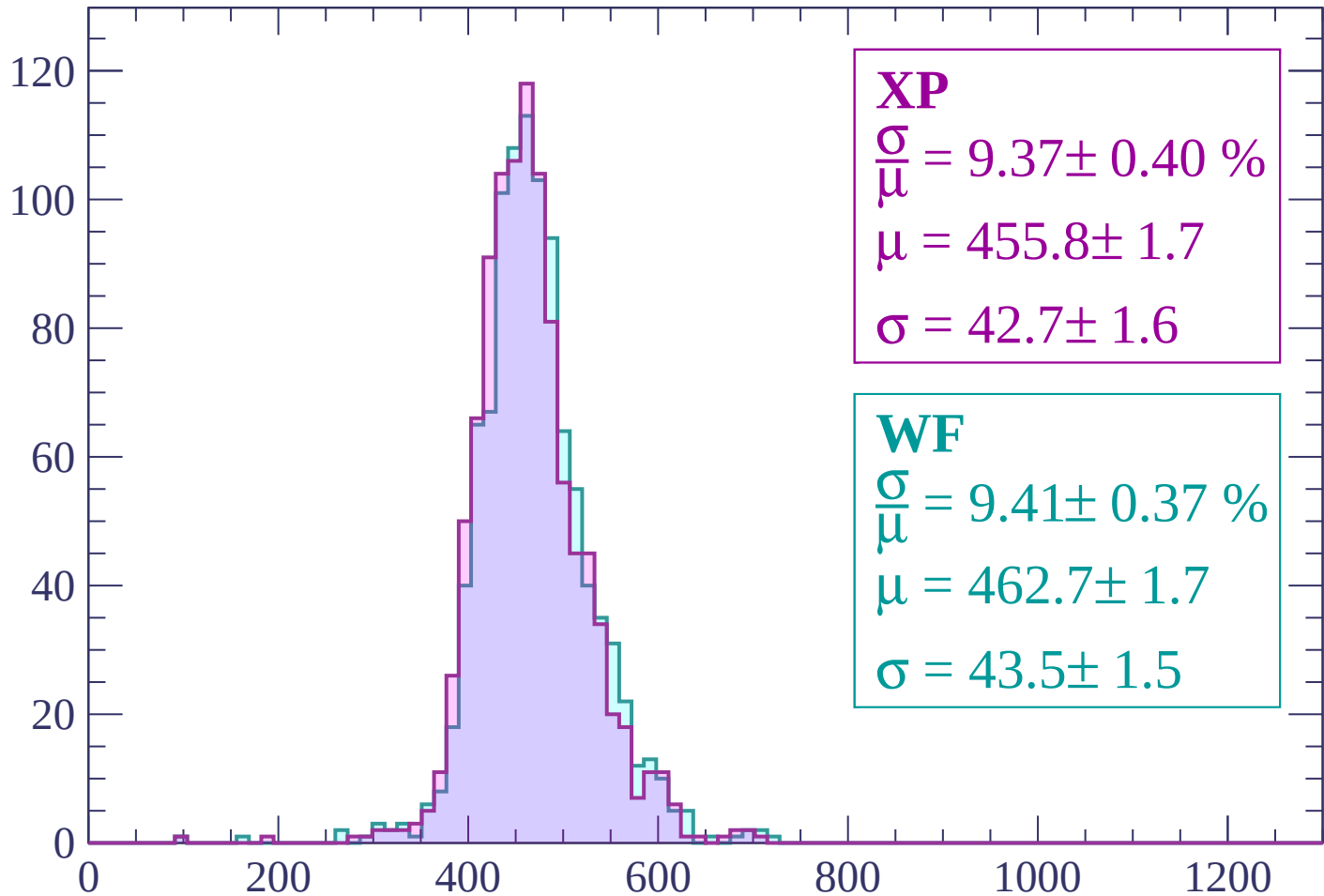
Energy loss | -1500 < p < -1440

Count



Energy loss | $-1440 < p < -1380$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.40 \%$$

$$\mu = 455.8 \pm 1.7$$

$$\sigma = 42.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.41 \pm 0.37 \%$$

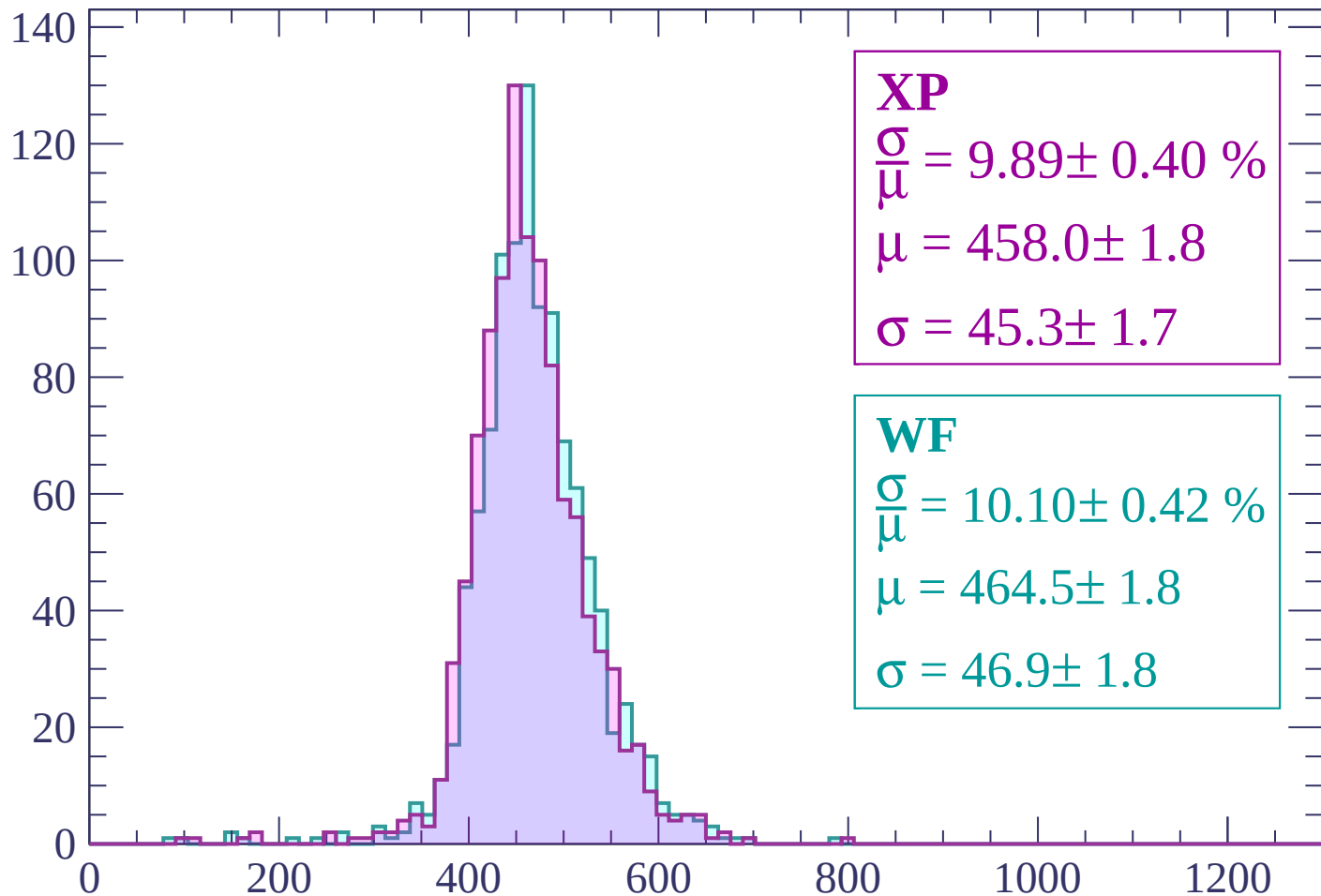
$$\mu = 462.7 \pm 1.7$$

$$\sigma = 43.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP

$$\frac{\sigma}{\mu} = 9.89 \pm 0.40 \%$$

$$\mu = 458.0 \pm 1.8$$

$$\sigma = 45.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.42 \%$$

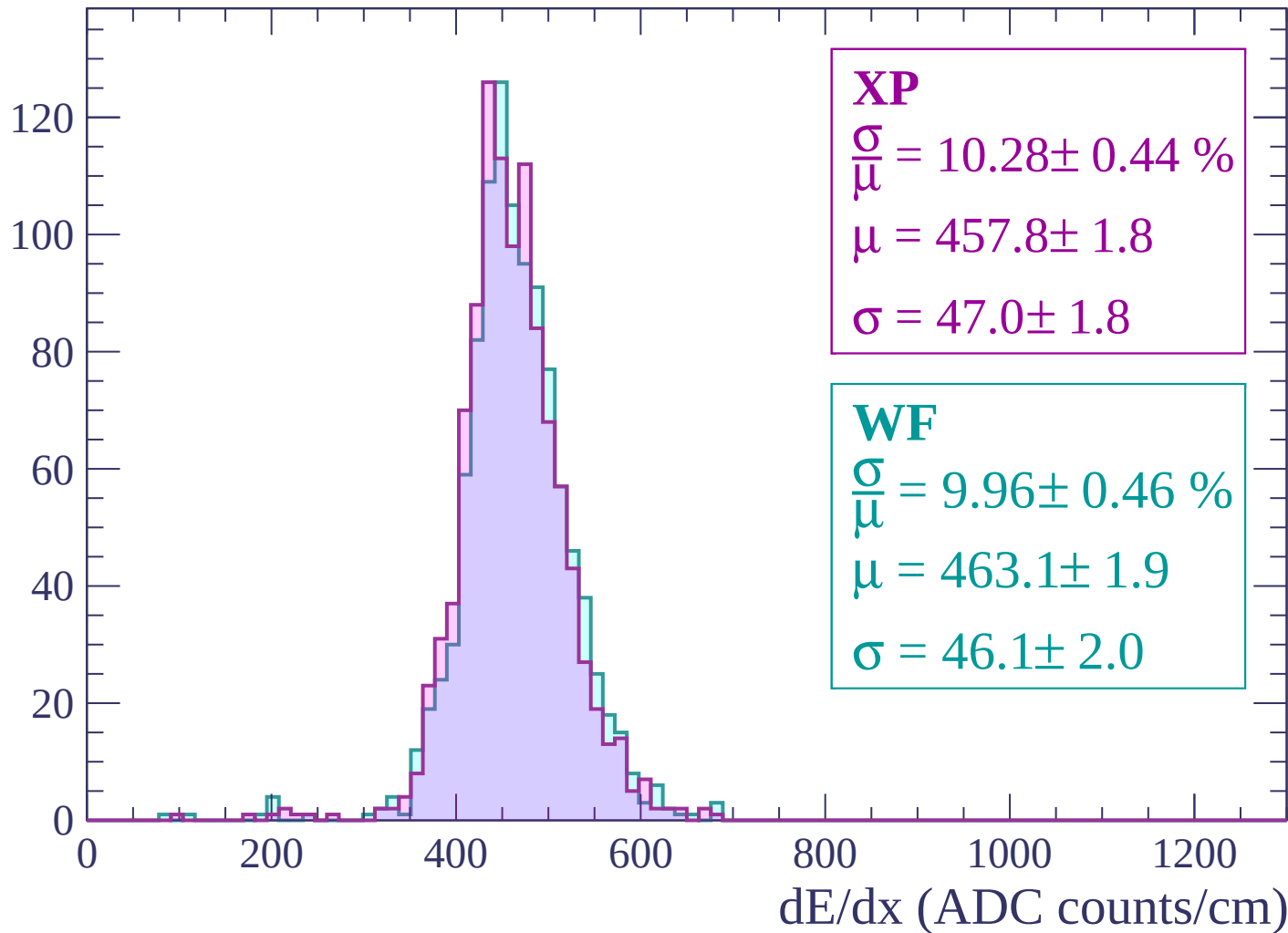
$$\mu = 464.5 \pm 1.8$$

$$\sigma = 46.9 \pm 1.8$$

dE/dx (ADC counts/cm)

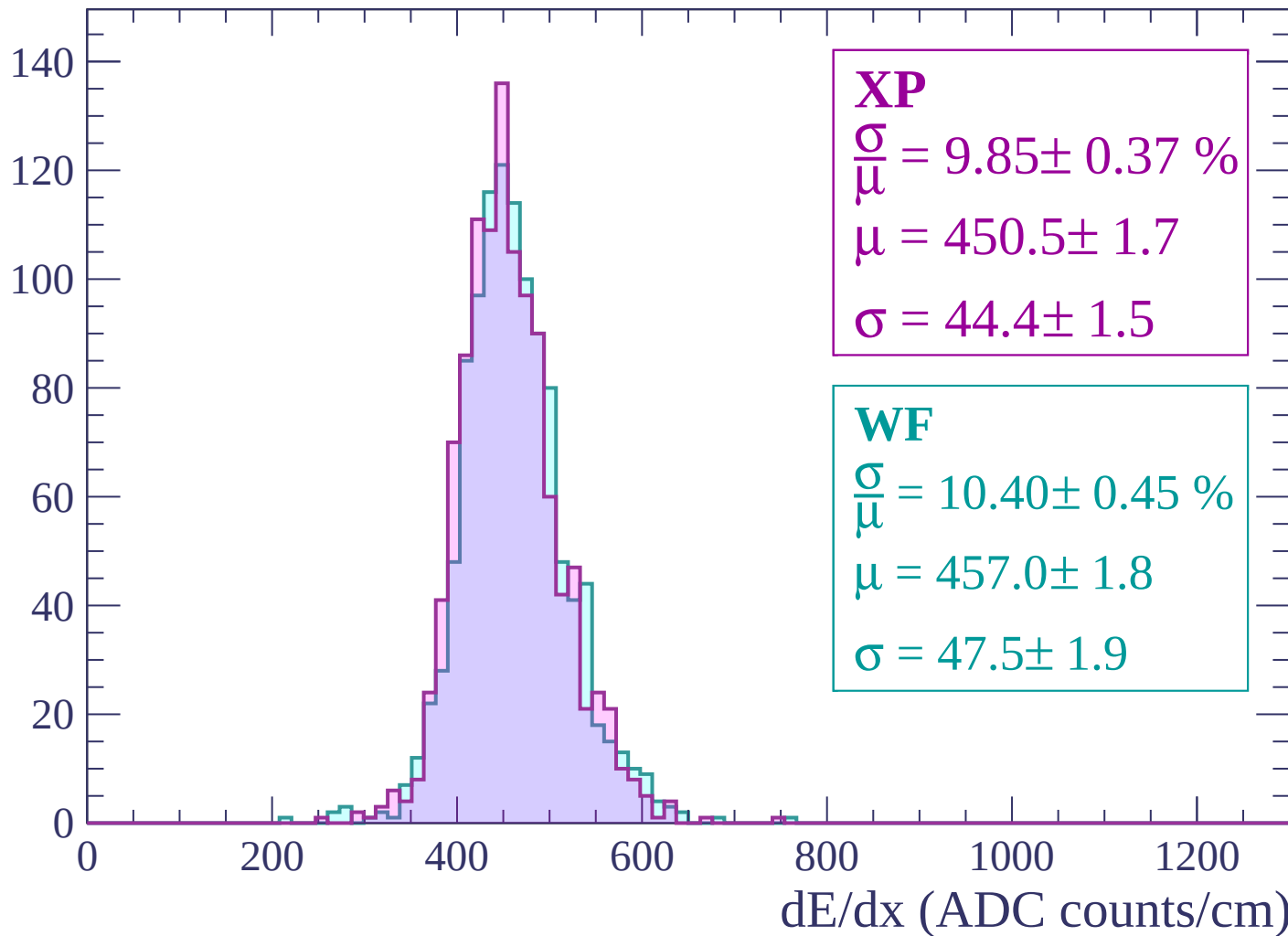
Energy loss | -1320 < p < -1260

Count



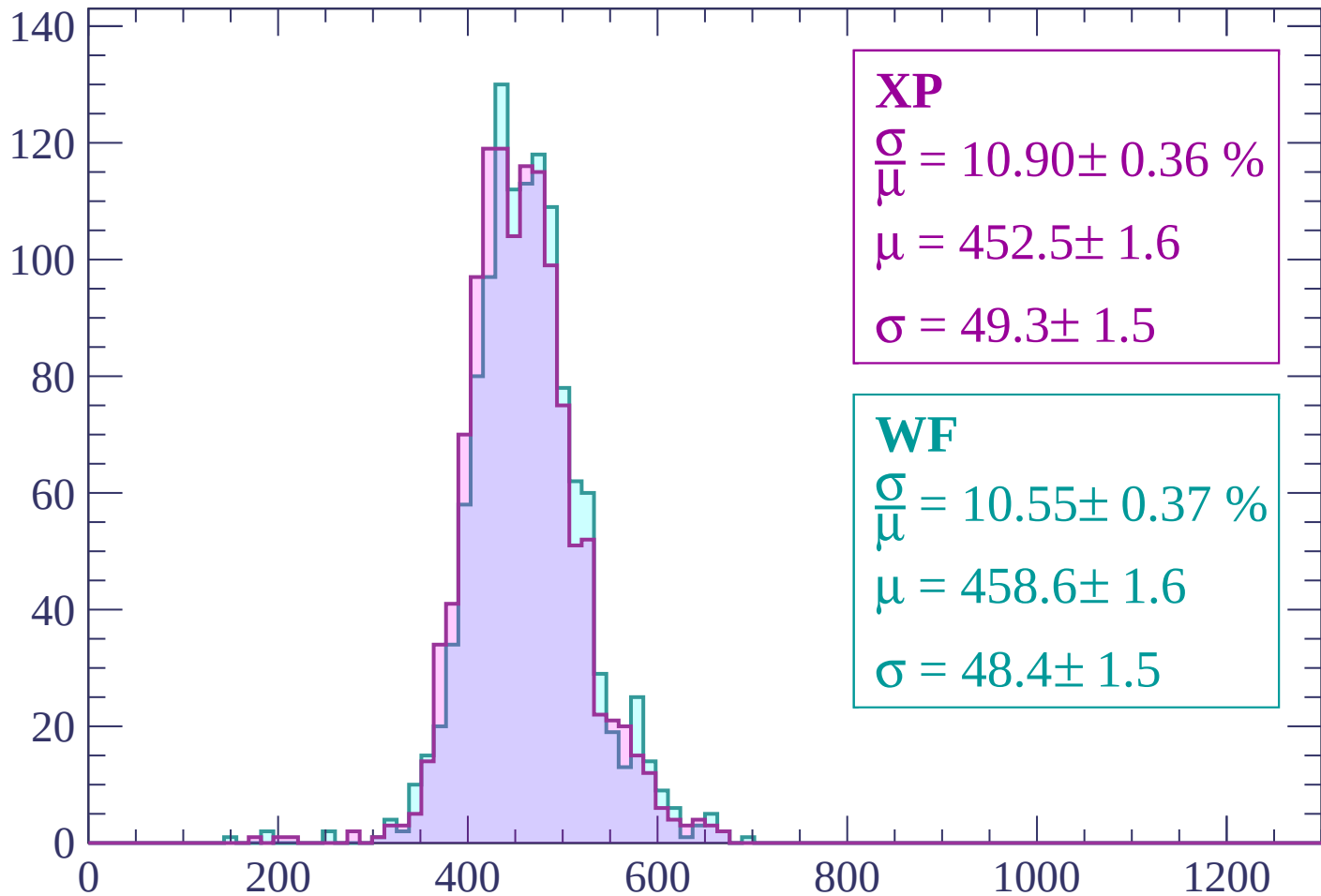
Energy loss | -1260 < p < -1200

Count



Energy loss | -1200 < p < -1140

Count



XP

$$\frac{\sigma}{\mu} = 10.90 \pm 0.36 \%$$

$$\mu = 452.5 \pm 1.6$$

$$\sigma = 49.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.55 \pm 0.37 \%$$

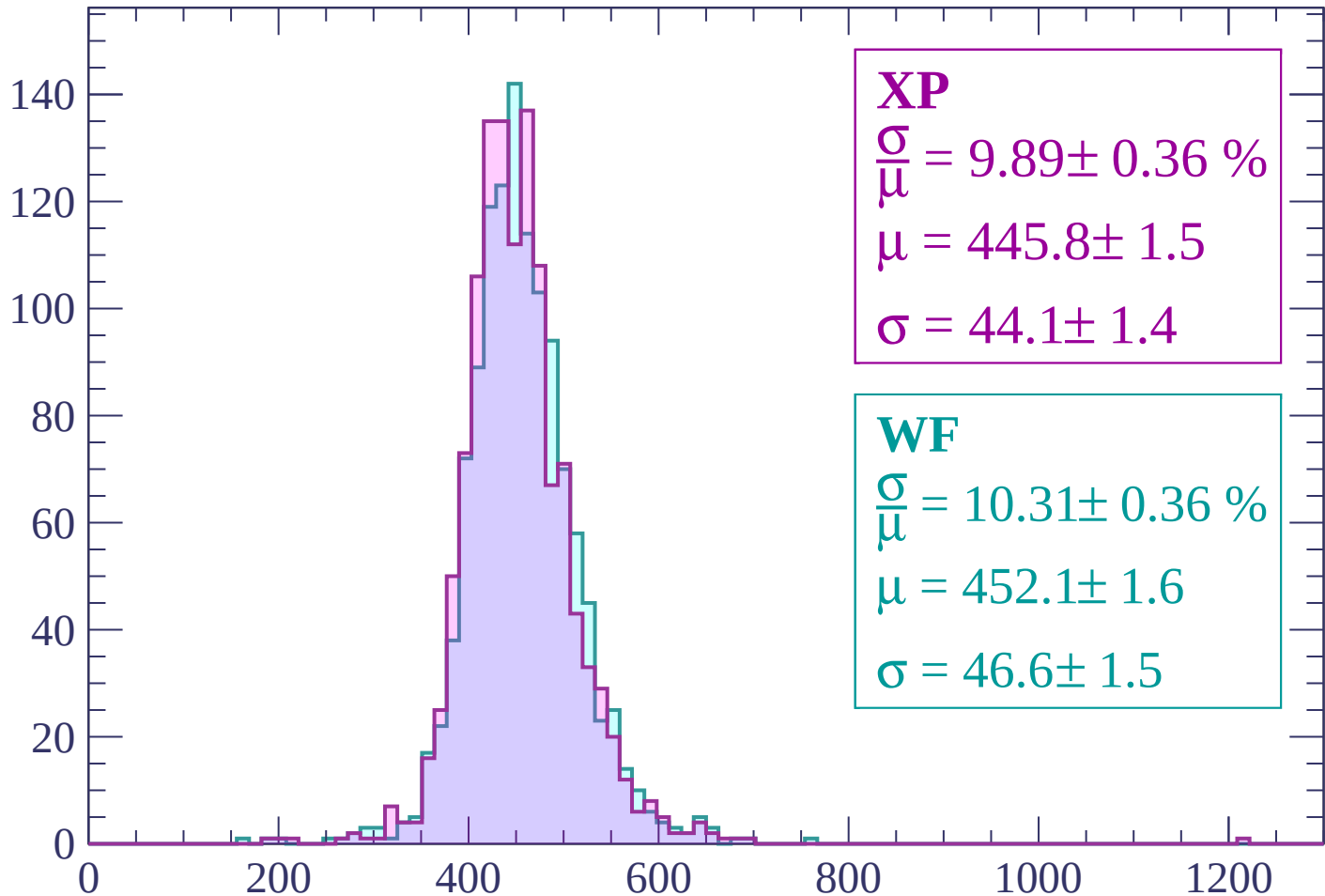
$$\mu = 458.6 \pm 1.6$$

$$\sigma = 48.4 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 9.89 \pm 0.36 \%$$

$$\mu = 445.8 \pm 1.5$$

$$\sigma = 44.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.31 \pm 0.36 \%$$

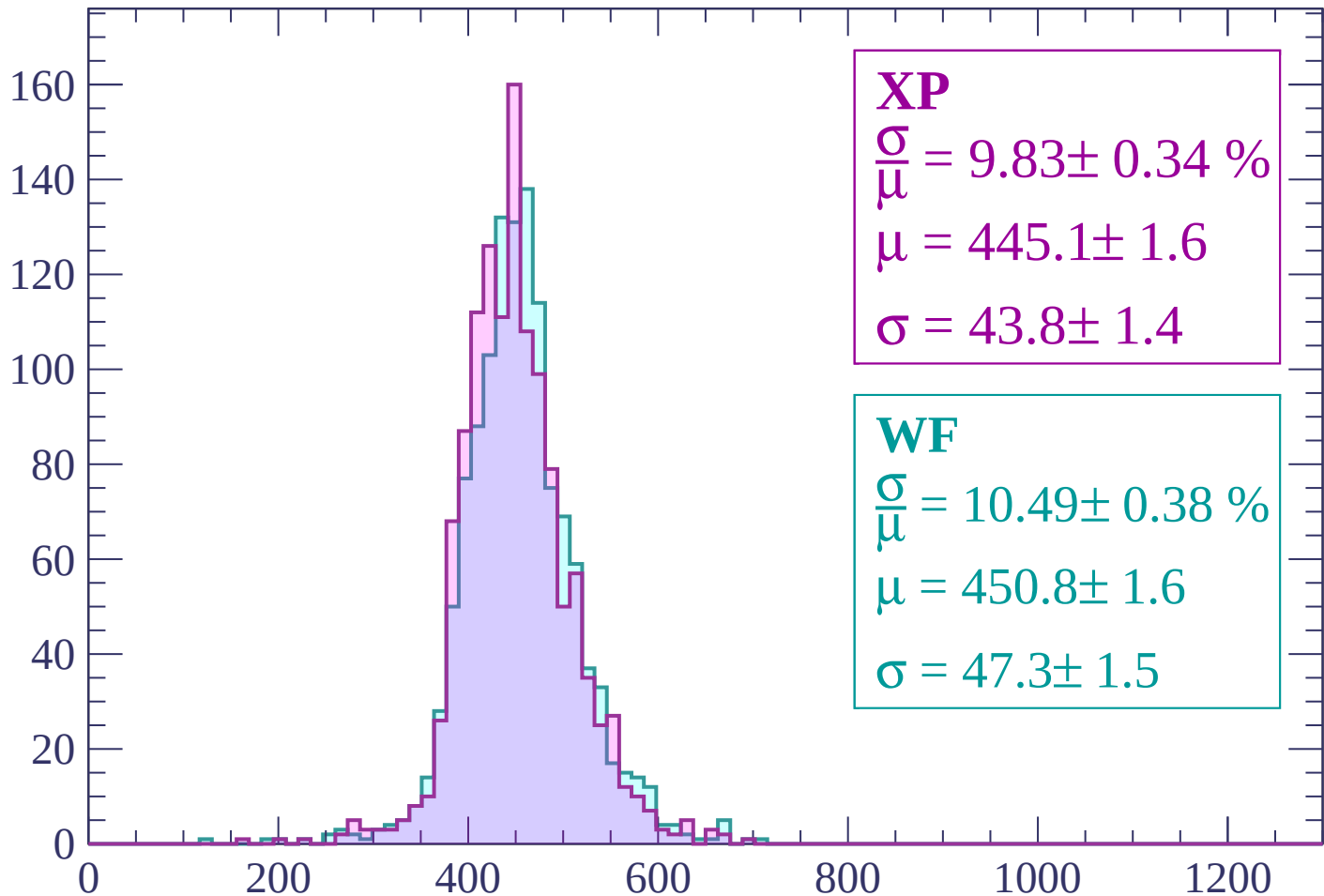
$$\mu = 452.1 \pm 1.6$$

$$\sigma = 46.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1020

Count



XP

$$\frac{\sigma}{\mu} = 9.83 \pm 0.34 \%$$

$$\mu = 445.1 \pm 1.6$$

$$\sigma = 43.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.49 \pm 0.38 \%$$

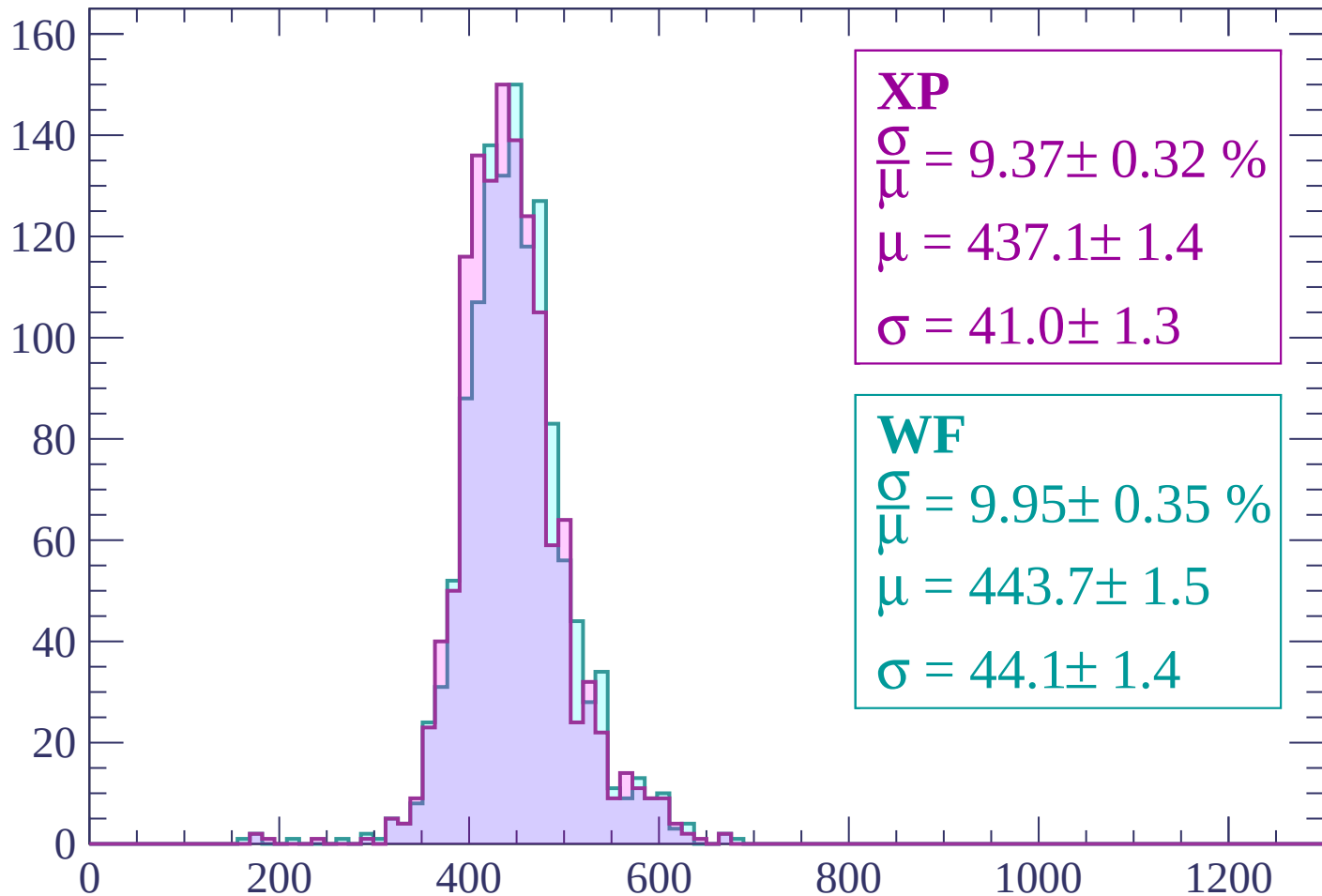
$$\mu = 450.8 \pm 1.6$$

$$\sigma = 47.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.32 \%$$

$$\mu = 437.1 \pm 1.4$$

$$\sigma = 41.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.95 \pm 0.35 \%$$

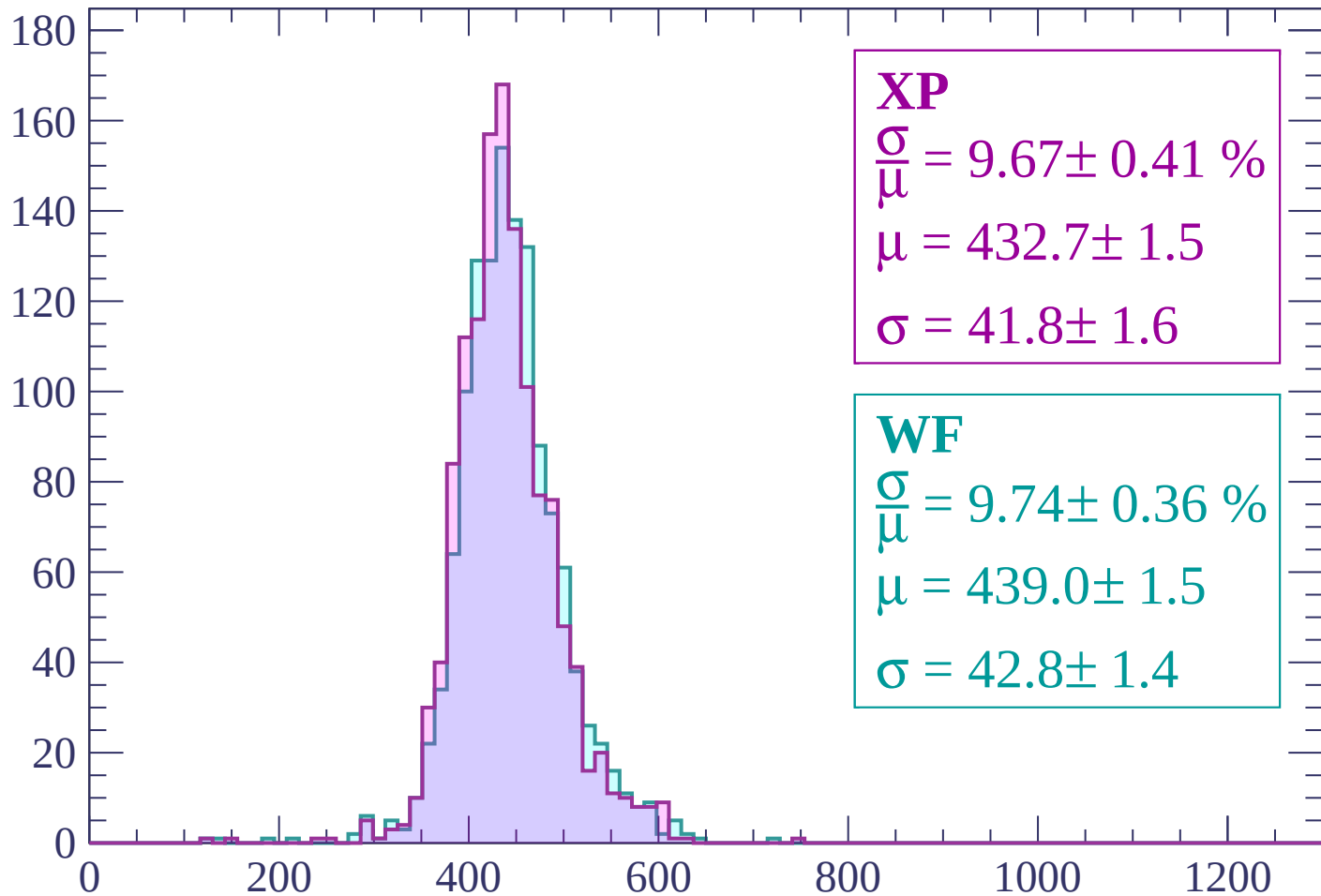
$$\mu = 443.7 \pm 1.5$$

$$\sigma = 44.1 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.41 \%$$

$$\mu = 432.7 \pm 1.5$$

$$\sigma = 41.8 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.74 \pm 0.36 \%$$

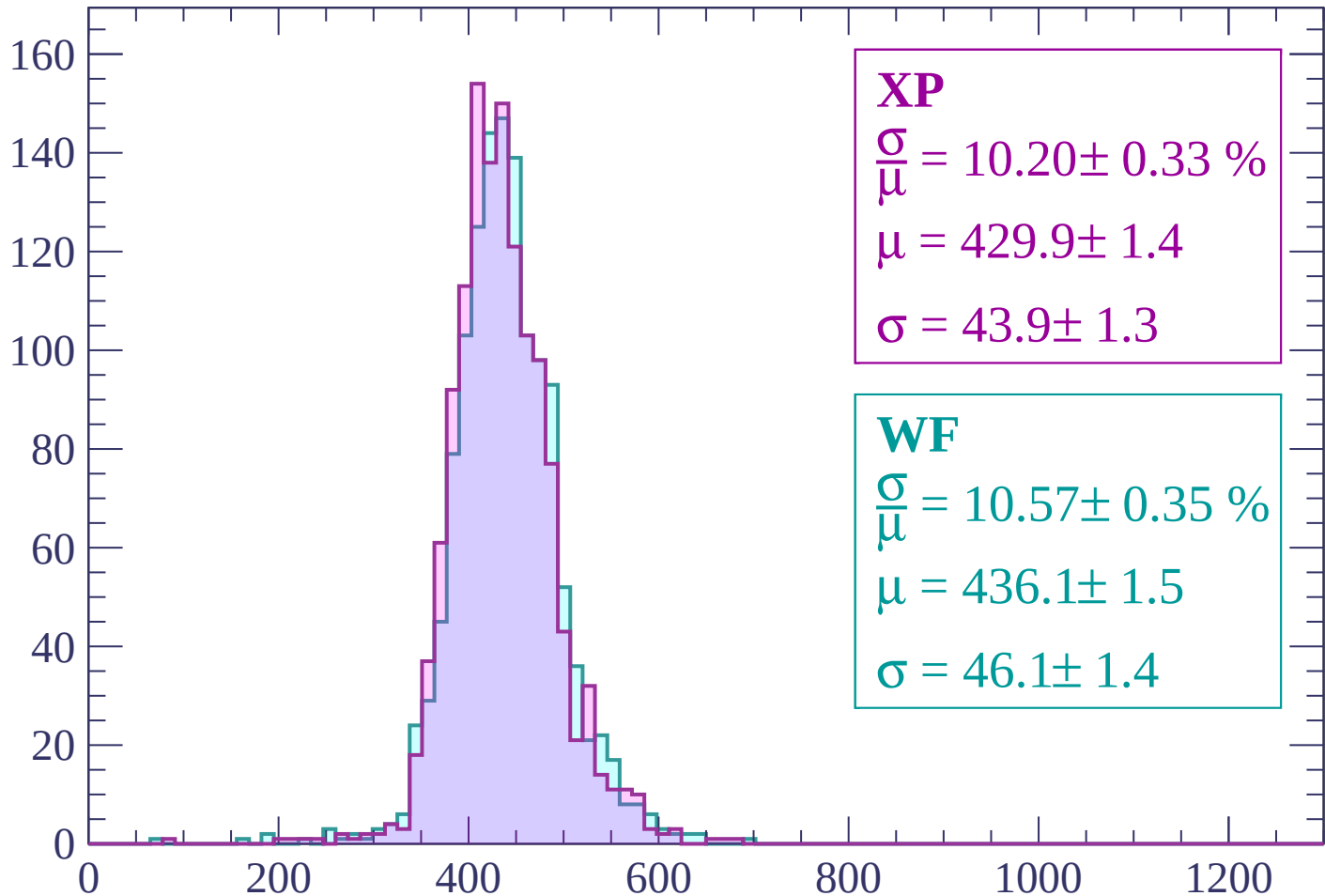
$$\mu = 439.0 \pm 1.5$$

$$\sigma = 42.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 10.20 \pm 0.33 \%$$

$$\mu = 429.9 \pm 1.4$$

$$\sigma = 43.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.57 \pm 0.35 \%$$

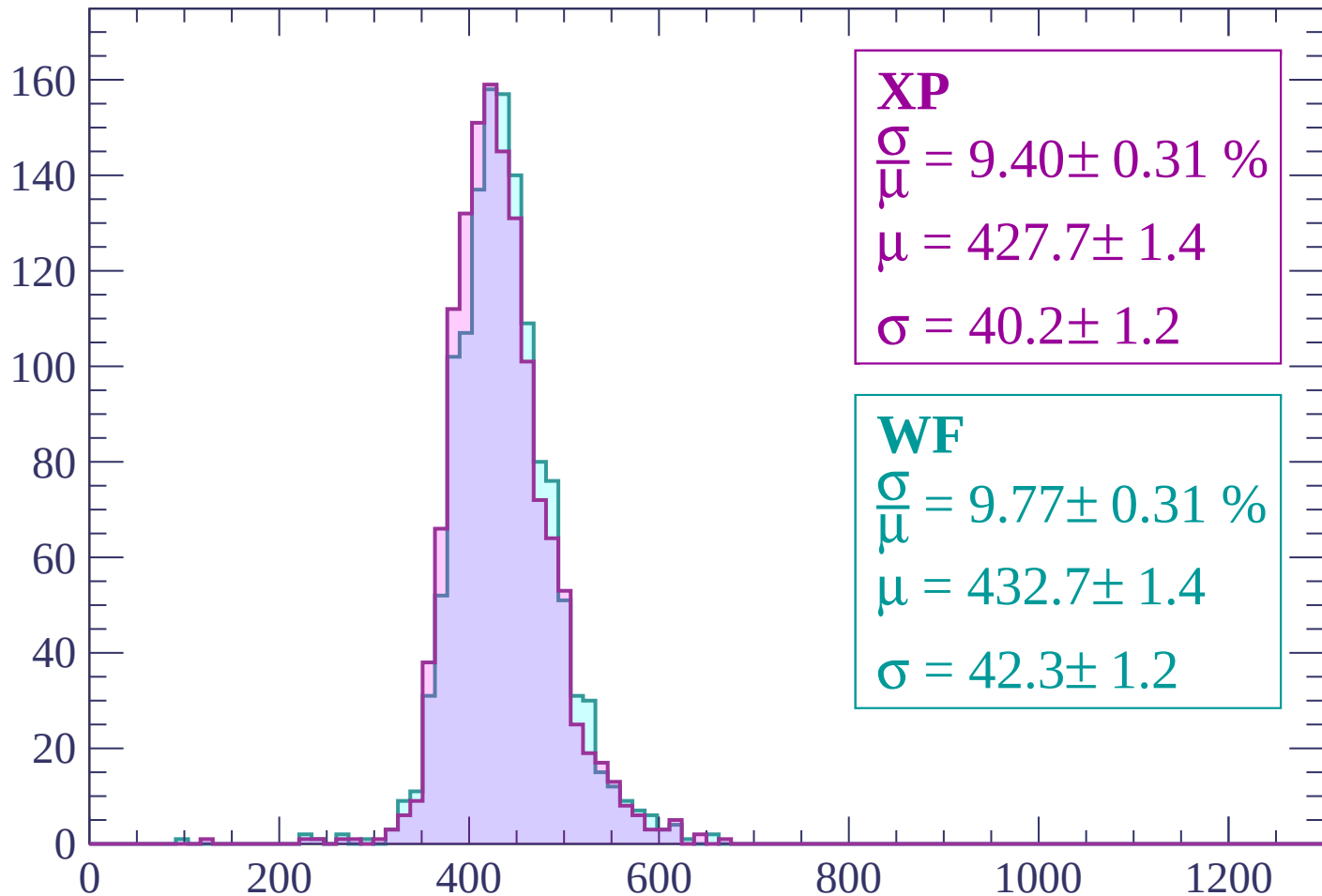
$$\mu = 436.1 \pm 1.5$$

$$\sigma = 46.1 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.31 \%$$

$$\mu = 427.7 \pm 1.4$$

$$\sigma = 40.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.77 \pm 0.31 \%$$

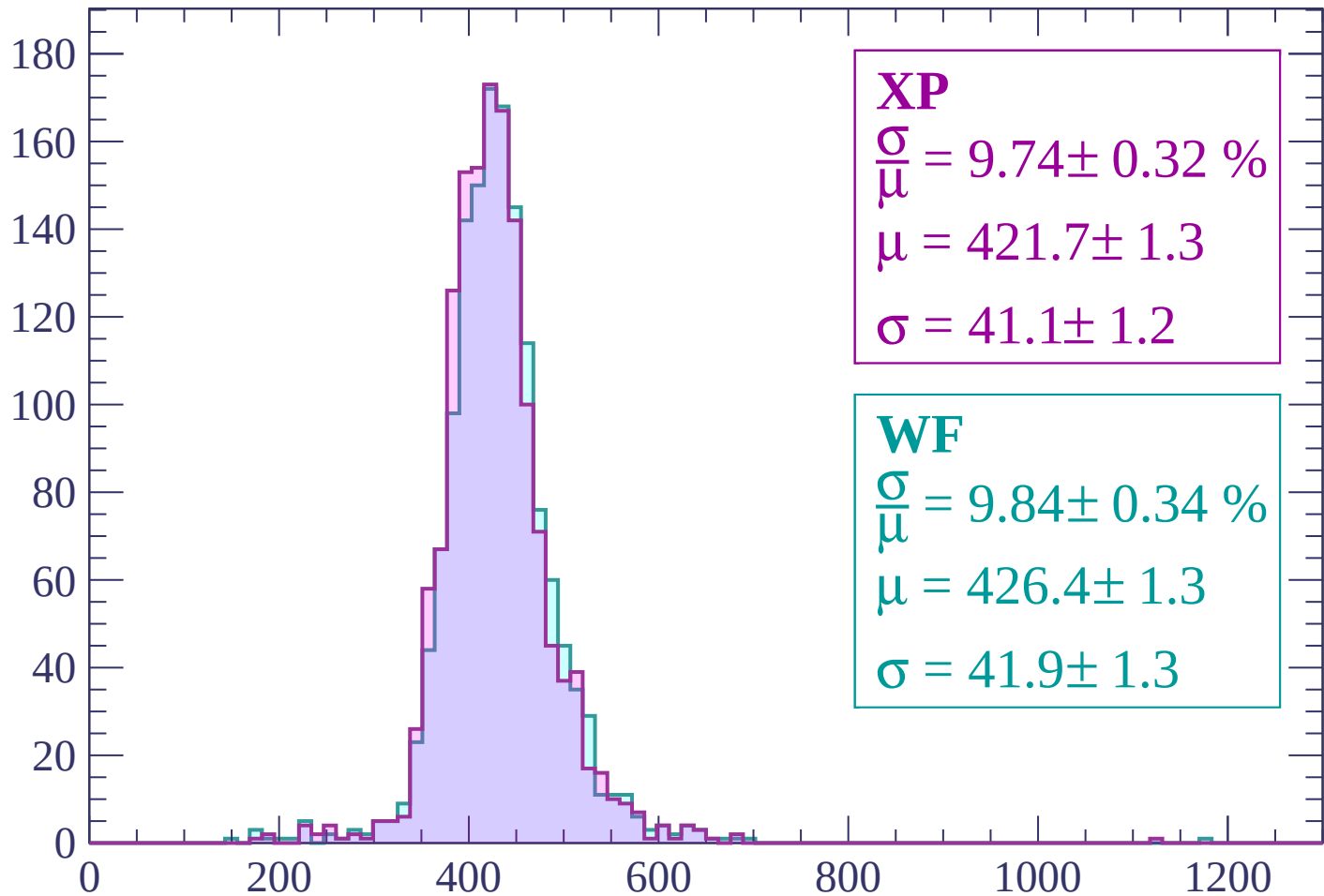
$$\mu = 432.7 \pm 1.4$$

$$\sigma = 42.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.74 \pm 0.32 \%$$

$$\mu = 421.7 \pm 1.3$$

$$\sigma = 41.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.84 \pm 0.34 \%$$

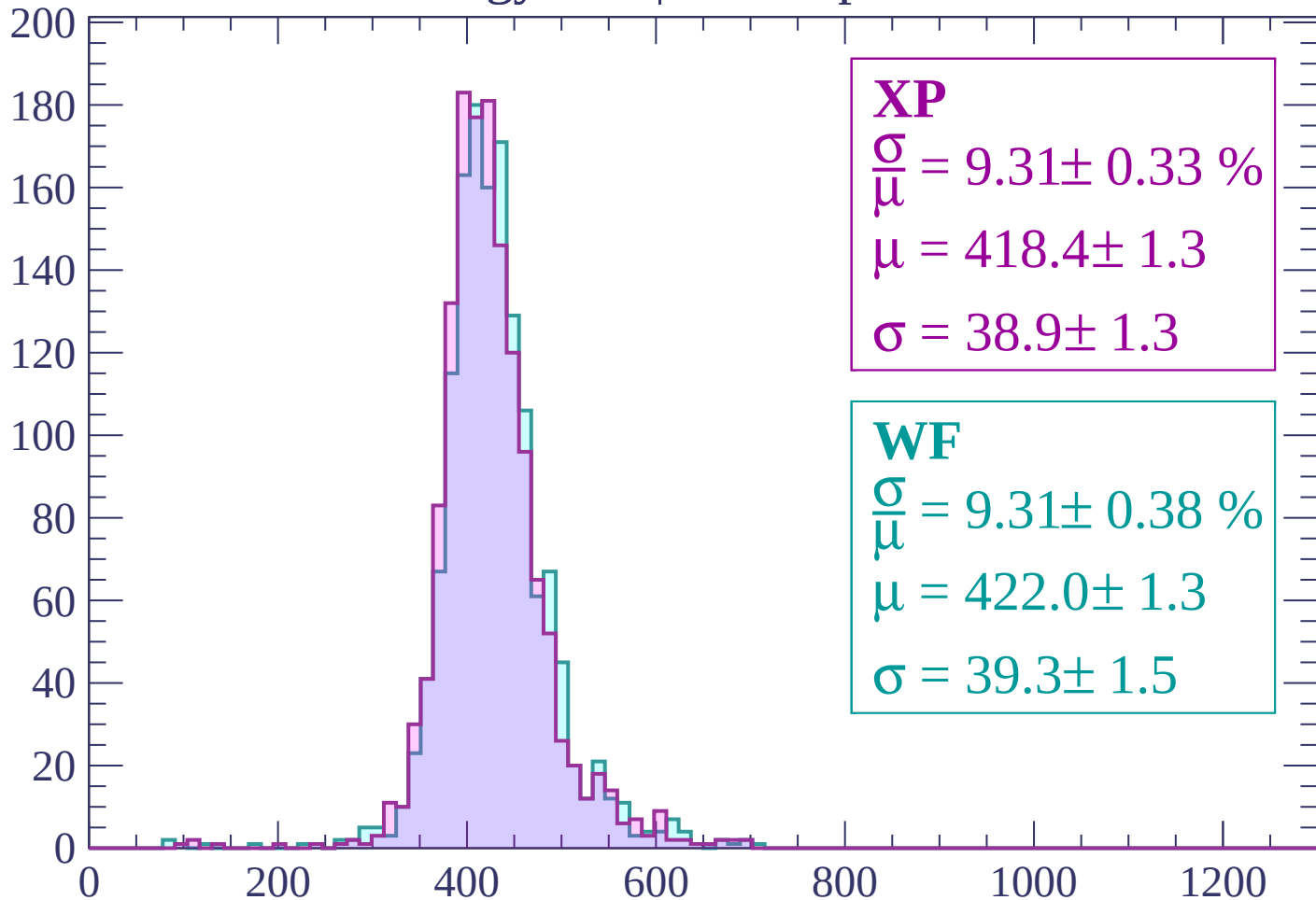
$$\mu = 426.4 \pm 1.3$$

$$\sigma = 41.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP

$$\frac{\sigma}{\mu} = 9.31 \pm 0.33 \%$$

$$\mu = 418.4 \pm 1.3$$

$$\sigma = 38.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.31 \pm 0.38 \%$$

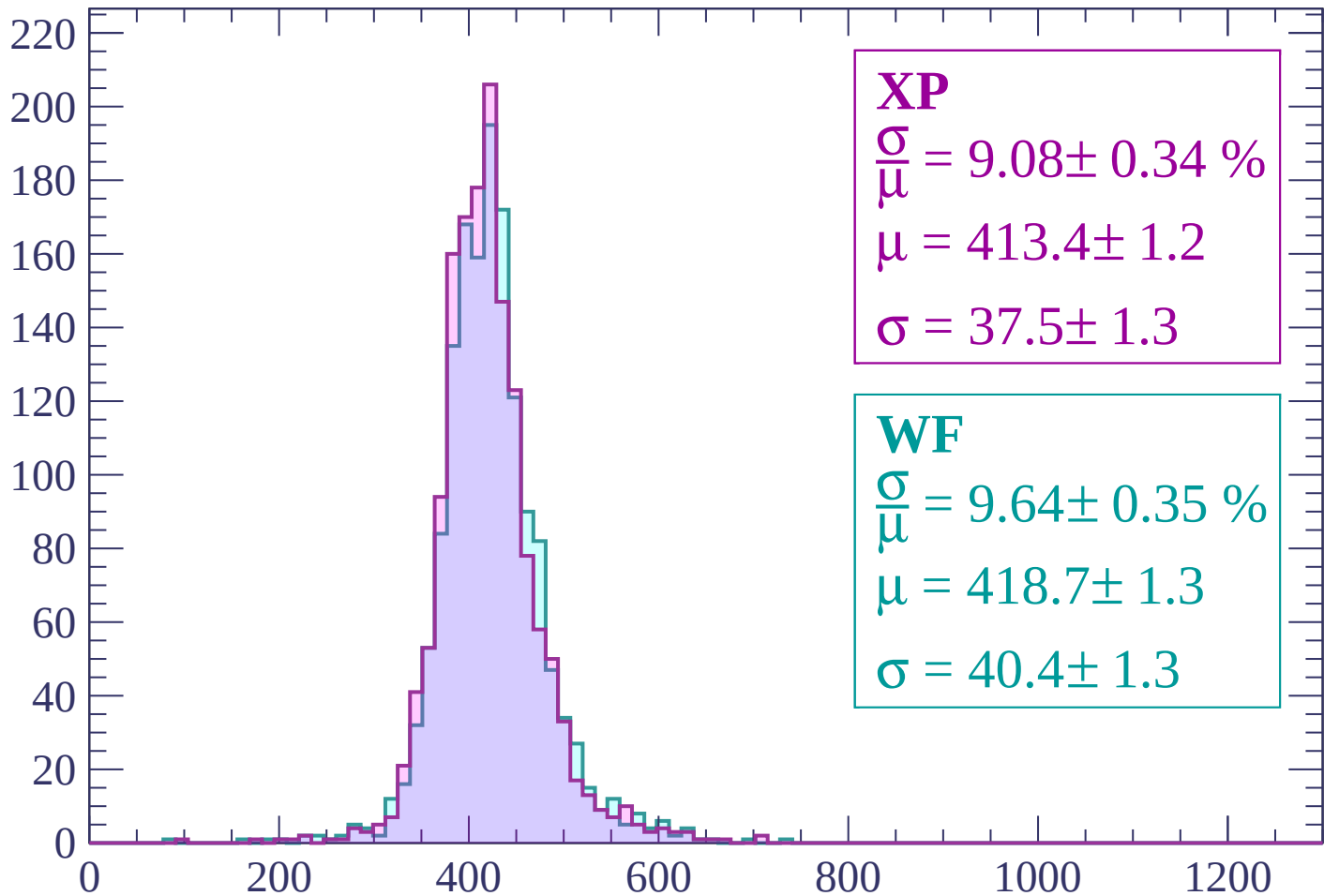
$$\mu = 422.0 \pm 1.3$$

$$\sigma = 39.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -660 < p < -600

Count



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.34 \%$$

$$\mu = 413.4 \pm 1.2$$

$$\sigma = 37.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.35 \%$$

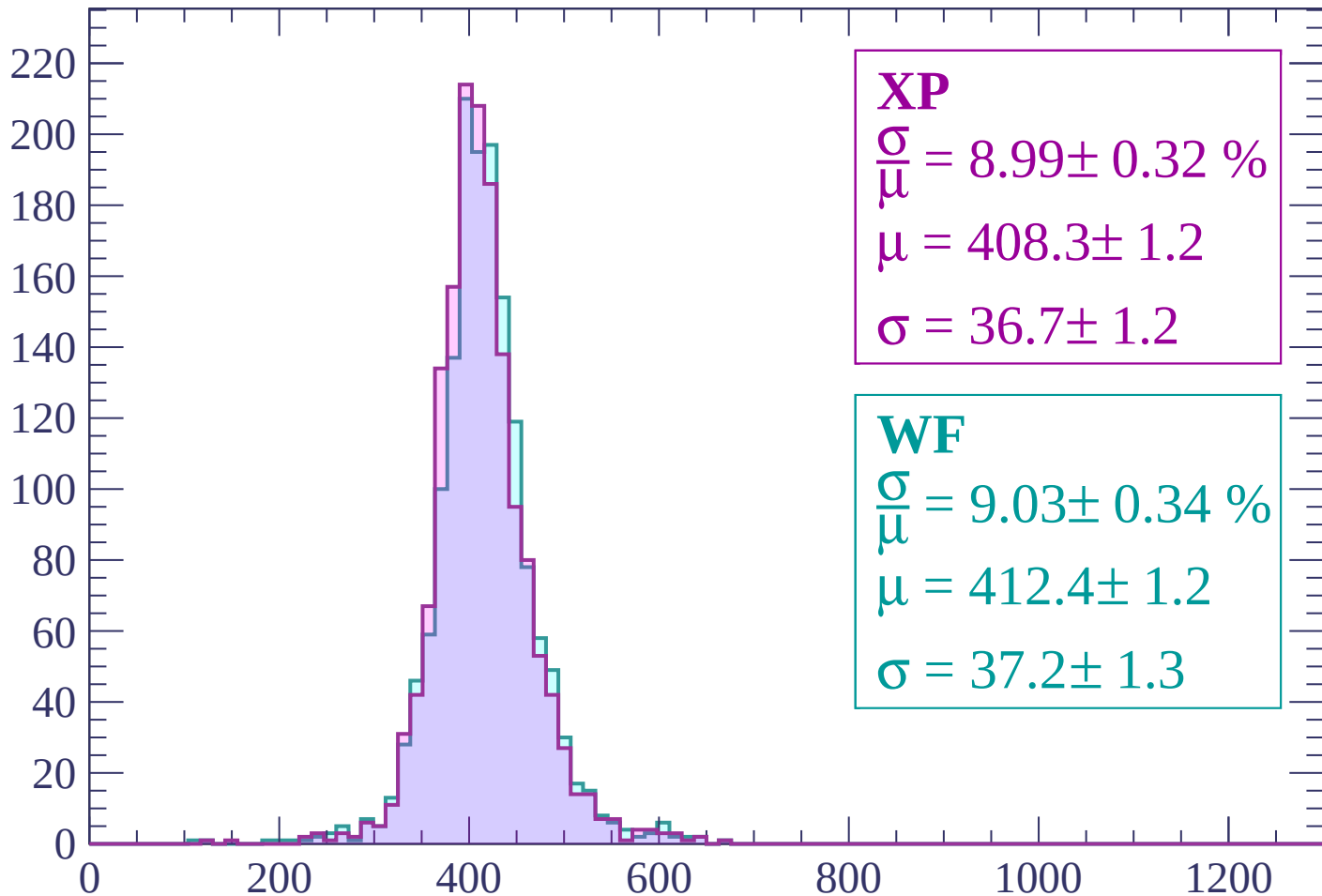
$$\mu = 418.7 \pm 1.3$$

$$\sigma = 40.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.32 \%$$

$$\mu = 408.3 \pm 1.2$$

$$\sigma = 36.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.34 \%$$

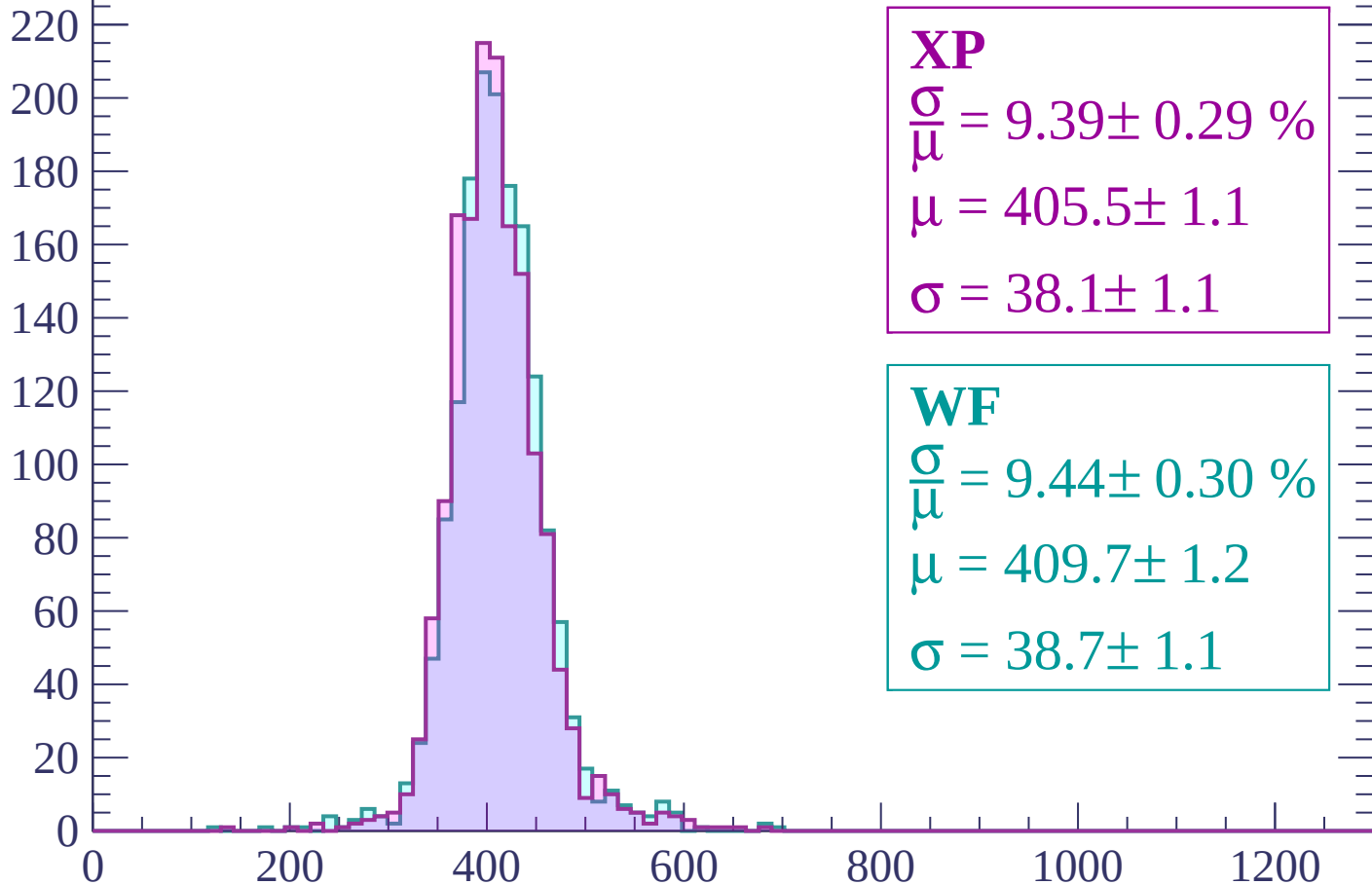
$$\mu = 412.4 \pm 1.2$$

$$\sigma = 37.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.29 \%$$

$$\mu = 405.5 \pm 1.1$$

$$\sigma = 38.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.30 \%$$

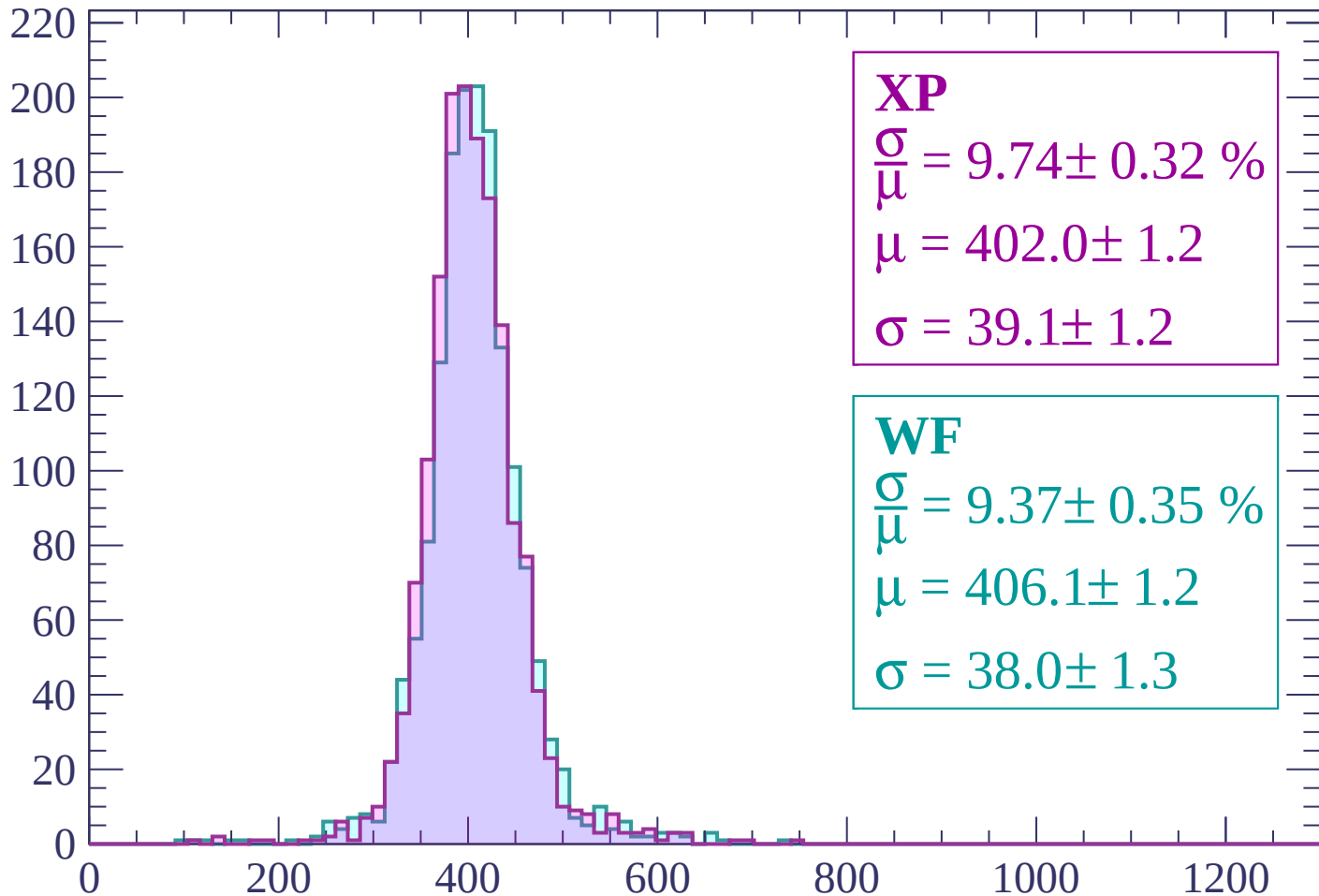
$$\mu = 409.7 \pm 1.2$$

$$\sigma = 38.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP

$$\frac{\sigma}{\mu} = 9.74 \pm 0.32 \%$$

$$\mu = 402.0 \pm 1.2$$

$$\sigma = 39.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.35 \%$$

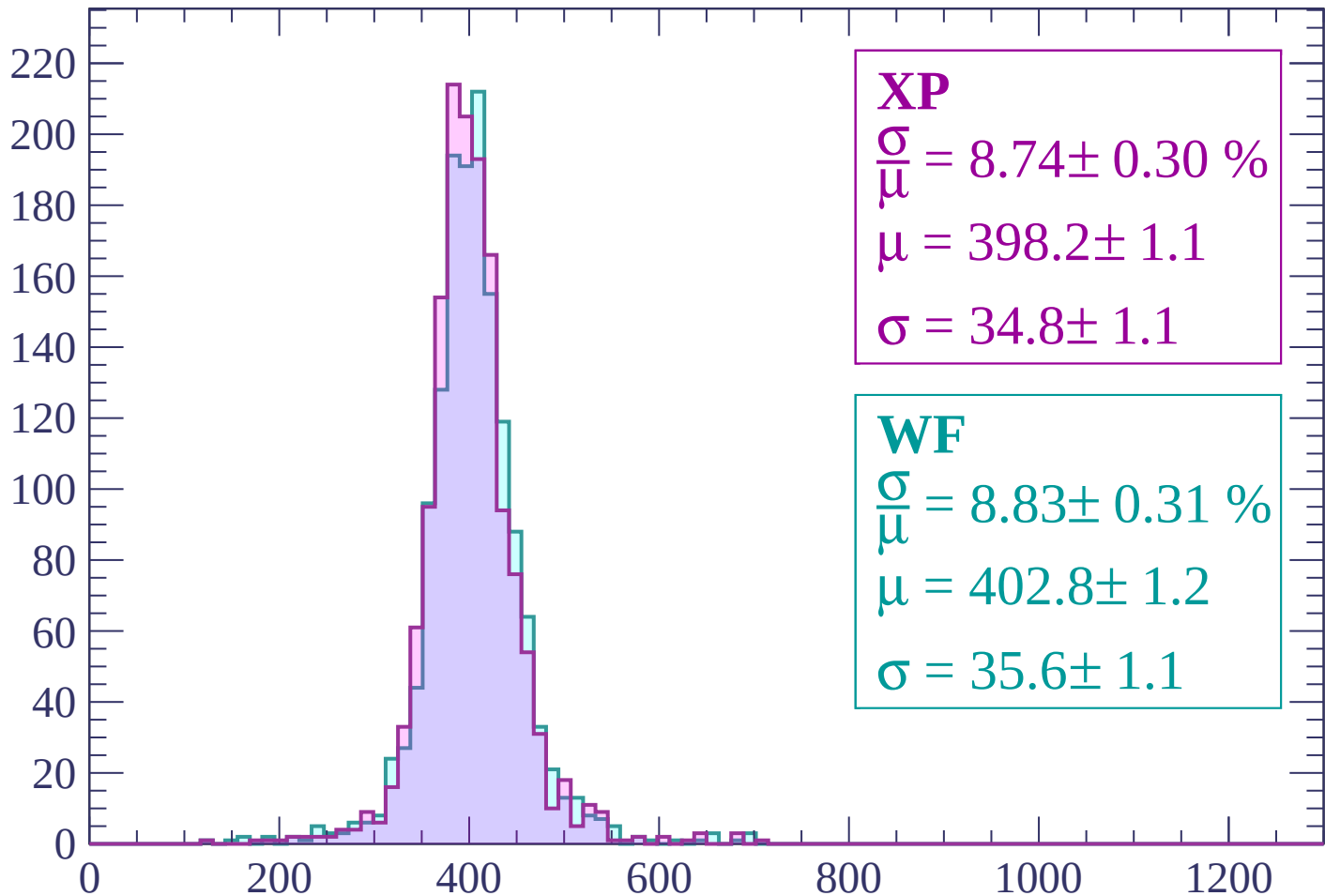
$$\mu = 406.1 \pm 1.2$$

$$\sigma = 38.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.30 \%$$

$$\mu = 398.2 \pm 1.1$$

$$\sigma = 34.8 \pm 1.1$$

WF

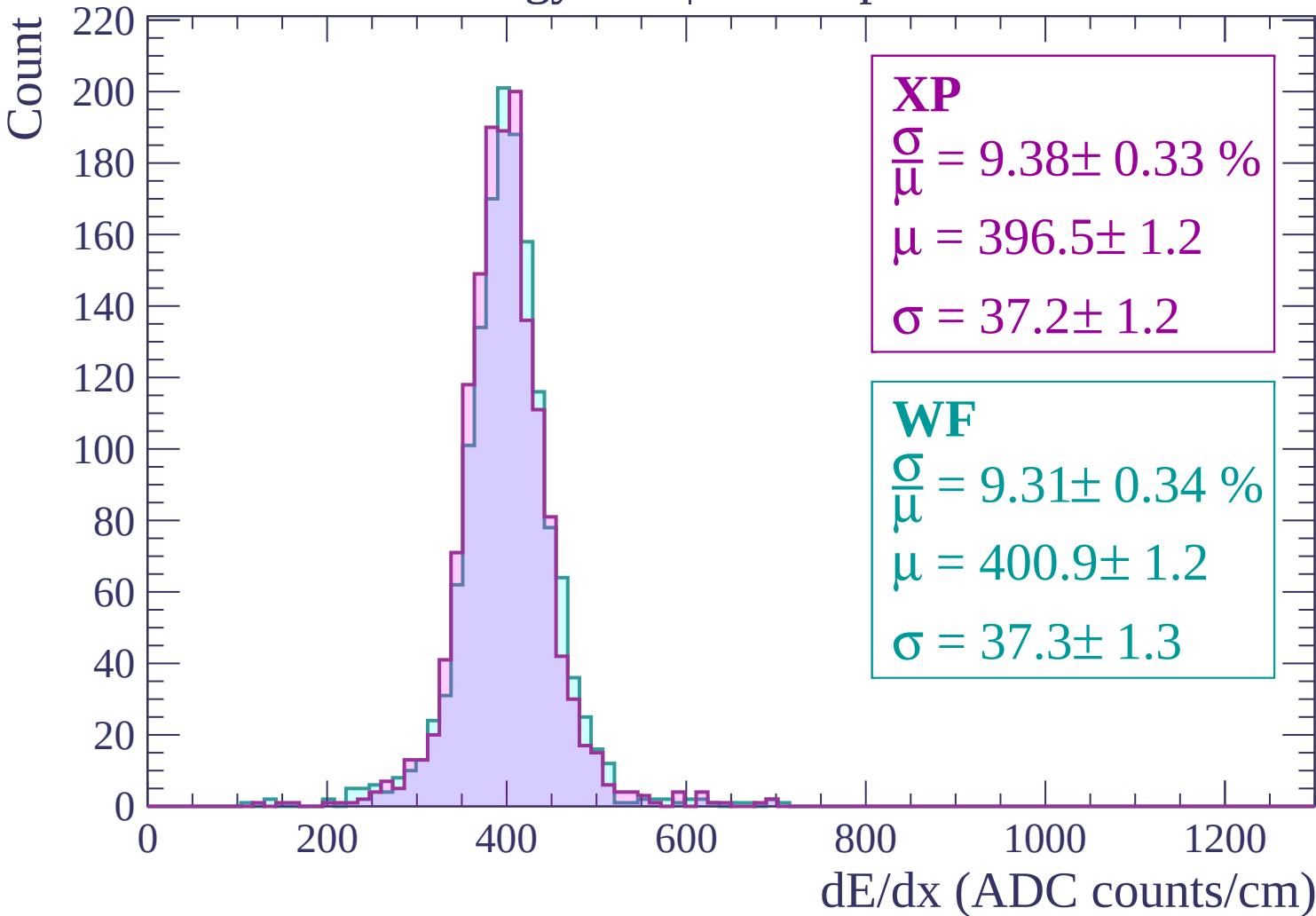
$$\frac{\sigma}{\mu} = 8.83 \pm 0.31 \%$$

$$\mu = 402.8 \pm 1.2$$

$$\sigma = 35.6 \pm 1.1$$

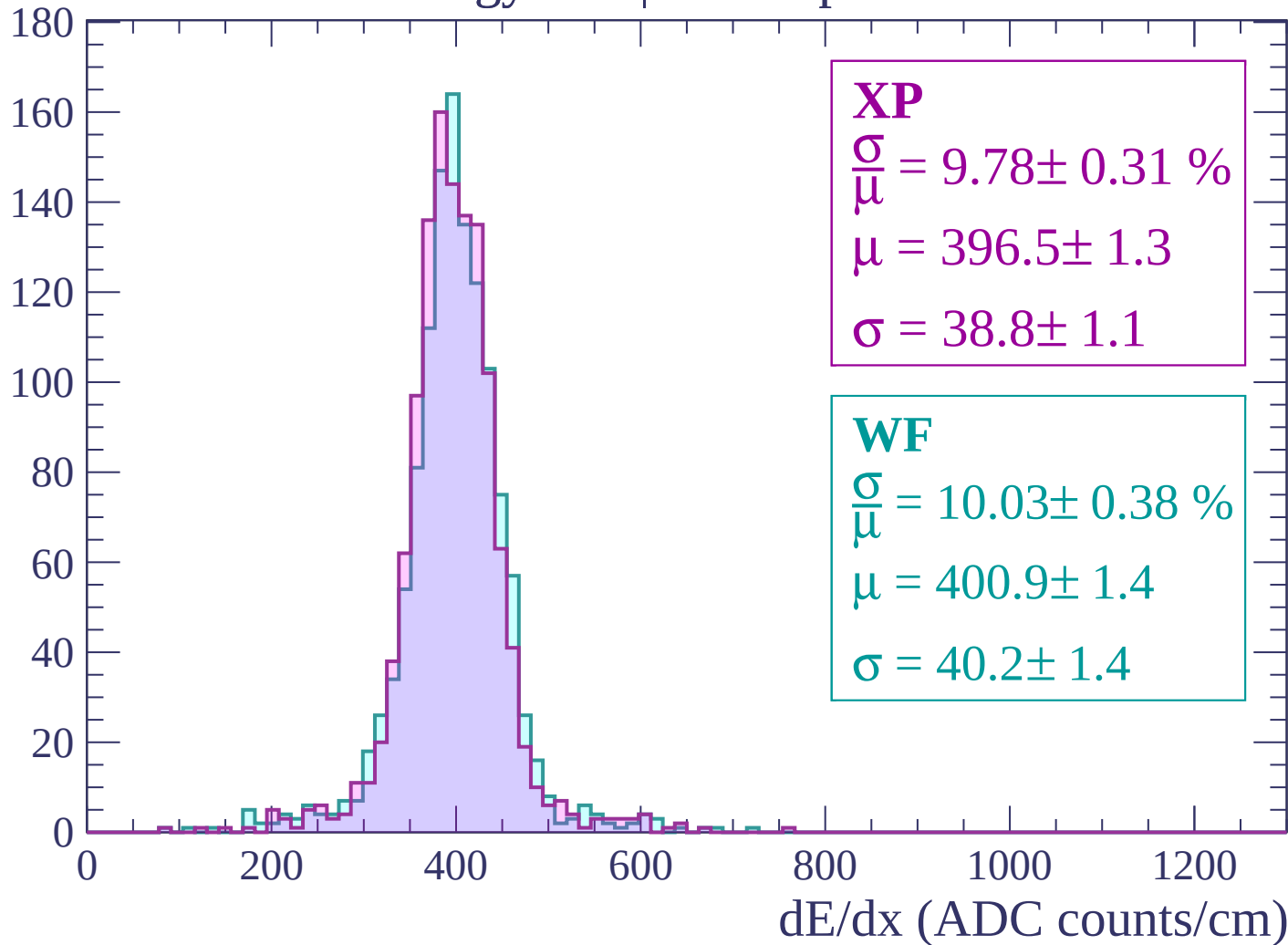
dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$



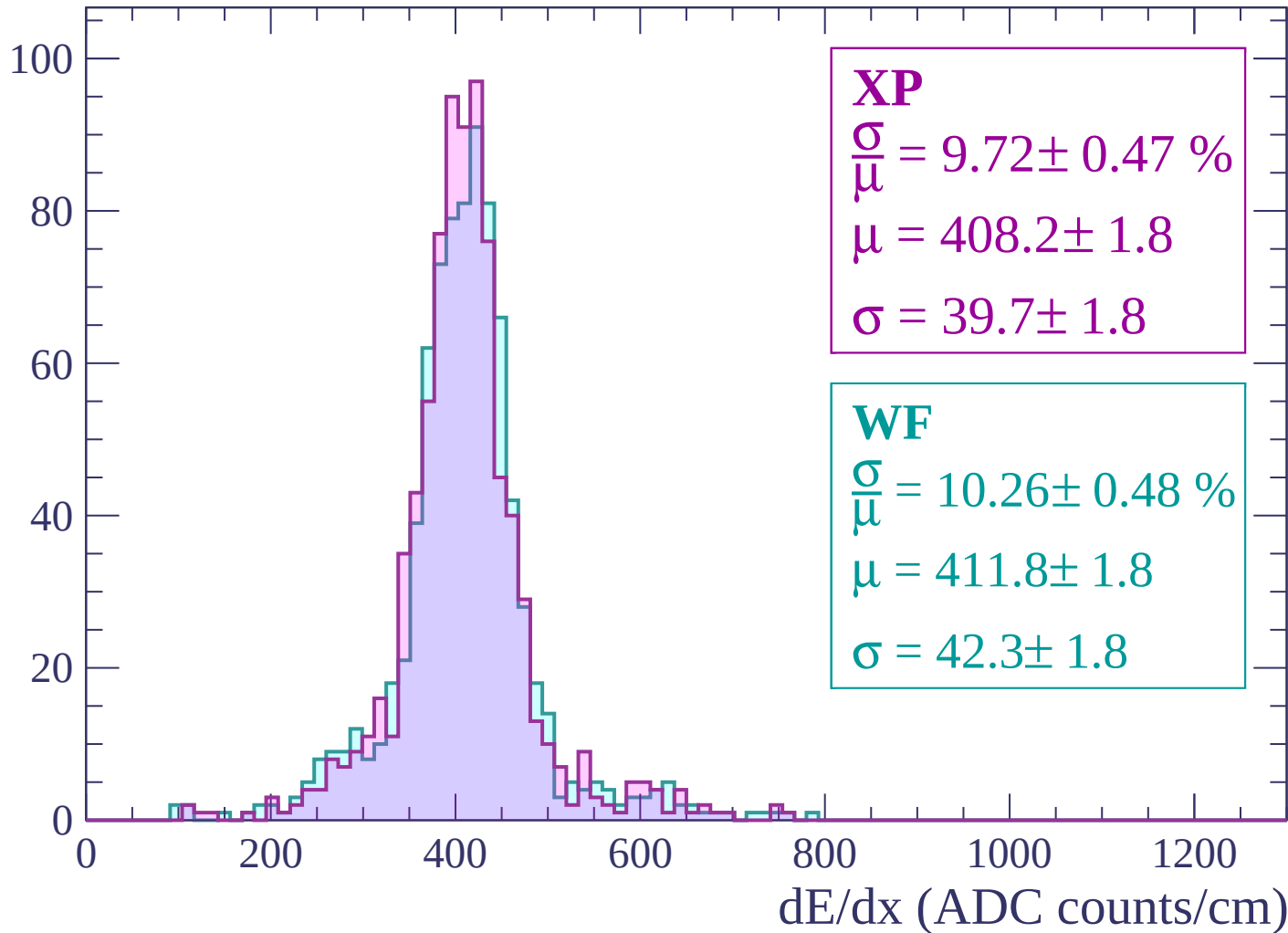
Energy loss | $-300 < p < -240$

Count



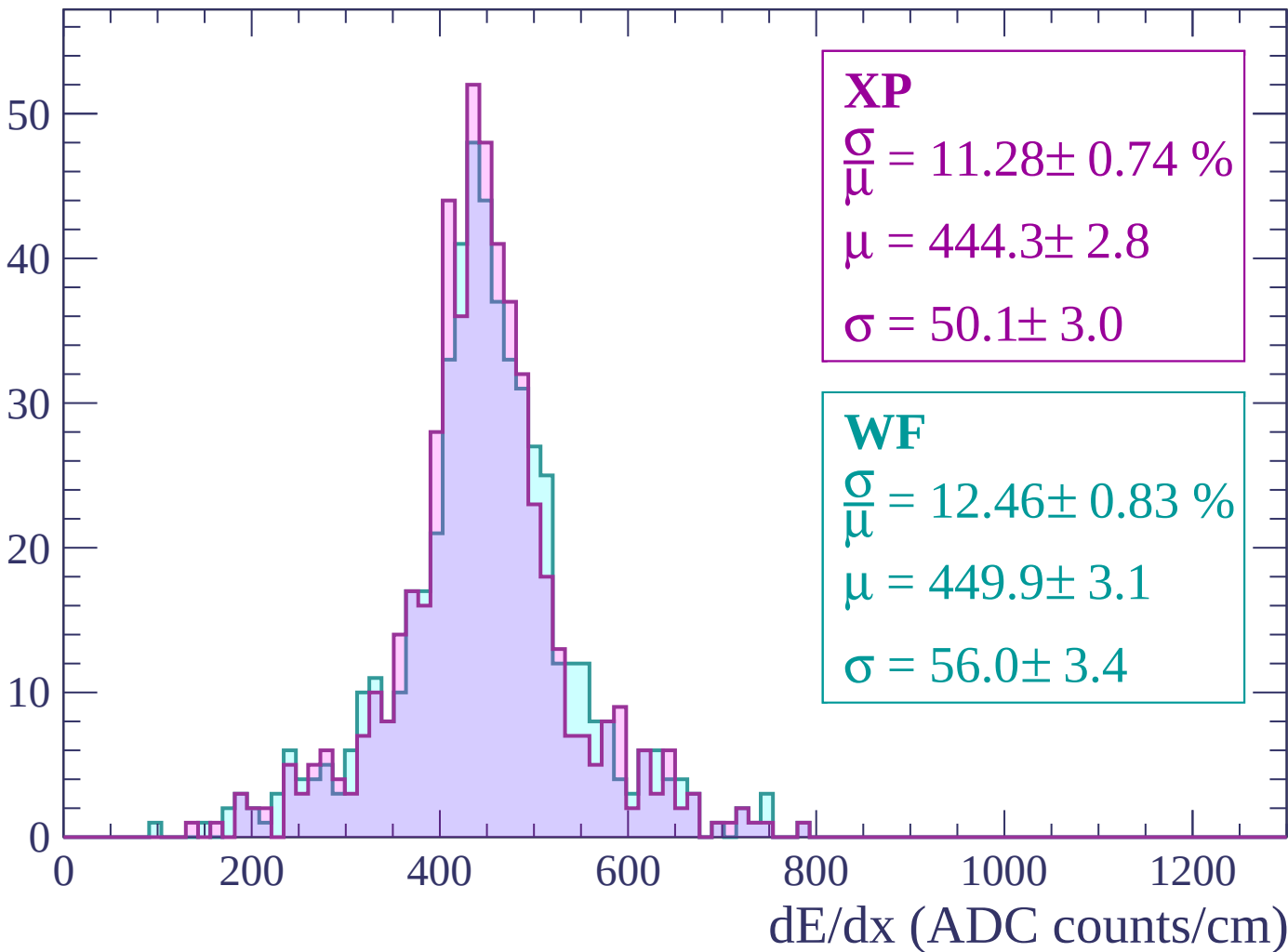
Energy loss | $-240 < p < -180$

Count



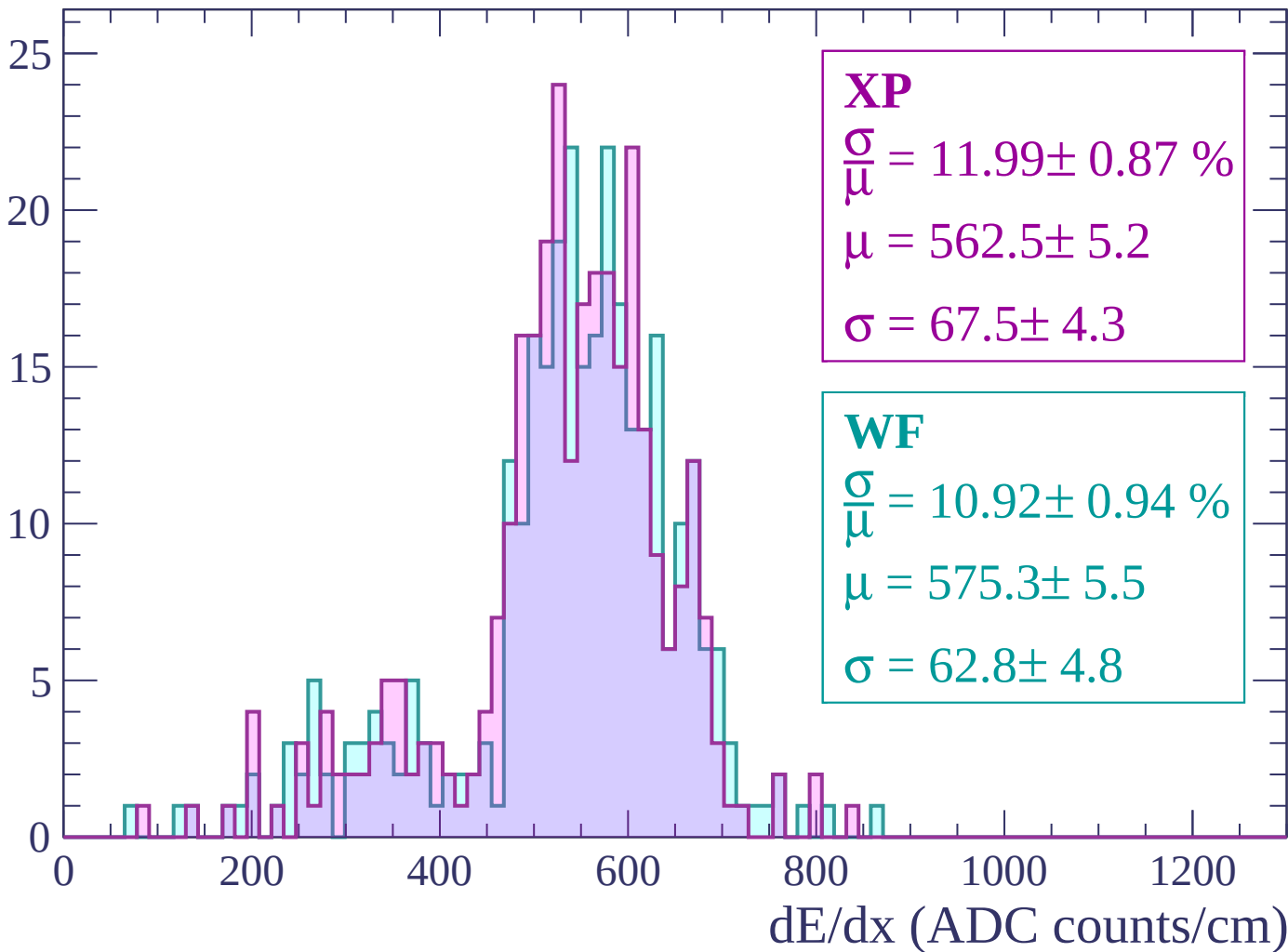
Energy loss | $-180 < p < -120$

Count



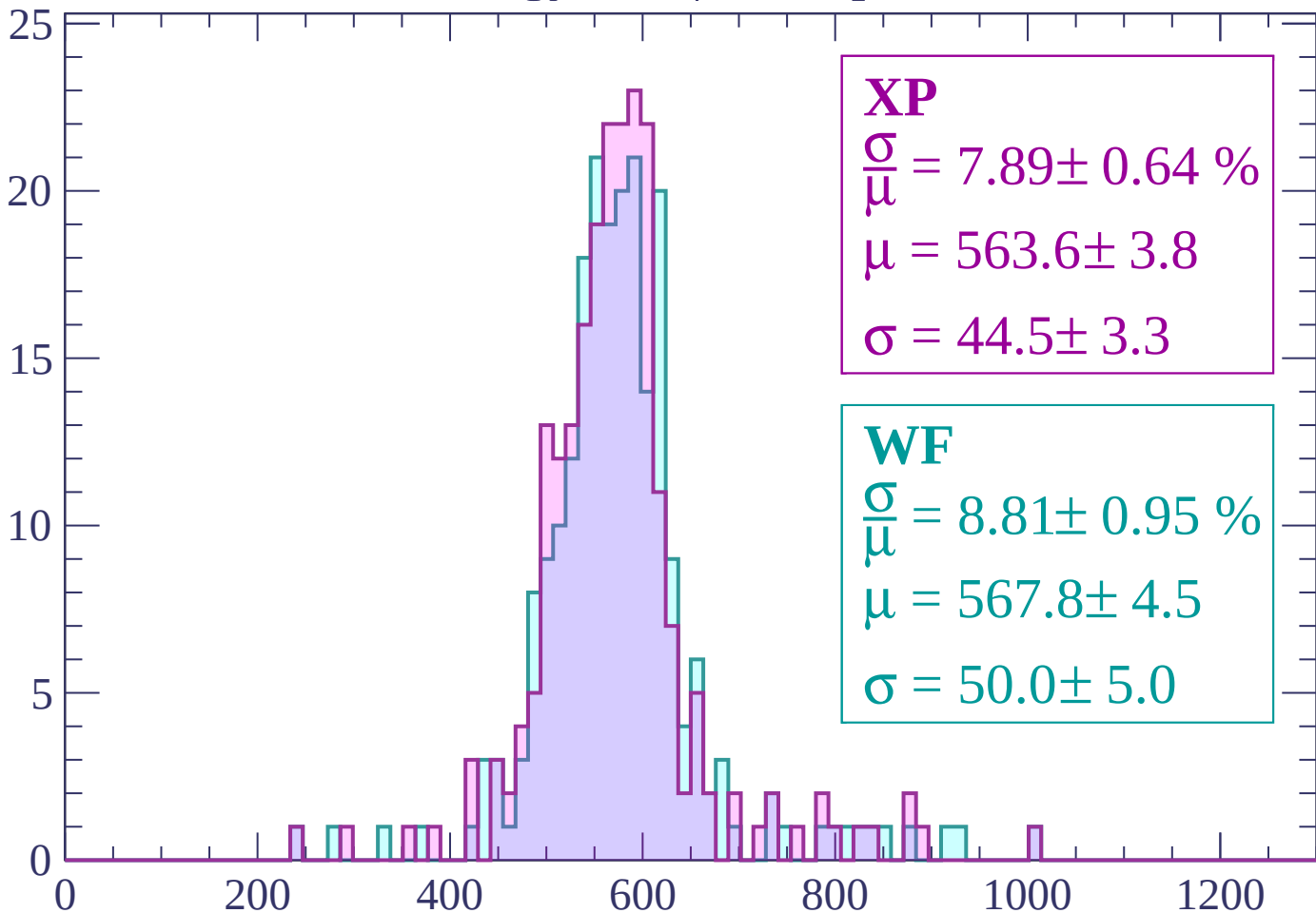
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

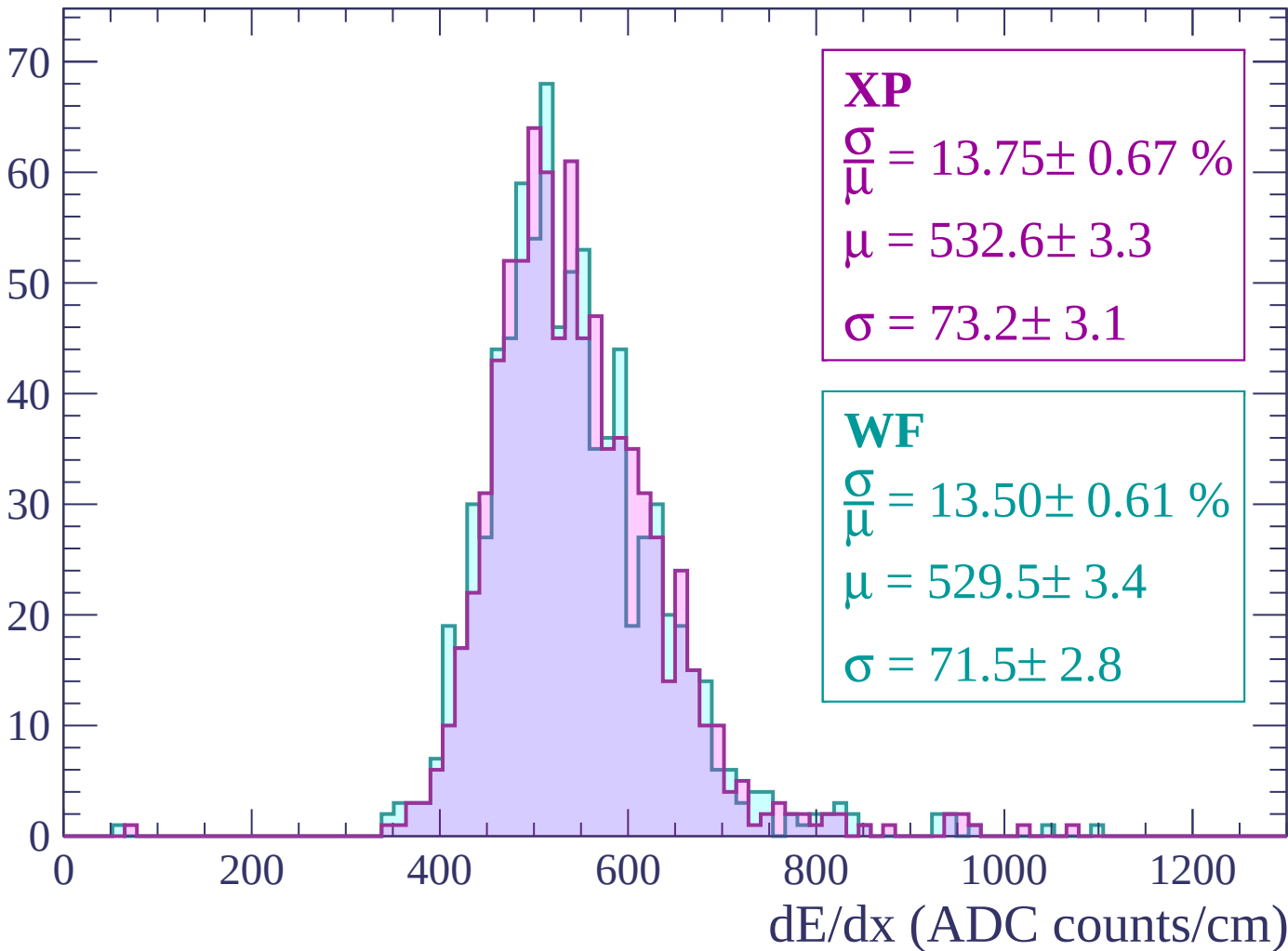
Count



dE/dx (ADC counts/cm)

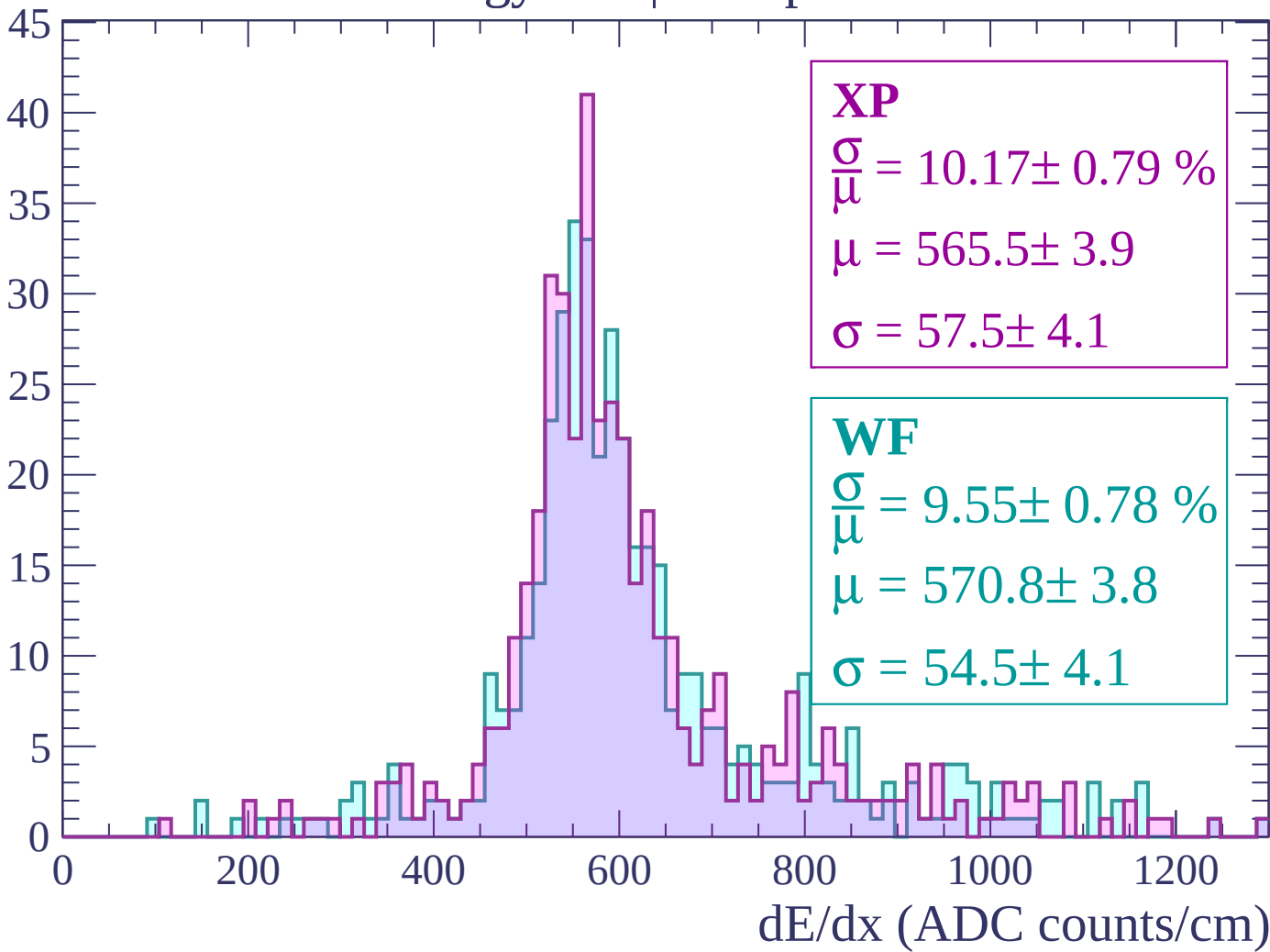
Energy loss | $0 < p < 60$

Count



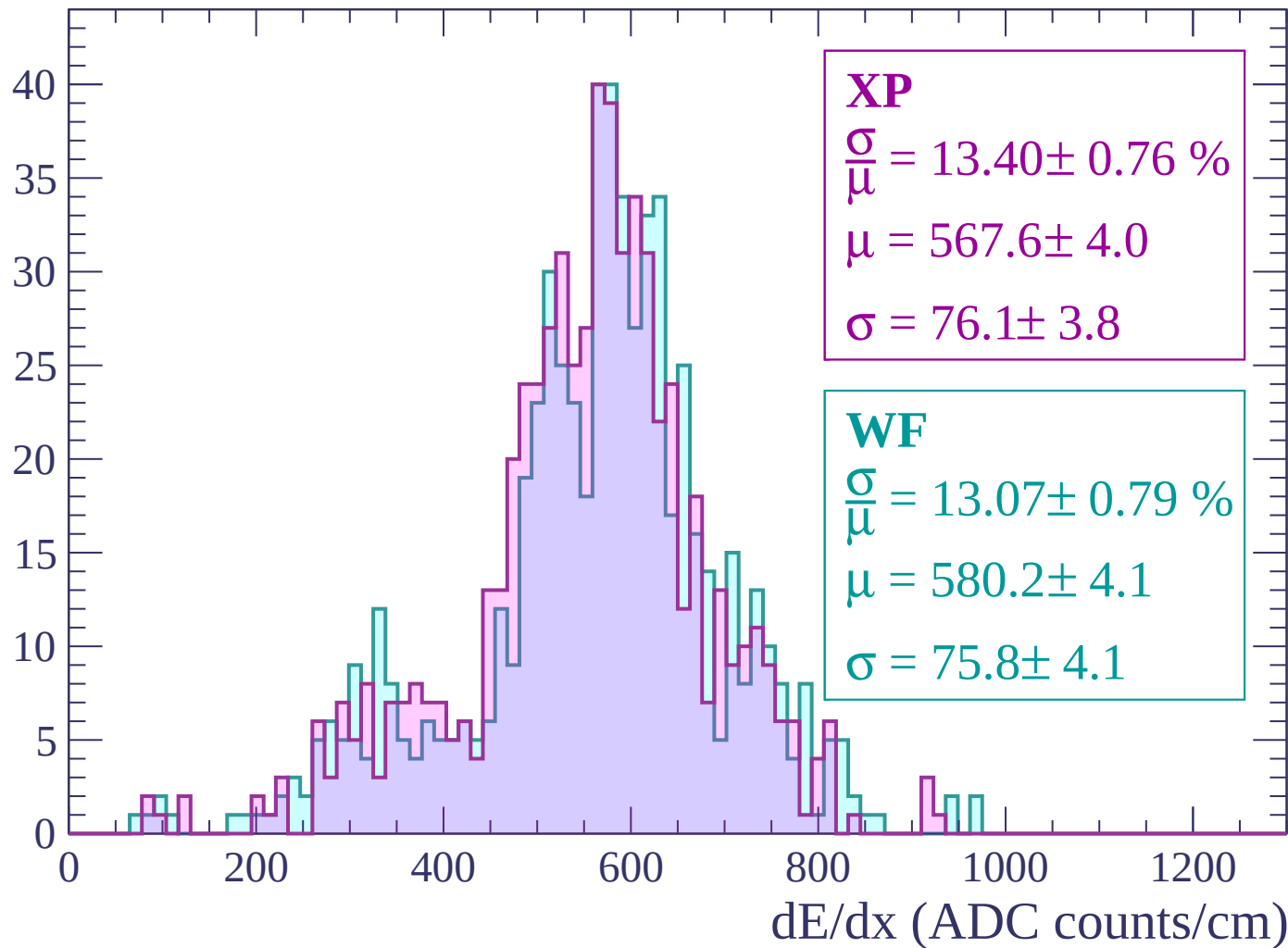
Energy loss | $60 < p < 120$

Count



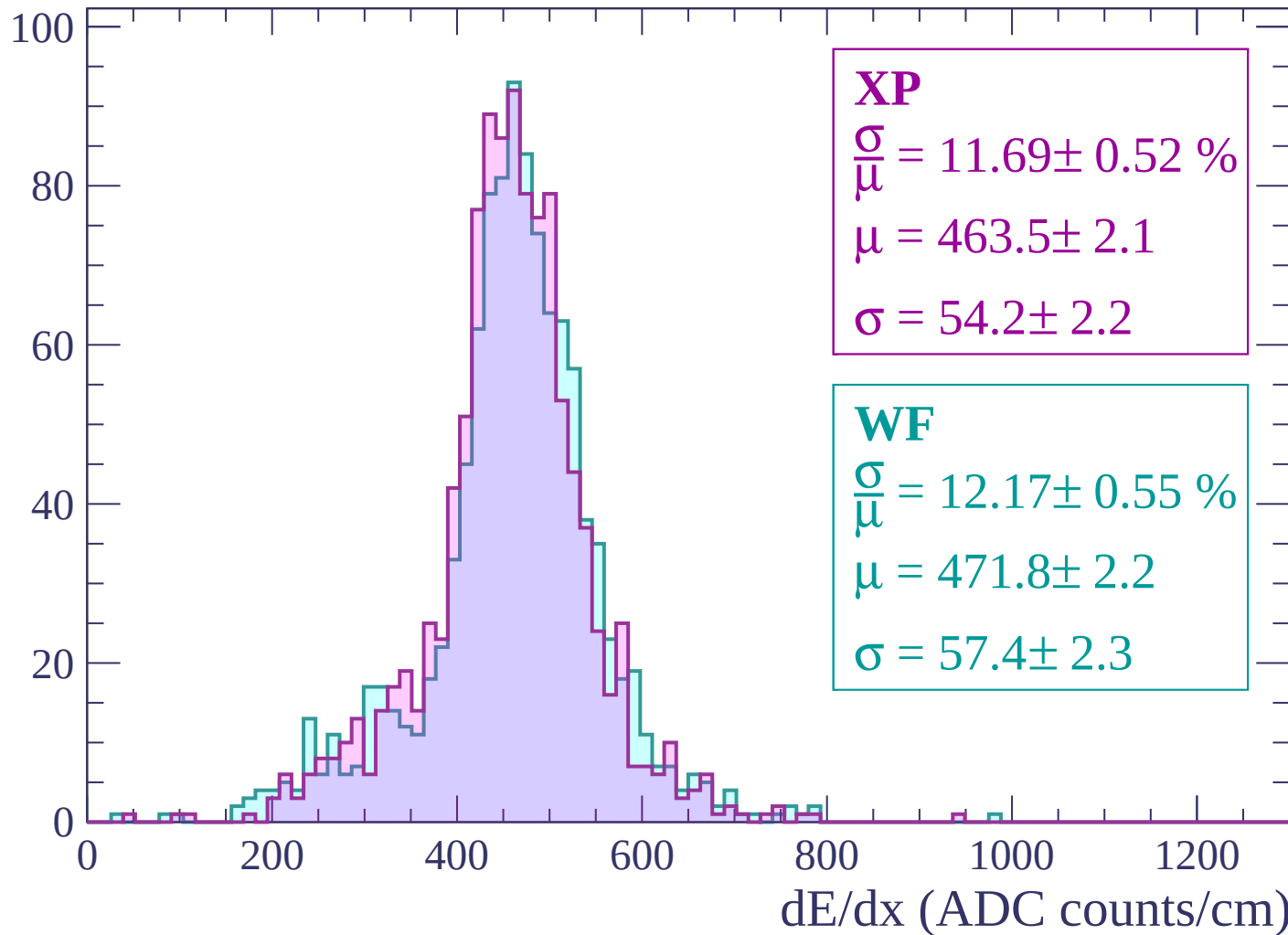
Energy loss | $120 < p < 180$

Count



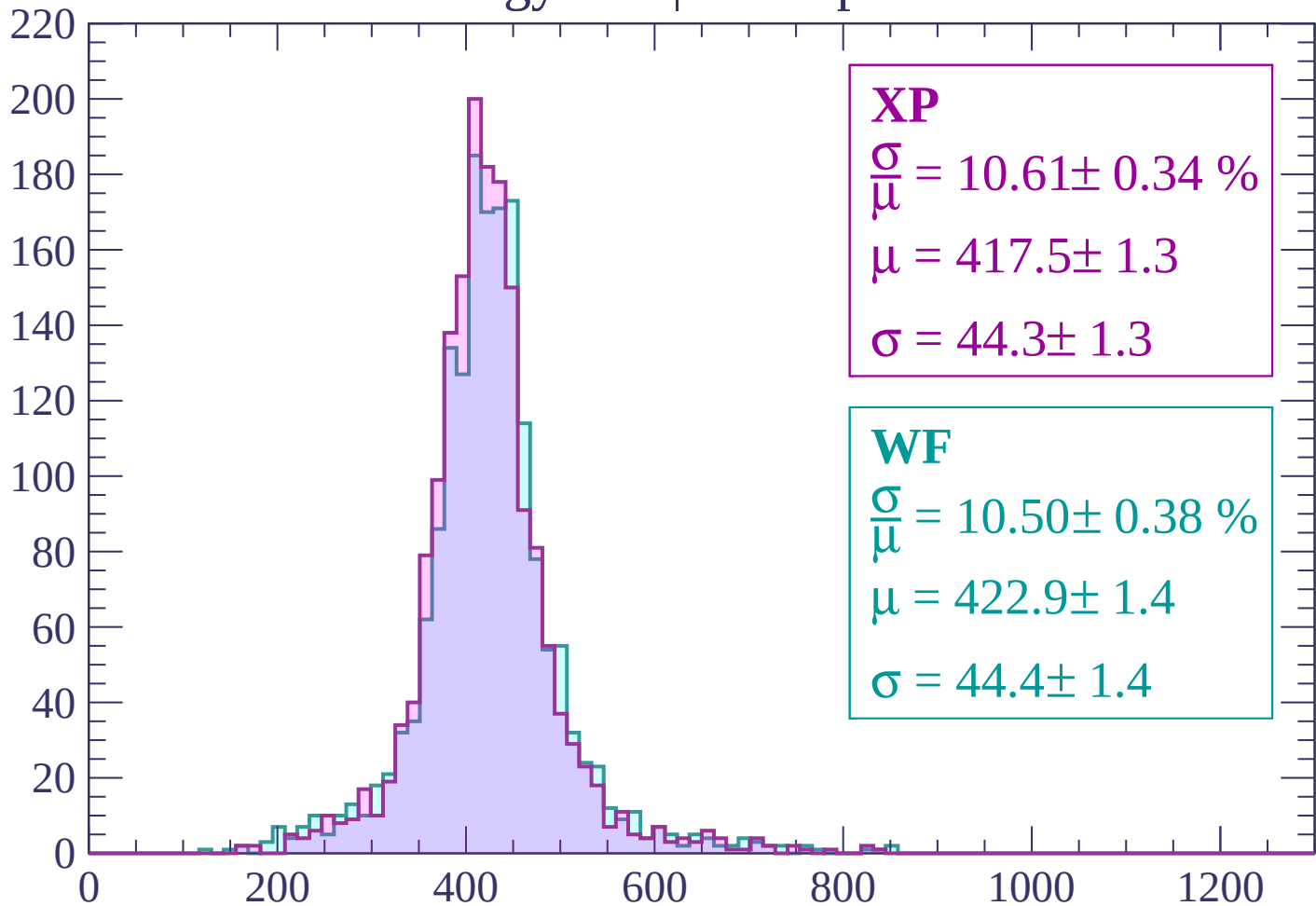
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP

$$\frac{\sigma}{\mu} = 10.61 \pm 0.34 \%$$

$$\mu = 417.5 \pm 1.3$$

$$\sigma = 44.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.50 \pm 0.38 \%$$

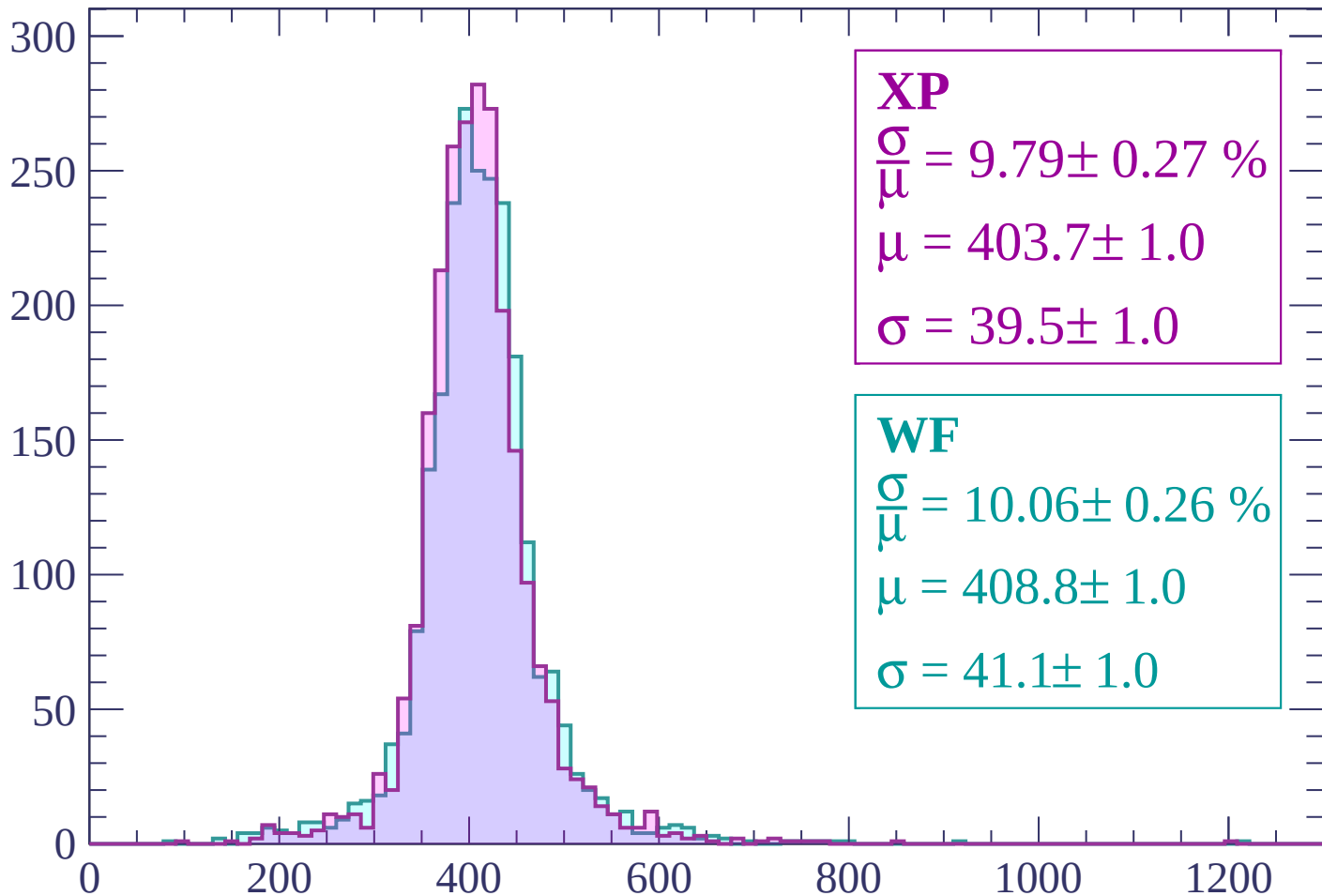
$$\mu = 422.9 \pm 1.4$$

$$\sigma = 44.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 9.79 \pm 0.27 \%$$

$$\mu = 403.7 \pm 1.0$$

$$\sigma = 39.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.06 \pm 0.26 \%$$

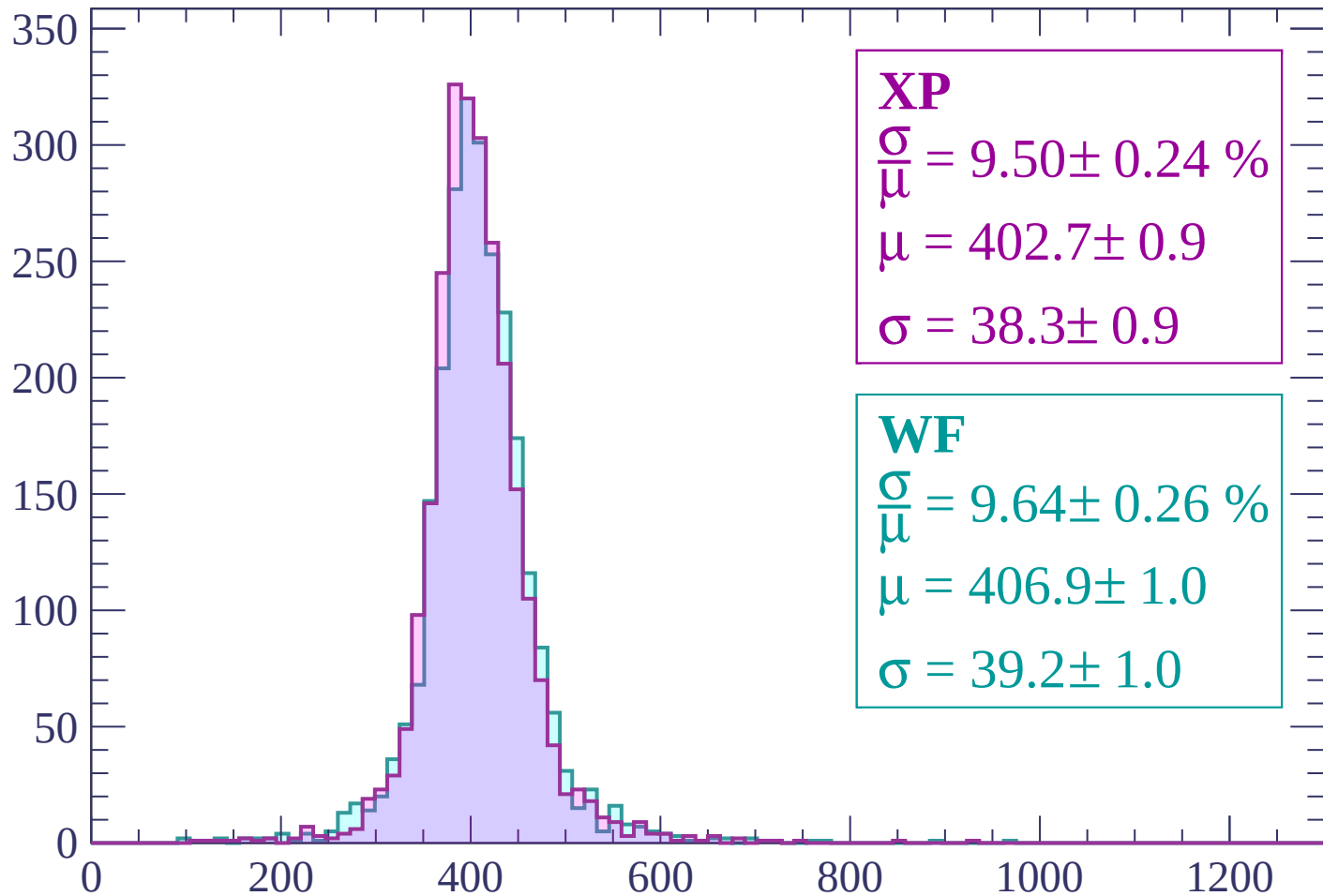
$$\mu = 408.8 \pm 1.0$$

$$\sigma = 41.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.24 \%$$

$$\mu = 402.7 \pm 0.9$$

$$\sigma = 38.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.26 \%$$

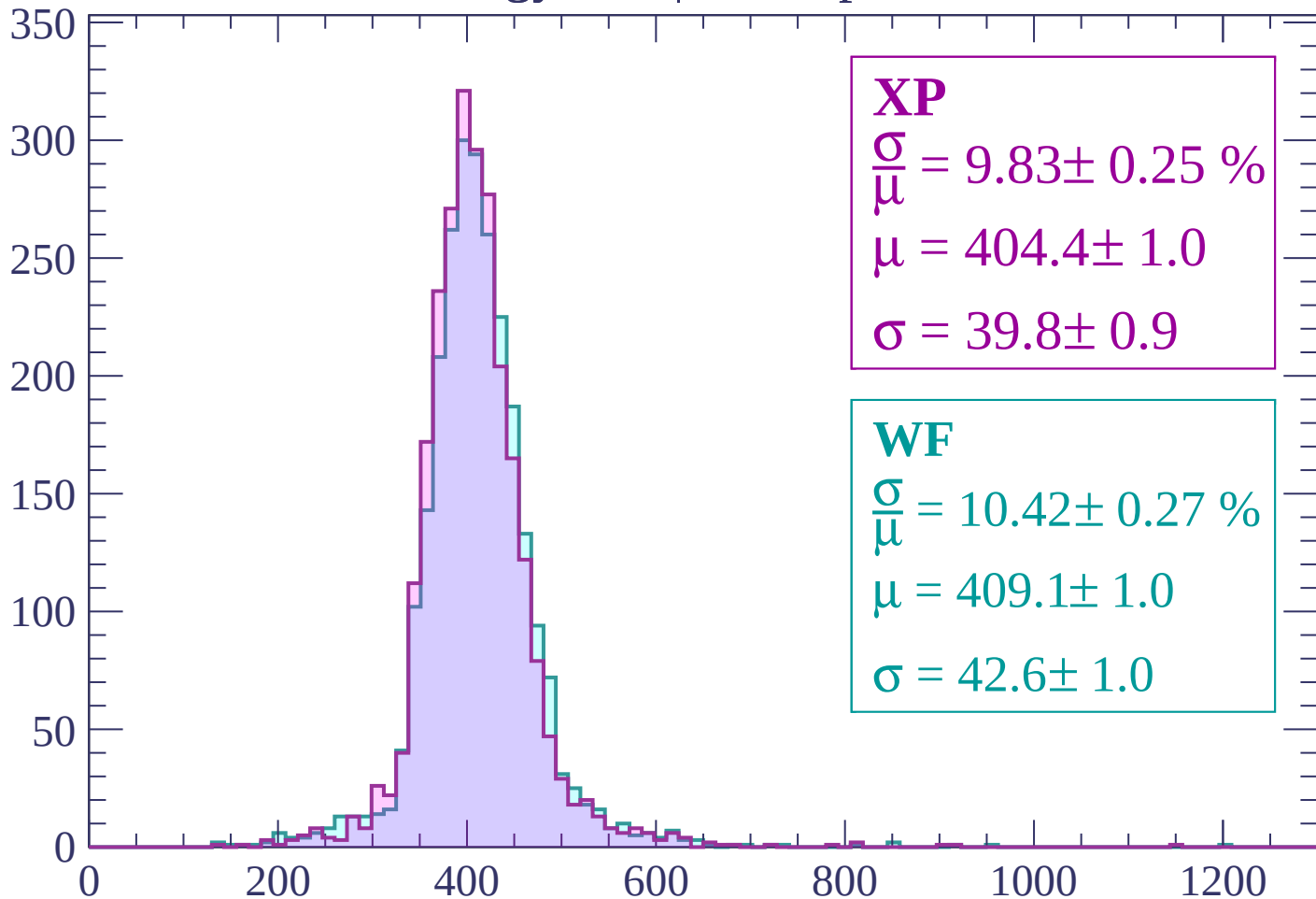
$$\mu = 406.9 \pm 1.0$$

$$\sigma = 39.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 9.83 \pm 0.25 \%$$

$$\mu = 404.4 \pm 1.0$$

$$\sigma = 39.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.42 \pm 0.27 \%$$

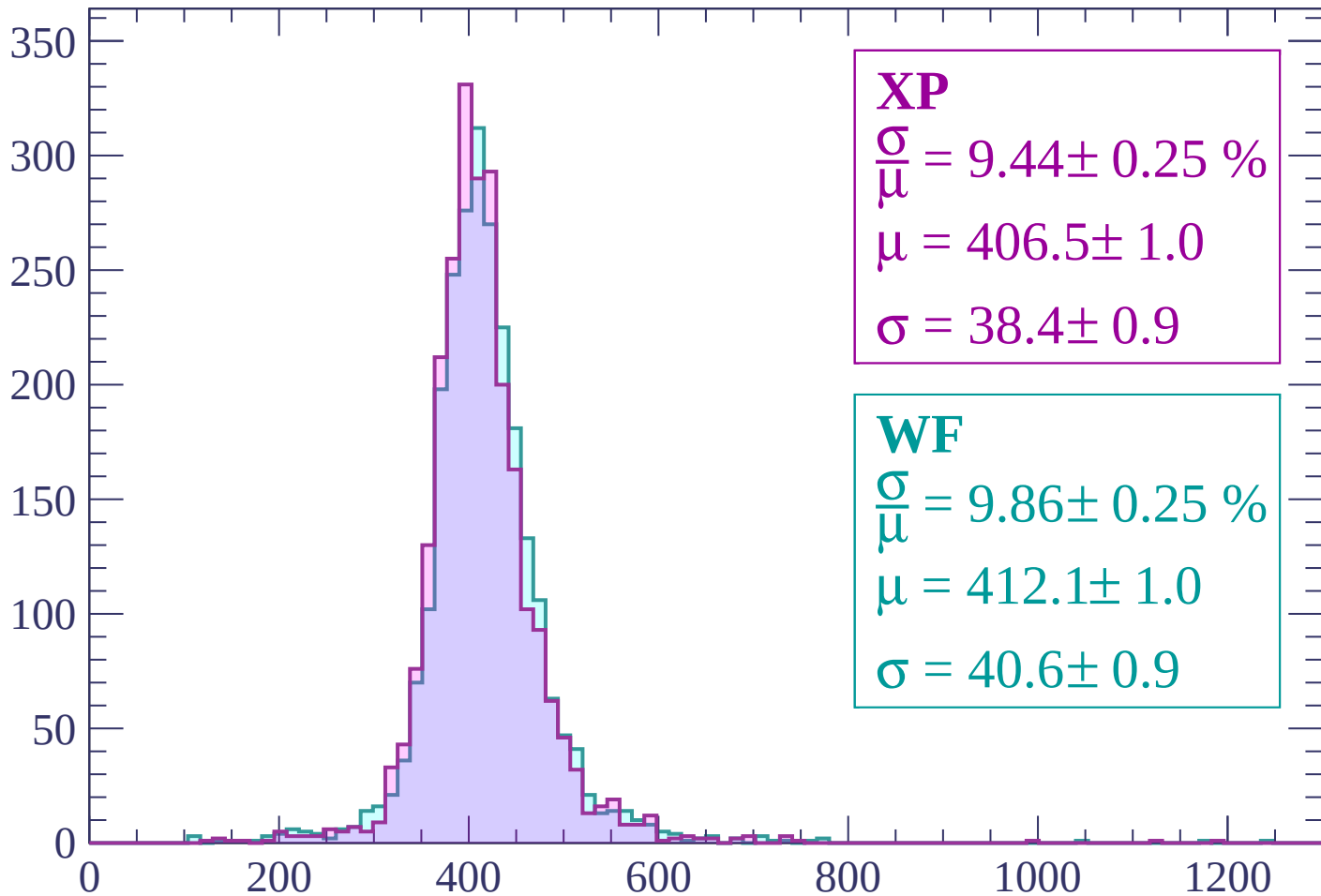
$$\mu = 409.1 \pm 1.0$$

$$\sigma = 42.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP

$$\frac{\sigma}{\mu} = 9.44 \pm 0.25 \%$$

$$\mu = 406.5 \pm 1.0$$

$$\sigma = 38.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.25 \%$$

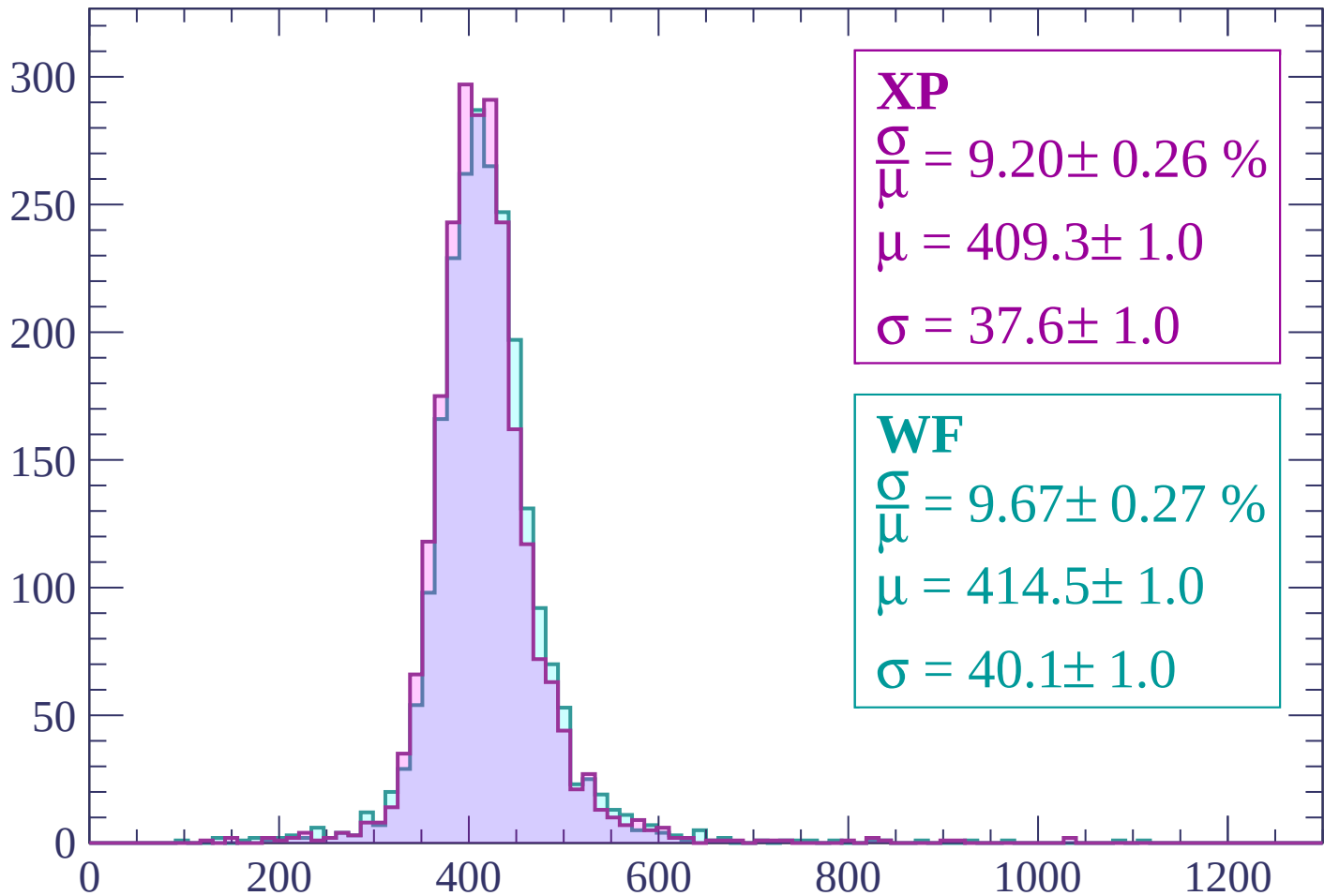
$$\mu = 412.1 \pm 1.0$$

$$\sigma = 40.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.26 \%$$

$$\mu = 409.3 \pm 1.0$$

$$\sigma = 37.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.27 \%$$

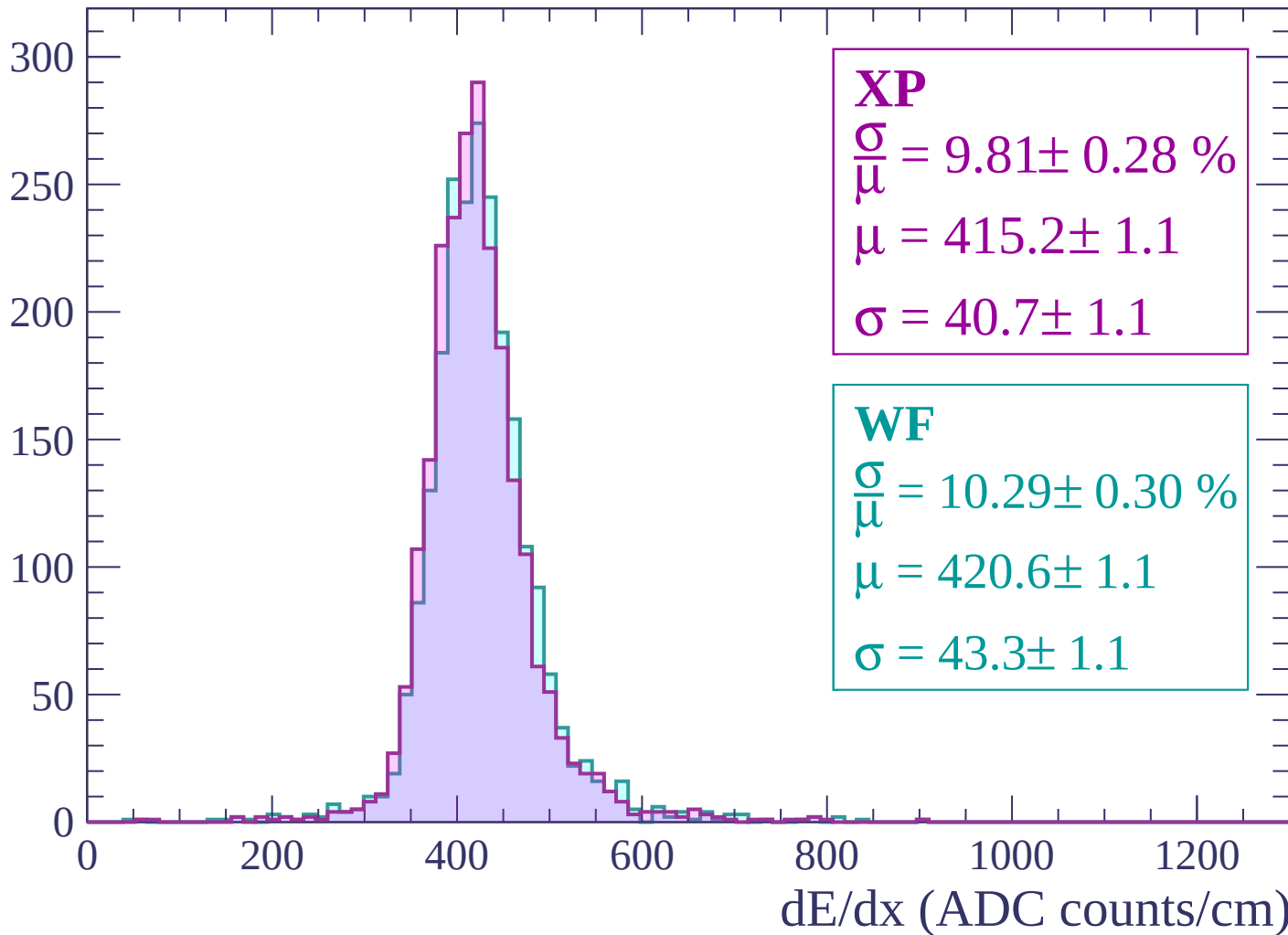
$$\mu = 414.5 \pm 1.0$$

$$\sigma = 40.1 \pm 1.0$$

dE/dx (ADC counts/cm)

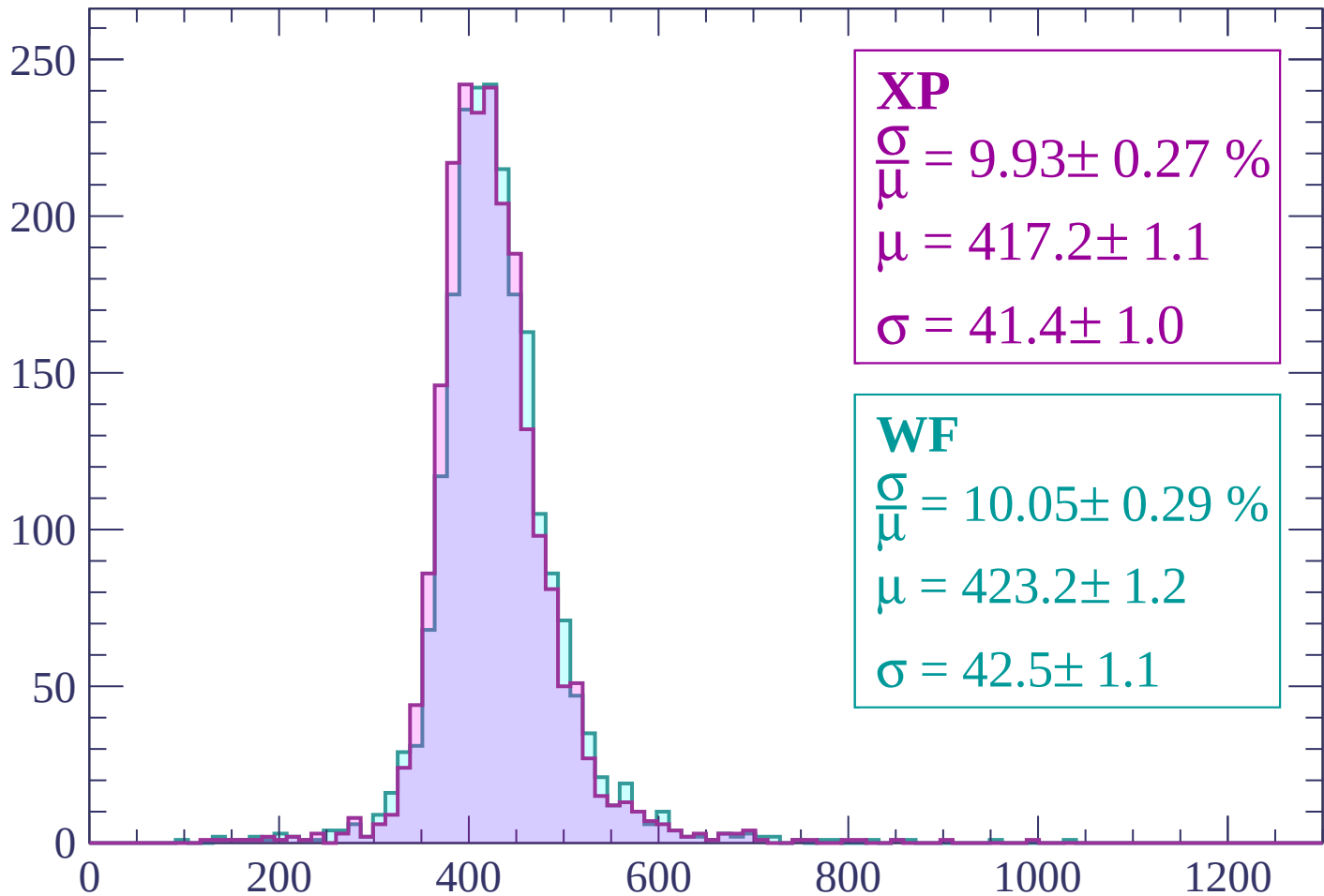
Energy loss | $600 < p < 660$

Count



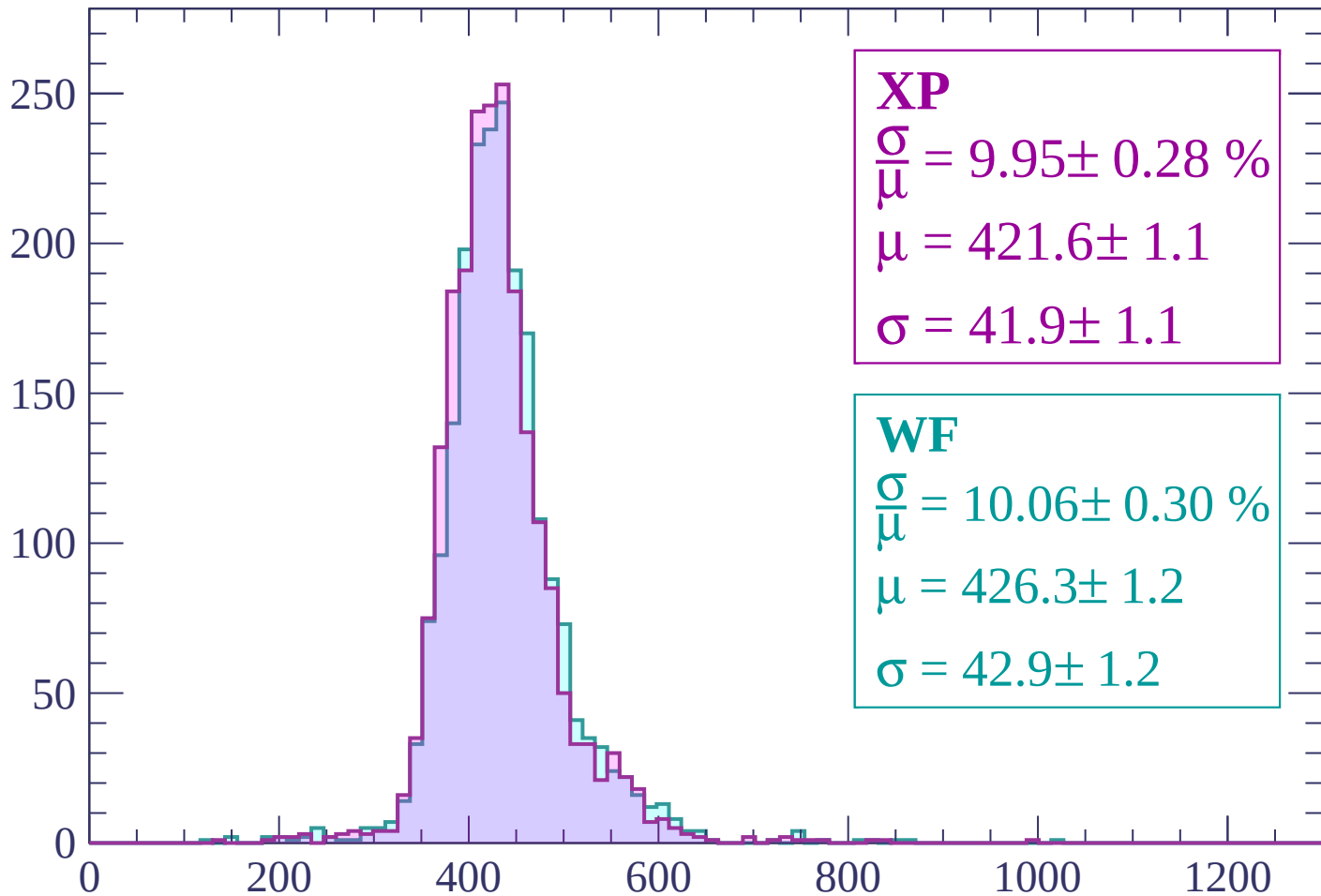
Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count



XP

$$\frac{\sigma}{\mu} = 9.95 \pm 0.28 \%$$

$$\mu = 421.6 \pm 1.1$$

$$\sigma = 41.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.06 \pm 0.30 \%$$

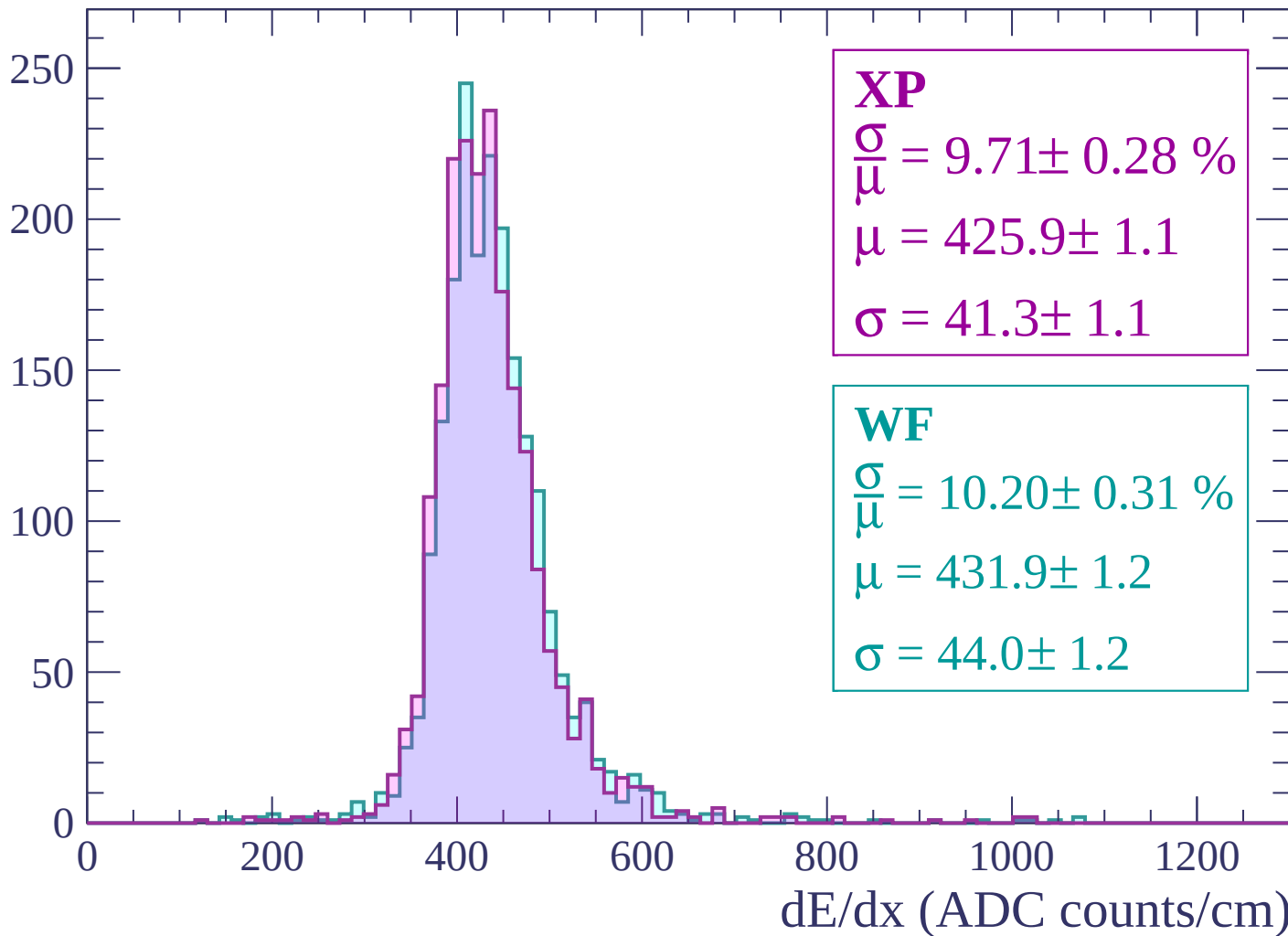
$$\mu = 426.3 \pm 1.2$$

$$\sigma = 42.9 \pm 1.2$$

dE/dx (ADC counts/cm)

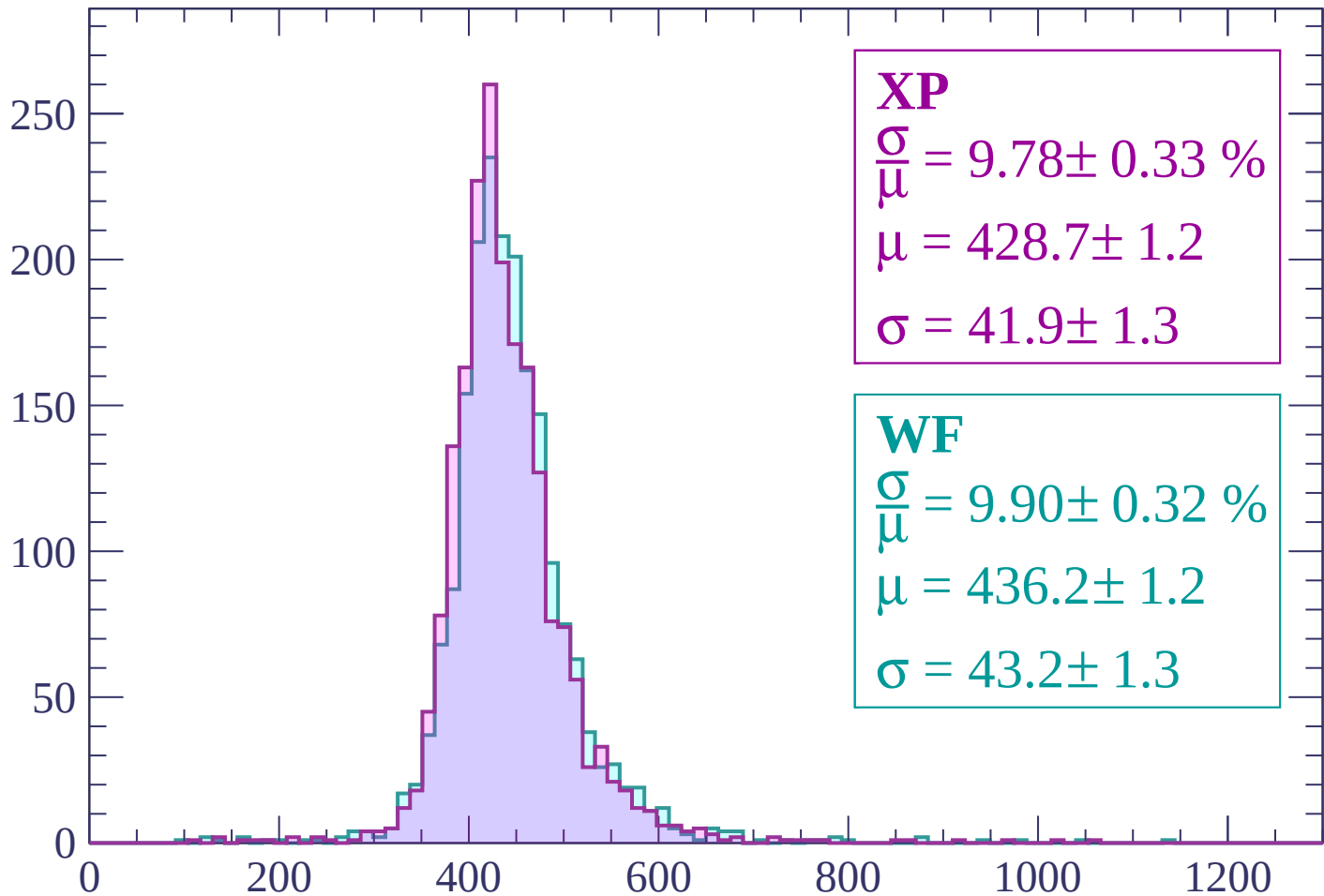
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP

$$\frac{\sigma}{\mu} = 9.78 \pm 0.33 \%$$

$$\mu = 428.7 \pm 1.2$$

$$\sigma = 41.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.90 \pm 0.32 \%$$

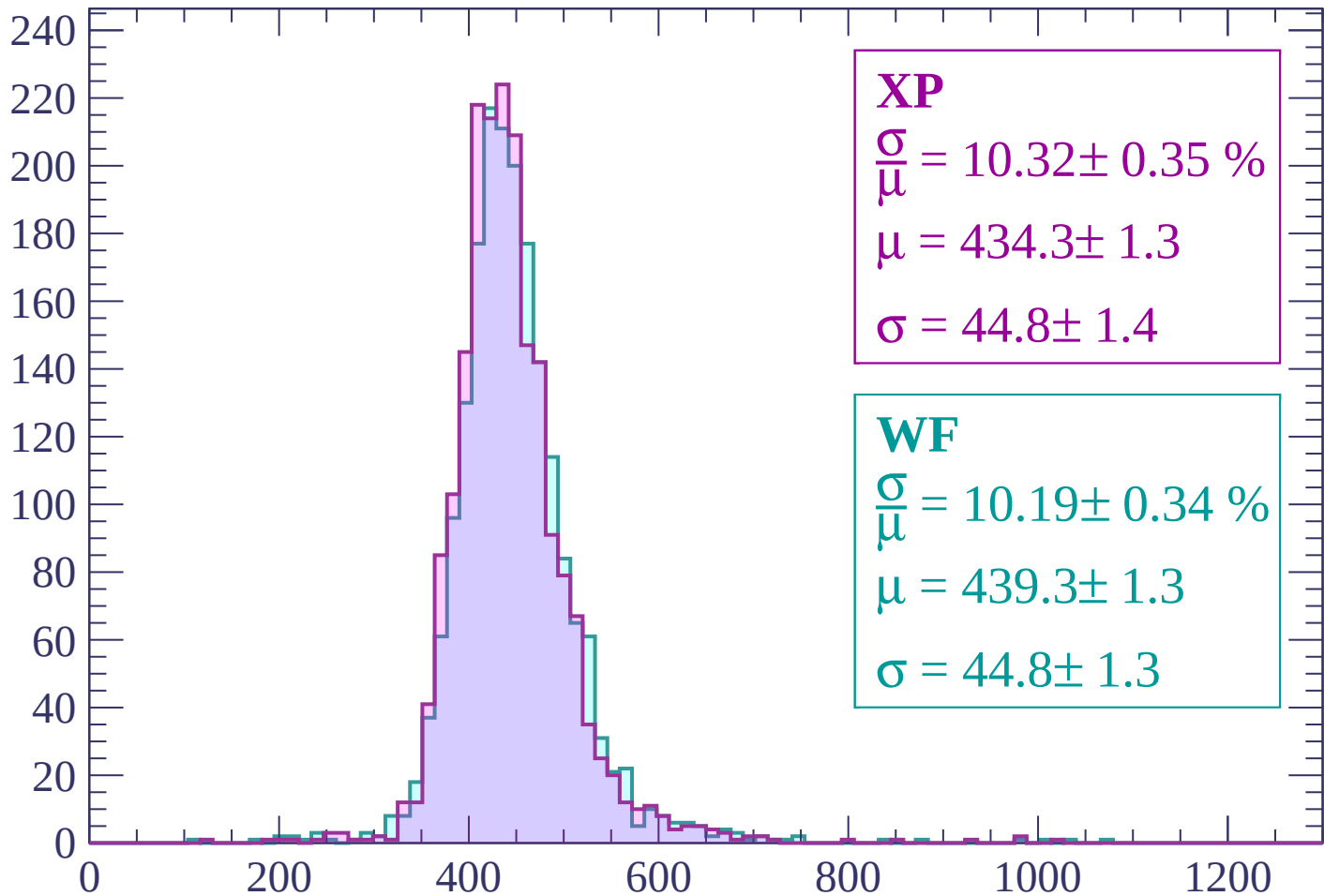
$$\mu = 436.2 \pm 1.2$$

$$\sigma = 43.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 10.32 \pm 0.35 \%$$

$$\mu = 434.3 \pm 1.3$$

$$\sigma = 44.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.19 \pm 0.34 \%$$

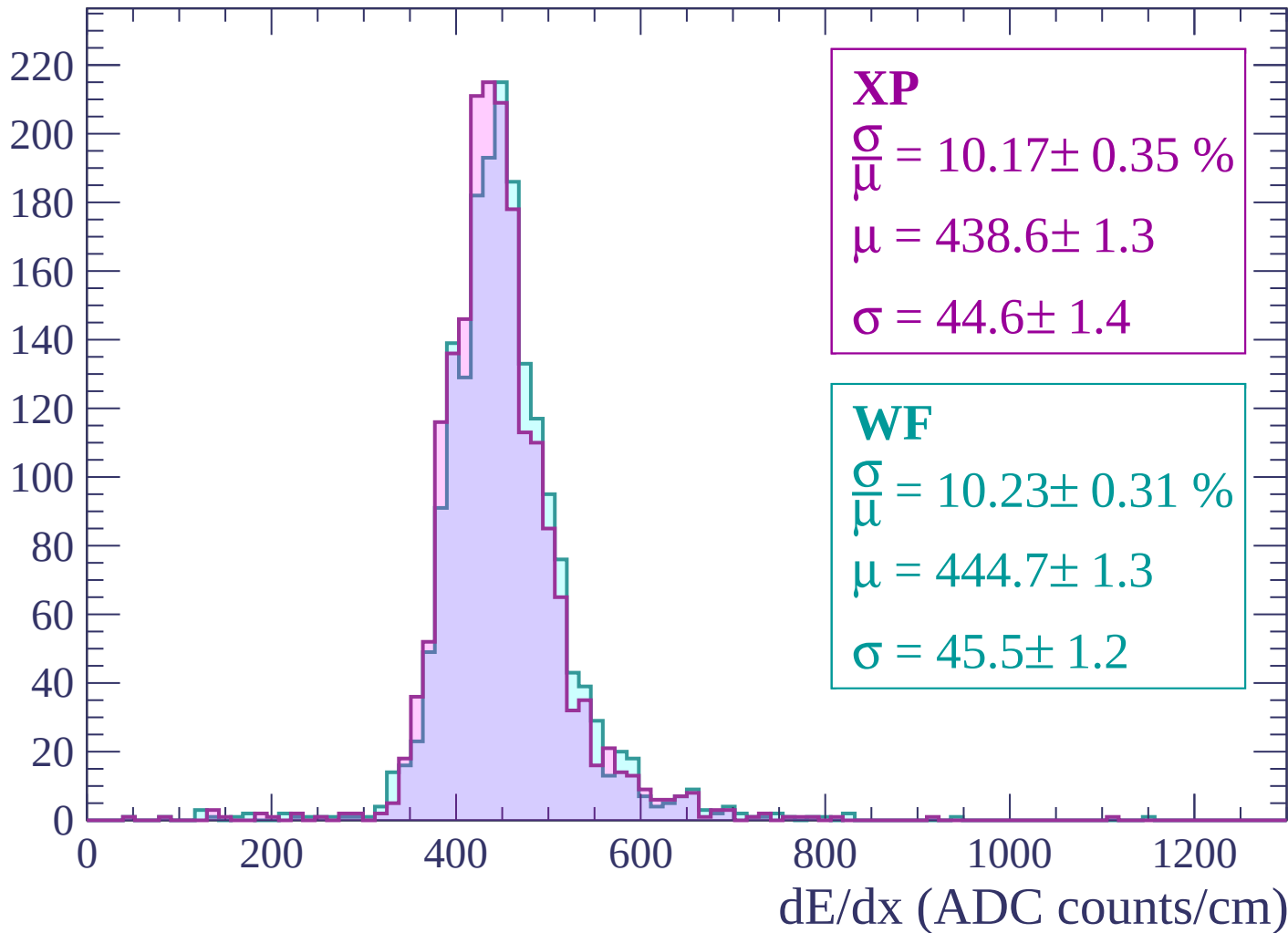
$$\mu = 439.3 \pm 1.3$$

$$\sigma = 44.8 \pm 1.3$$

dE/dx (ADC counts/cm)

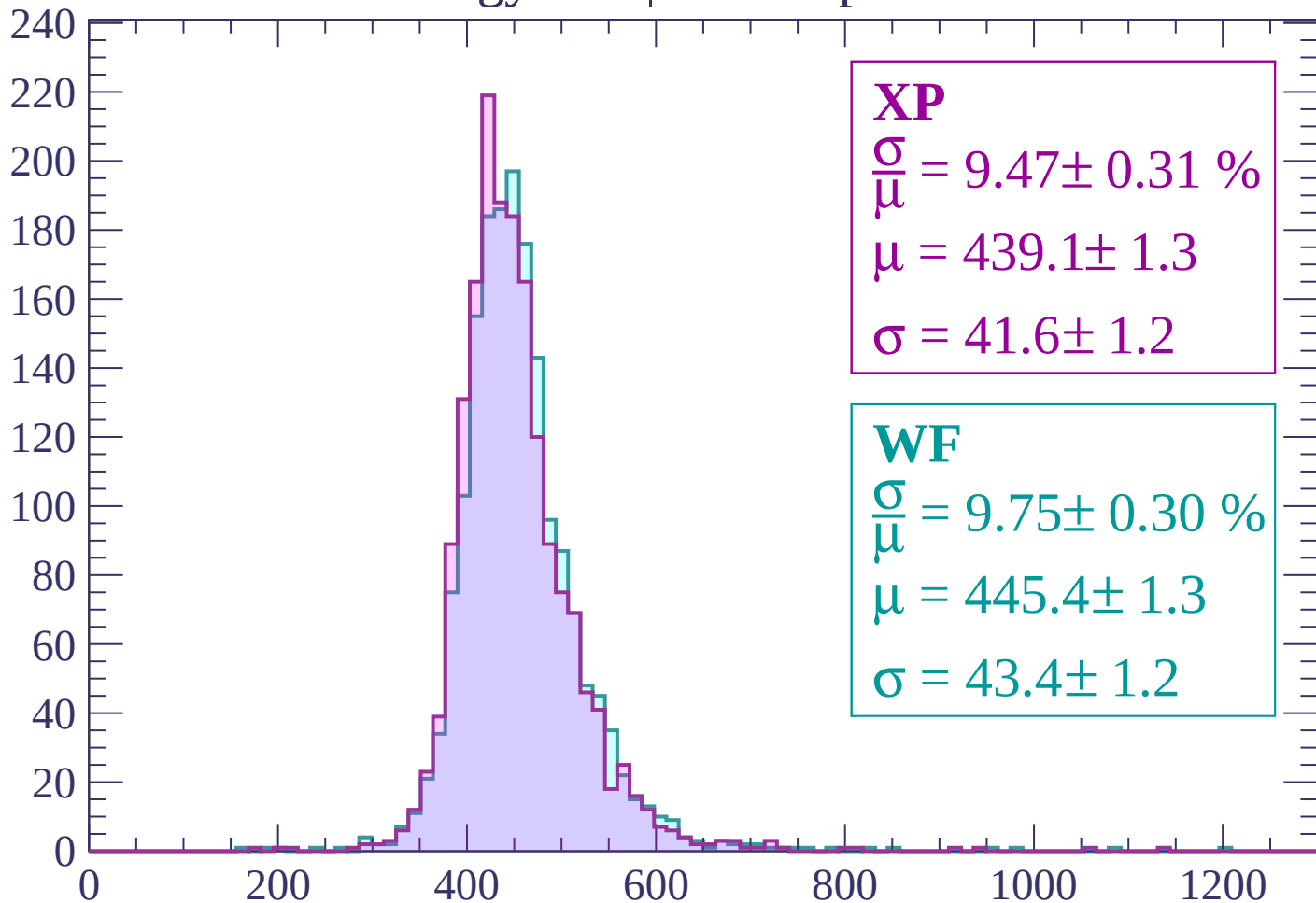
Energy loss | $960 < p < 1020$

Count



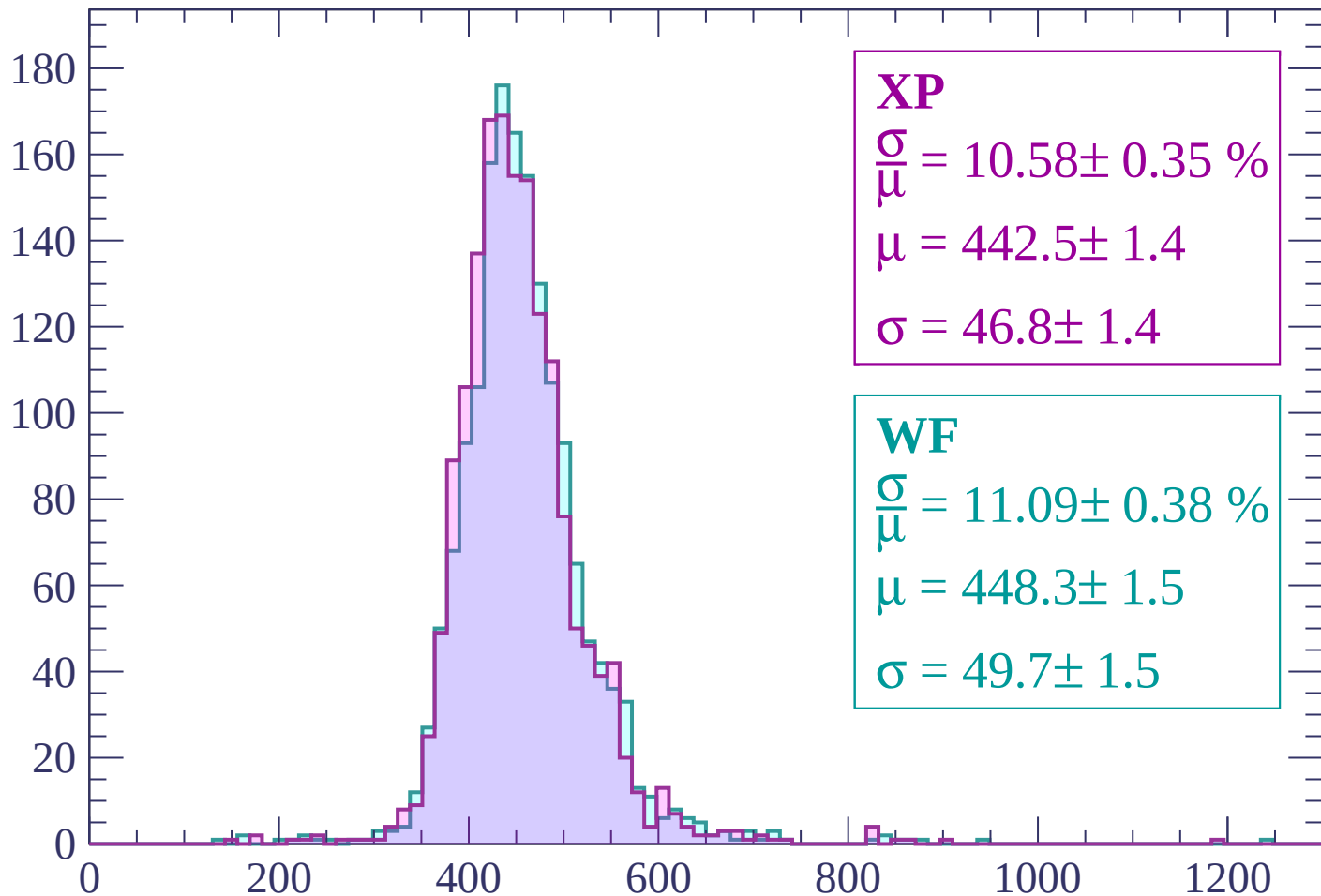
Energy loss | $1020 < p < 1080$

Count



Energy loss | $1080 < p < 1140$

Count



XP

$$\frac{\sigma}{\mu} = 10.58 \pm 0.35 \%$$

$$\mu = 442.5 \pm 1.4$$

$$\sigma = 46.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.09 \pm 0.38 \%$$

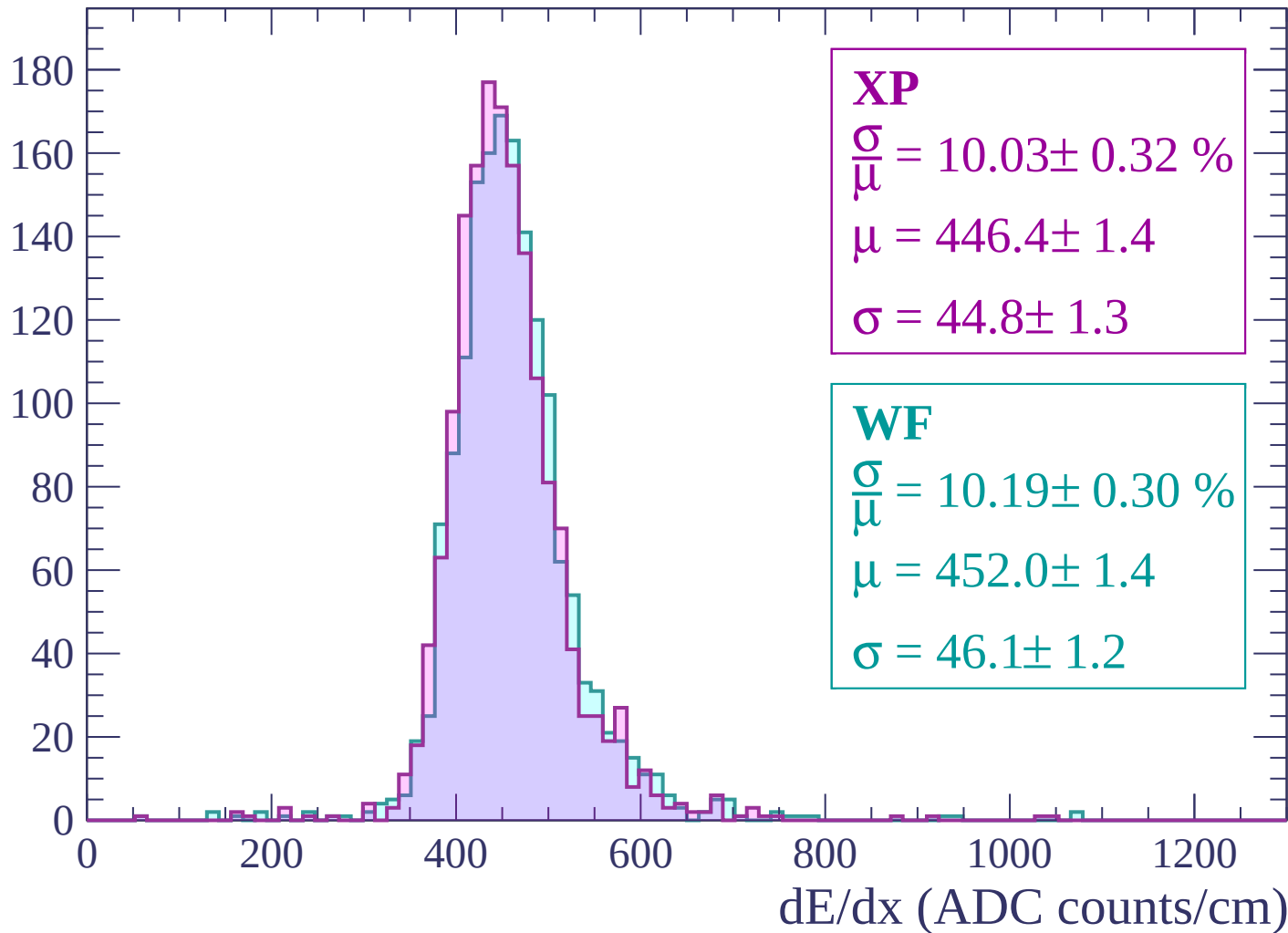
$$\mu = 448.3 \pm 1.5$$

$$\sigma = 49.7 \pm 1.5$$

dE/dx (ADC counts/cm)

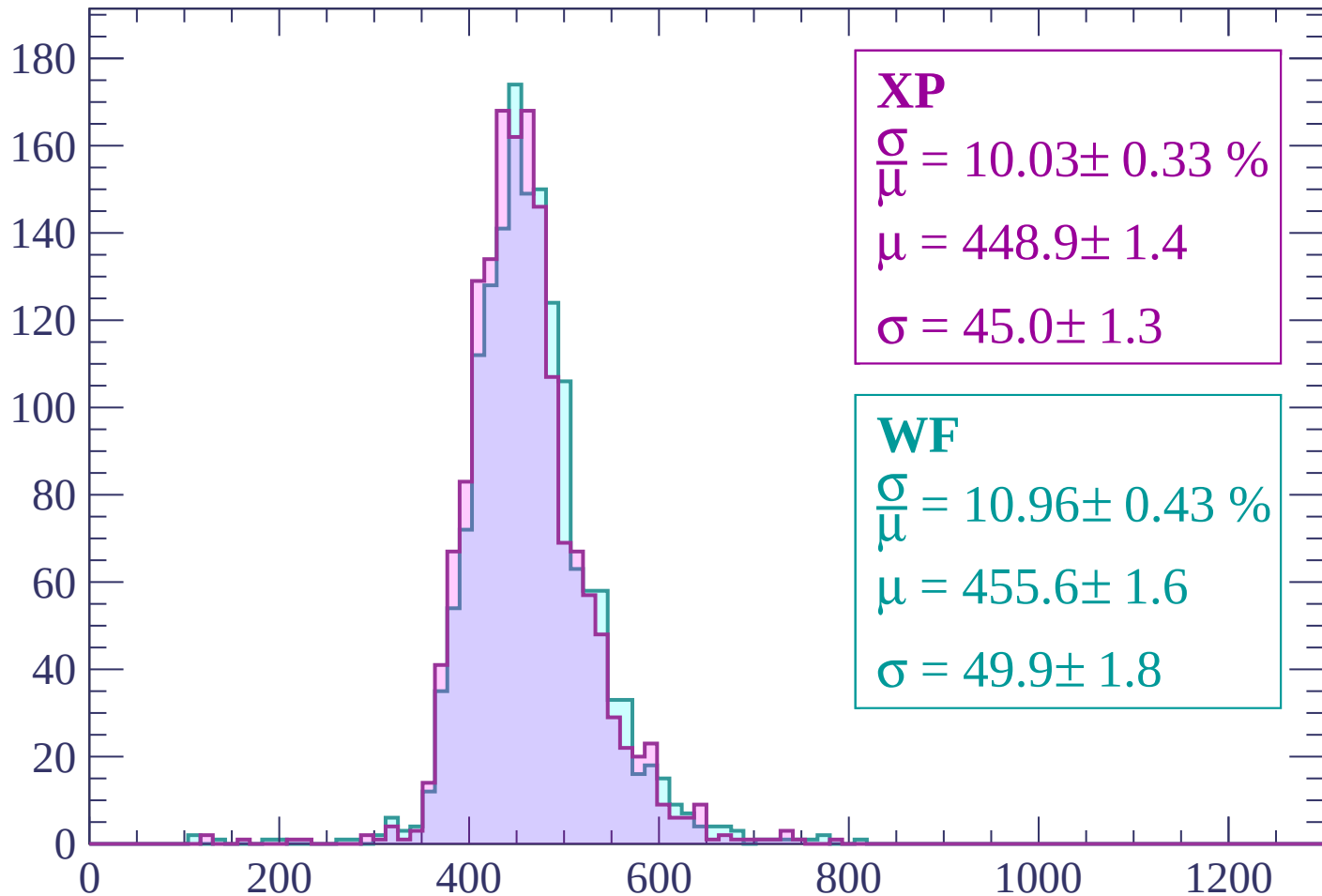
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP

$$\frac{\sigma}{\mu} = 10.03 \pm 0.33 \%$$

$$\mu = 448.9 \pm 1.4$$

$$\sigma = 45.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.96 \pm 0.43 \%$$

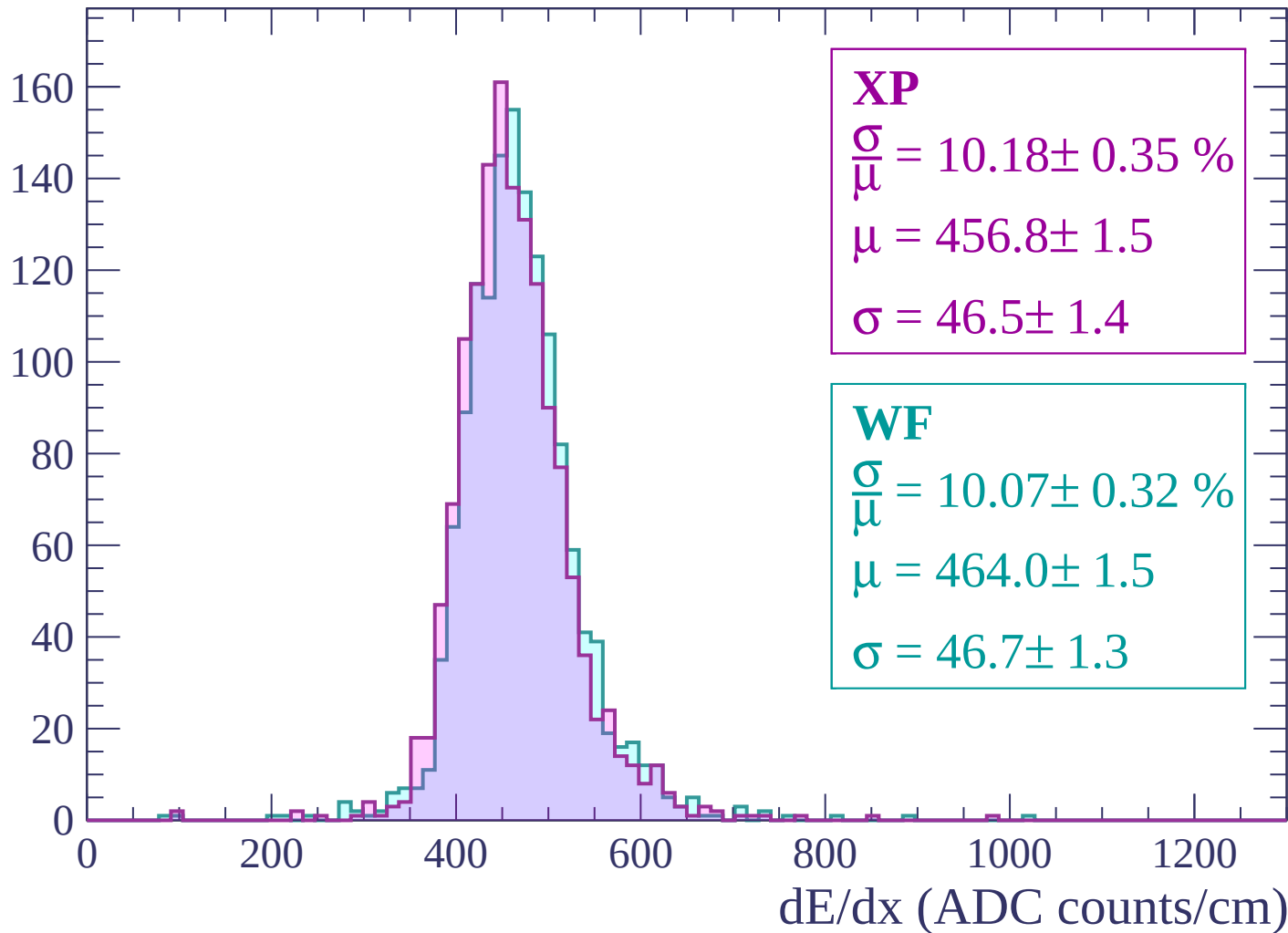
$$\mu = 455.6 \pm 1.6$$

$$\sigma = 49.9 \pm 1.8$$

dE/dx (ADC counts/cm)

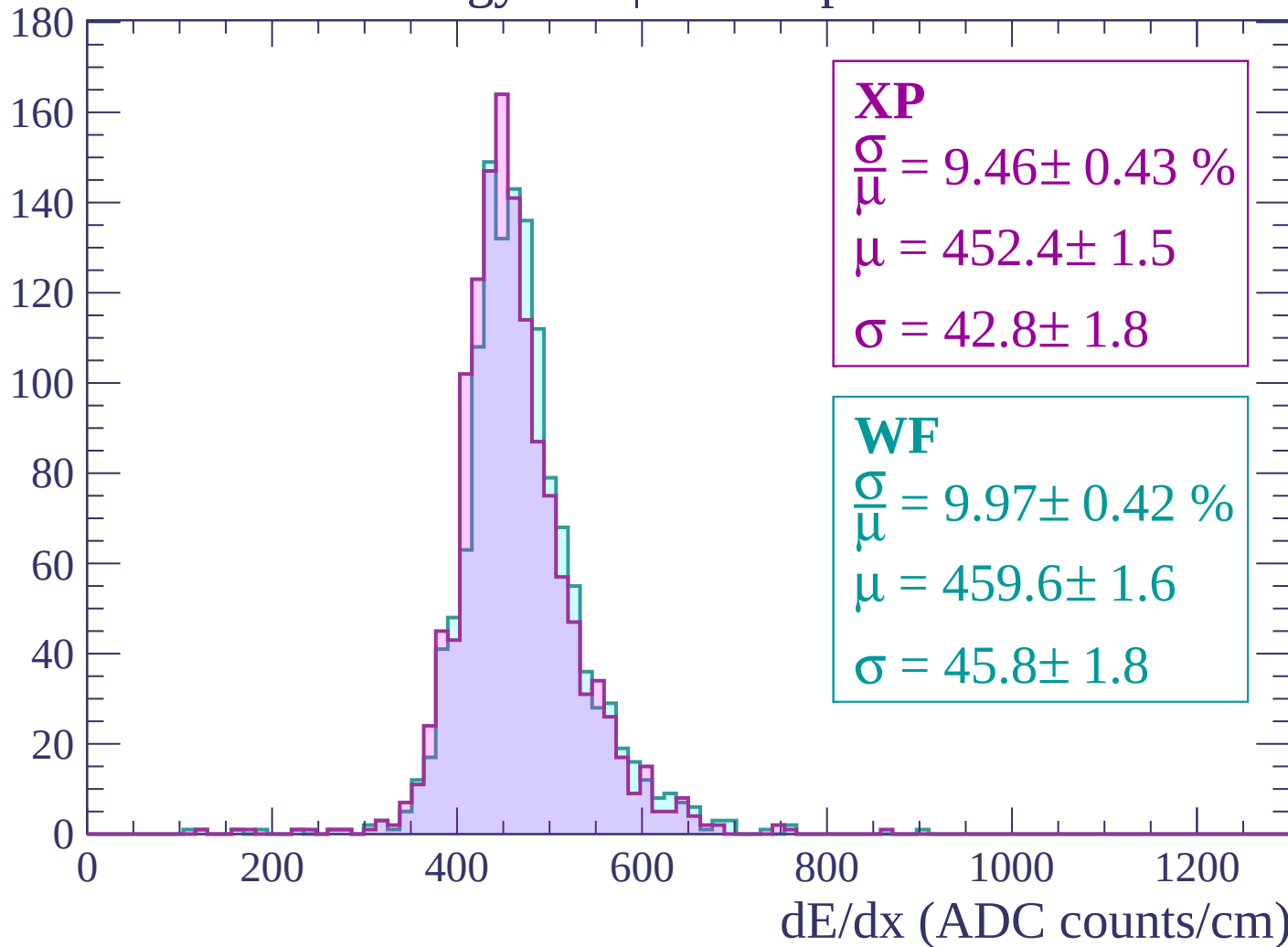
Energy loss | $1260 < p < 1320$

Count



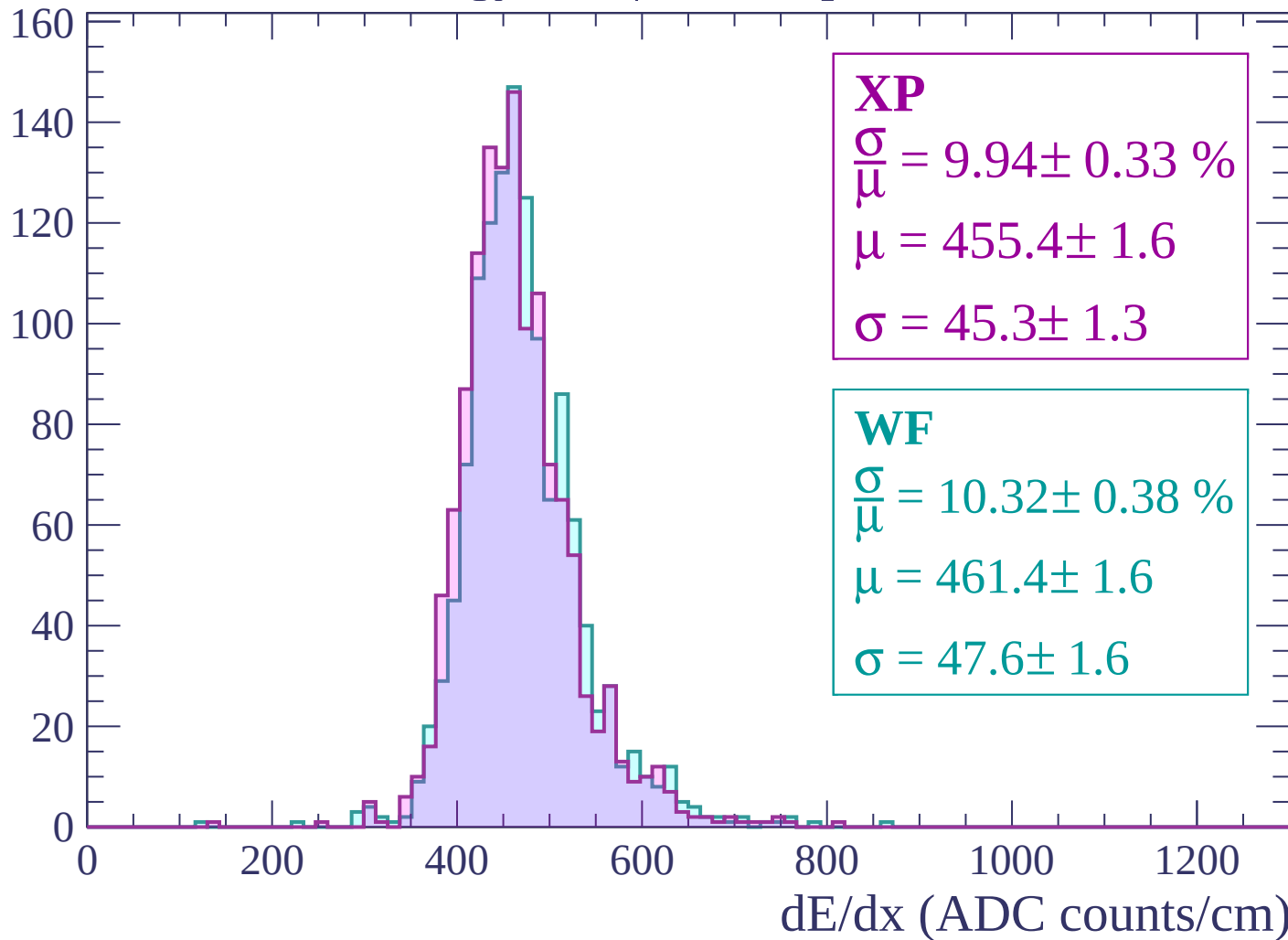
Energy loss | $1320 < p < 1380$

Count



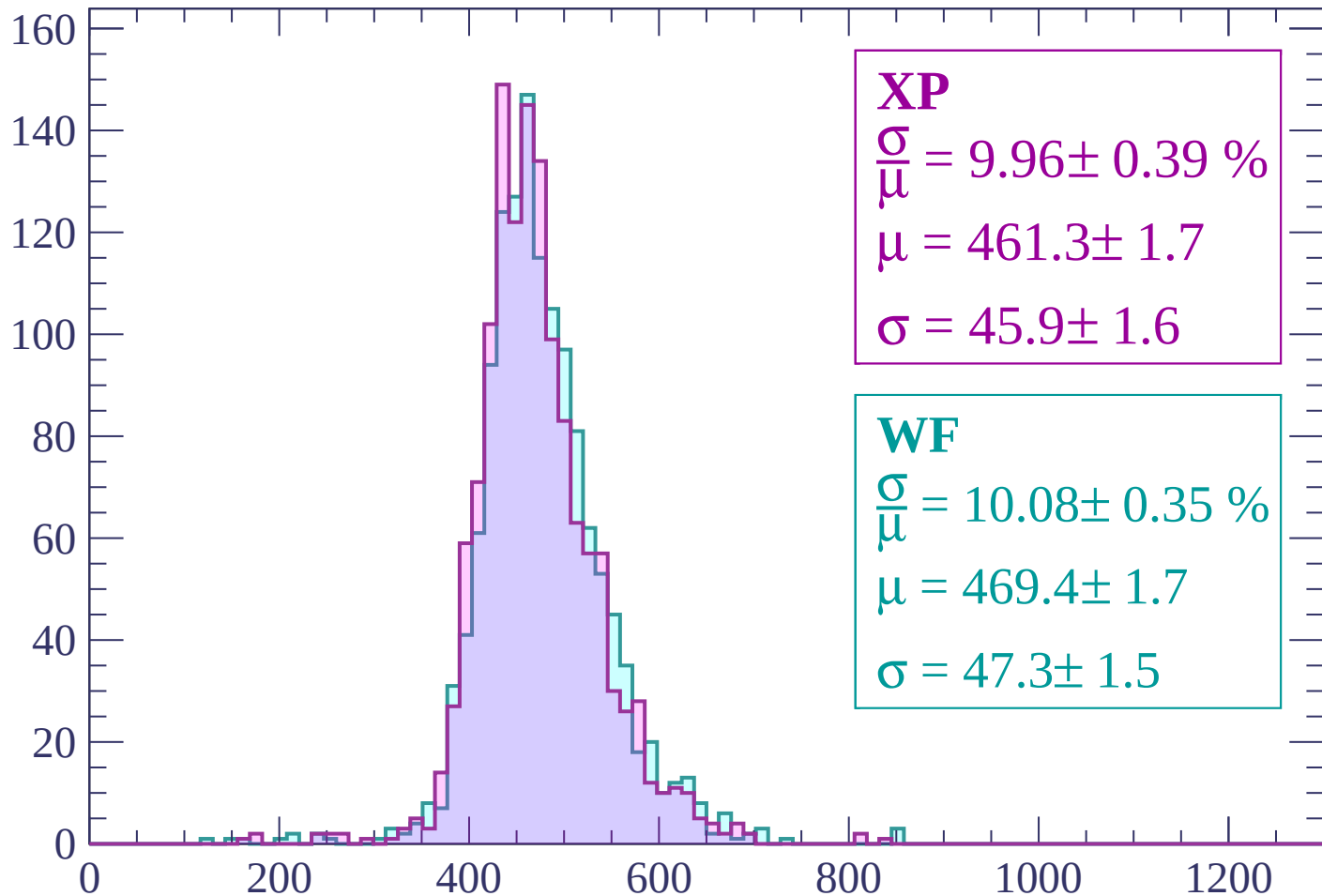
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.39 \%$$

$$\mu = 461.3 \pm 1.7$$

$$\sigma = 45.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 10.08 \pm 0.35 \%$$

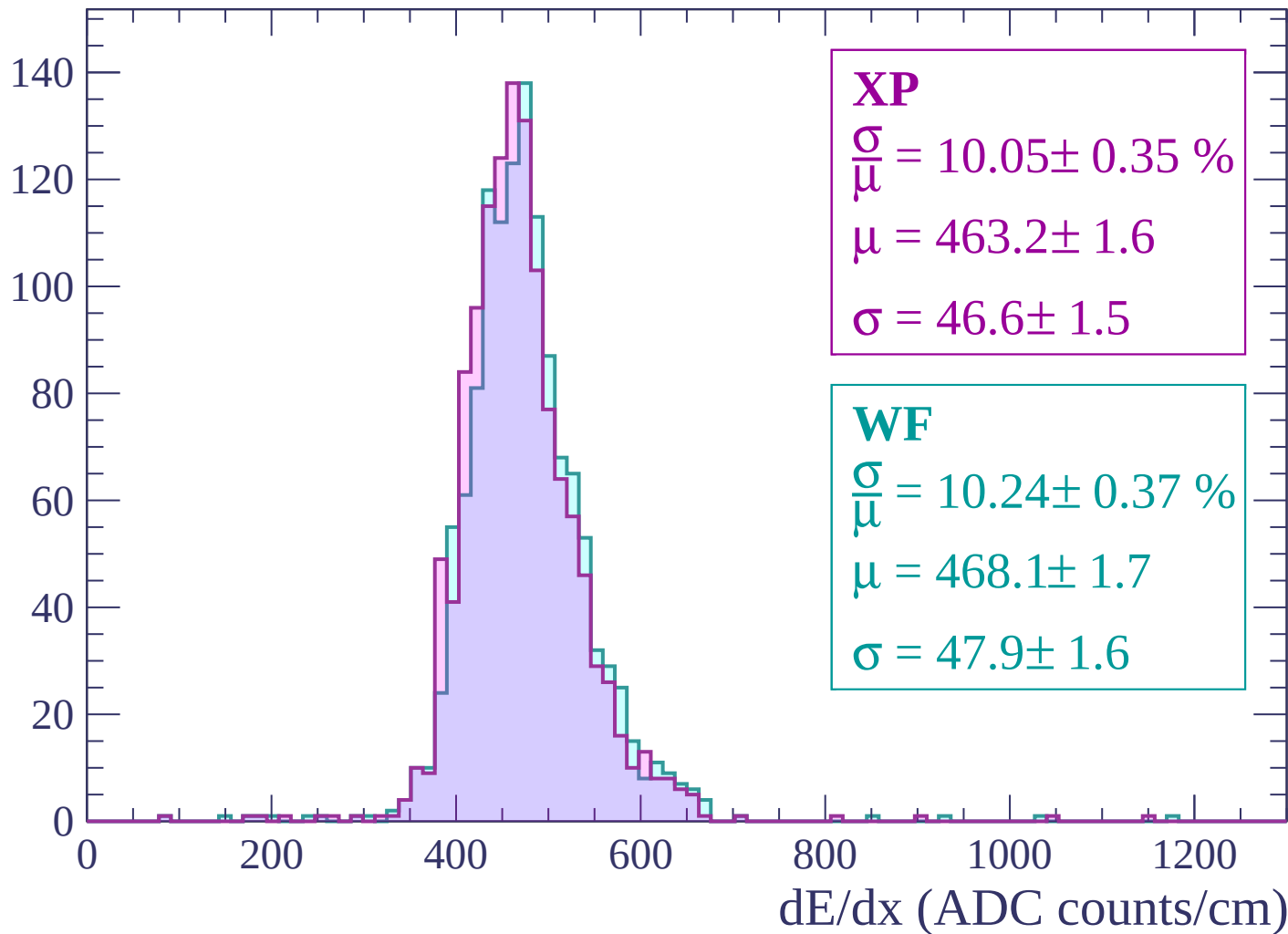
$$\mu = 469.4 \pm 1.7$$

$$\sigma = 47.3 \pm 1.5$$

dE/dx (ADC counts/cm)

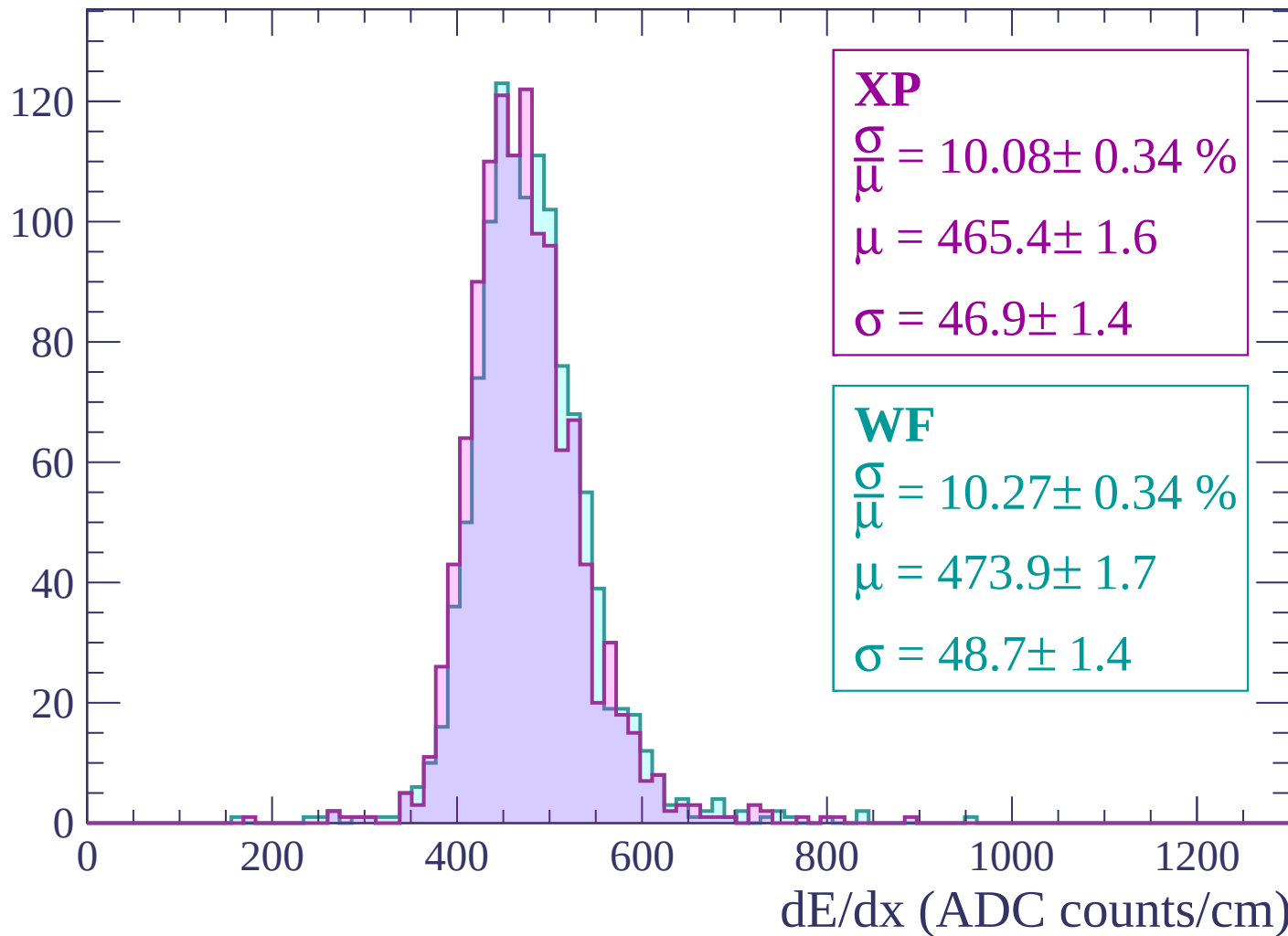
Energy loss | $1500 < p < 1560$

Count



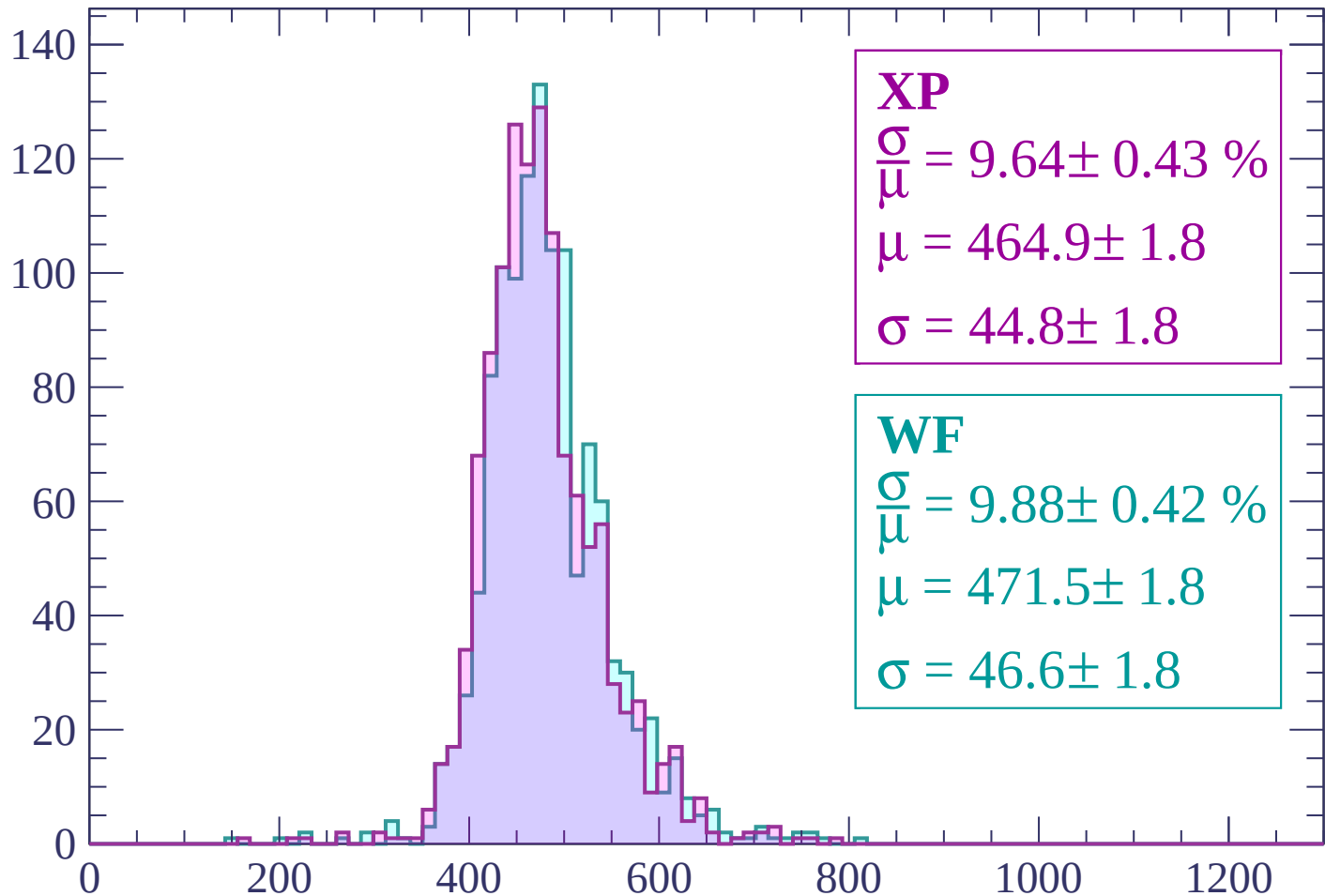
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.43 \%$$

$$\mu = 464.9 \pm 1.8$$

$$\sigma = 44.8 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 9.88 \pm 0.42 \%$$

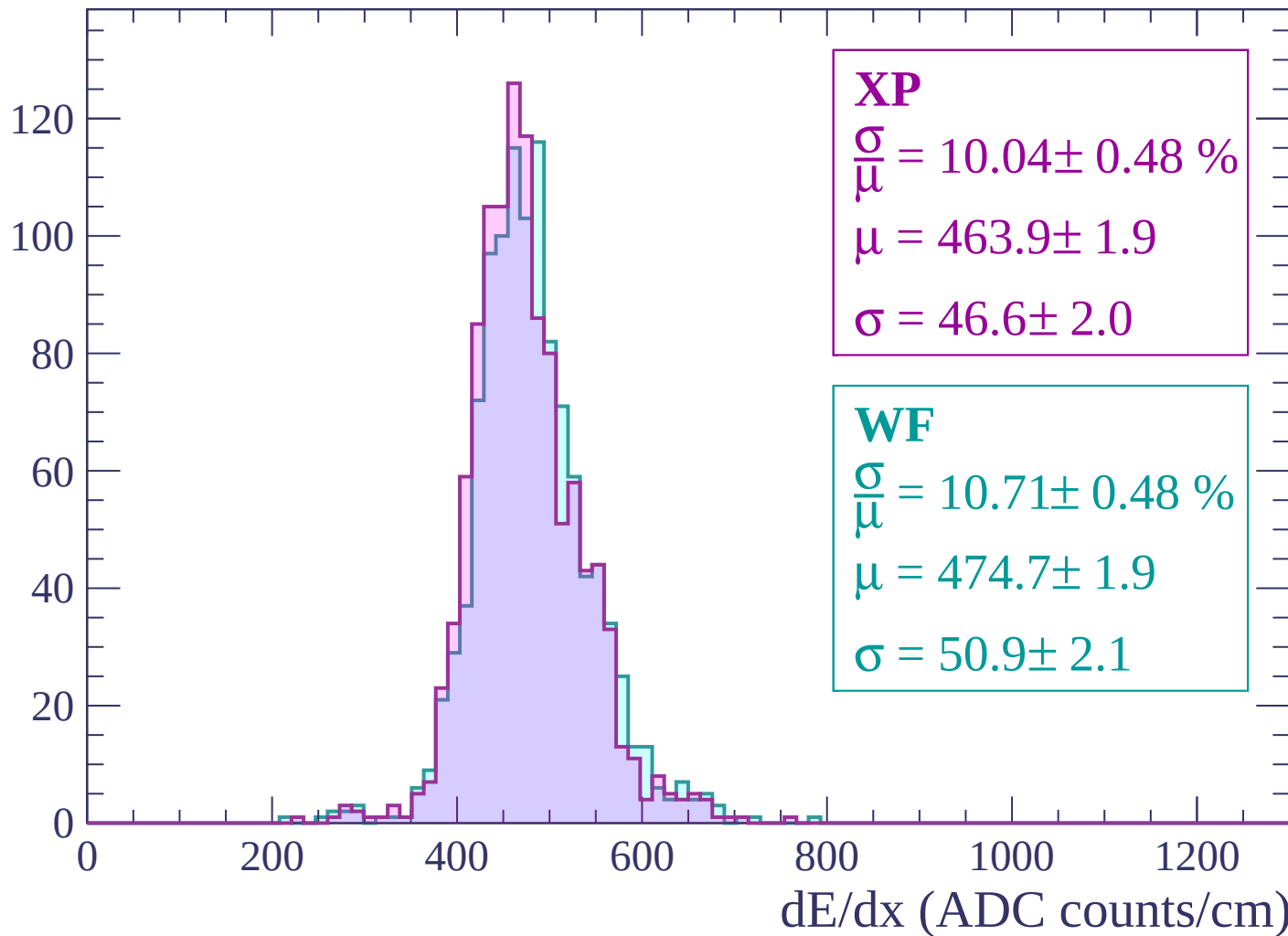
$$\mu = 471.5 \pm 1.8$$

$$\sigma = 46.6 \pm 1.8$$

dE/dx (ADC counts/cm)

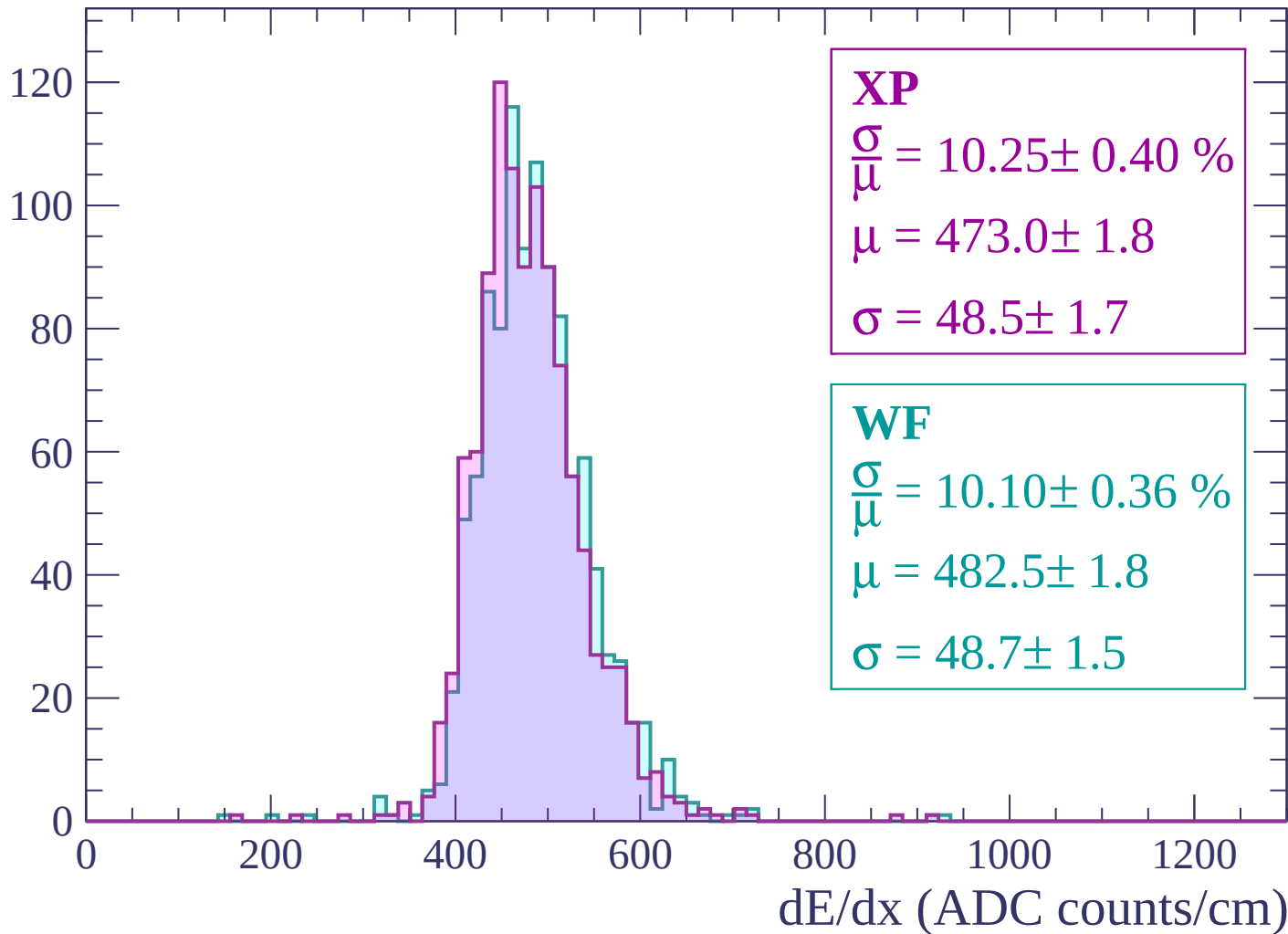
Energy loss | $1680 < p < 1740$

Count



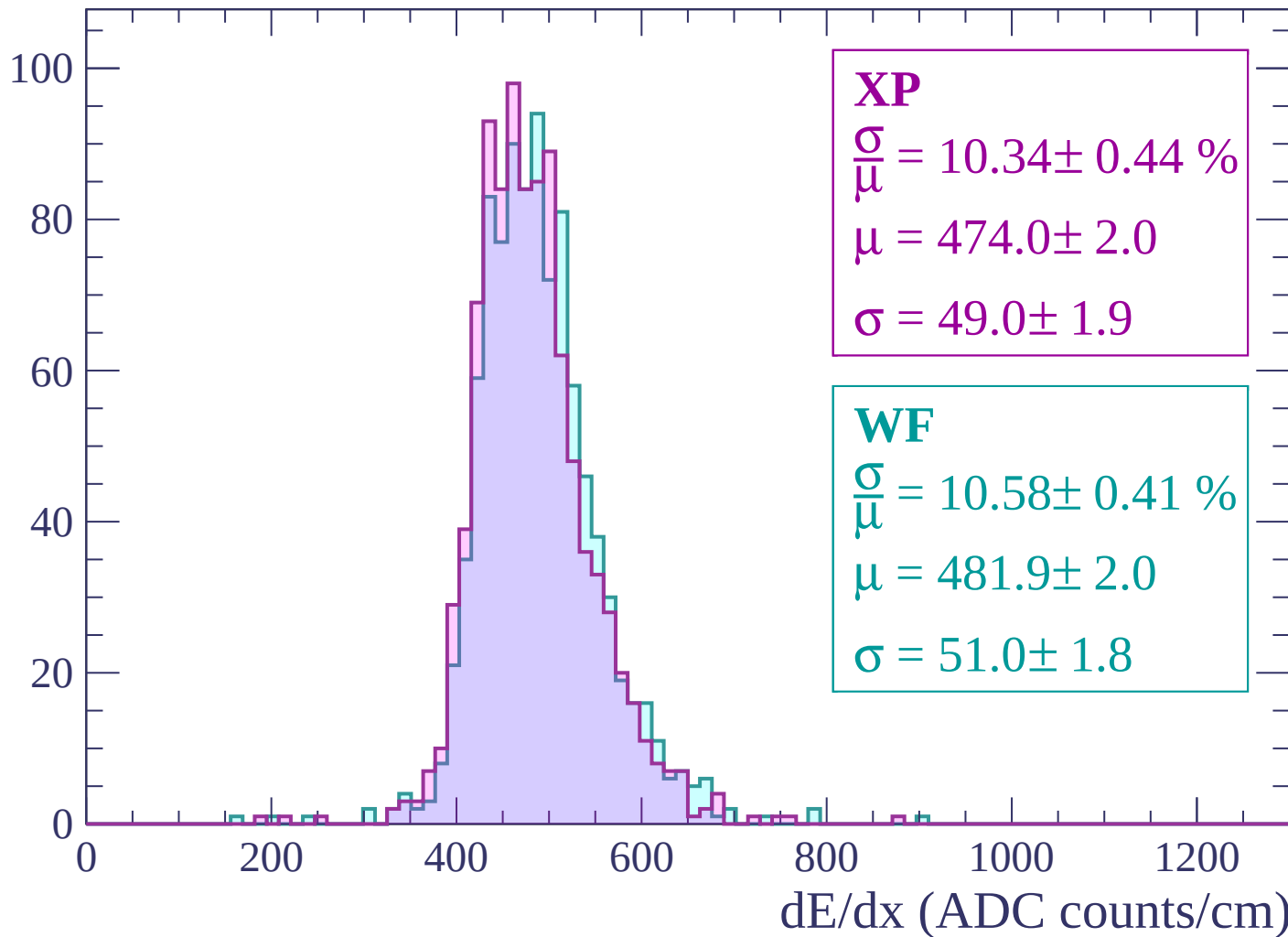
Energy loss | $1740 < p < 1800$

Count



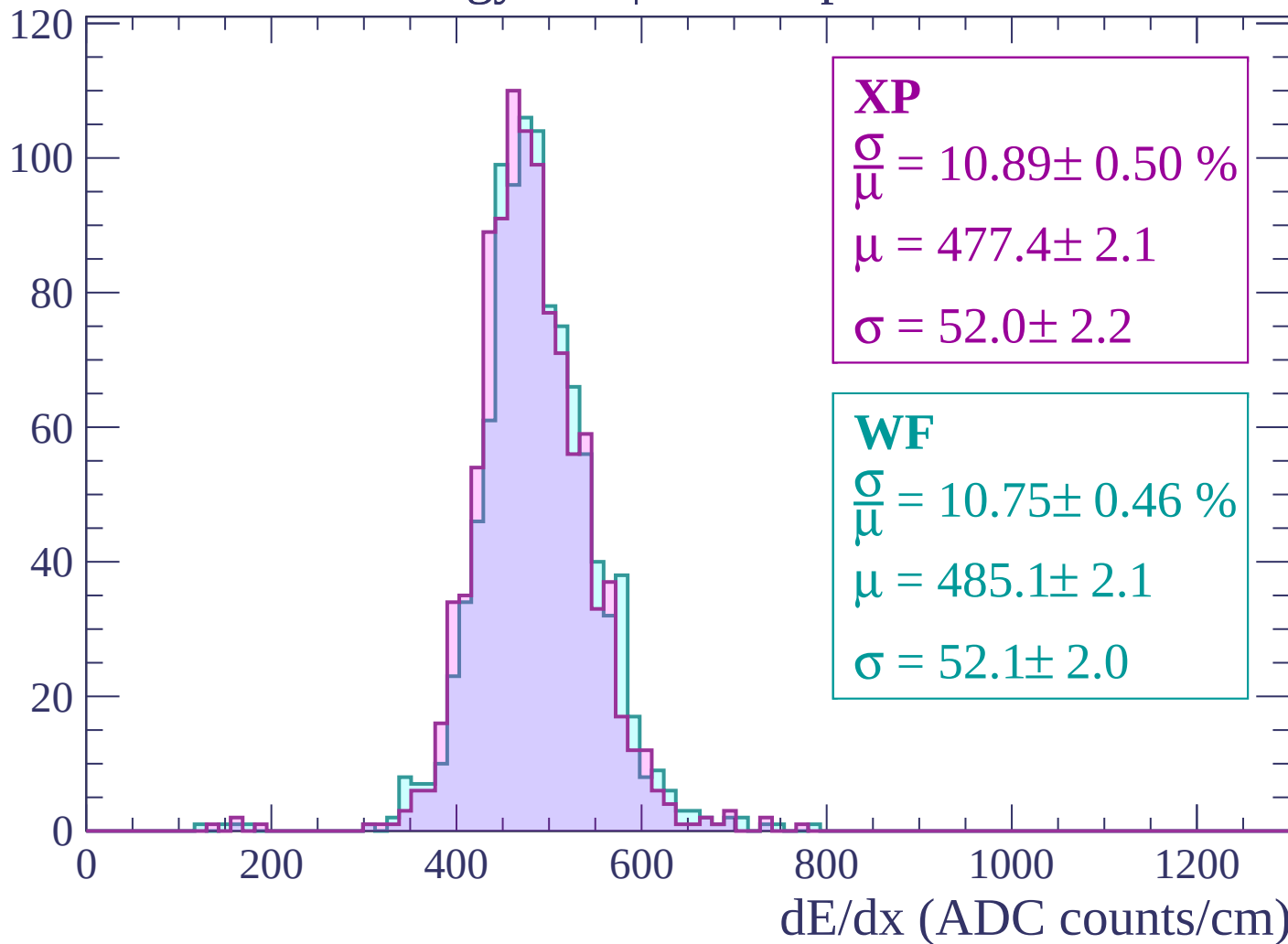
Energy loss | $1800 < p < 1860$

Count



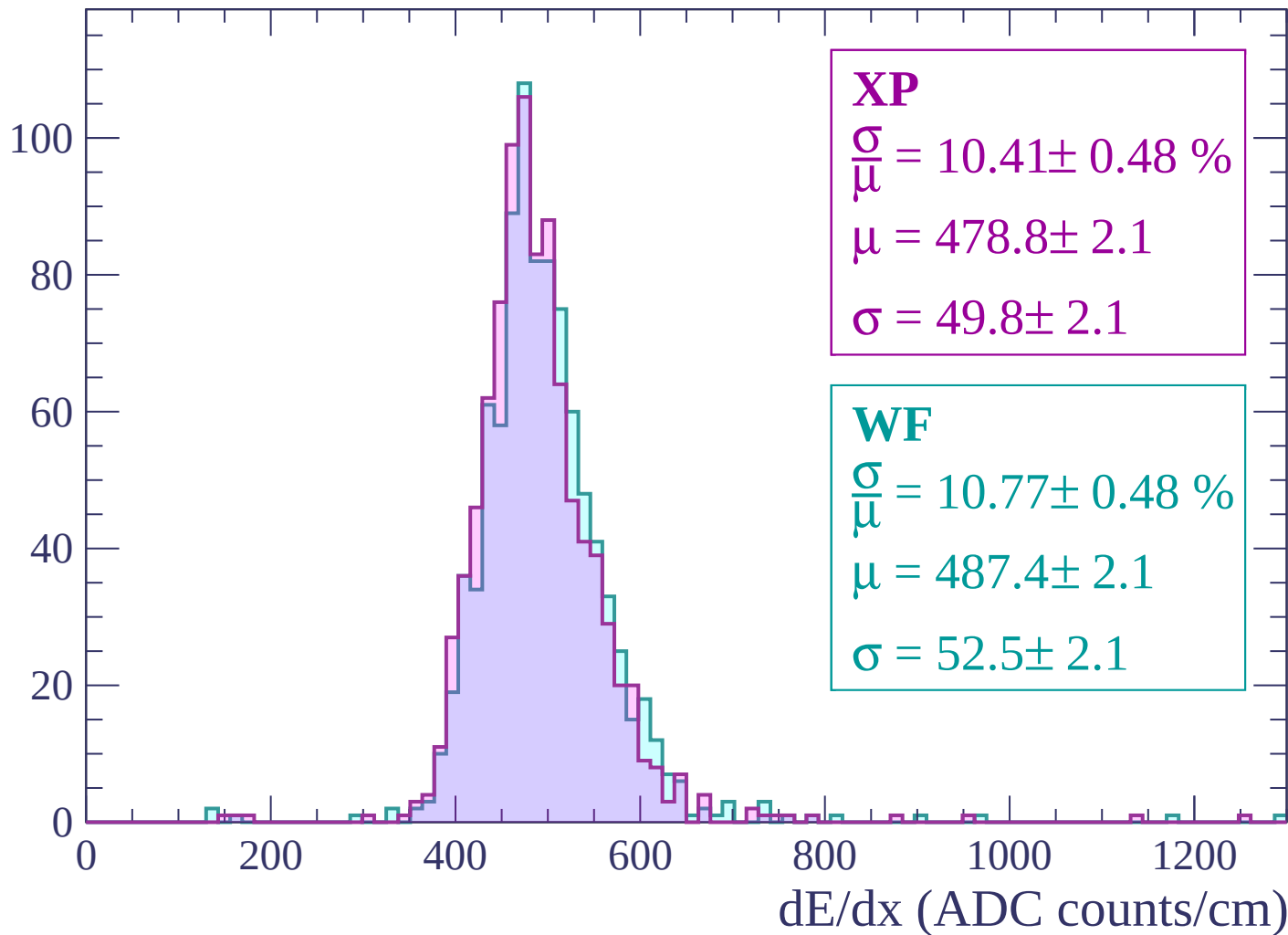
Energy loss | $1860 < p < 1920$

Count



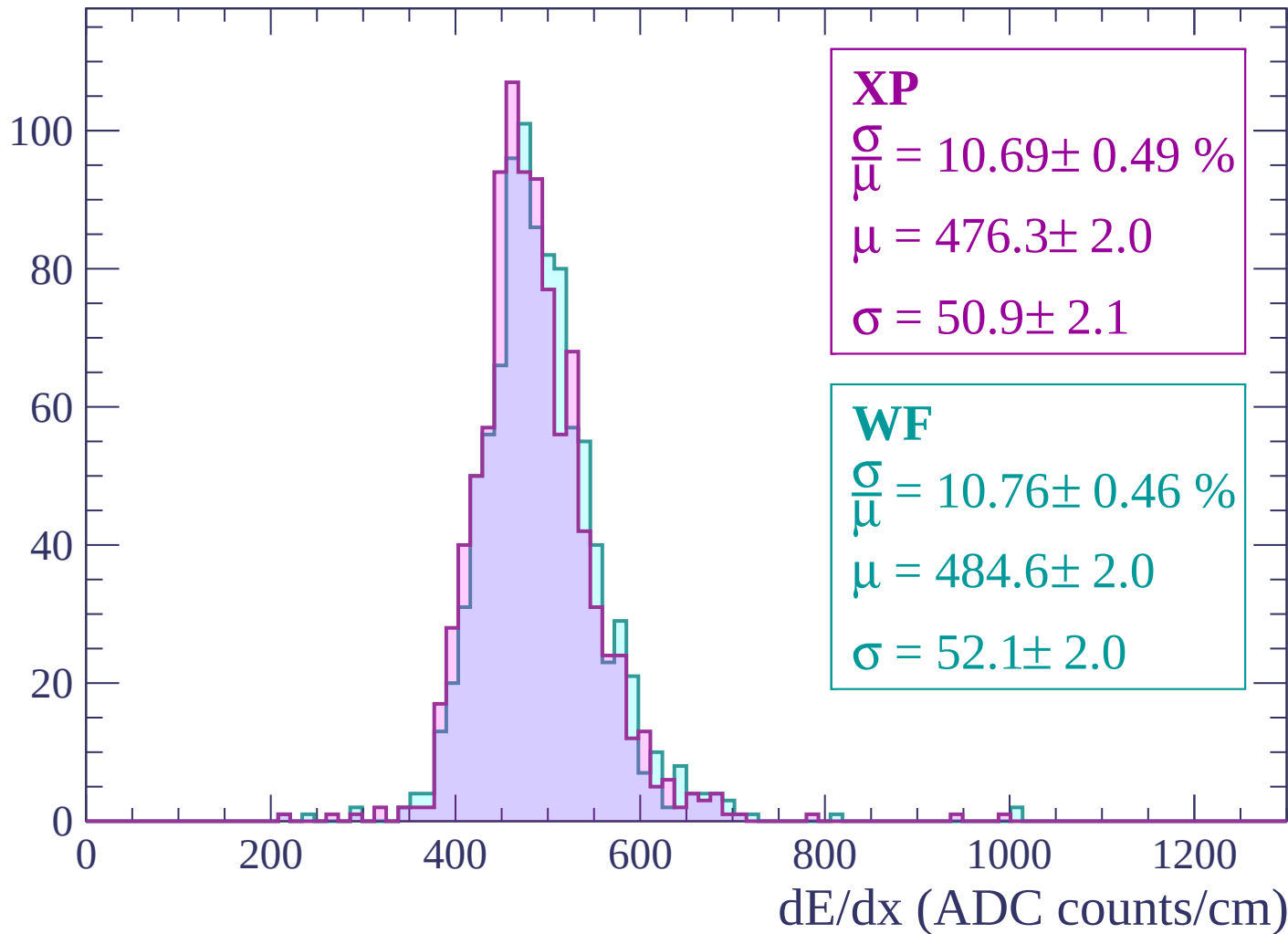
Energy loss | $1920 < p < 1980$

Count



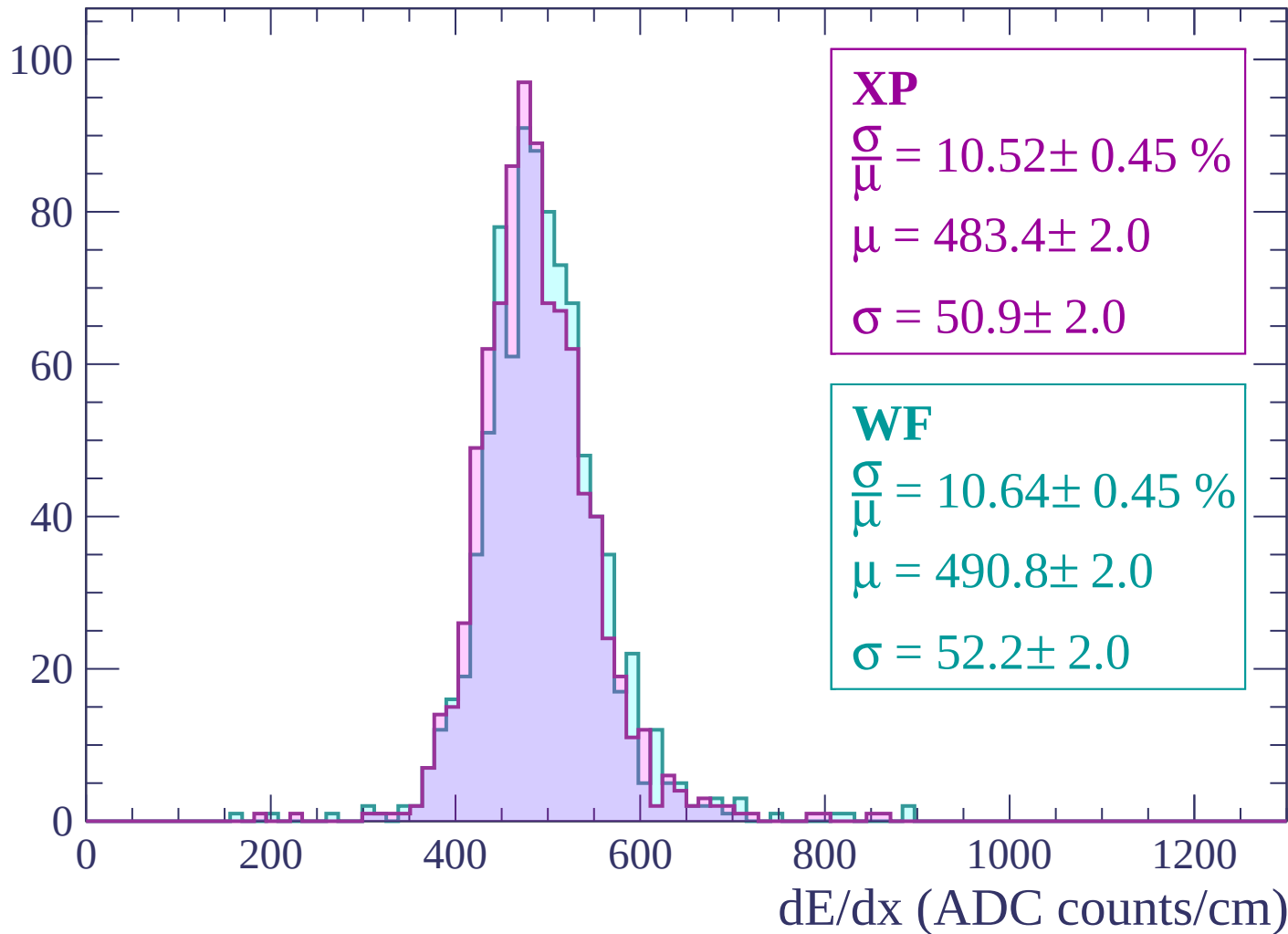
Energy loss | $1980 < p < 2040$

Count



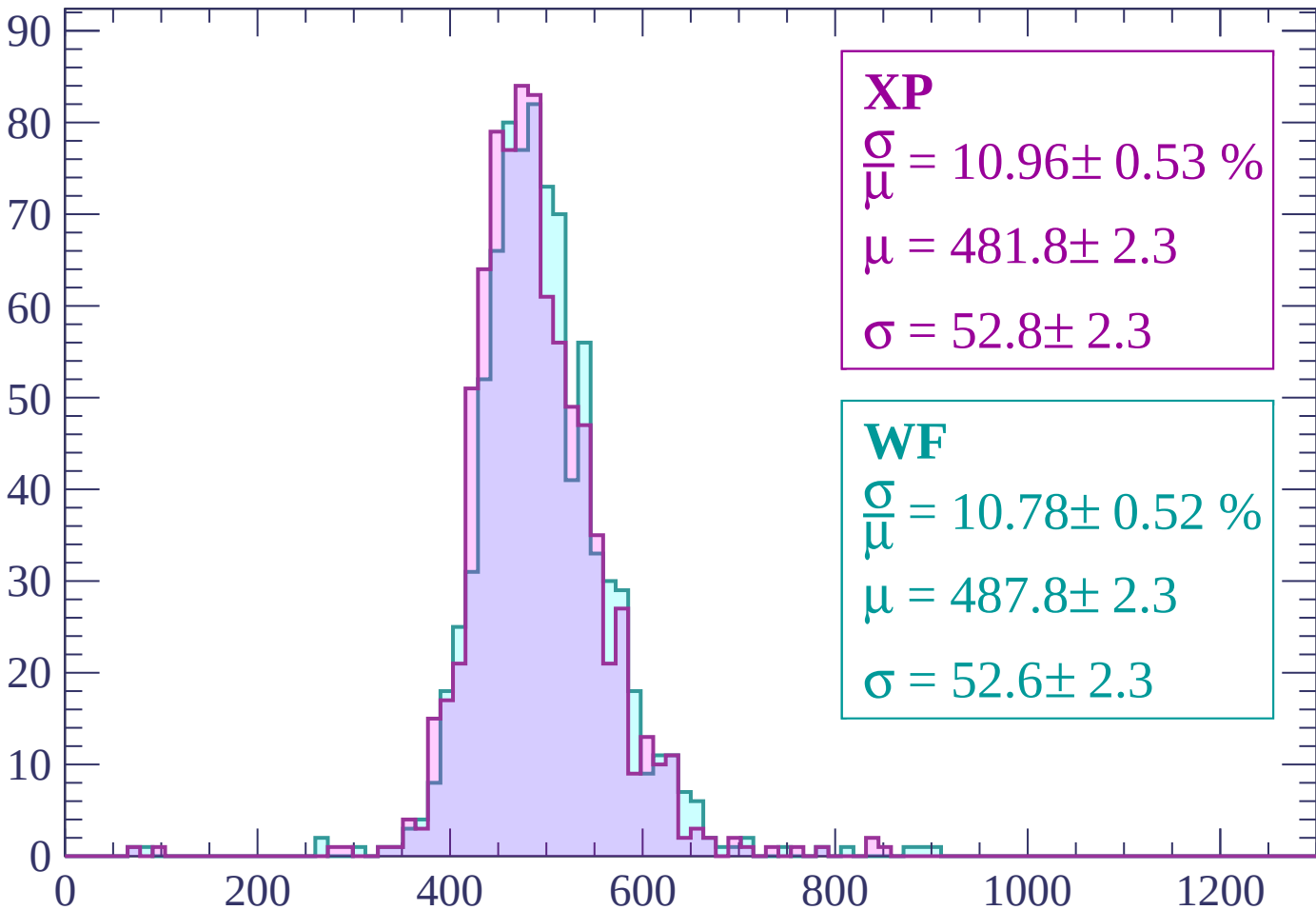
Energy loss | $2040 < p < 2100$

Count



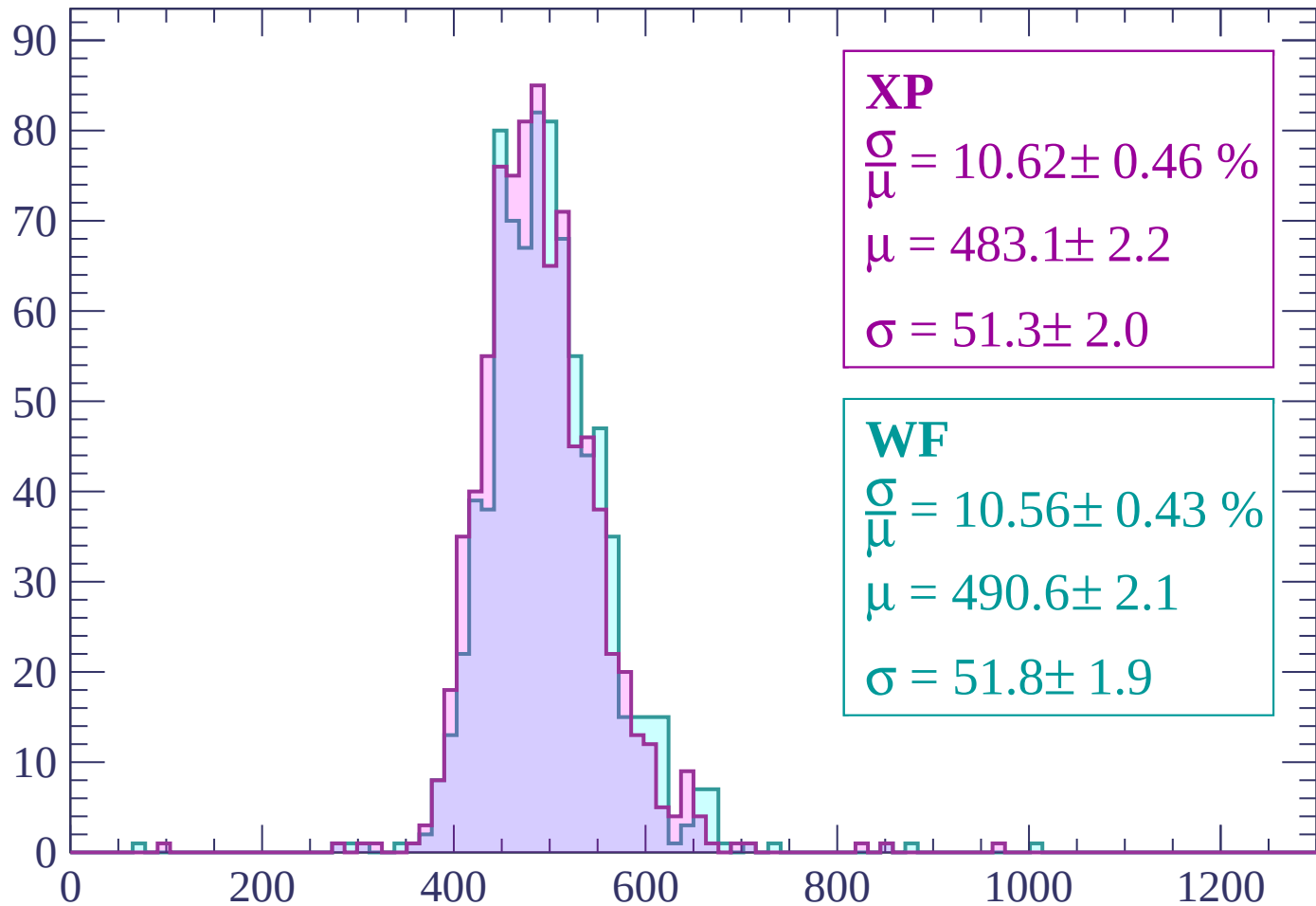
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.62 \pm 0.46 \%$$

$$\mu = 483.1 \pm 2.2$$

$$\sigma = 51.3 \pm 2.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.56 \pm 0.43 \%$$

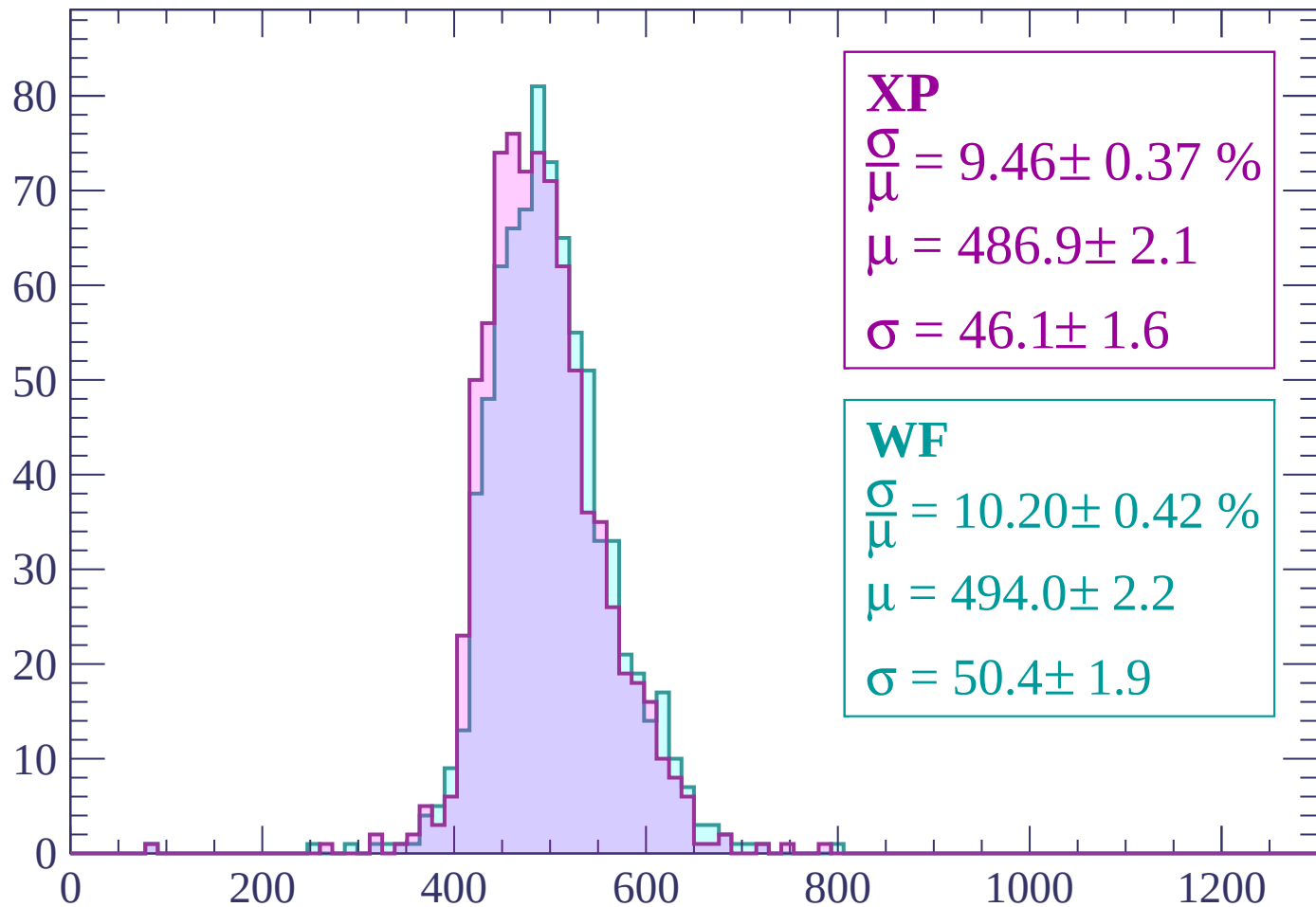
$$\mu = 490.6 \pm 2.1$$

$$\sigma = 51.8 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.37 \%$$

$$\mu = 486.9 \pm 2.1$$

$$\sigma = 46.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 10.20 \pm 0.42 \%$$

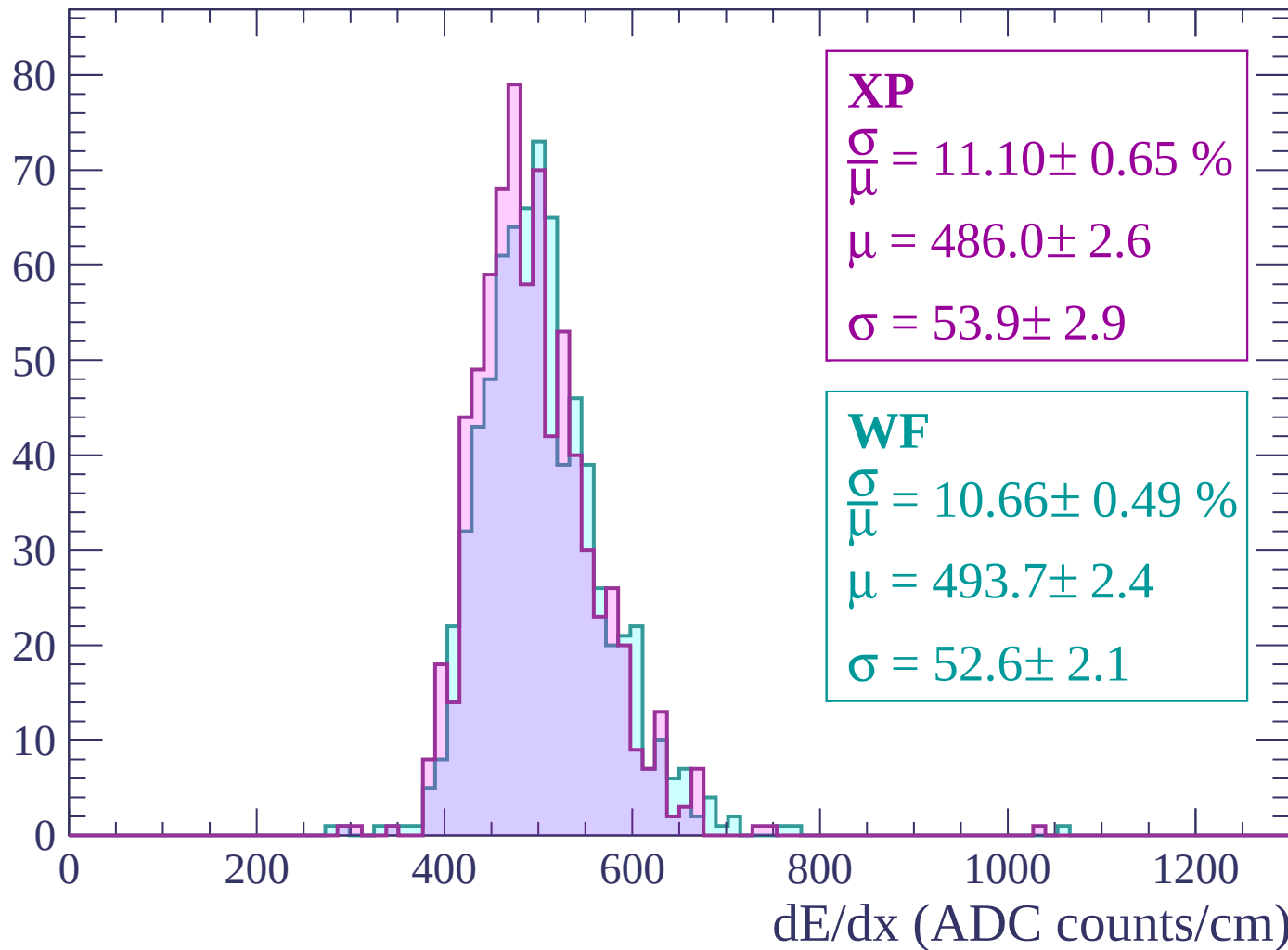
$$\mu = 494.0 \pm 2.2$$

$$\sigma = 50.4 \pm 1.9$$

dE/dx (ADC counts/cm)

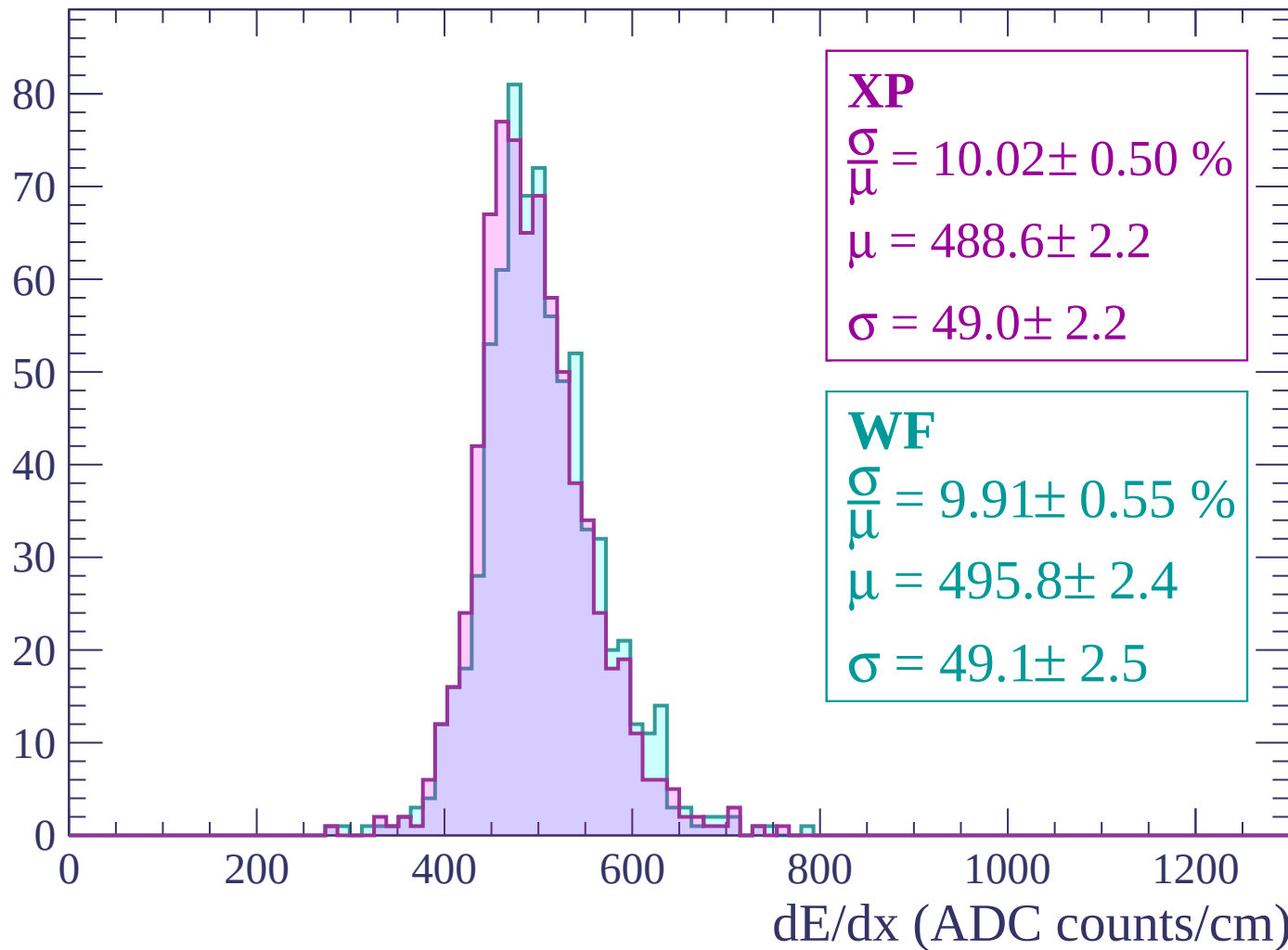
Energy loss | $2280 < p < 2340$

Count



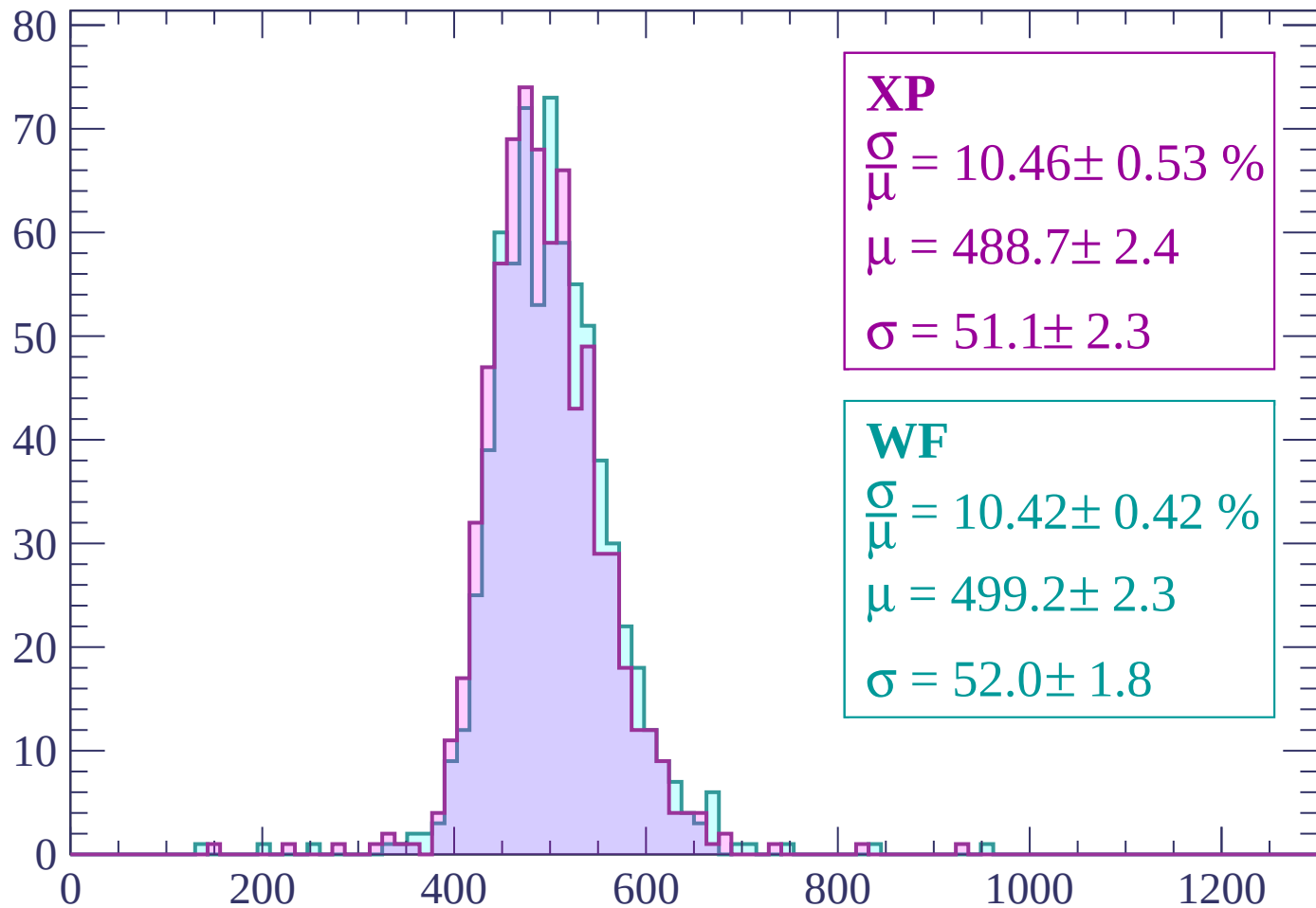
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP

$$\frac{\sigma}{\mu} = 10.46 \pm 0.53 \%$$

$$\mu = 488.7 \pm 2.4$$

$$\sigma = 51.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 10.42 \pm 0.42 \%$$

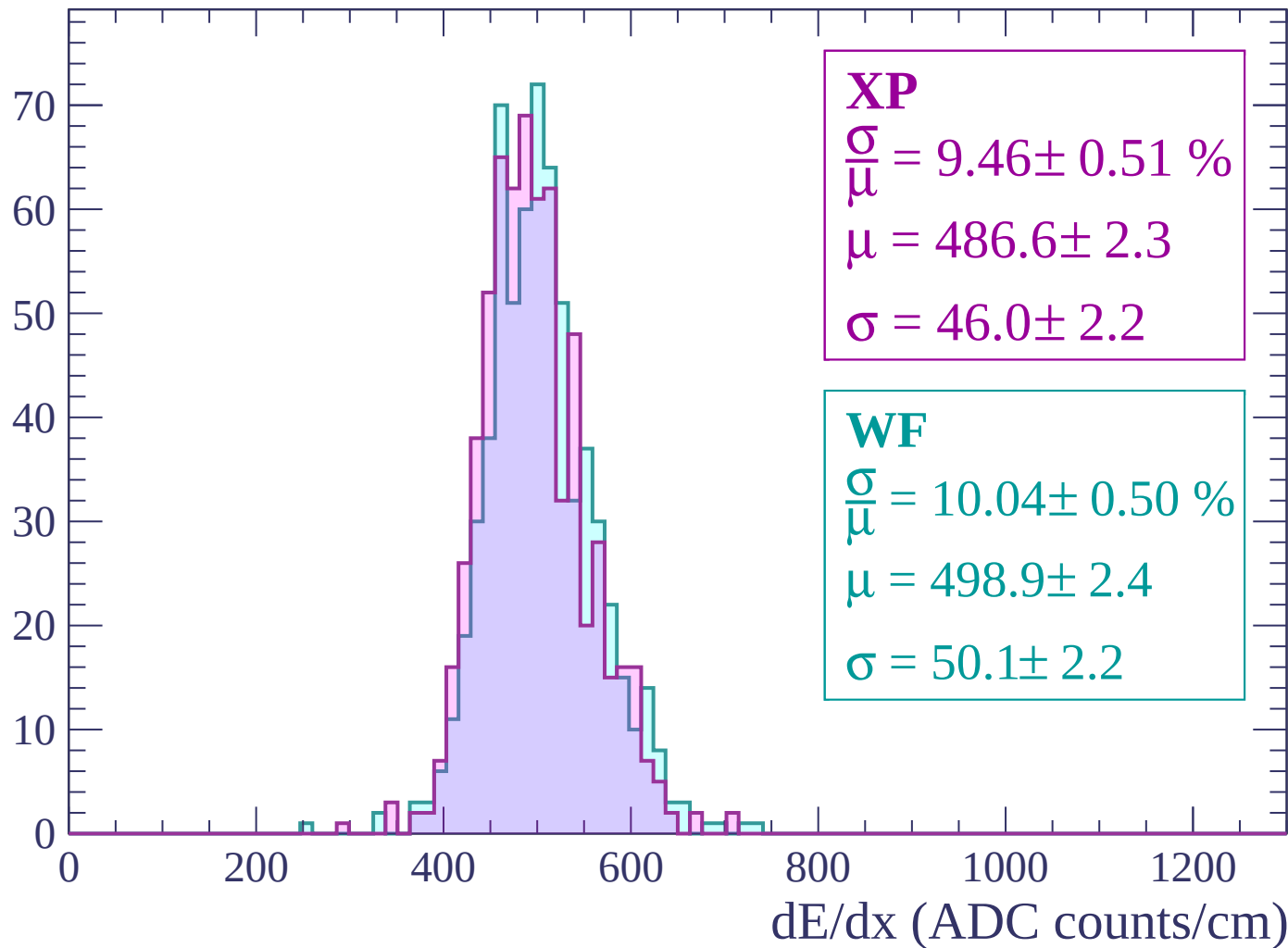
$$\mu = 499.2 \pm 2.3$$

$$\sigma = 52.0 \pm 1.8$$

dE/dx (ADC counts/cm)

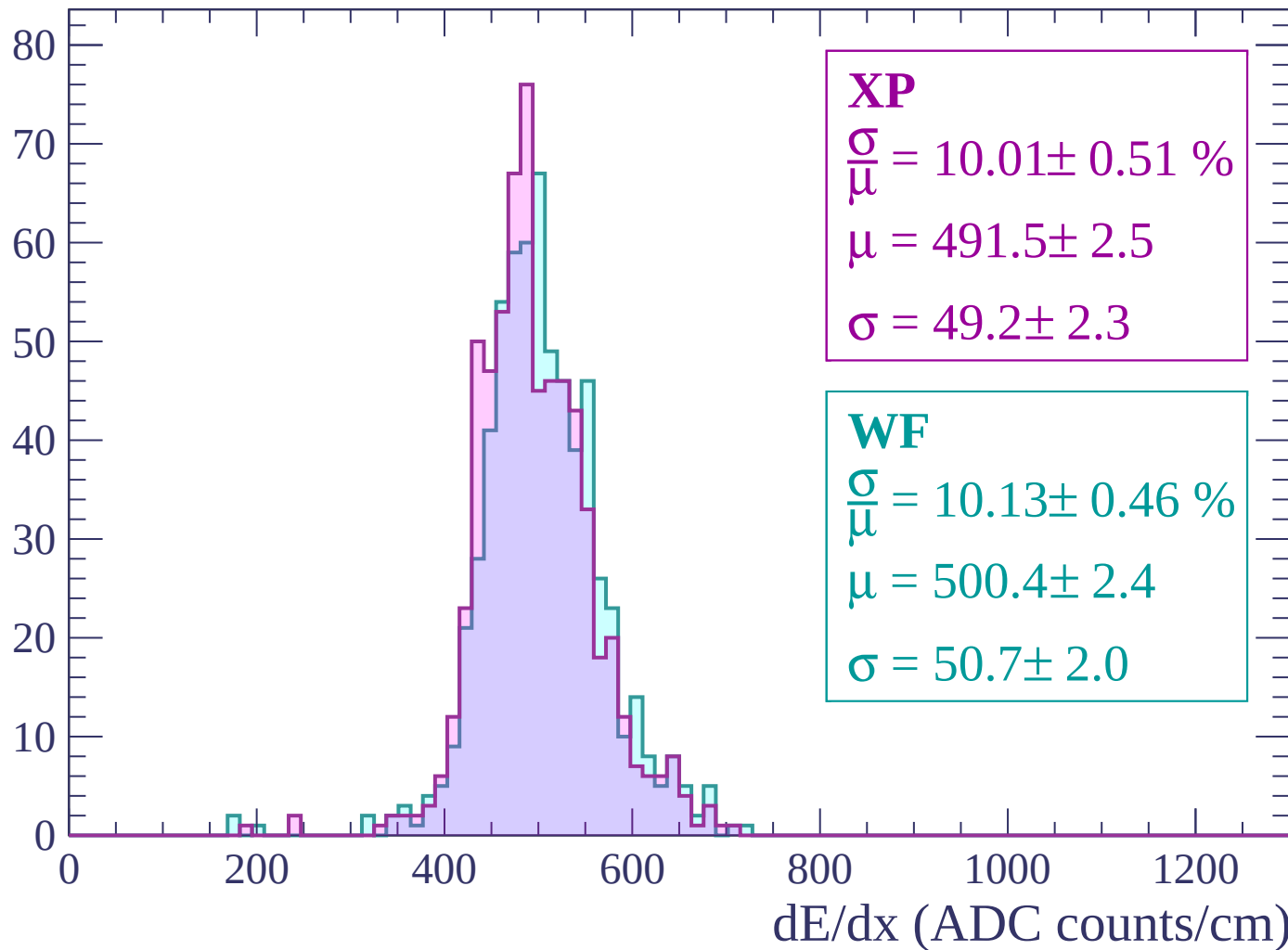
Energy loss | $2460 < p < 2520$

Count



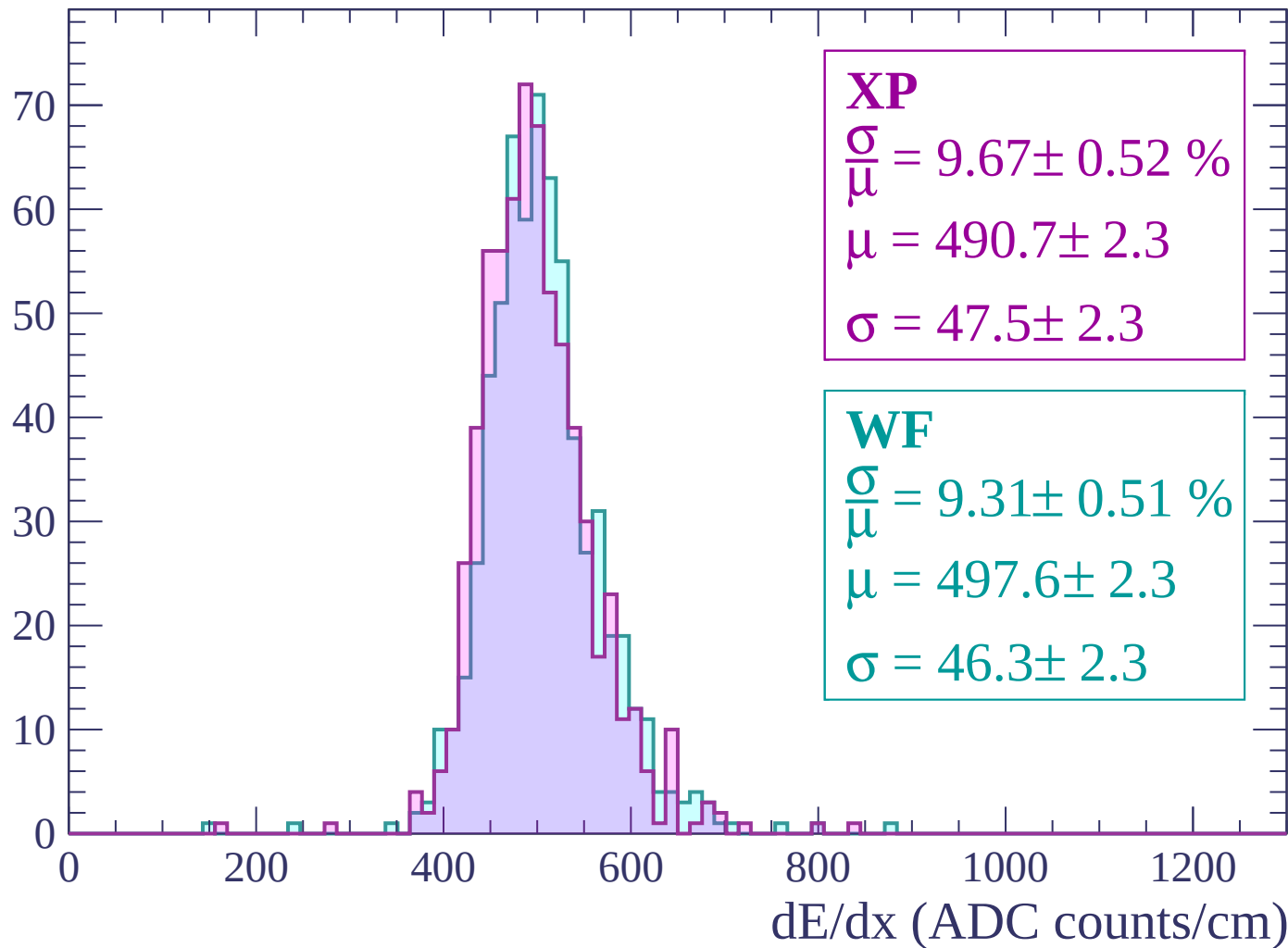
Energy loss | $2520 < p < 2580$

Count



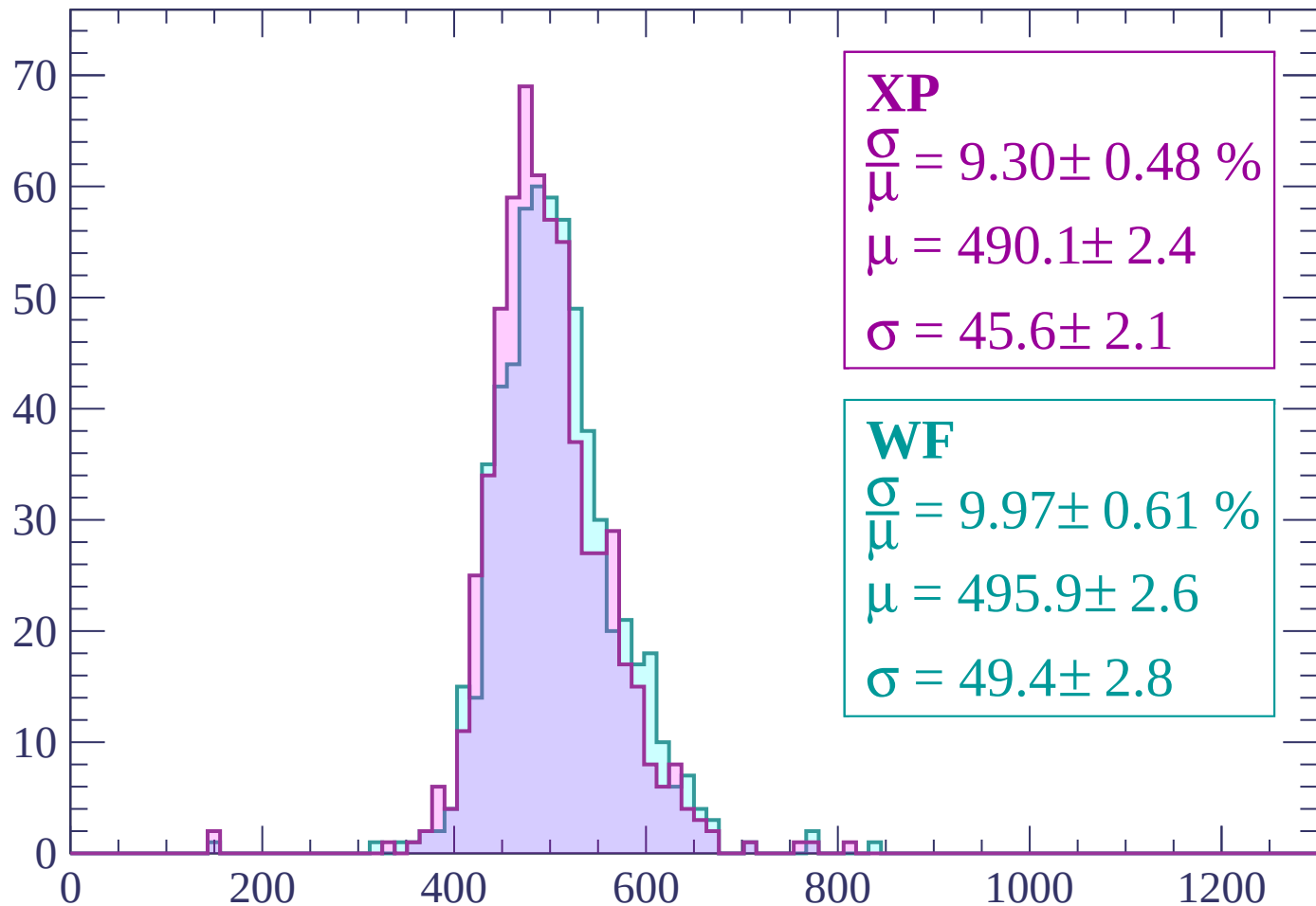
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.48 \%$$

$$\mu = 490.1 \pm 2.4$$

$$\sigma = 45.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 9.97 \pm 0.61 \%$$

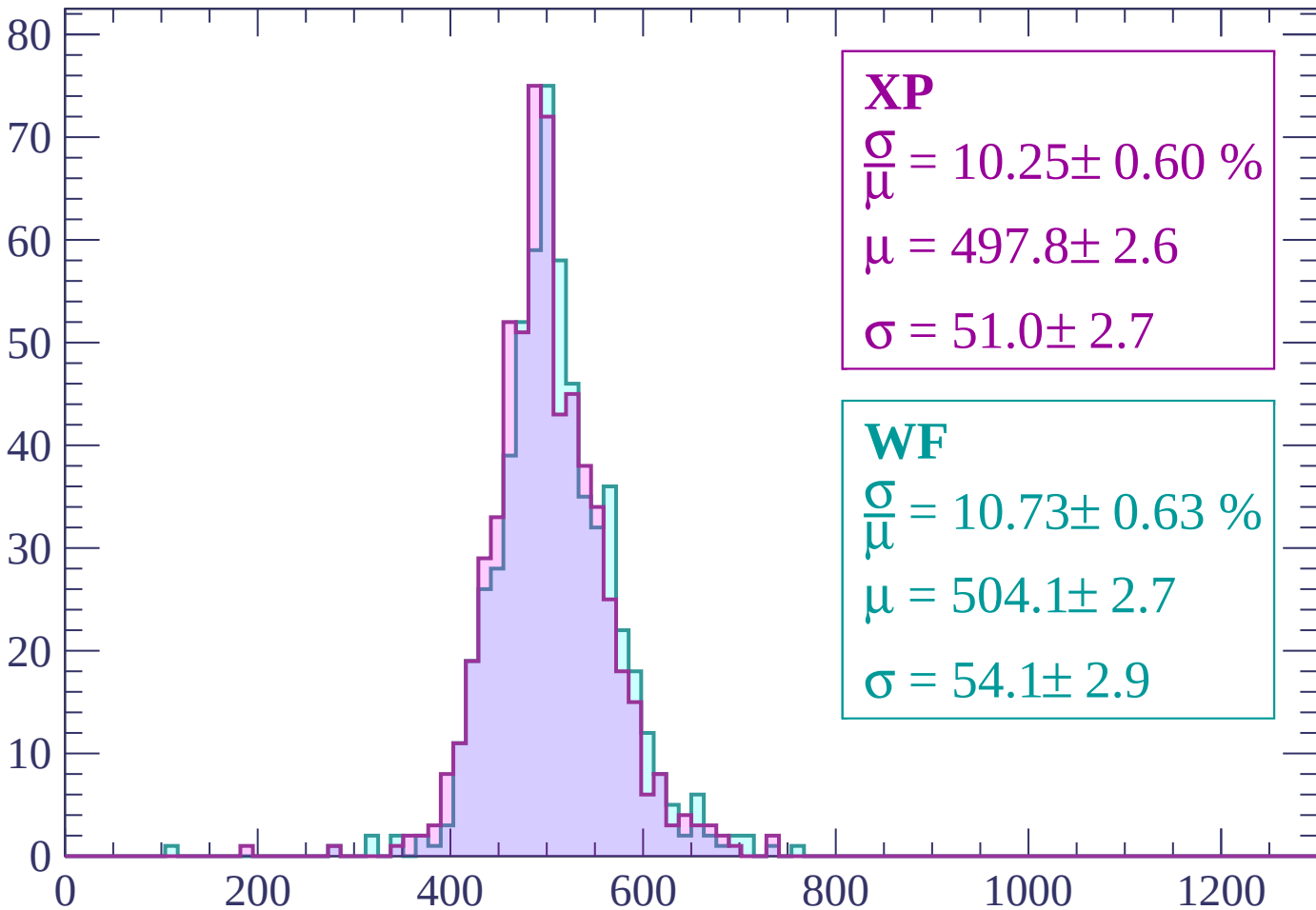
$$\mu = 495.9 \pm 2.6$$

$$\sigma = 49.4 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP

$$\frac{\sigma}{\mu} = 10.25 \pm 0.60 \%$$

$$\mu = 497.8 \pm 2.6$$

$$\sigma = 51.0 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 10.73 \pm 0.63 \%$$

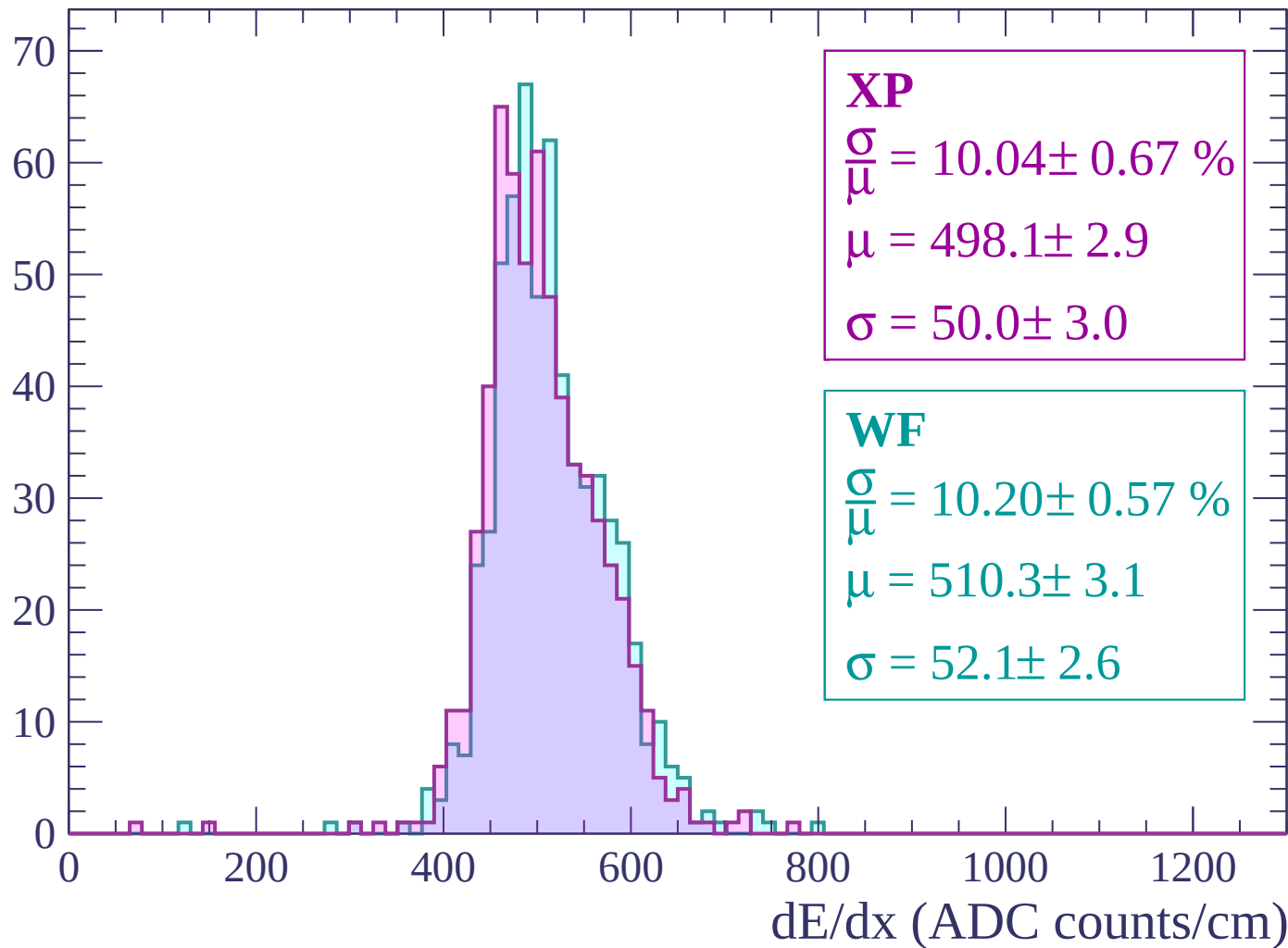
$$\mu = 504.1 \pm 2.7$$

$$\sigma = 54.1 \pm 2.9$$

dE/dx (ADC counts/cm)

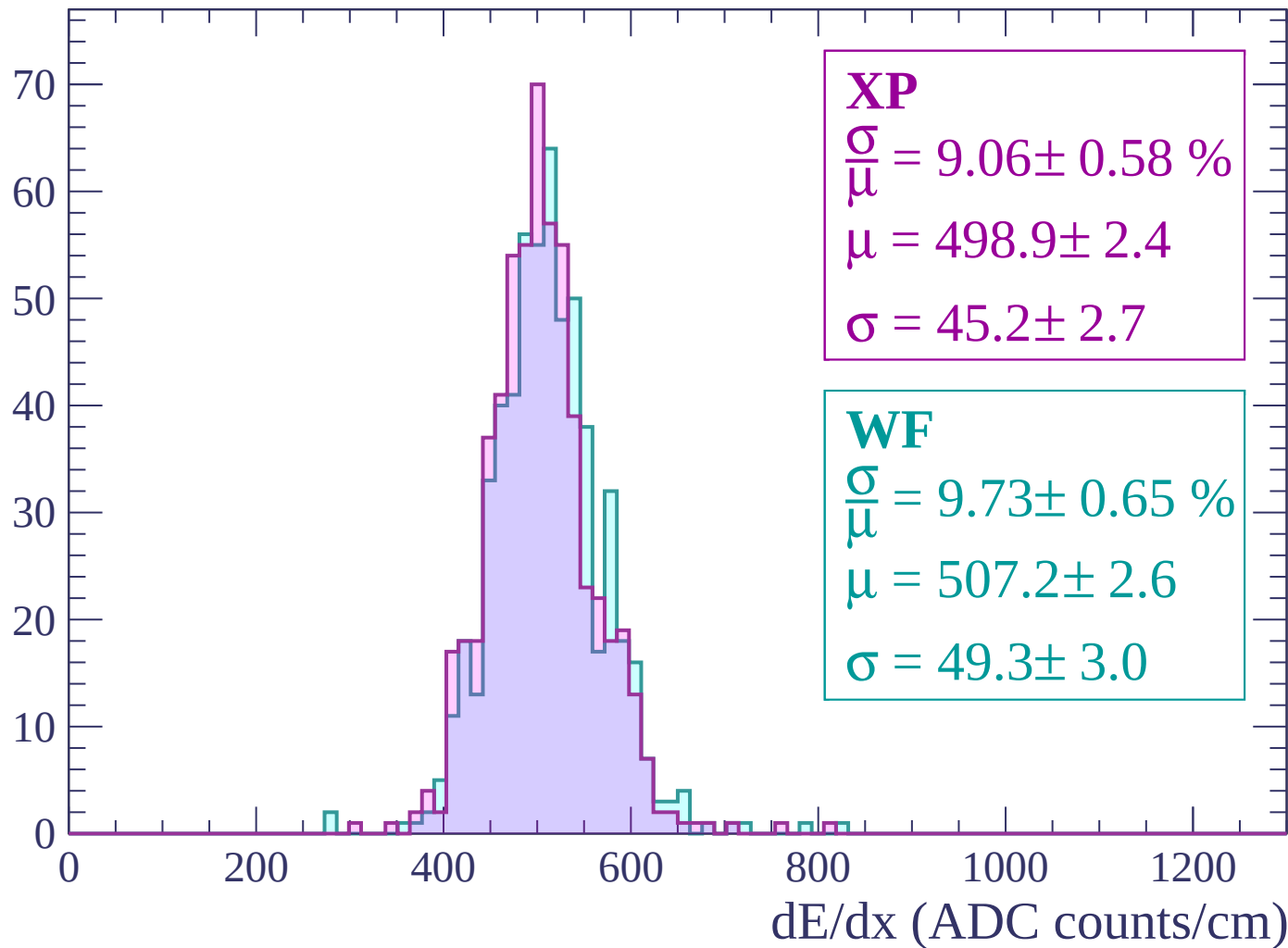
Energy loss | $2760 < p < 2820$

Count



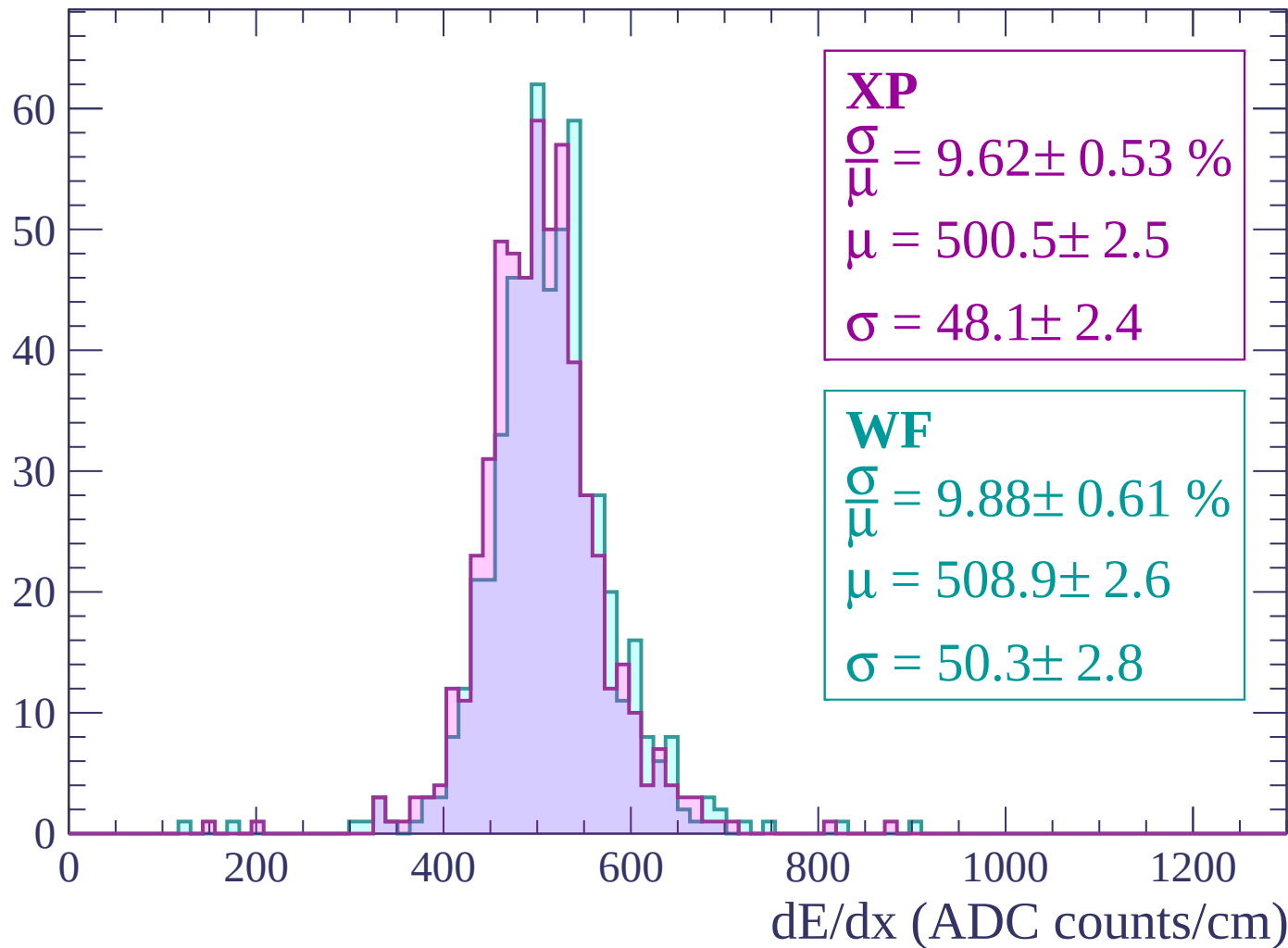
Energy loss | $2820 < p < 2880$

Count



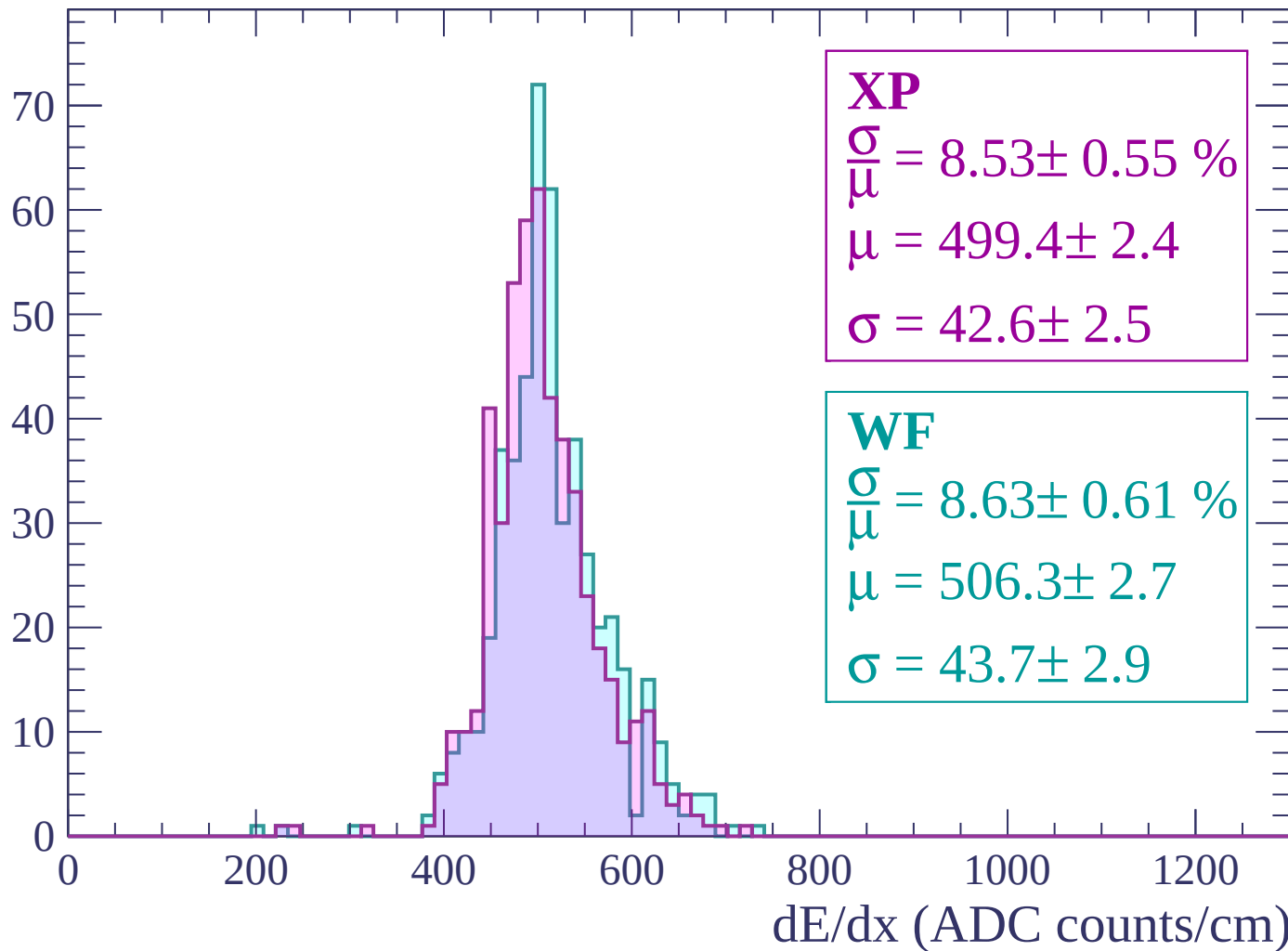
Energy loss | $2880 < p < 2940$

Count



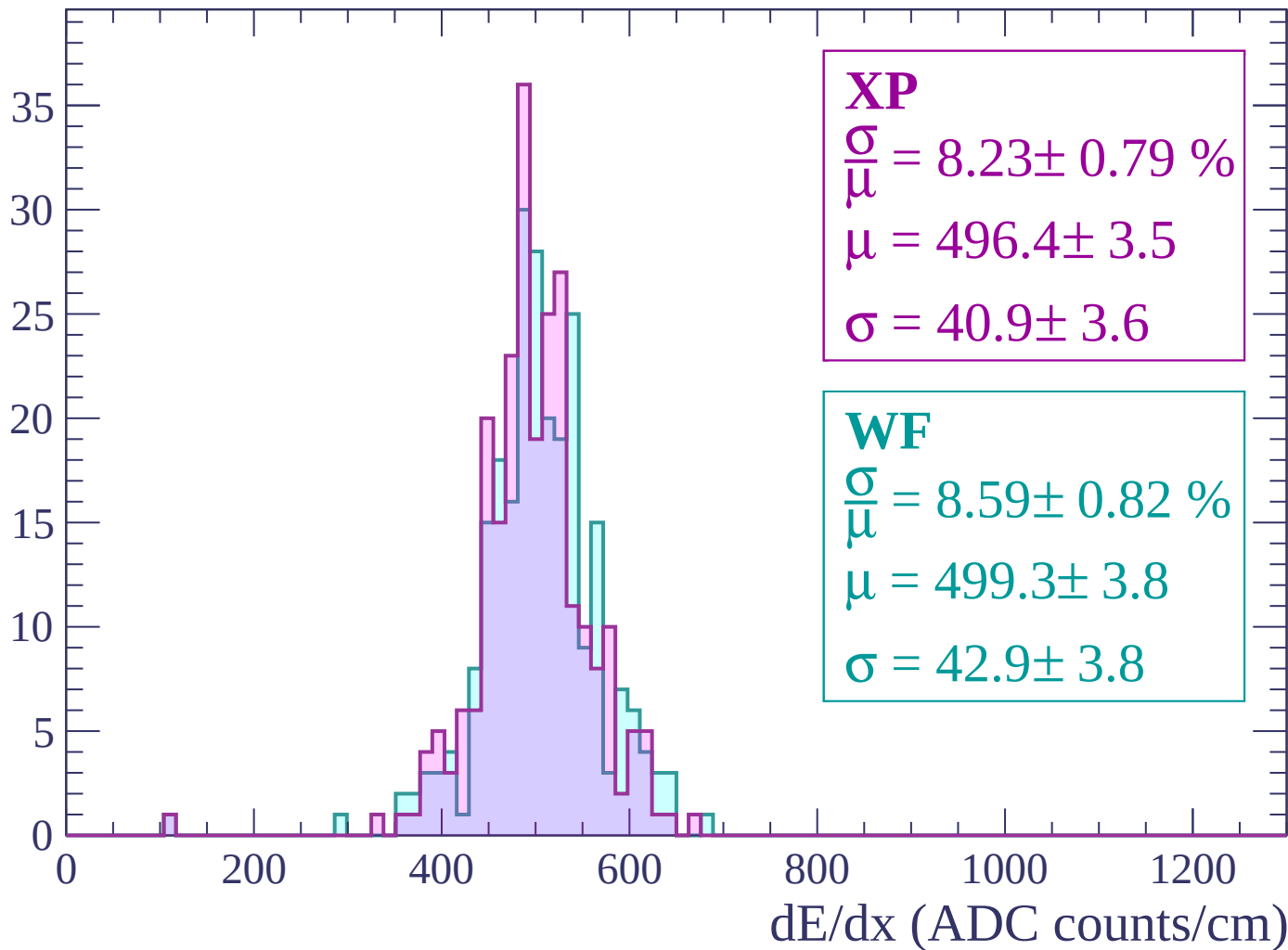
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

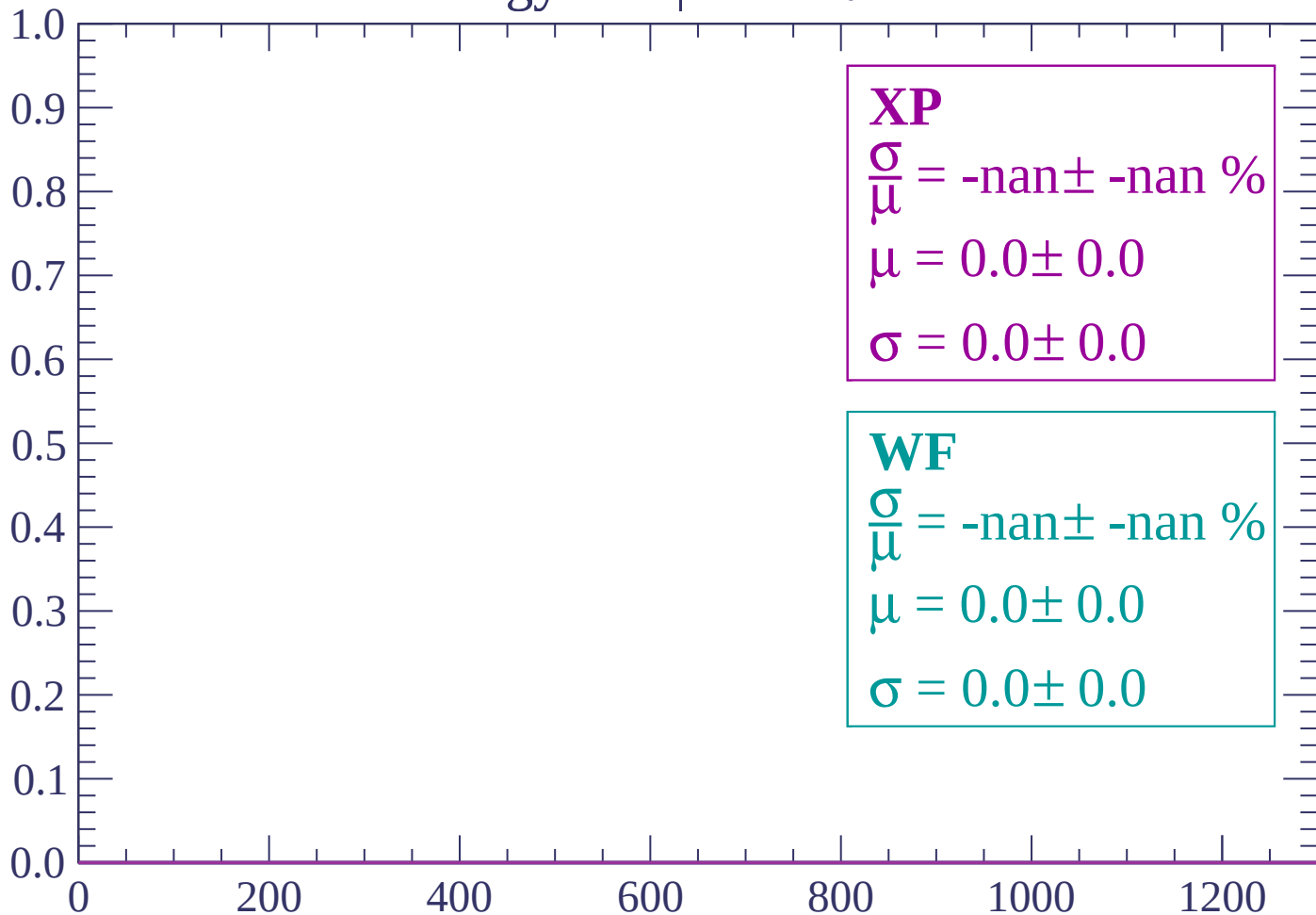
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

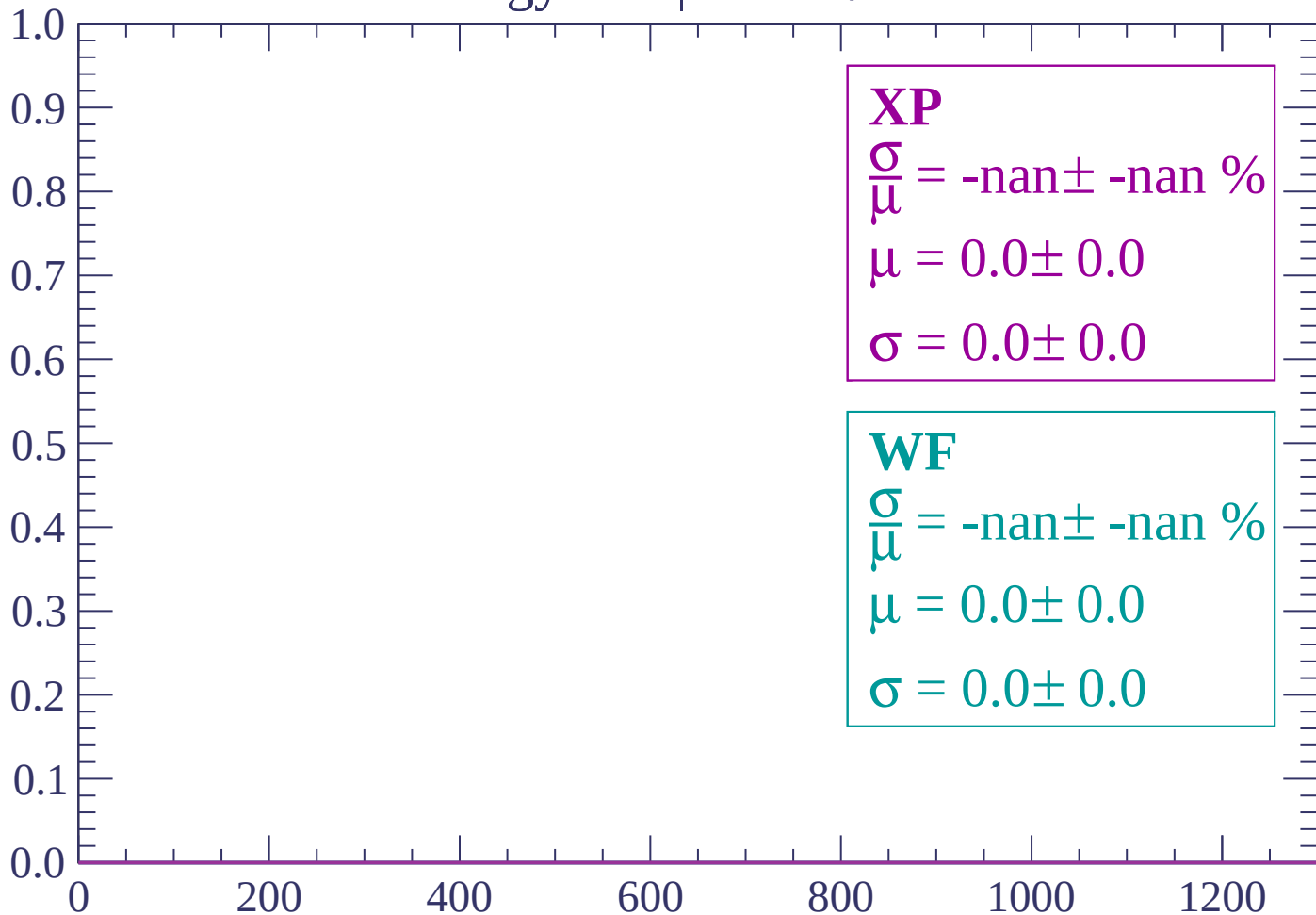
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-74 < \theta < -72$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-72 < \theta < -70$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-70 < \theta < -68$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-68 < \theta < -66$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-66 < \theta < -64$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

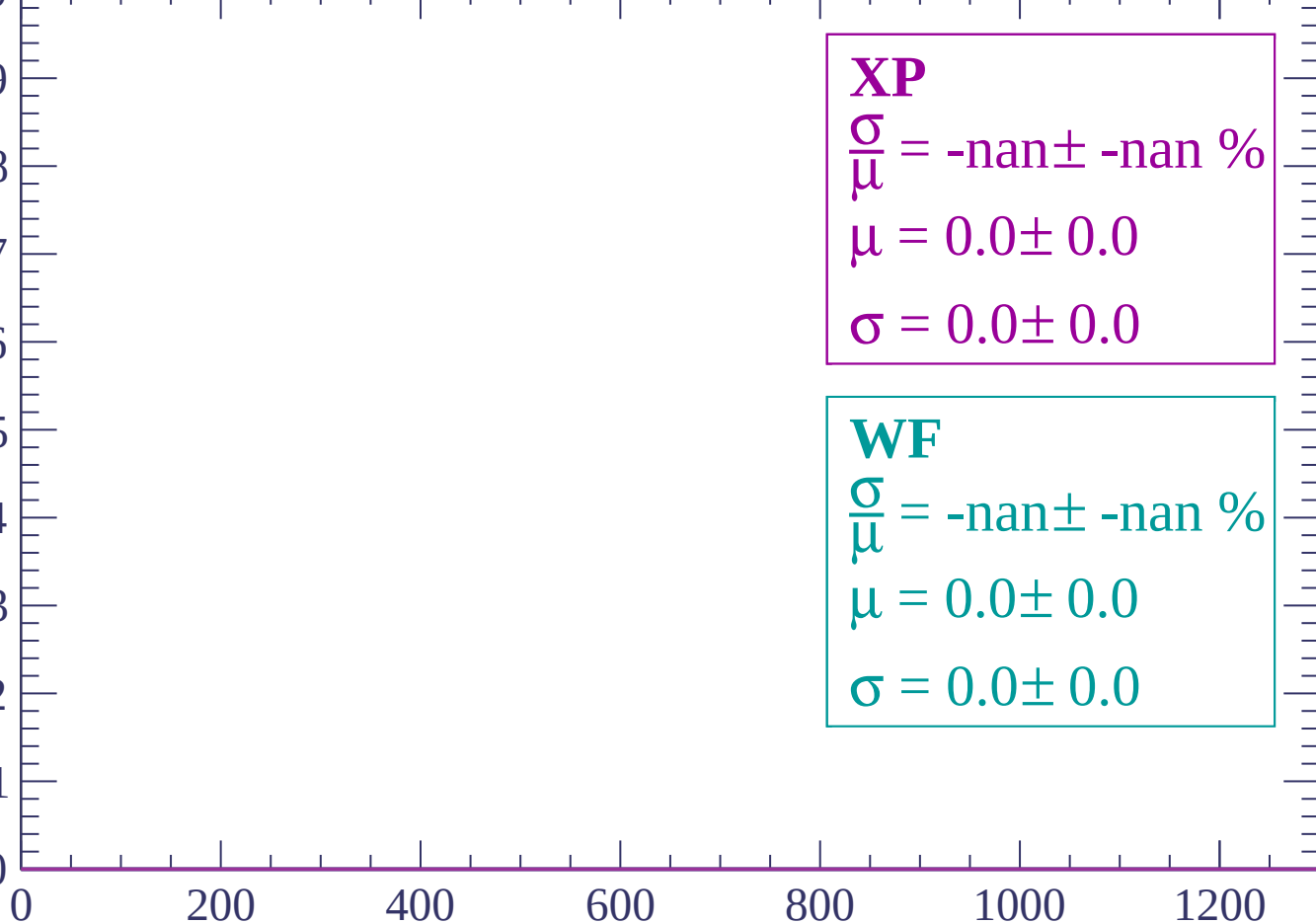
1200

dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-60 < \theta < -58$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

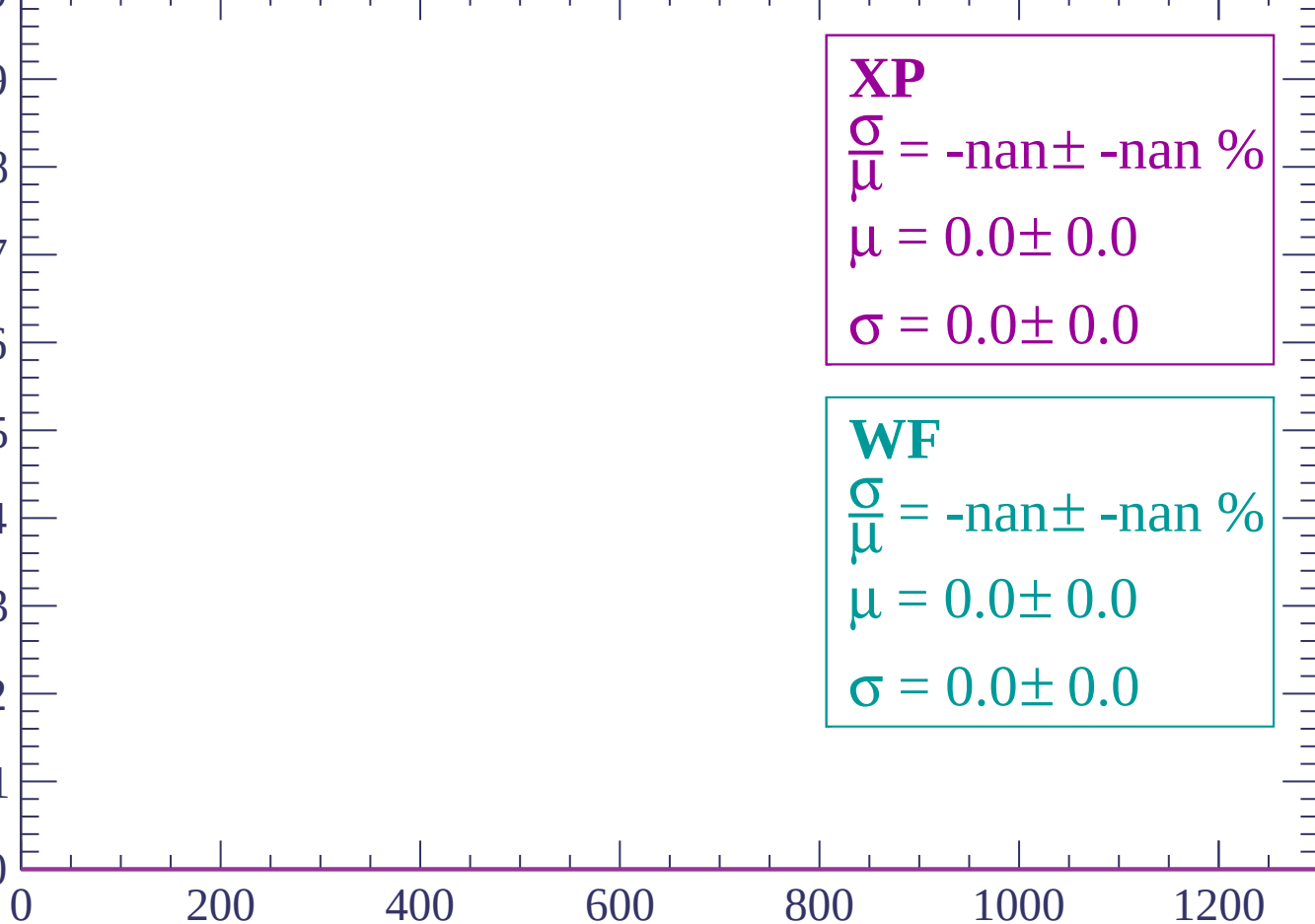
1200

dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-54 < \theta < -52$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-52 < \theta < -50$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

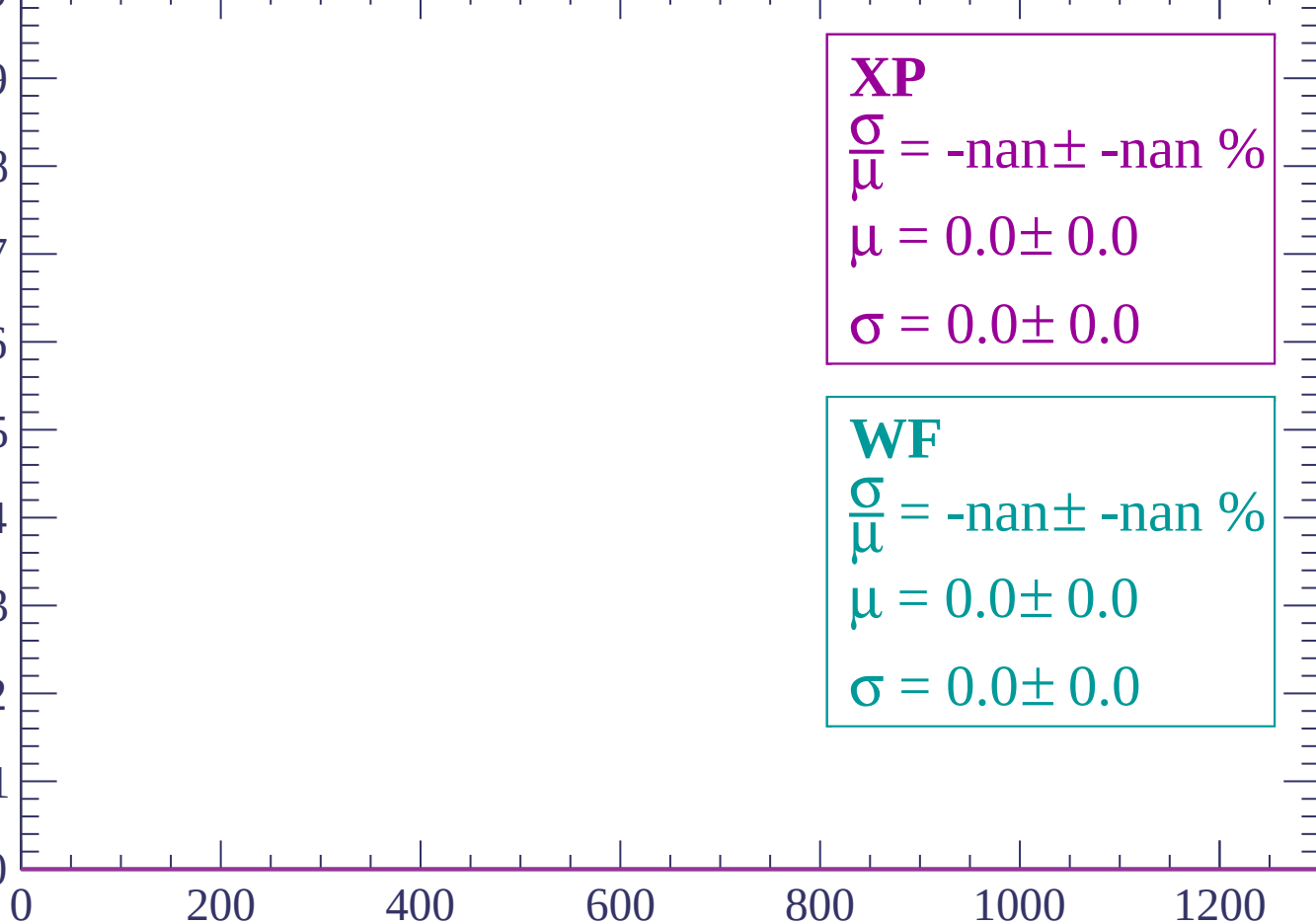
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-50 < \theta < -48$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



dE/dx (ADC counts/cm)

Energy loss | $-48 < \theta < -46$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

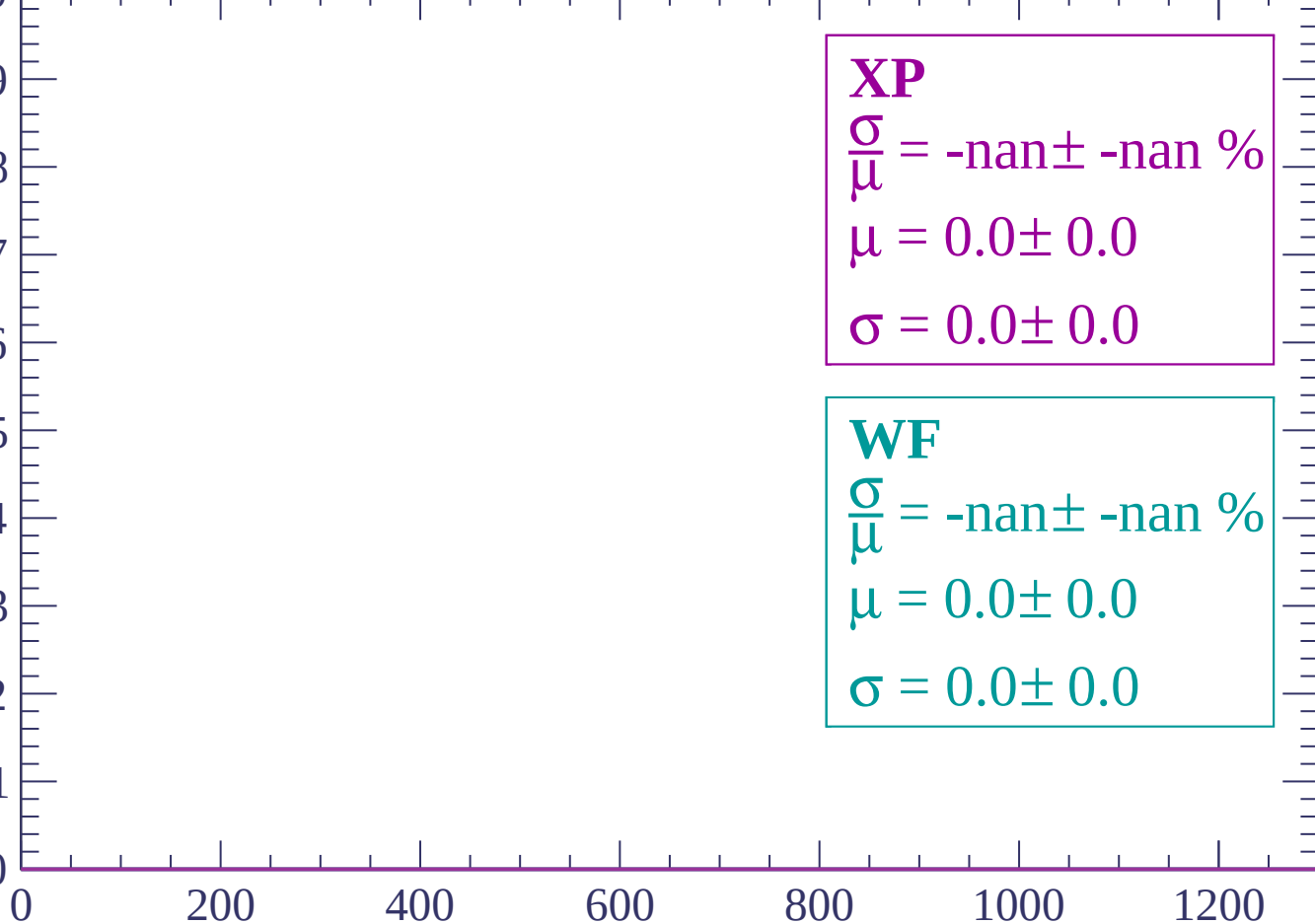
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-46 < \theta < -44$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



dE/dx (ADC counts/cm)

Energy loss | $-44 < \theta < -42$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-42 < \theta < -40$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-40 < \theta < -38$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

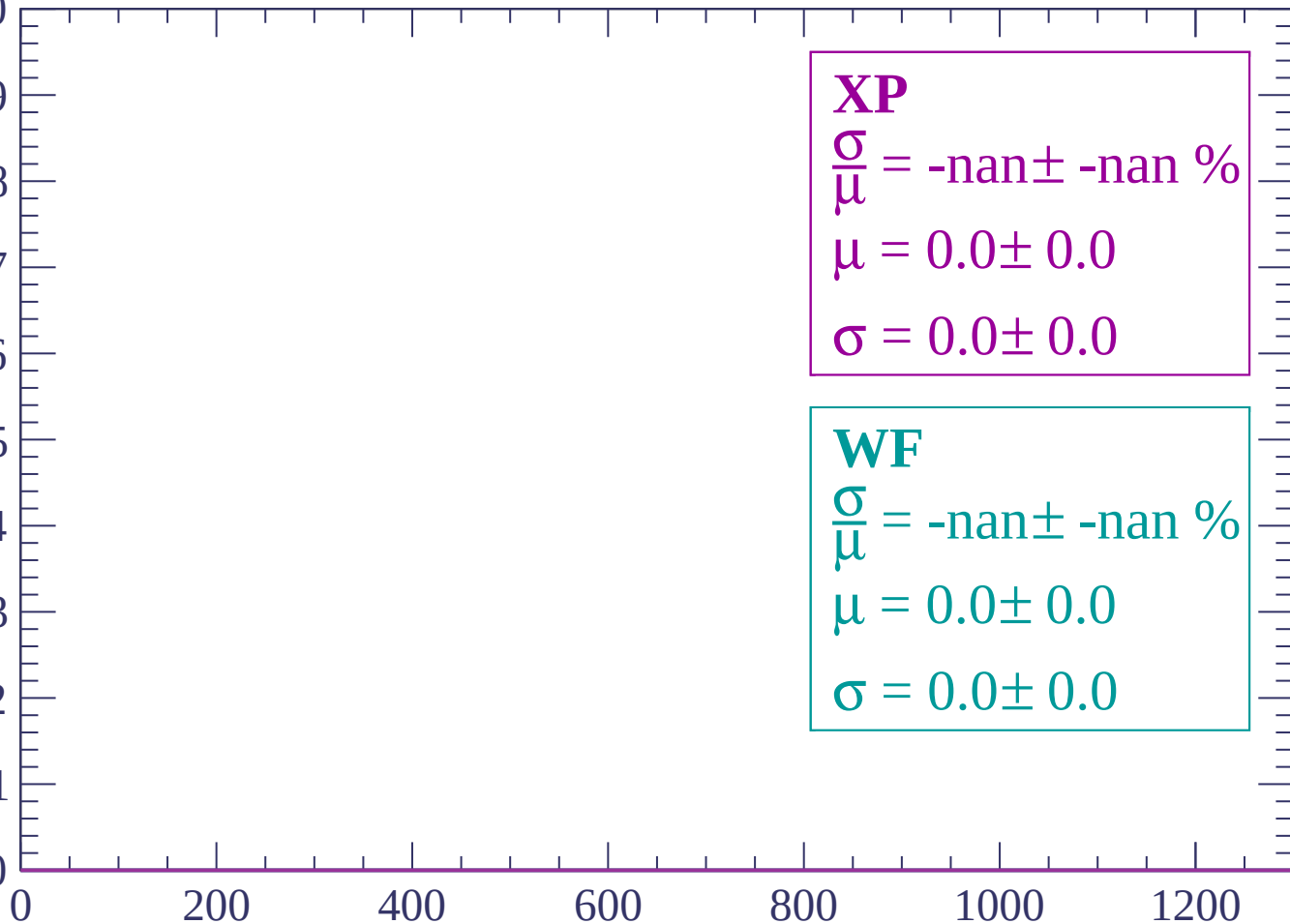
1200

dE/dx (ADC counts/cm)

Energy loss | $-38 < \theta < -36$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



dE/dx (ADC counts/cm)

Energy loss | $-36 < \theta < -34$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

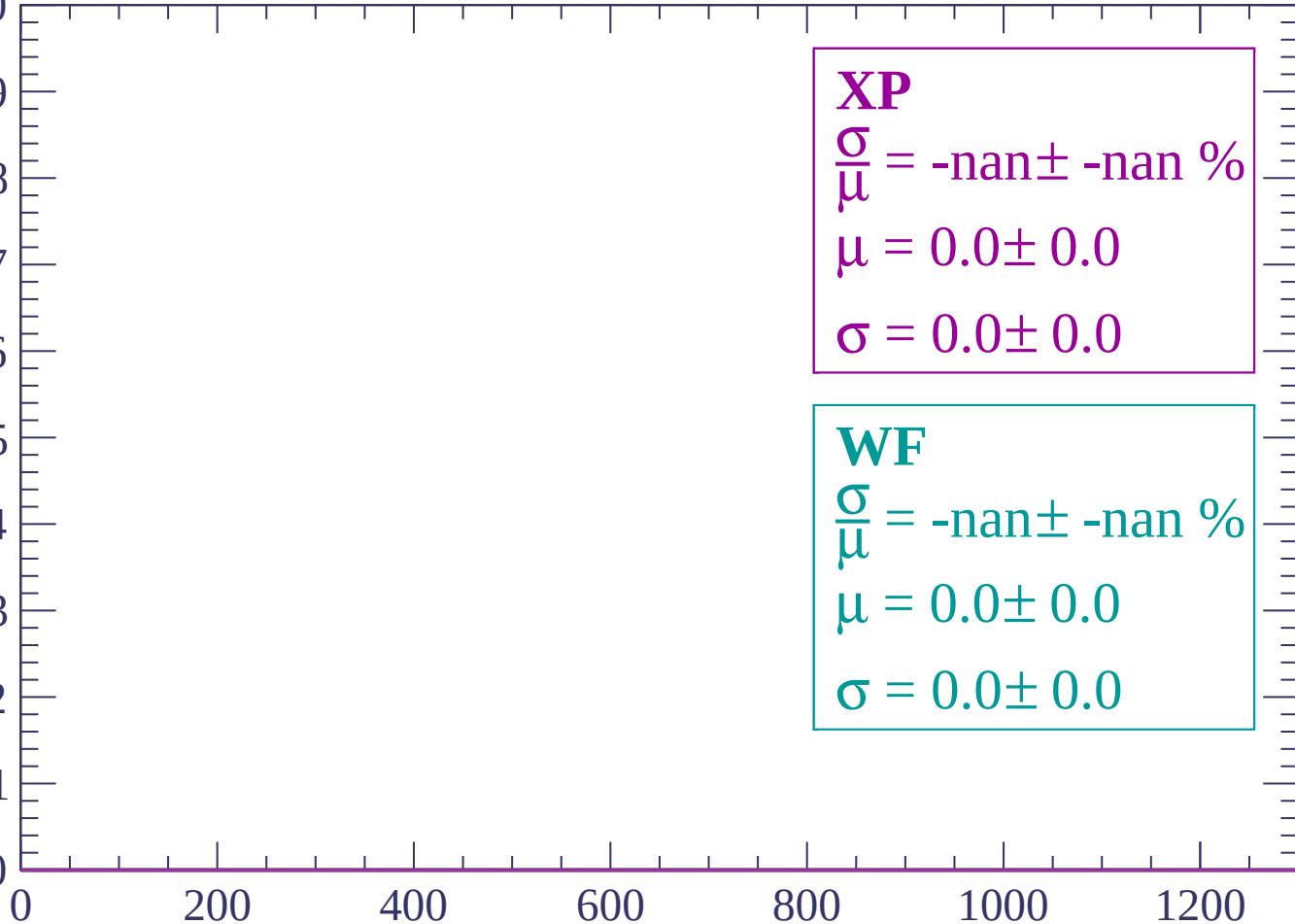
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-34 < \theta < -32$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



dE/dx (ADC counts/cm)

Energy loss | $-32 < \theta < -30$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

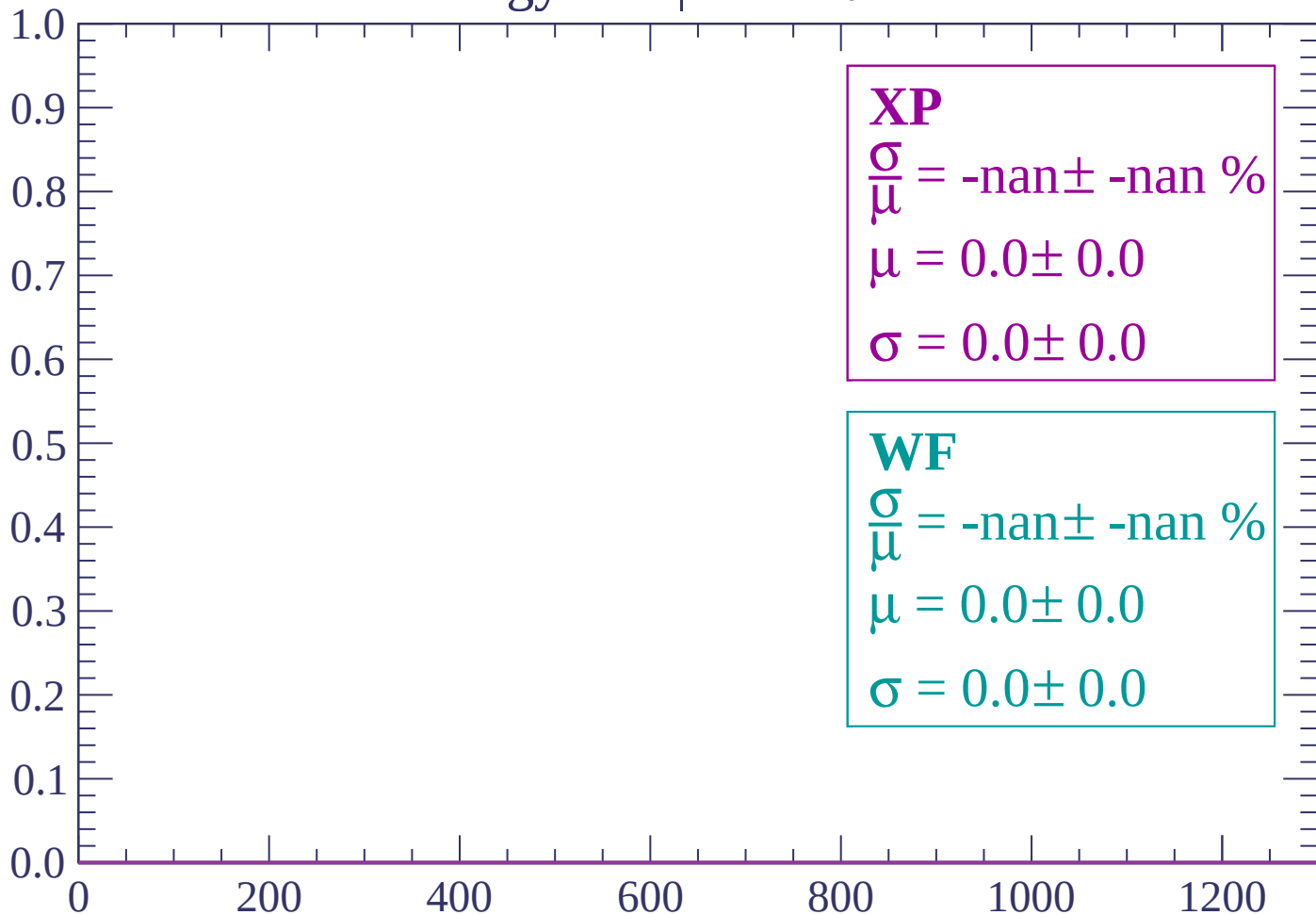
1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -28$

Count



dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

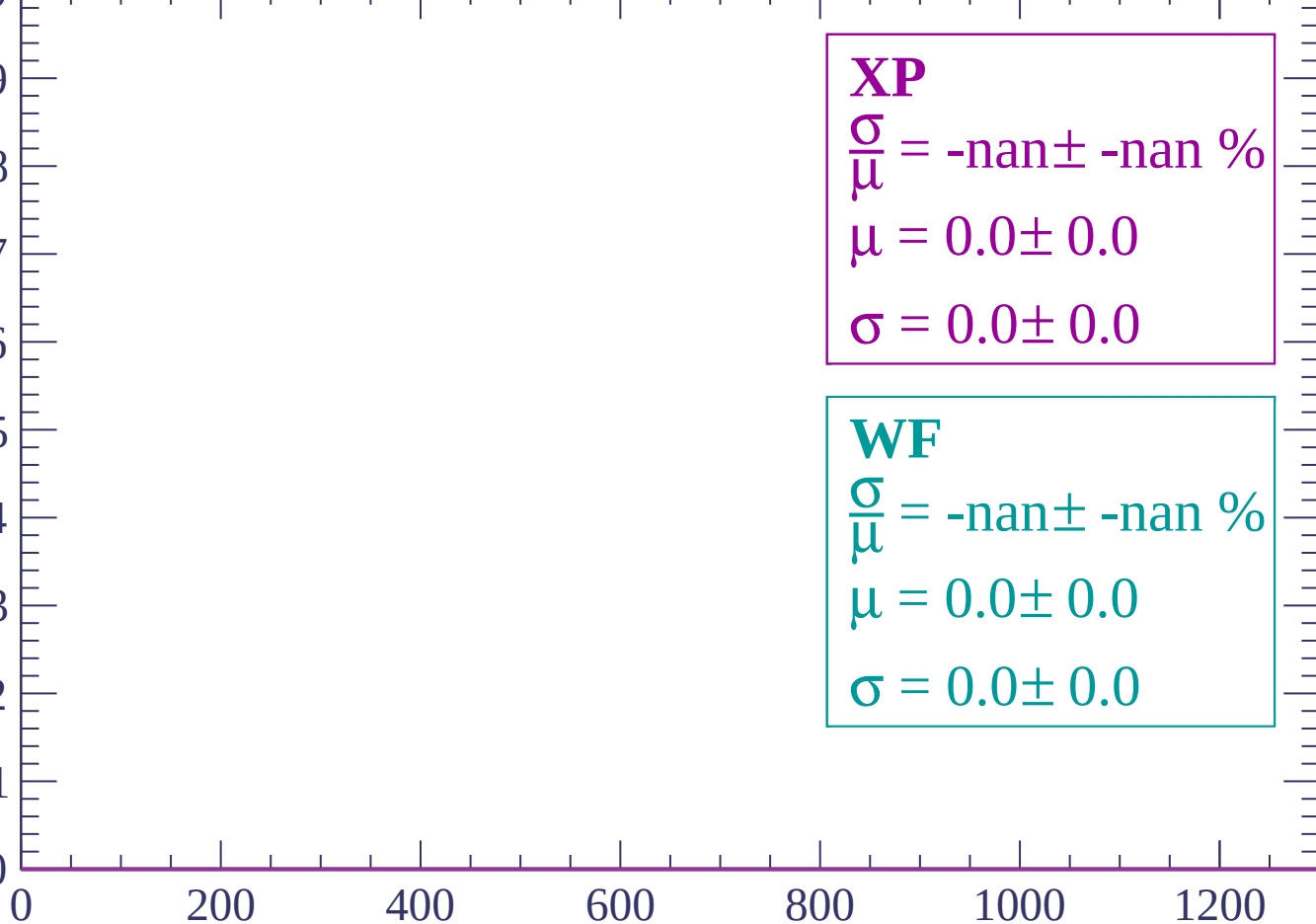
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-26 < \theta < -24$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-22 < \theta < -20$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-20 < \theta < -18$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-18 < \theta < -16$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-16 < \theta < -14$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-14 < \theta < -12$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-12 < \theta < -10$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

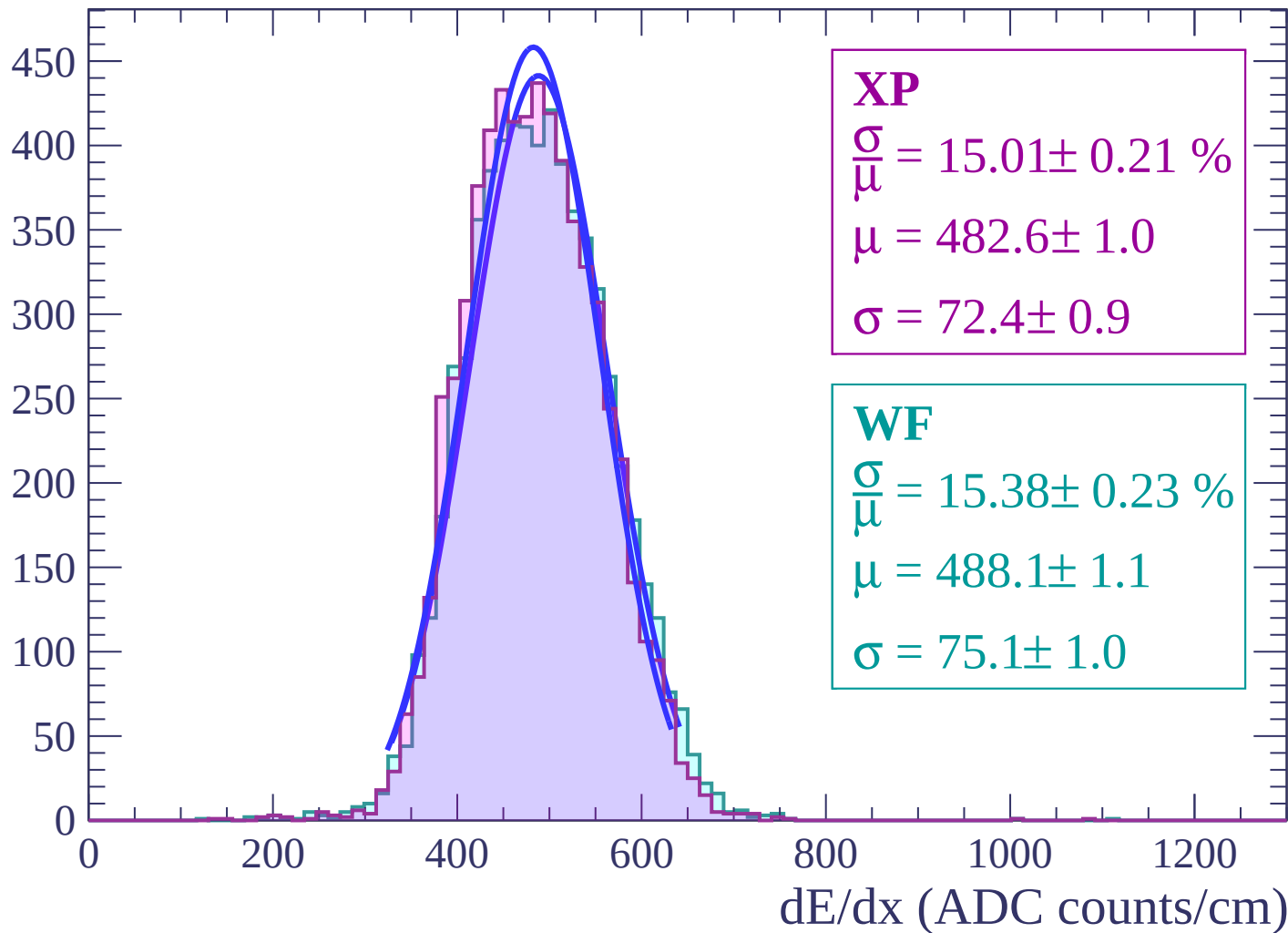
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

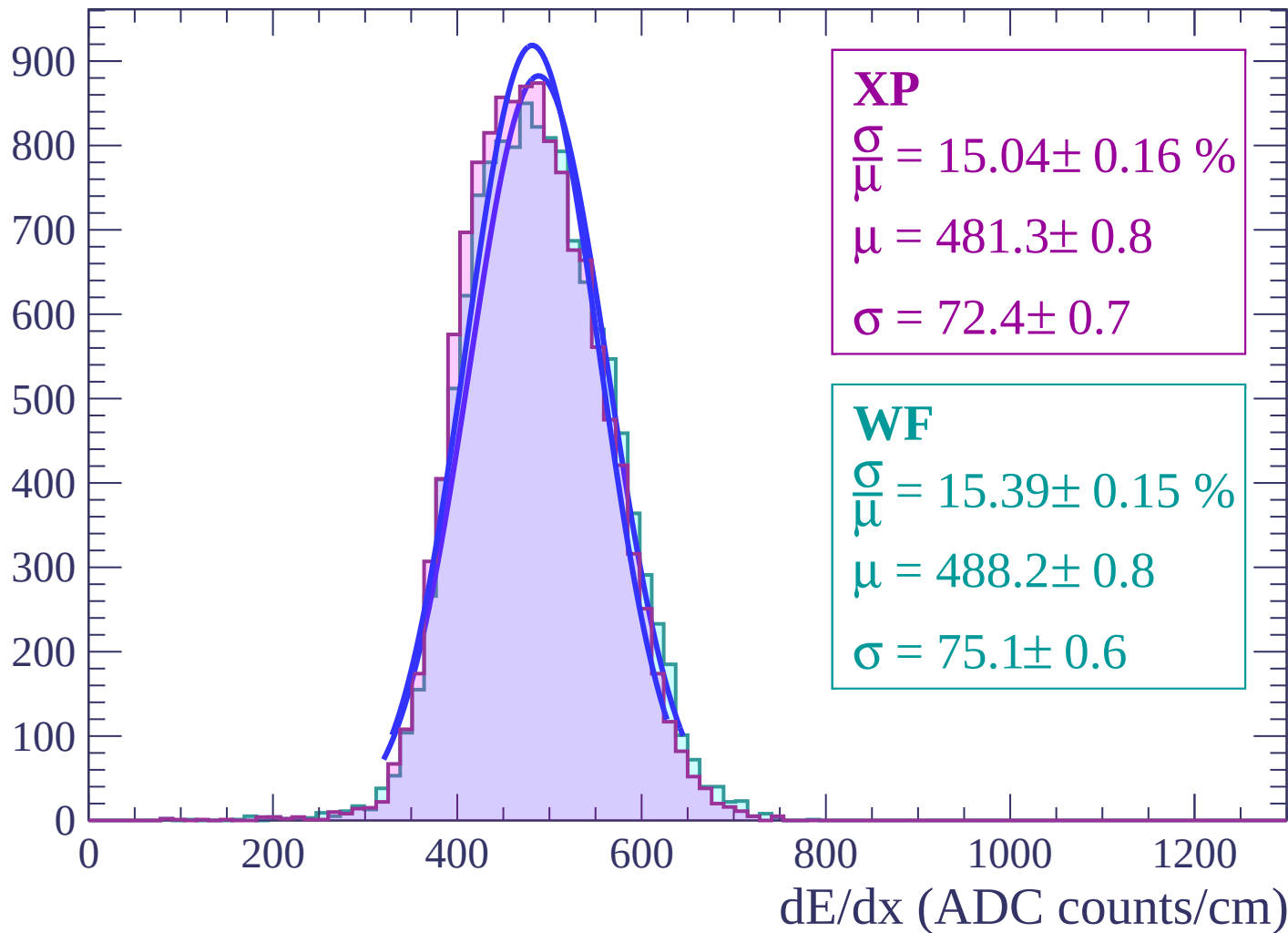
Energy loss | $-10 < \theta < -8$

Count



Energy loss | $-8 < \theta < -6$

Count



Energy loss | $-6 < \theta < -4$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 14.79 \pm 0.14 \%$$

$$\mu = 480.8 \pm 0.7$$

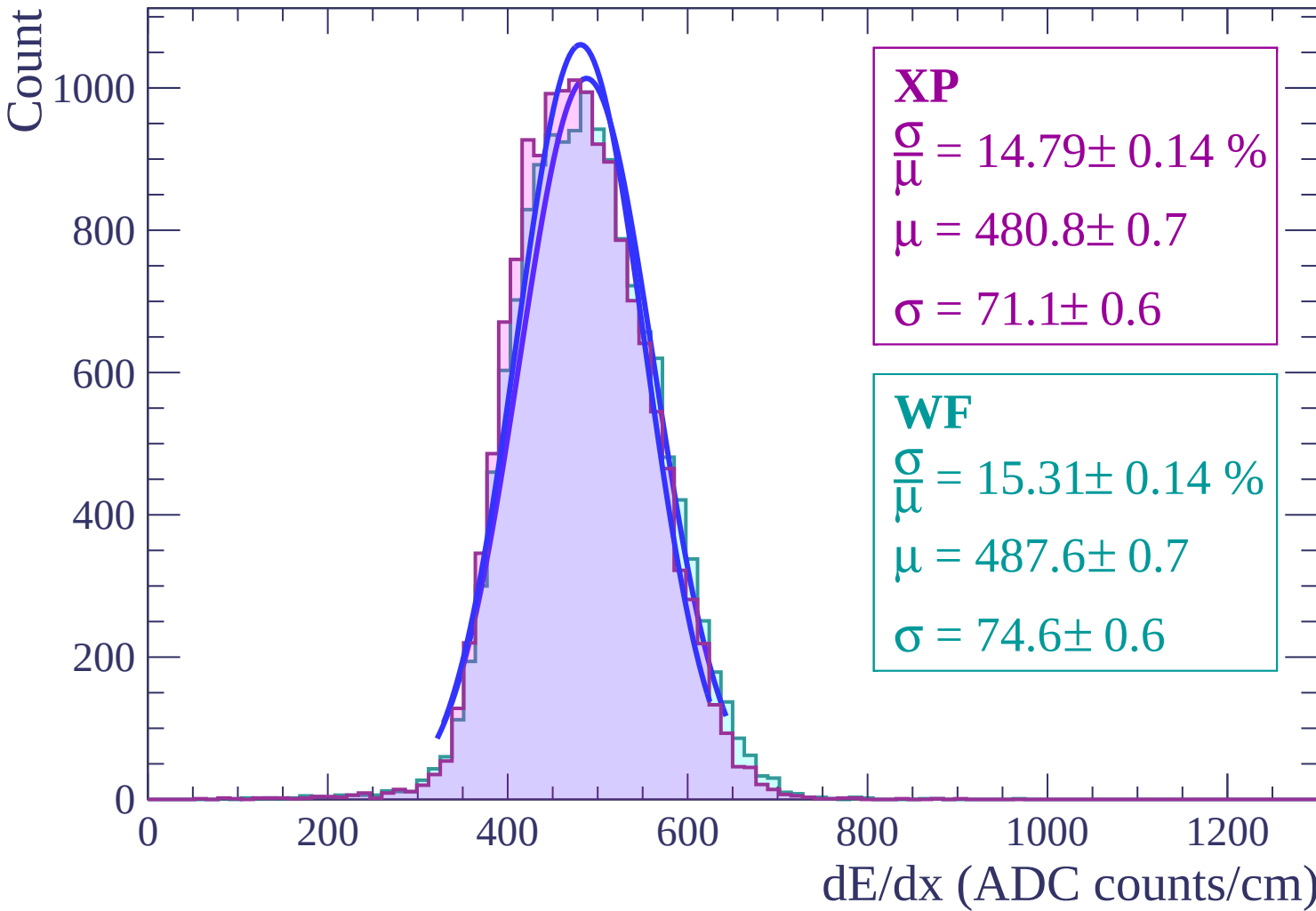
$$\sigma = 71.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.31 \pm 0.14 \%$$

$$\mu = 487.6 \pm 0.7$$

$$\sigma = 74.6 \pm 0.6$$



Energy loss | $-4 < \theta < -2$

Count

1000

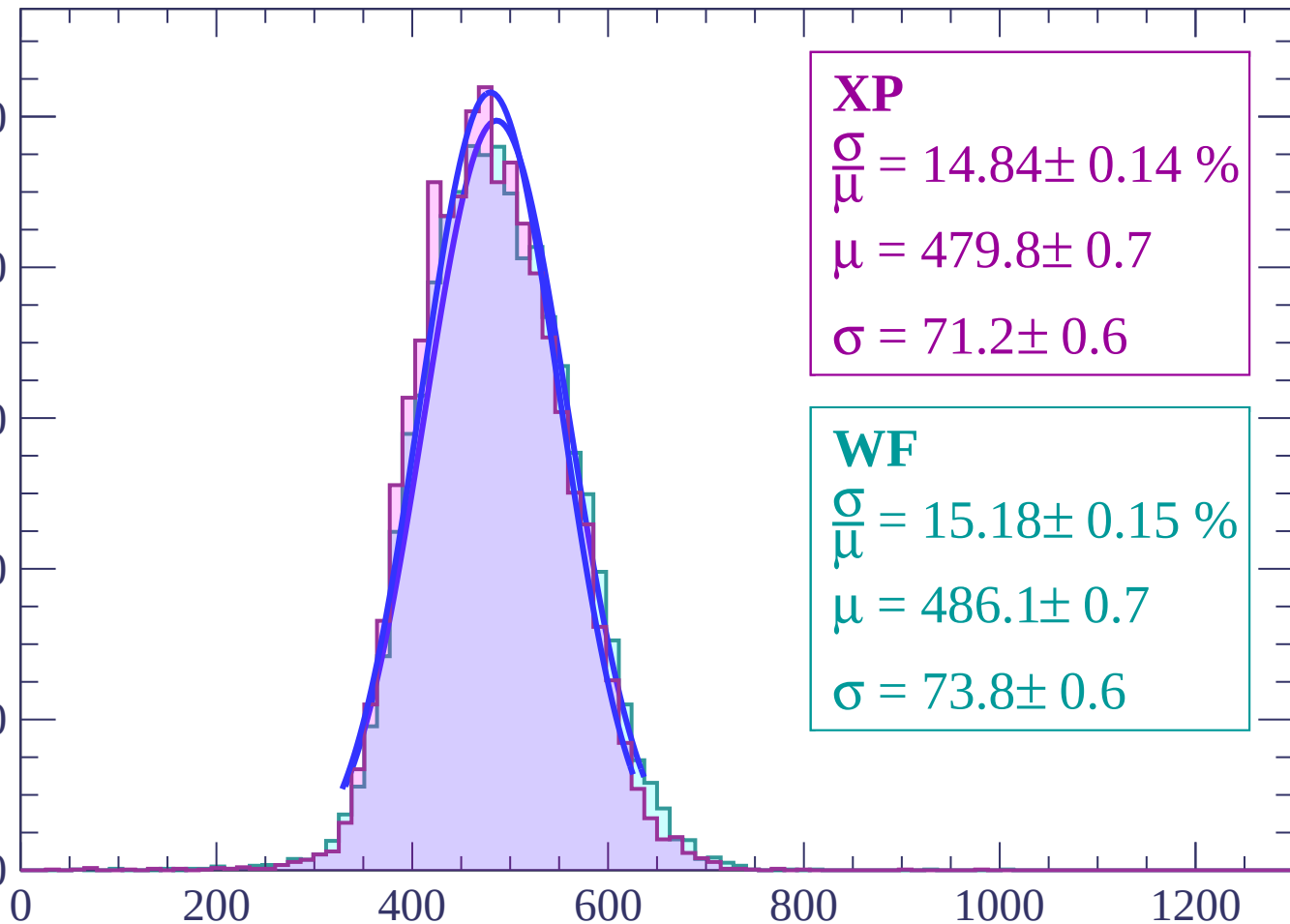
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.84 \pm 0.14 \%$$

$$\mu = 479.8 \pm 0.7$$

$$\sigma = 71.2 \pm 0.6$$

WF

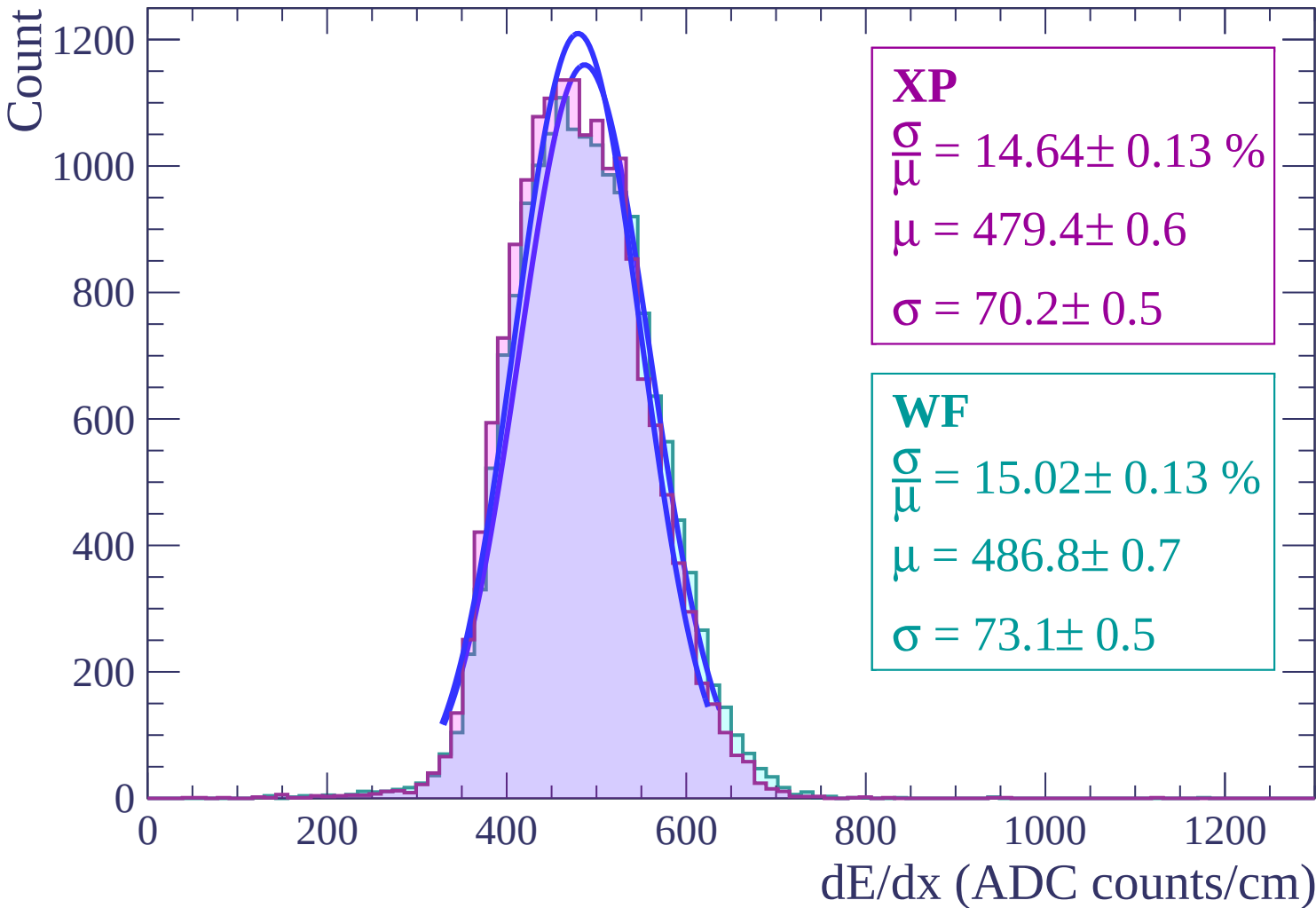
$$\frac{\sigma}{\mu} = 15.18 \pm 0.15 \%$$

$$\mu = 486.1 \pm 0.7$$

$$\sigma = 73.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss $|-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

2500

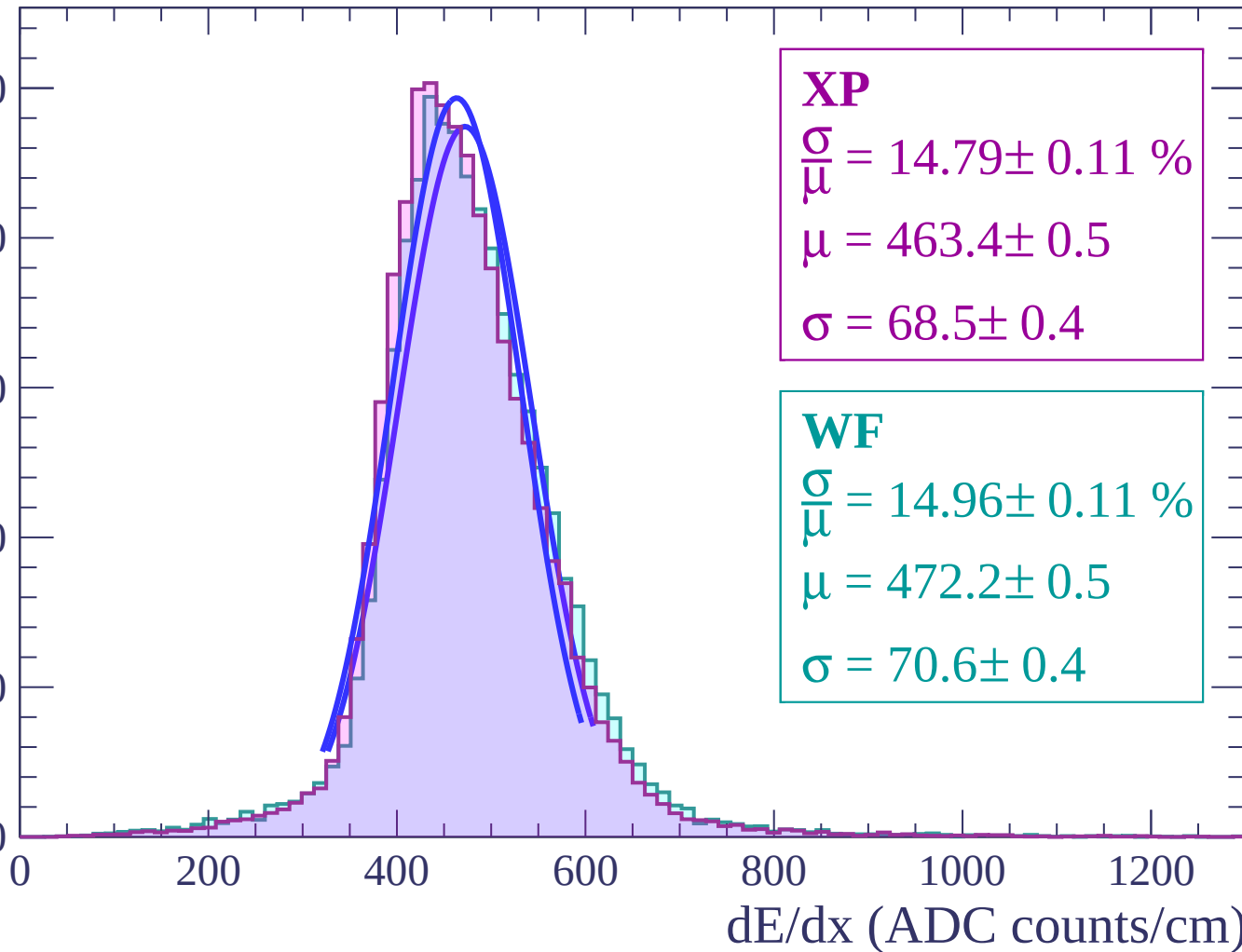
2000

1500

1000

500

0



Energy loss | $2 < \theta < 4$

Count

1200

1000

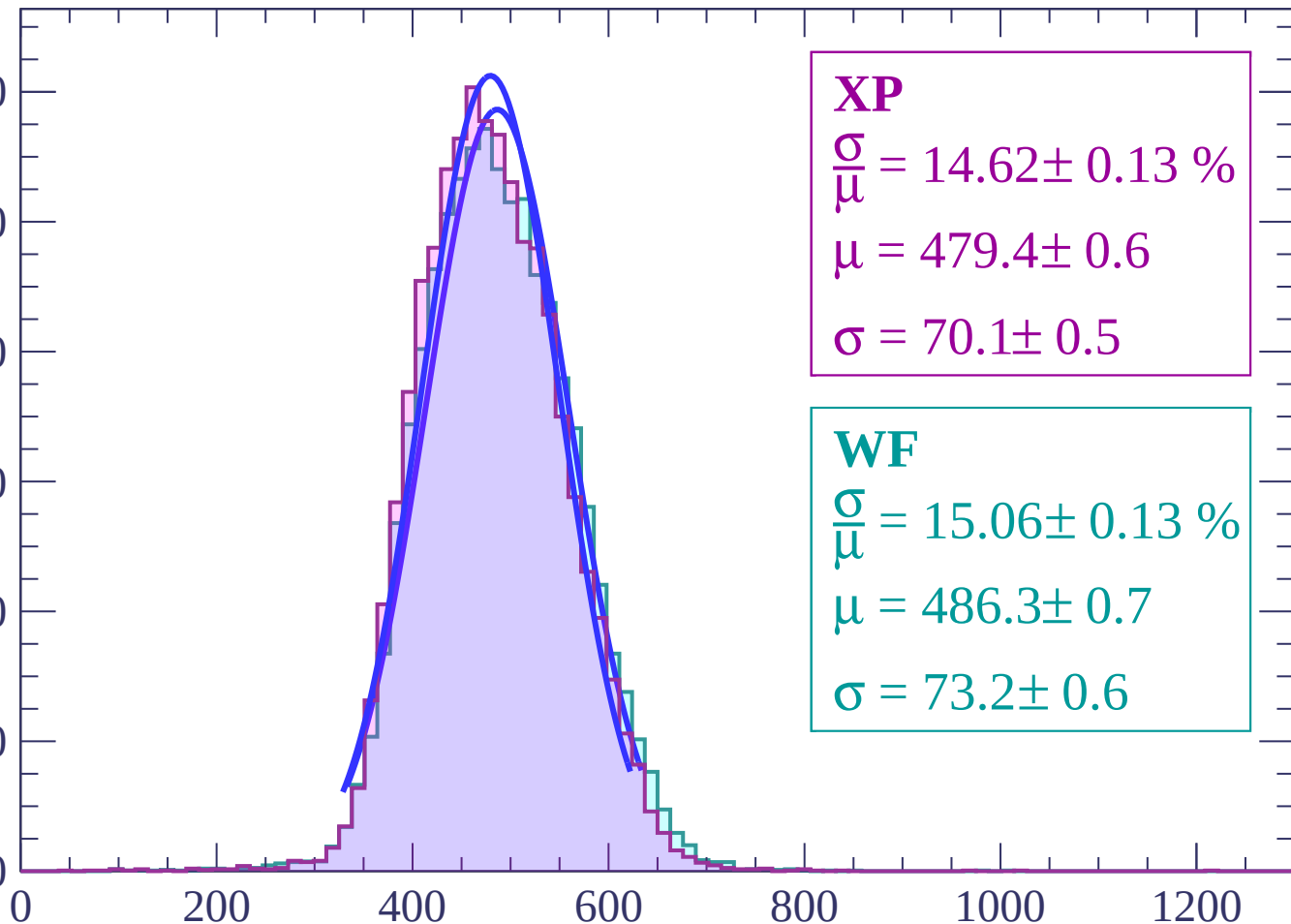
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.62 \pm 0.13 \%$$

$$\mu = 479.4 \pm 0.6$$

$$\sigma = 70.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 15.06 \pm 0.13 \%$$

$$\mu = 486.3 \pm 0.7$$

$$\sigma = 73.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$

Count

1000

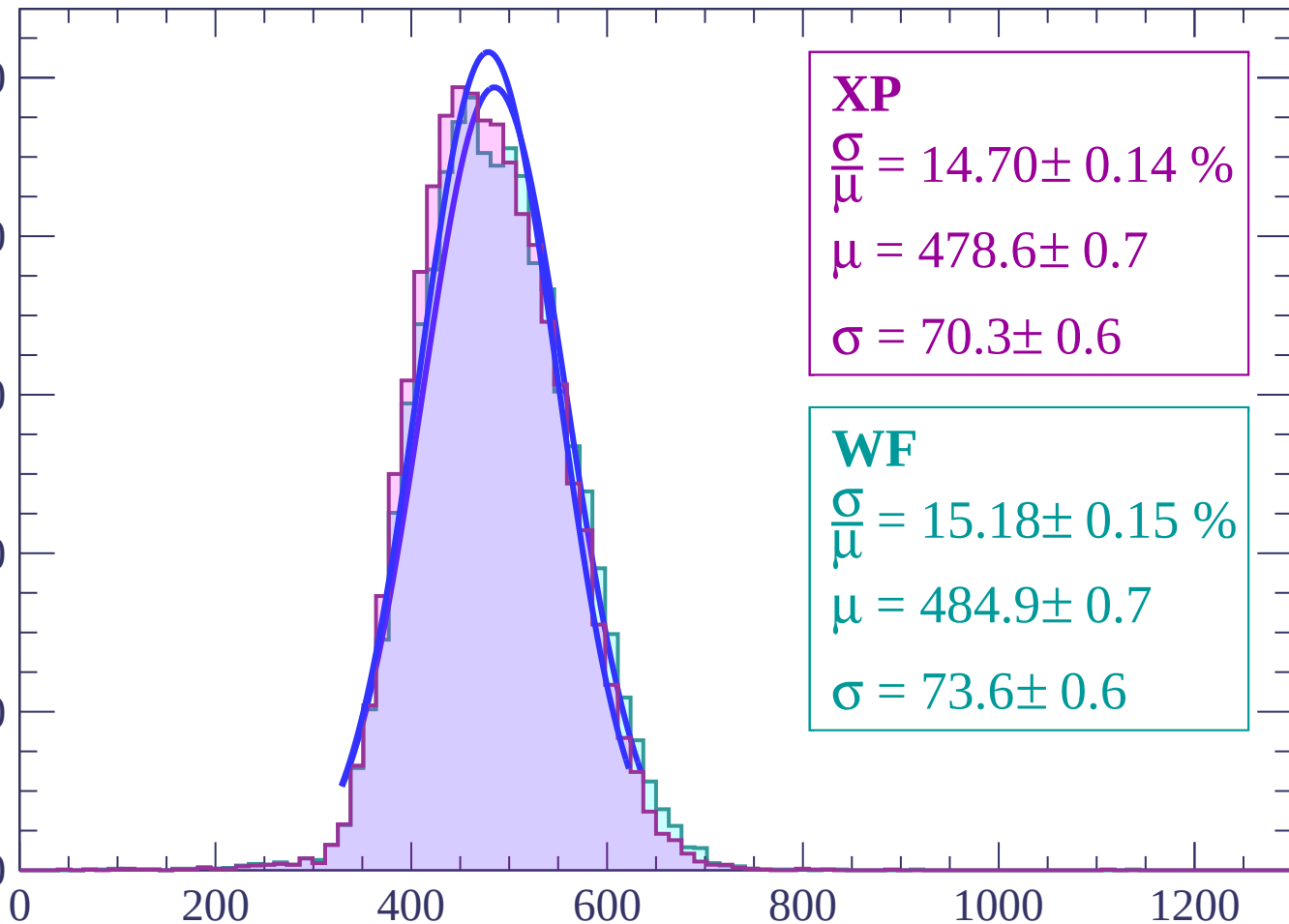
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.70 \pm 0.14 \%$$

$$\mu = 478.6 \pm 0.7$$

$$\sigma = 70.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.18 \pm 0.15 \%$$

$$\mu = 484.9 \pm 0.7$$

$$\sigma = 73.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

Count

1000

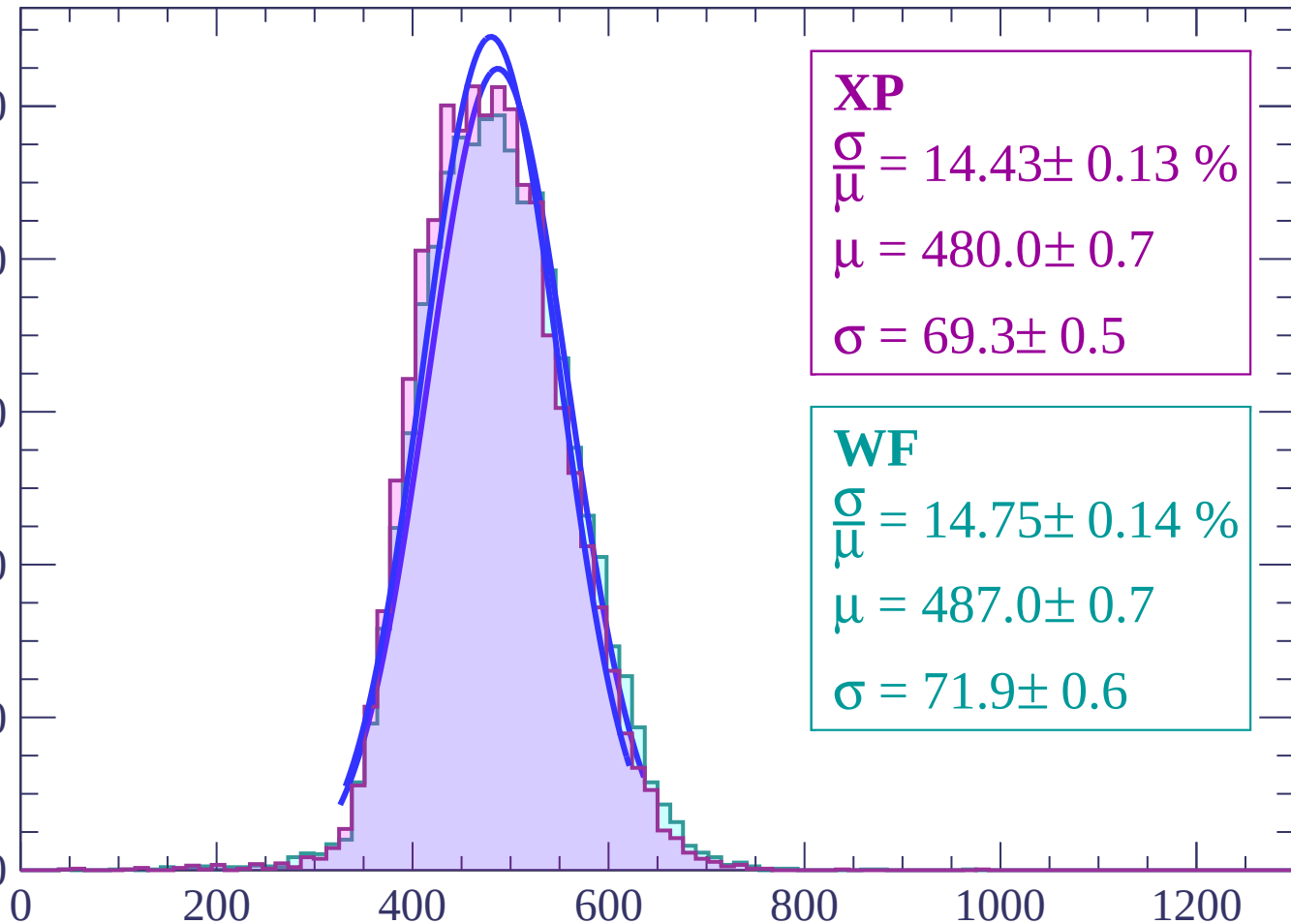
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.43 \pm 0.13 \%$$

$$\mu = 480.0 \pm 0.7$$

$$\sigma = 69.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 14.75 \pm 0.14 \%$$

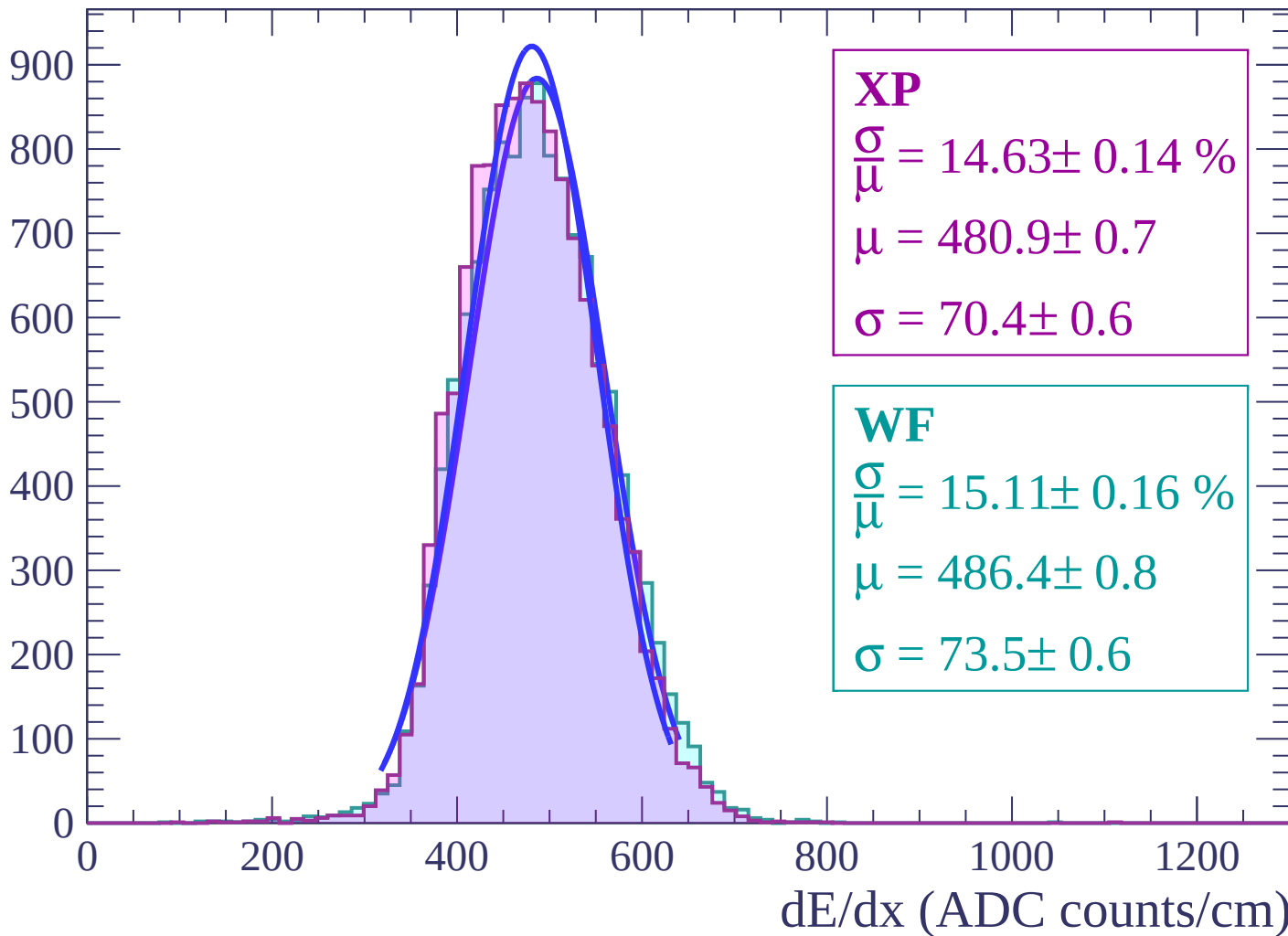
$$\mu = 487.0 \pm 0.7$$

$$\sigma = 71.9 \pm 0.6$$

dE/dx (ADC counts/cm)

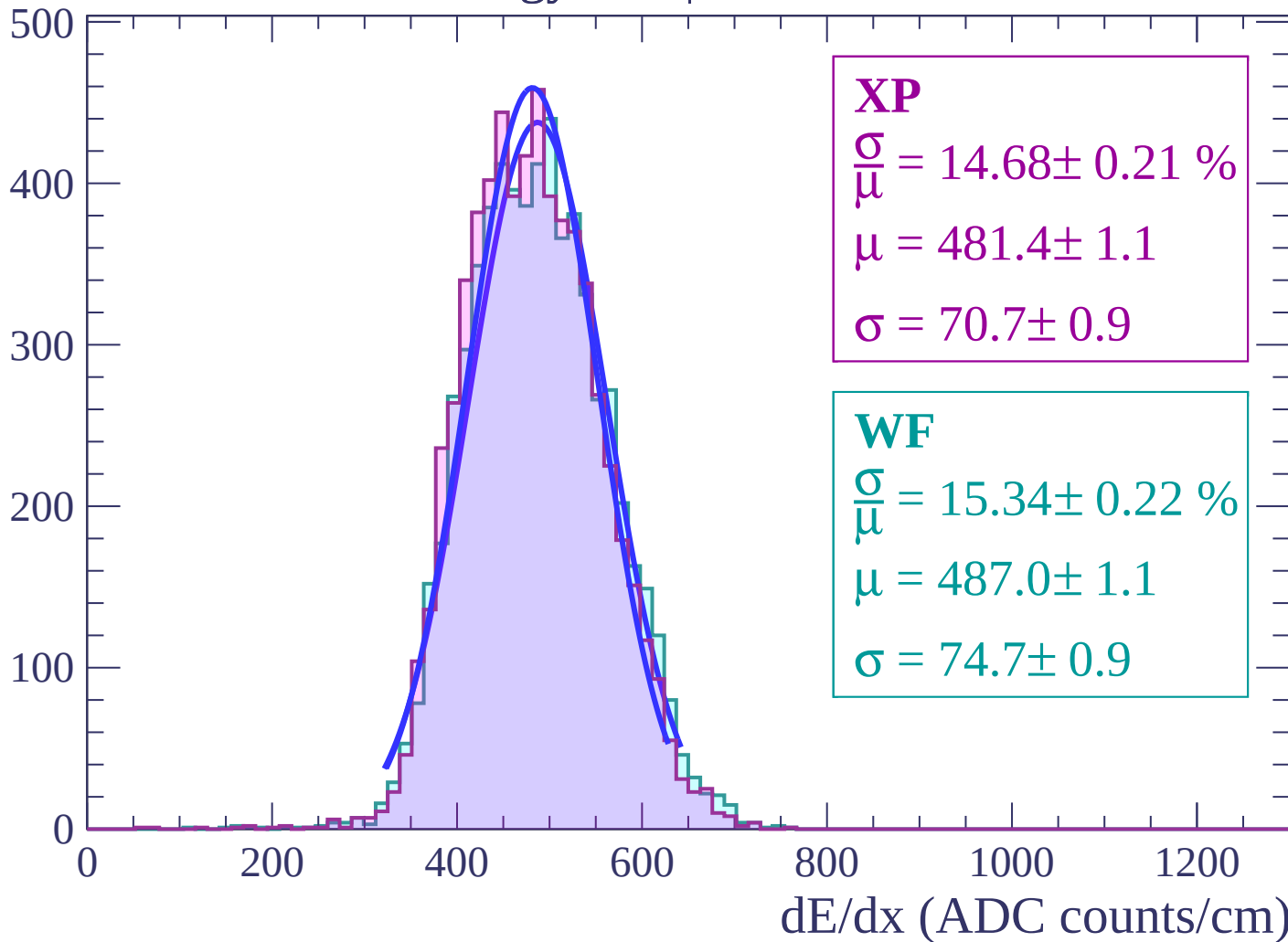
Energy loss | $8 < \theta < 10$

Count



Energy loss | $10 < \theta < 12$

Count



Energy loss | $12 < \theta < 14$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $14 < \theta < 16$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

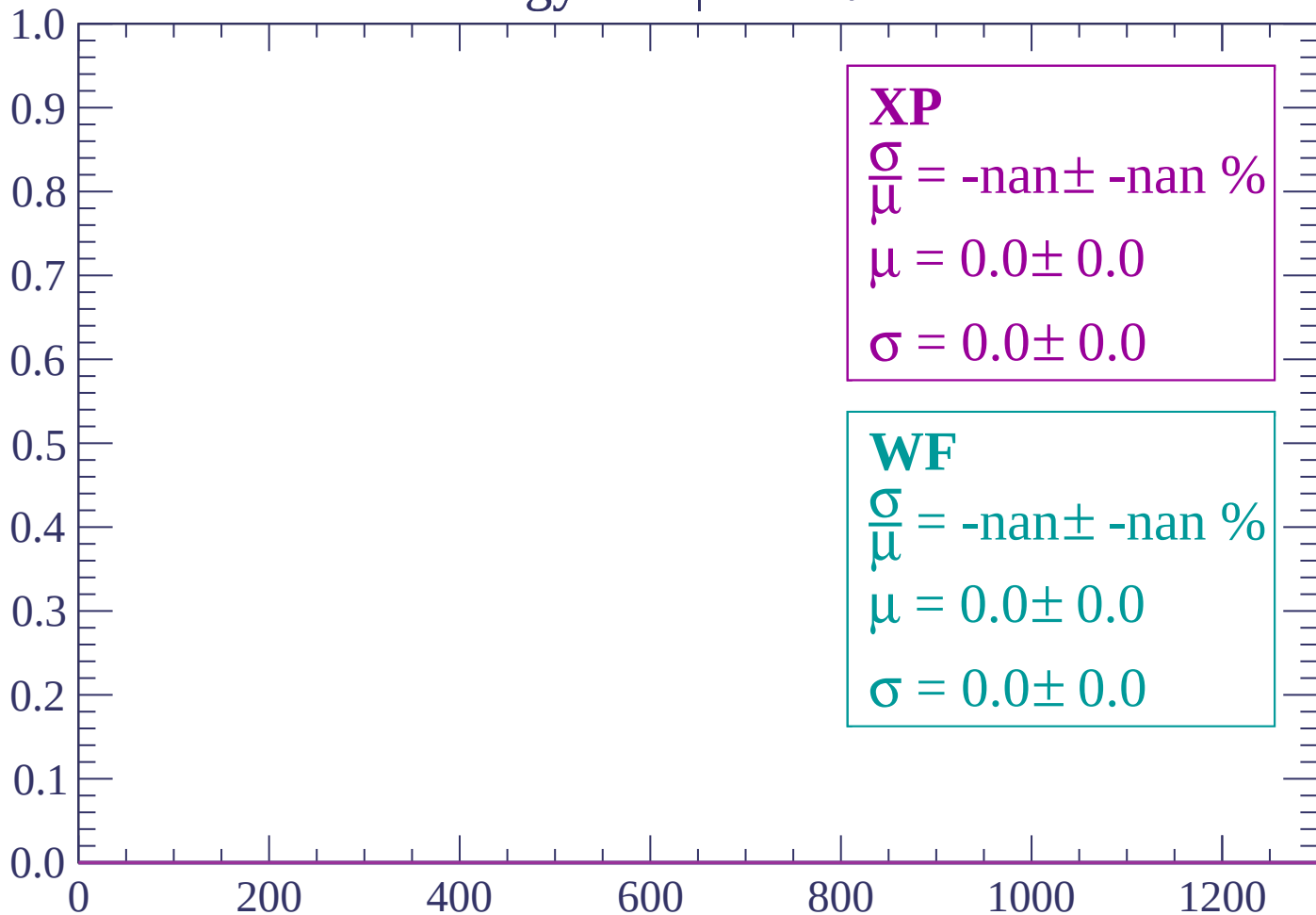
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $16 < \theta < 18$

Count



dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

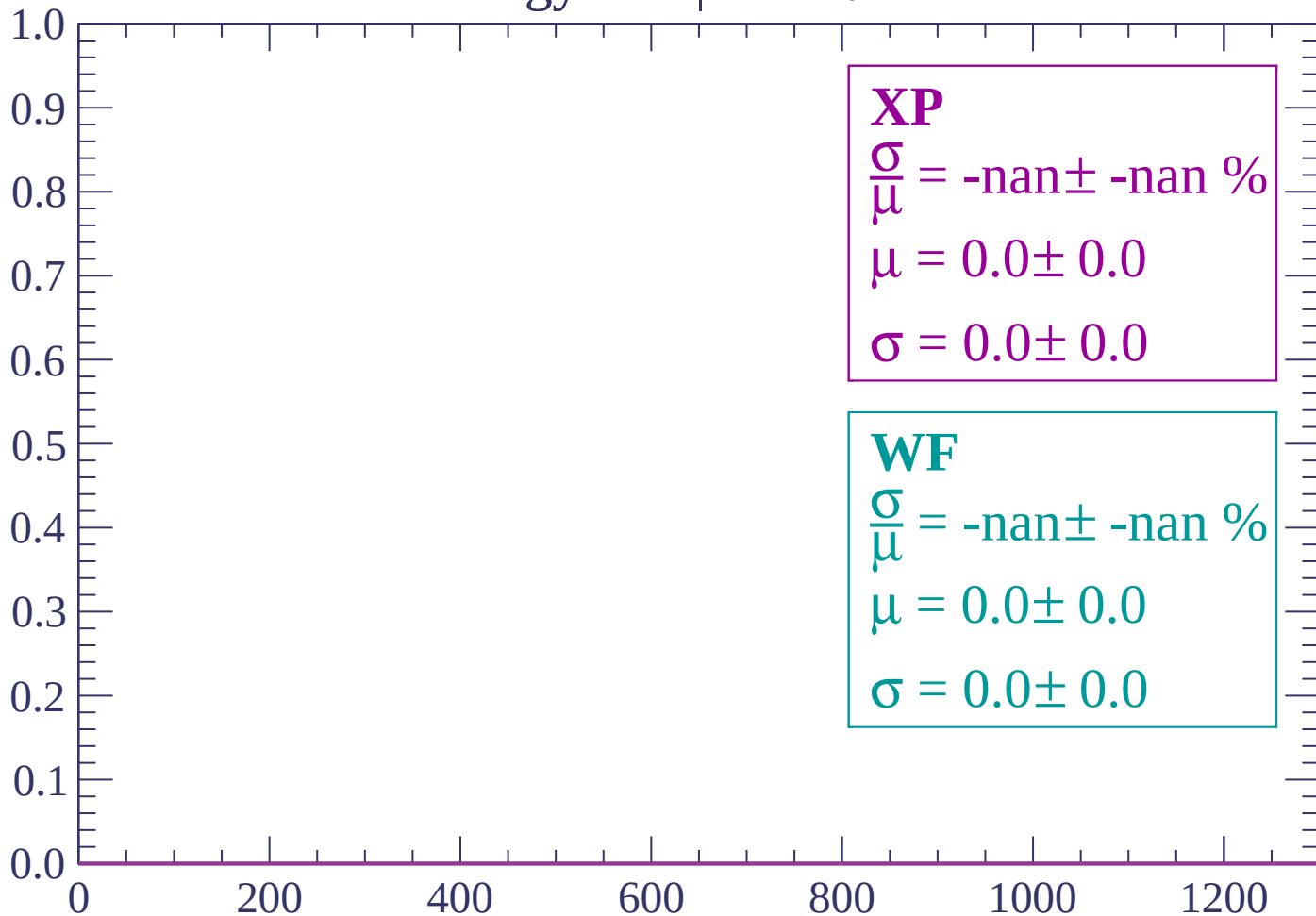
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $20 < \theta < 22$

Count



dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

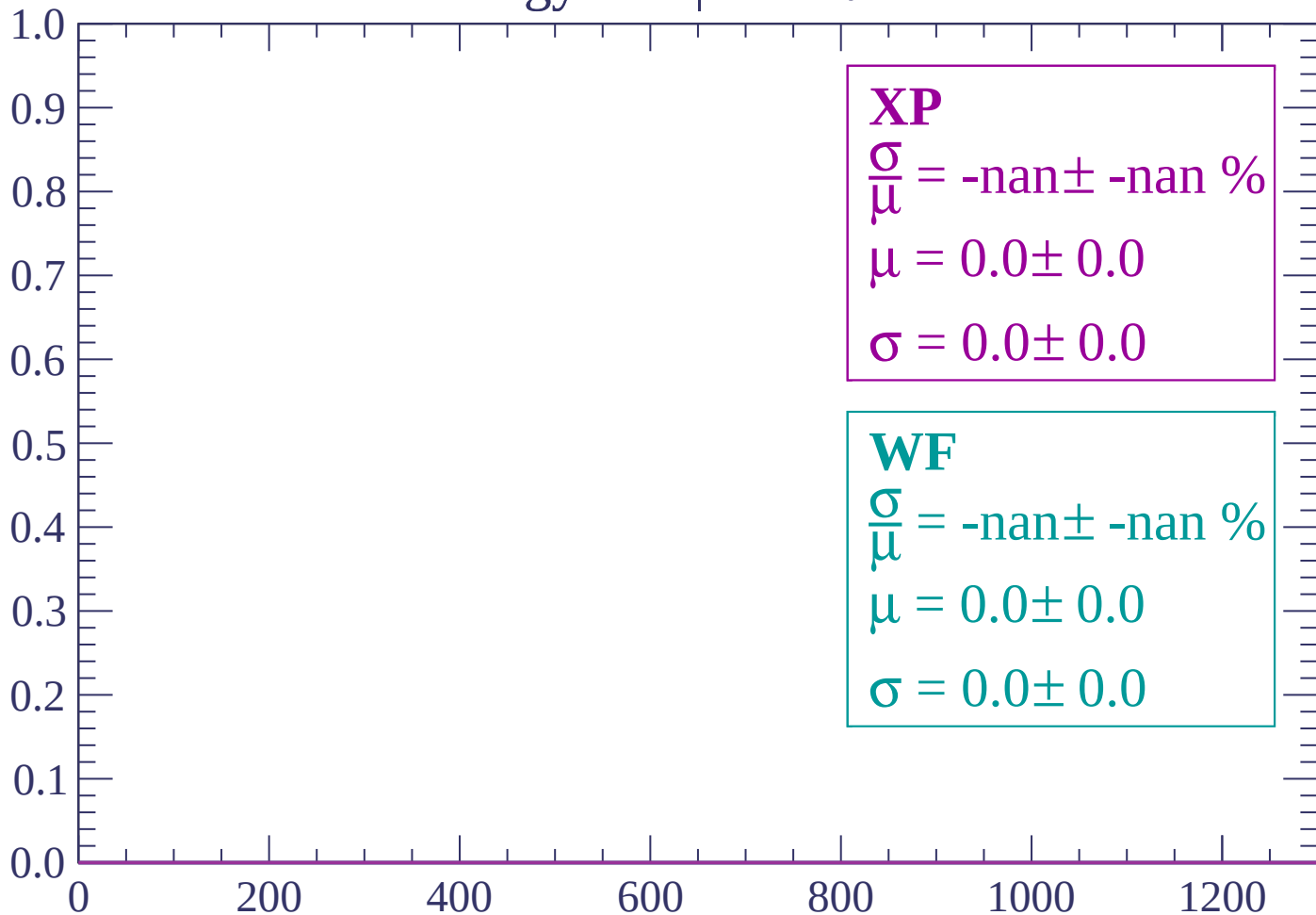
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $24 < \theta < 26$

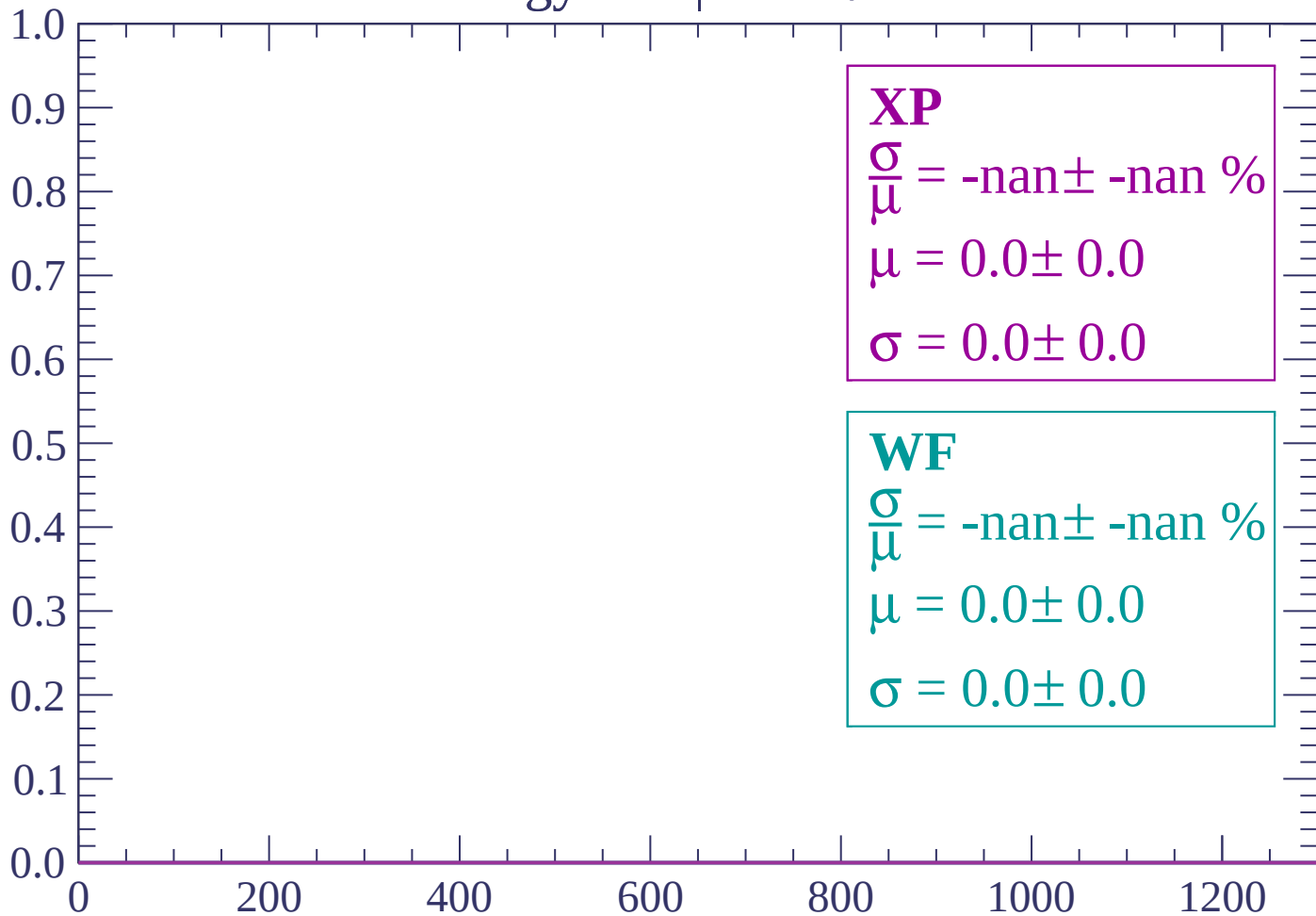
Count



dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

Count



dE/dx (ADC counts/cm)

Energy loss | $28 < \theta < 30$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $30 < \theta < 32$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $32 < \theta < 34$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $34 < \theta < 36$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

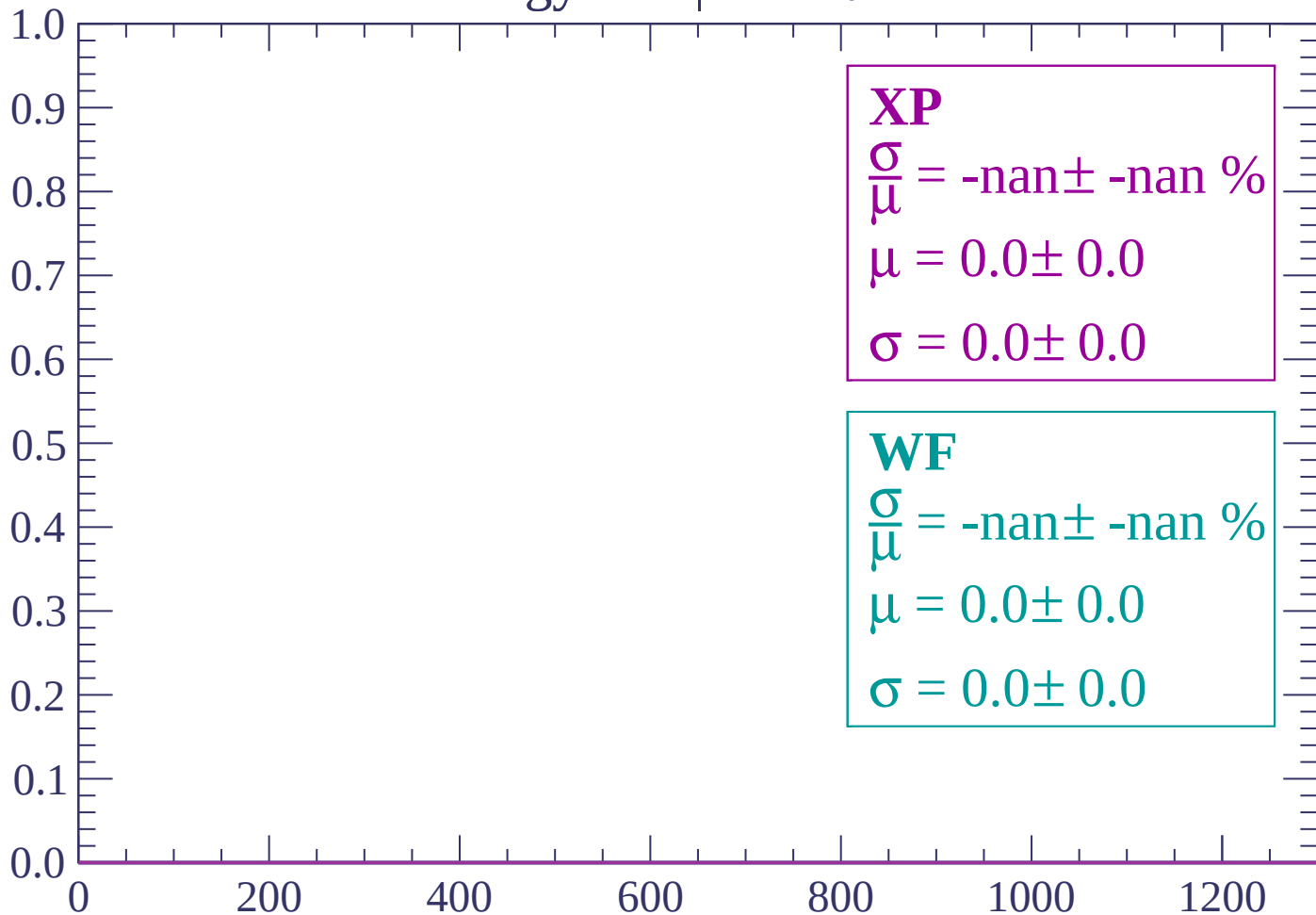
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $36 < \theta < 38$

Count



dE/dx (ADC counts/cm)

Energy loss | $38 < \theta < 40$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

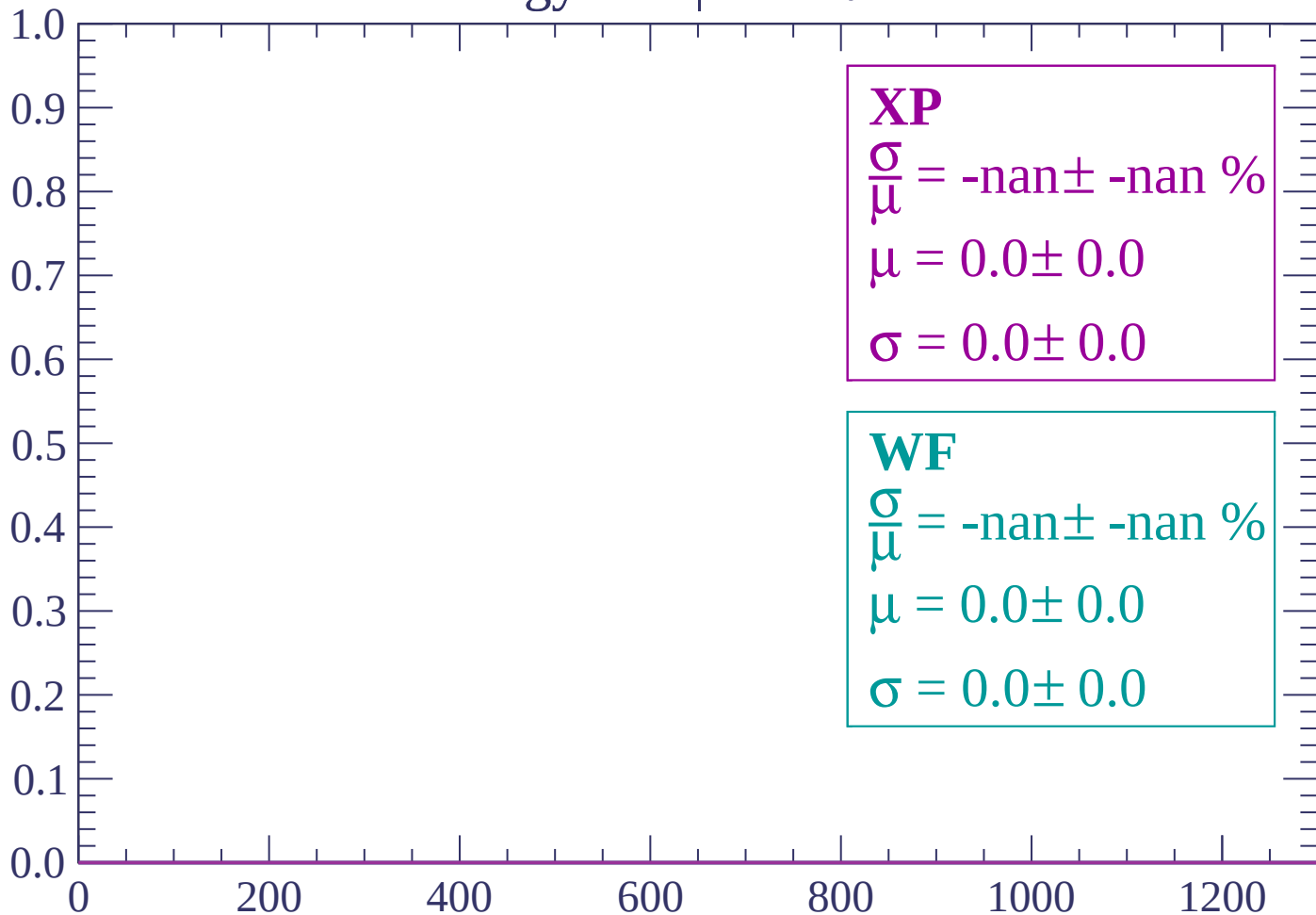
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $40 < \theta < 42$

Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \theta < 44$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $44 < \theta < 46$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $46 < \theta < 48$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $48 < \theta < 50$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

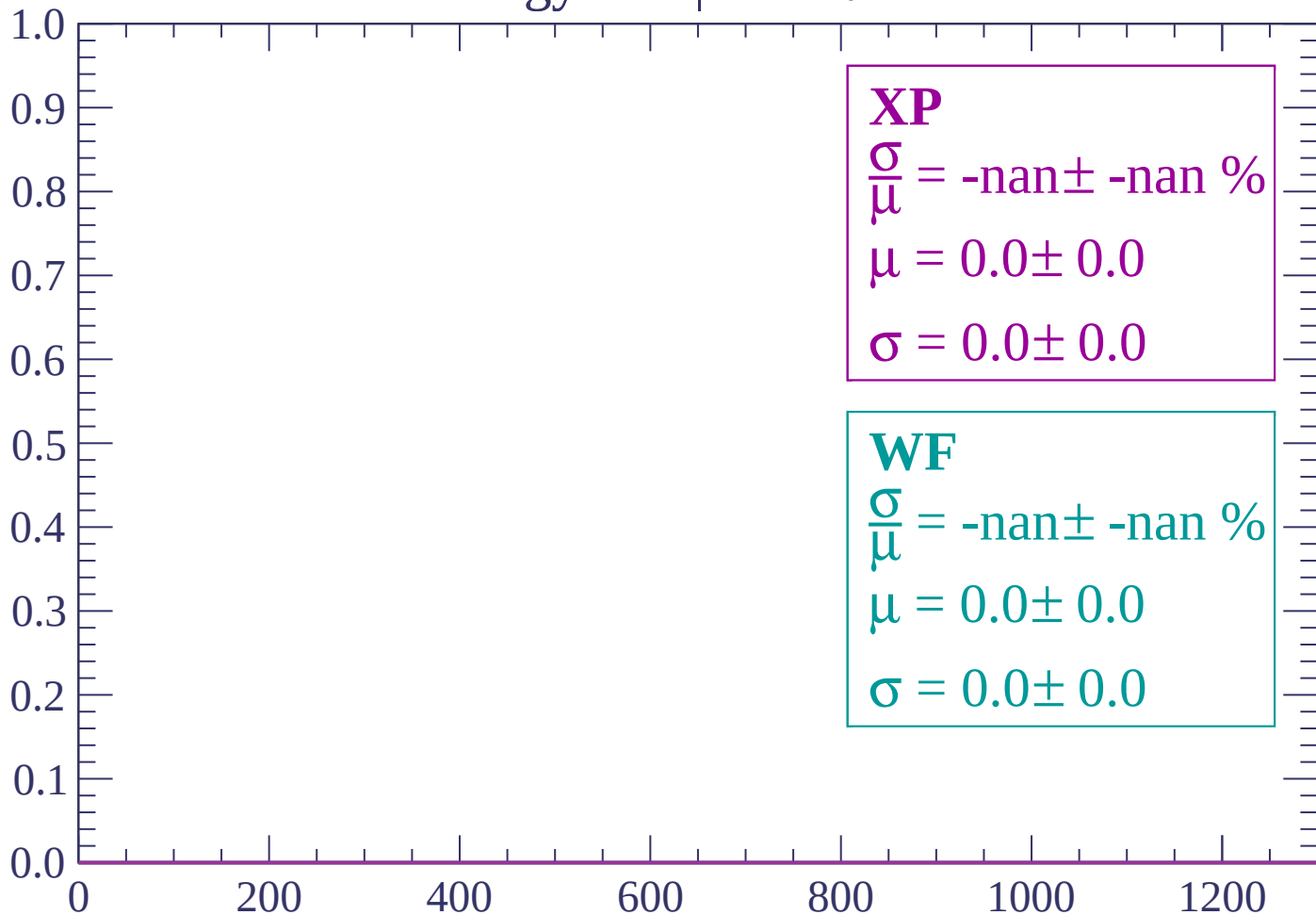
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $50 < \theta < 52$

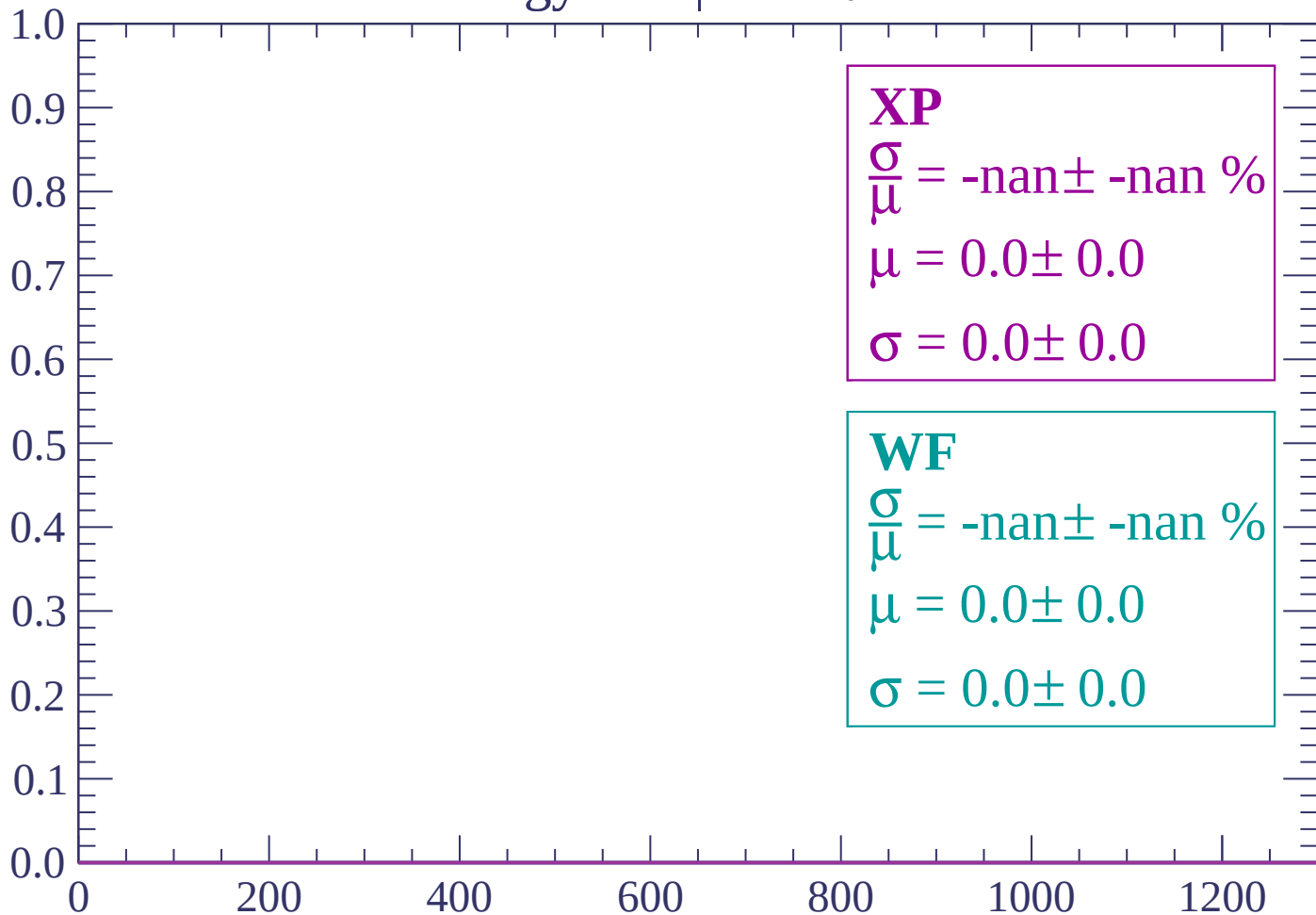
Count



dE/dx (ADC counts/cm)

Energy loss | $52 < \theta < 54$

Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $56 < \theta < 58$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $58 < \theta < 60$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $60 < \theta < 62$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $62 < \theta < 64$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $64 < \theta < 66$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $66 < \theta < 68$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $68 < \theta < 70$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

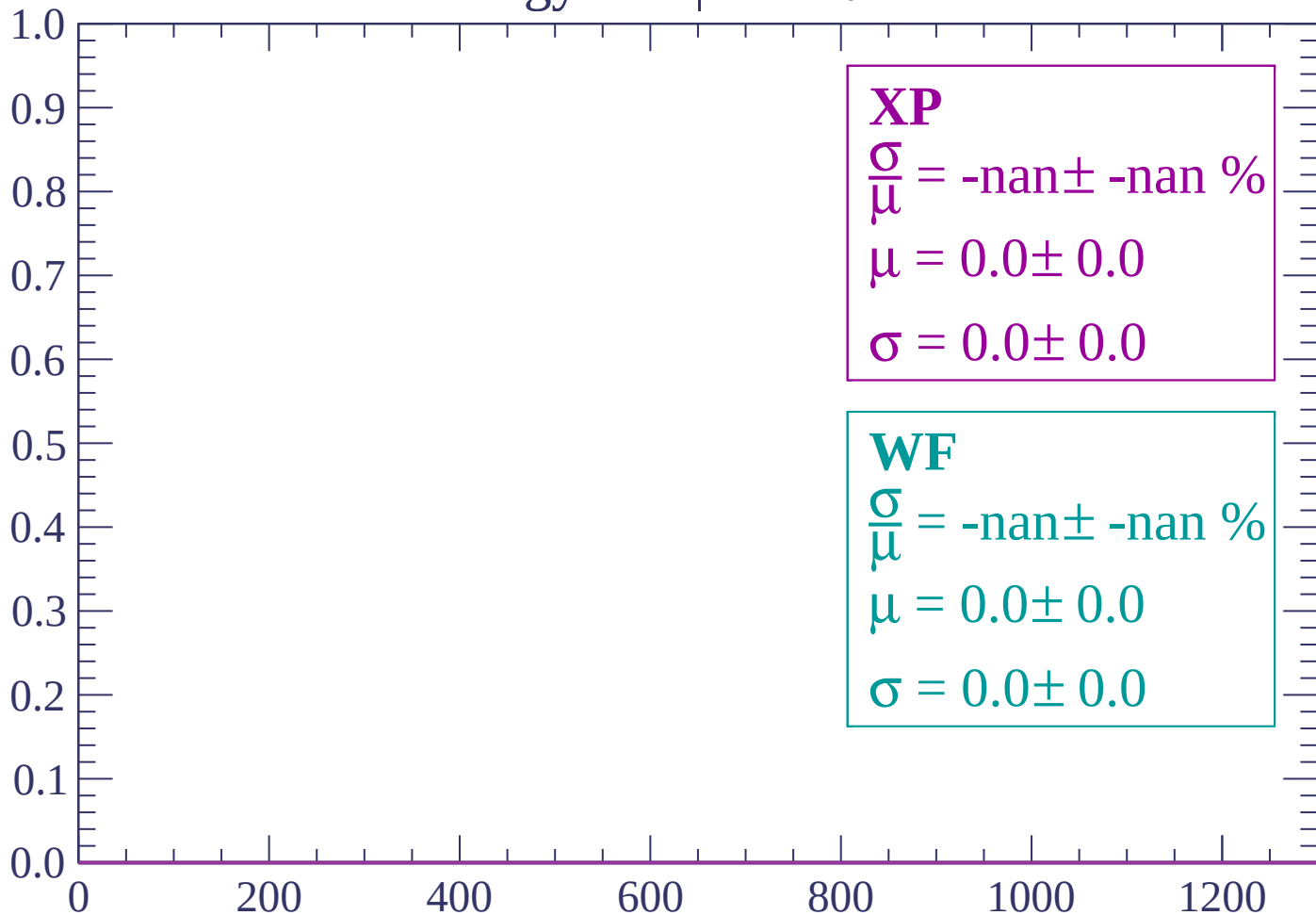
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $70 < \theta < 72$

Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

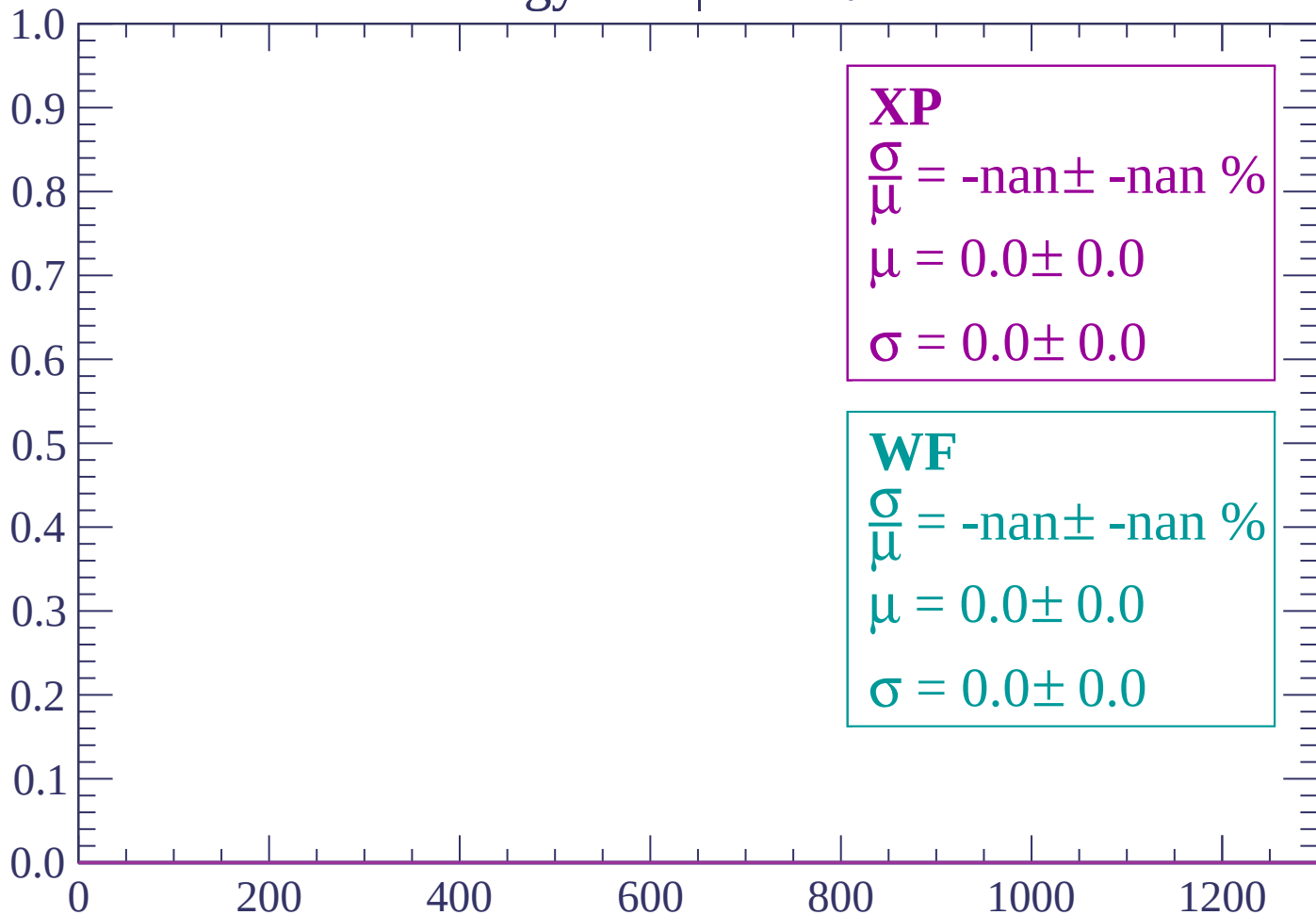
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $74 < \theta < 76$

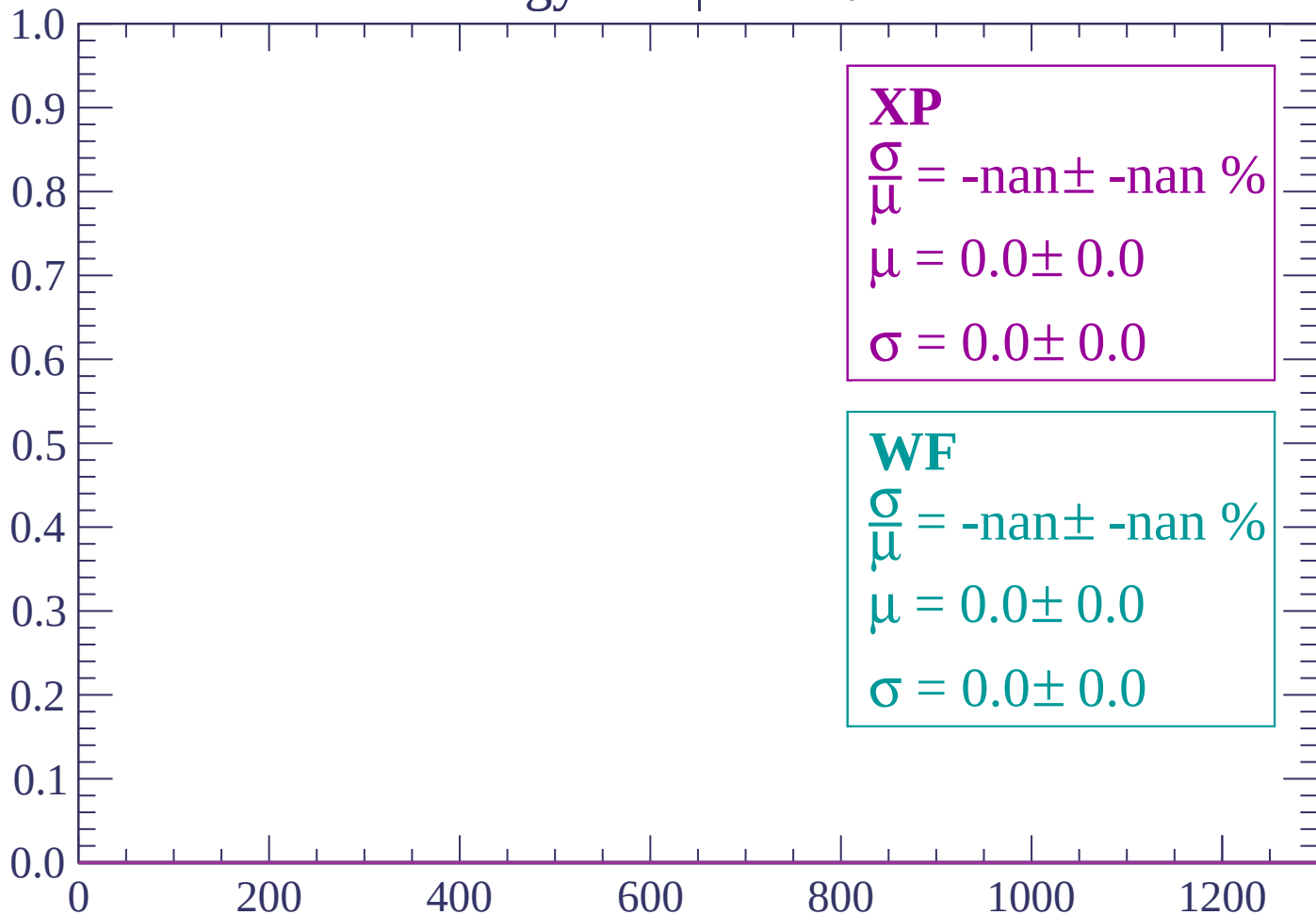
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

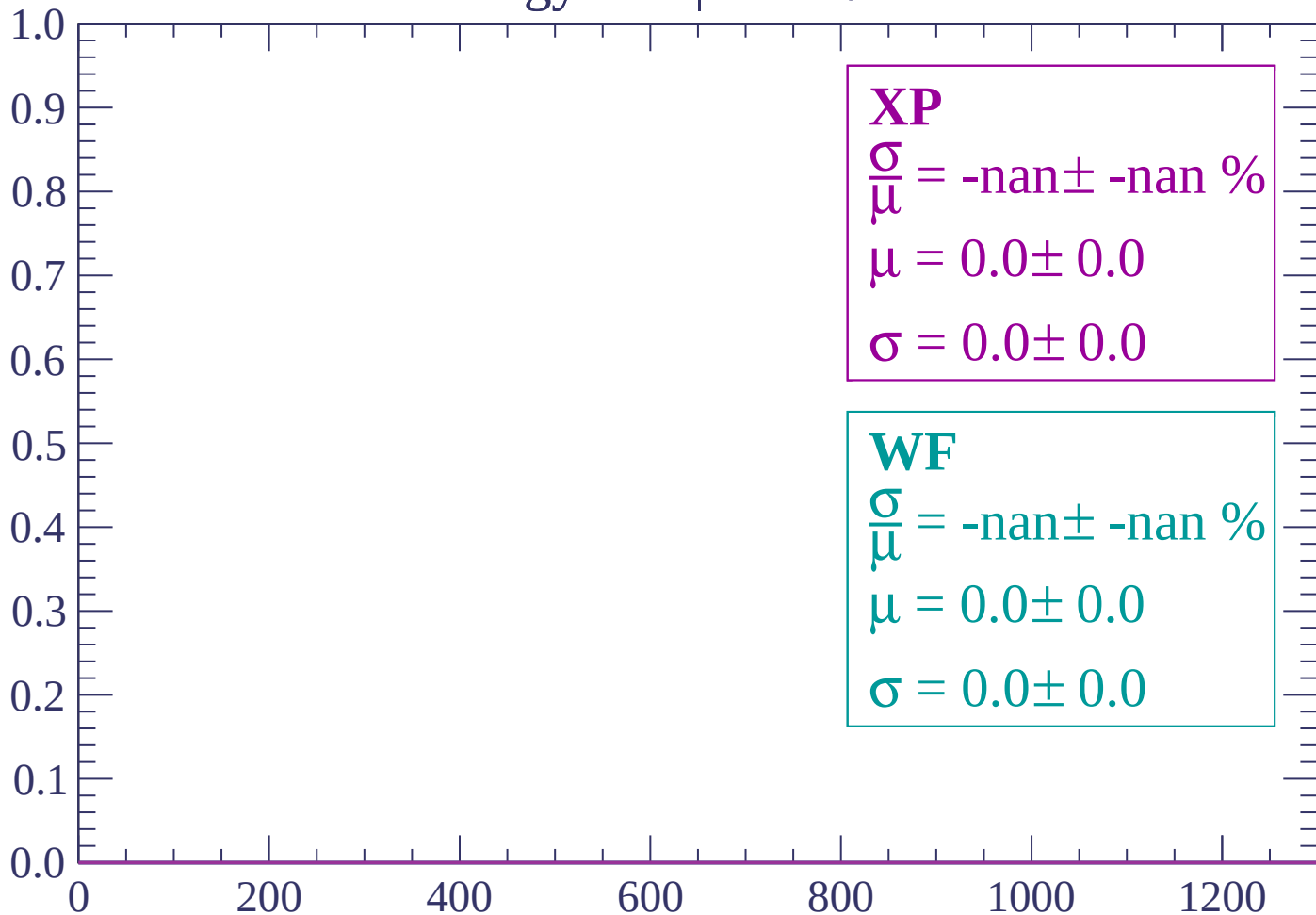
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

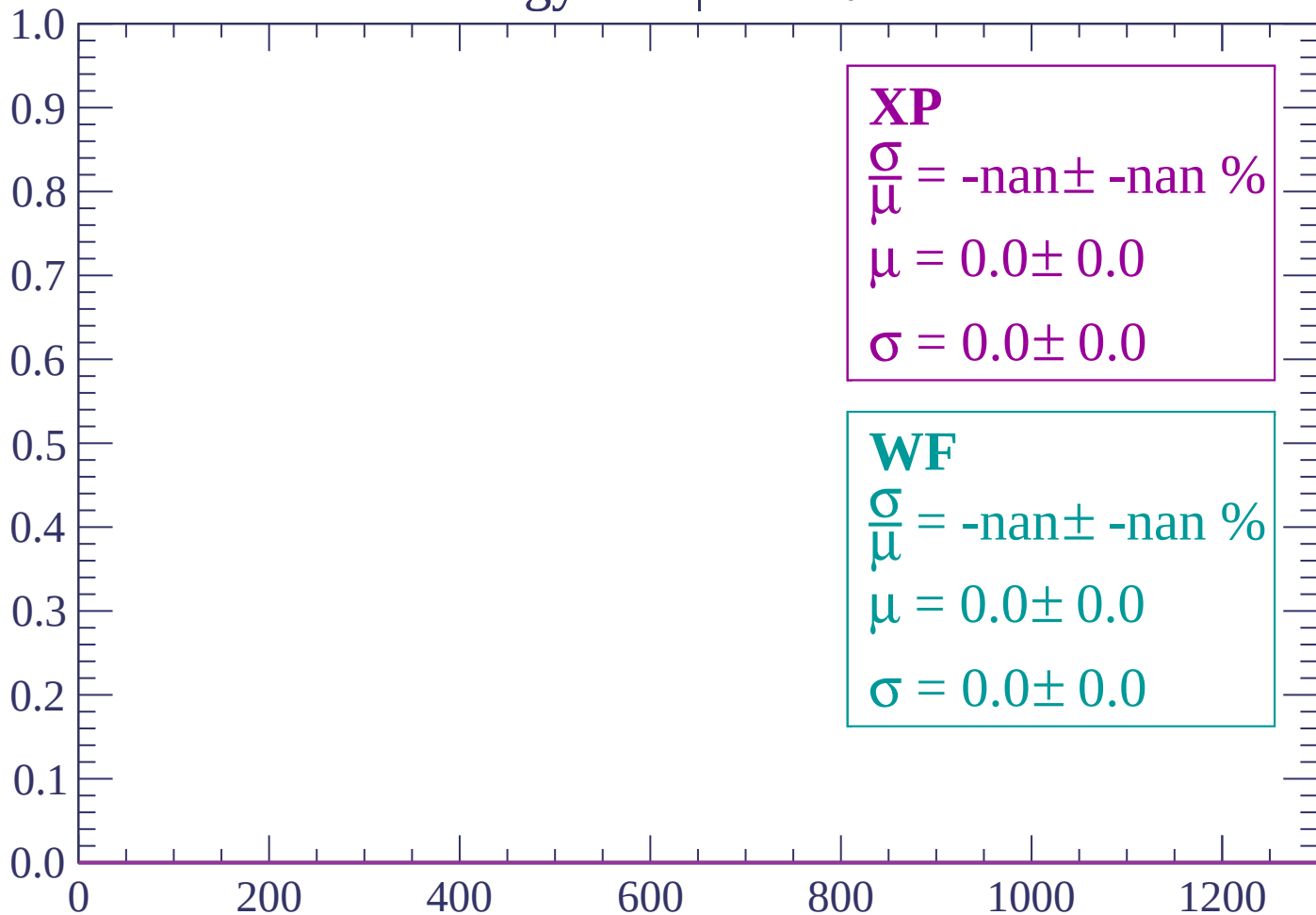
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

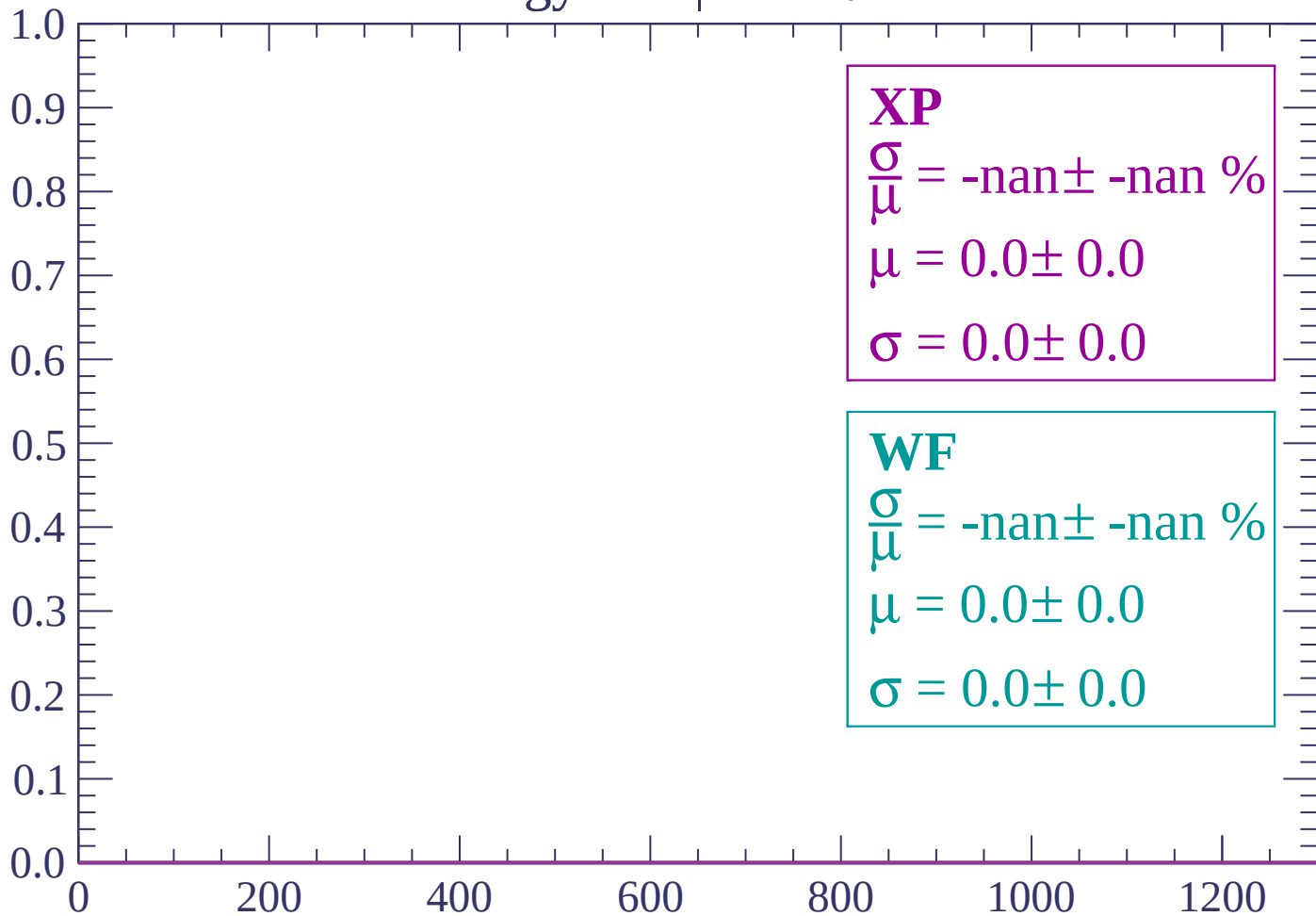
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

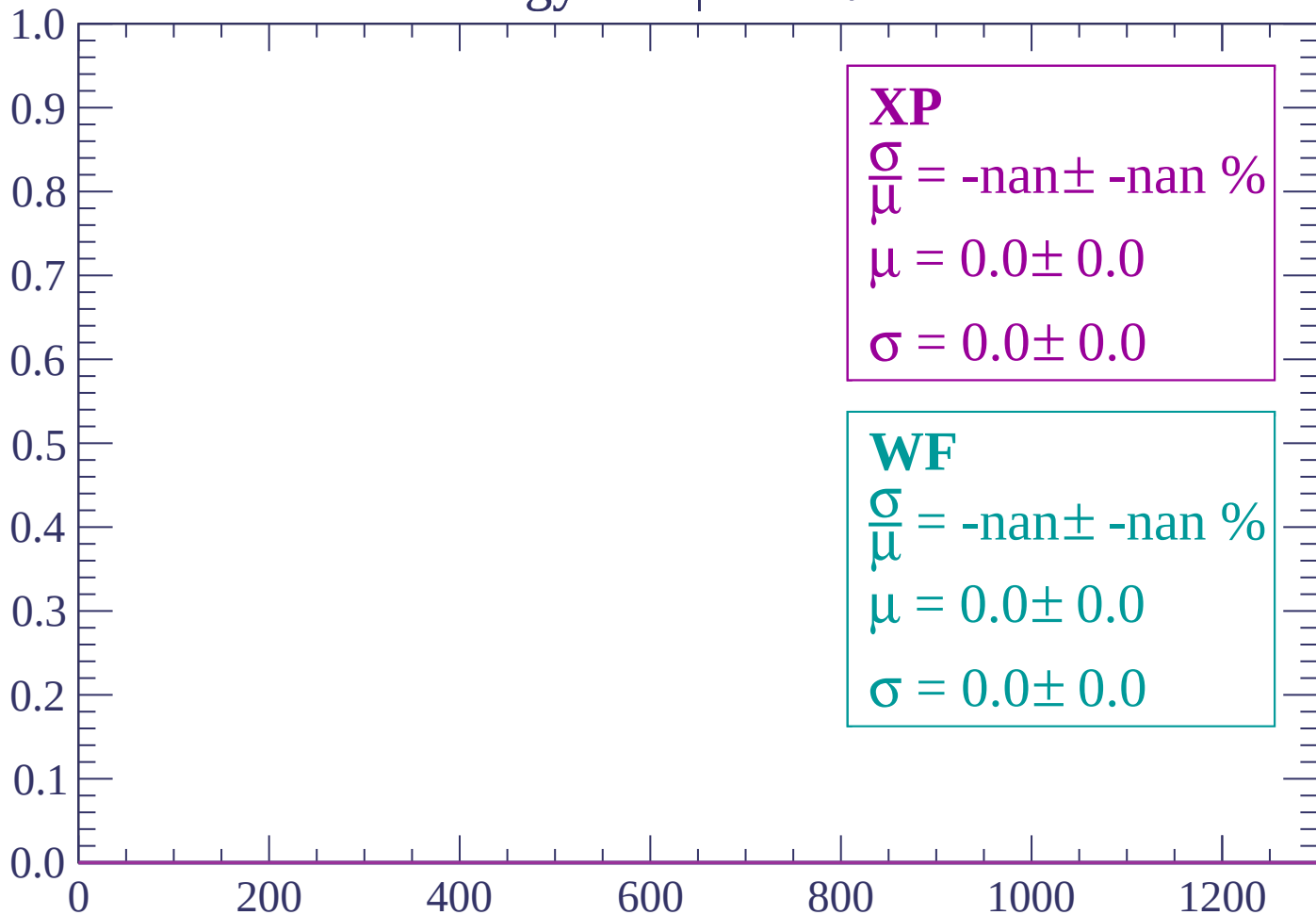
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

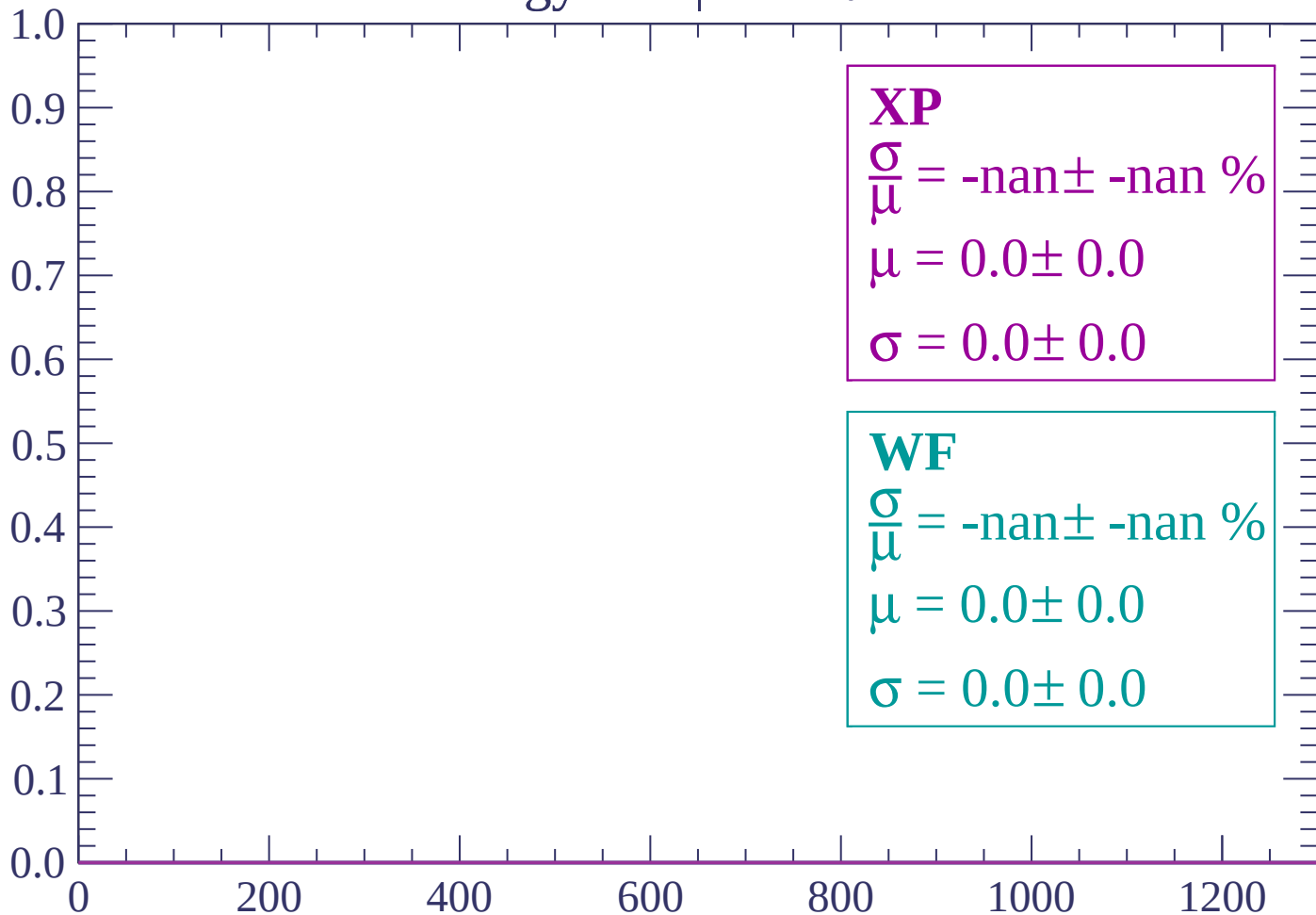
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

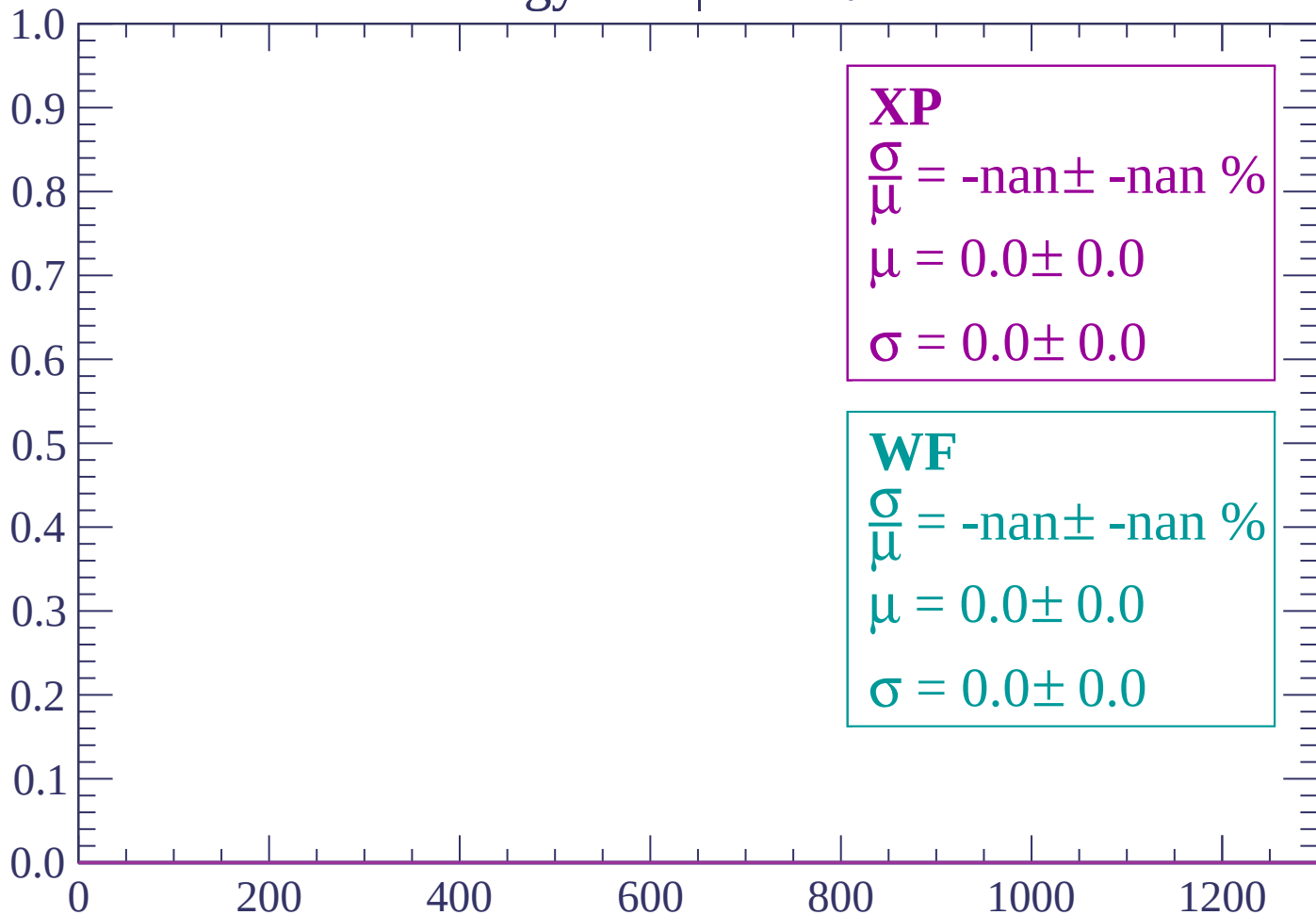
Count



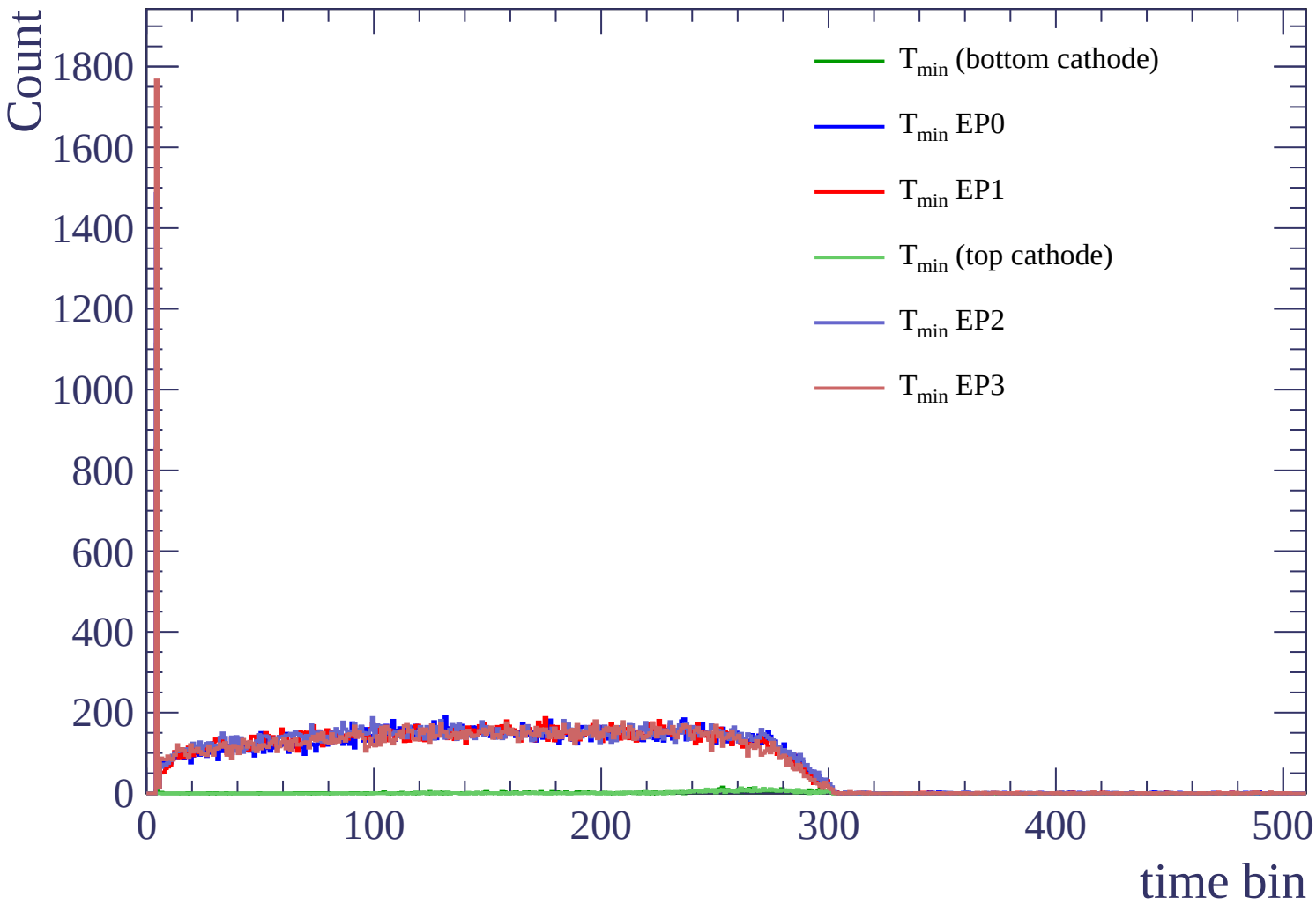
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

